The Science of Winning

Planning, Periodizing and Optimizing Swim Training

J. Olbrecht

FG partners

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Based on refined outcomes following a re-analysis of the characteristics of anaerobic capacity exercises, some changes have been made regarding their

CONTENTS

ACKNOWLEDGMENTS

Many great people, too many to name them all, have helped me along the road to this handbook. In thanking a few, I extend my thanks equally to everyone.

Special thanks to Prof. Dr. med. Dr. h. c. W. Hollmann, former head of the German Sports University of Cologne and Prof. Dr. A. Mader, my mentor, who inspired and helped me find the link between pure and applied science, between theory and practice.

I am grateful to all the coaches, throughout the world, who generously shared their information, experience, training concepts and ideas. A close collaboration and a good teamwork made this project possible. My gratitude extends to all the athletes I have had the great pleasure to work with throughout my career. I wish them all the best of success.

This English edition would not have been possible without the efforts and support of several key people, and I thank them; especially Ernest W. and Cheryl Maglischo, dear friends and colleagues, for their precious advice and mentoring. Without their help this handbook would never have sounded like English. Thanks to you, Jonathan Wiggins, ASA coach and dear friend, who kindly accepted to review my manuscript and to critically evaluate it.

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Finally many thanks go to my family who have showered me with love and support and especially to my wife, Pascale, for her inspiration, drive and above all her endless patience in trying to unravel my manuscript.

This work really is the product of all of them.

I hope you will take great pleasure in reading this handbook and I wish you a successful career as a coach.

FOREWORD

The training of competitive swimmers has not changed very much since the late 1970's. The last significant advance occurred in the Western world when information about blood testing and the anaerobic threshold came from the laboratory of the esteemed sports scientist, Alois Mader, from the Institute of Sport in Cologne, in what was then West Germany. Many of us embraced the anaerobic threshold concept of training and used blood testing to monitor the progress of our swimmers. In the process we encountered many difficulties in areas involving selection of testing protocols, interpreting the results of blood tests, and applying those results to the planning of training. One of the most confusing contradictions was that, in many cases, athletes who improved their anaerobic thresholds did not necessarily perform better while, at the same time, some athletes who appeared to lose aerobic endurance actually had lifetime best swims. We knew we lacked important information about this concept of training and the monitoring procedures we were using but we did not know what that information was.

Jan Olbrecht has written a most important book that fills many of the gaps in our knowledge. I am proud to endorse this book because it details the results of more than two decades of research on planning and monitoring the training of competitive swimmers that addresses many of the questions we had about the anaerobic threshold concept of training. This information will change the way coaches and athletes approach the planning of their training.

Let me talk a little about Jan Olbrecht the person and scientist before discussing his excellent text.

Jan Olbrecht was student of Dr. Mader's when I first met him in the early 1980's. Although he has served as a consultant to coaches and athletes from a variety of sports, Jan's first love is competitive swimming. Having been a national champion and record holding swimmer in his native Belgium, he is particularly interested in applying his research to the training of swimmers. When I met him, I was seeking information about training competitive swimmers in general and blood testing in particular and Jan was kind enough to share his knowledge with me. We have remained friends since that time.

Jan is a respected scientific professional whose research has been reported in many prestigious journals throughout the world. He is also much sought after as a speaker at scientific meetings. Despite his impressive scientific achievements, Jan remains a practical person who is interested, first and foremost, in communicating his research findings to coaches and athletes in order to help them perform better. For this reason, his research has always been specific to the training situation and his results have always been immediately useful to coaches like myself and others throughout the world.

Jan has spent his adult years developing procedures not only to investigate important aspects of training in the laboratory but also to developing monitoring procedures that can be used in the field to help athletes structure their training more effectively. Because of his athletic background, Jan has the ability to communicate the results of his studies to coaches and athletes in understandable and practical terms. For this reason, several coaches urged him to put his information into print, which led to the writing of this book.

Within these pages, Jan describes the complex relationship between aerobic and anaerobic training and makes it clear why interpreting the results of blood testing on aerobic endurance without also considering the effect that training had on anaerobic capacity will result in errors of interpretation that may cause coaches to make the wrong judgements about the training needs of their athletes.

He advances some new important concepts. Most notably that training at anaerobic threshold speeds is not the most effective way to improve aerobic endurance and that aerobic and anaerobic metabolism must be developed to optimum, not maximum, levels in order to perform well in competition. Also included are his ideas on the planning and periodization of training. These ideas have contributed considerably to the recent success of competitive swimmers and athletes from other sports throughout Europe but particularly in the Netherlands and Belgium. Jan presents this information with understandable writing that is replete with practical examples.

There is an excellent section on why training must be individualized even for athletes who compete in the same event. It also includes information on how to “spot” athletes with different physiological makeups and how to adapt training for them. In another important portion of this text he discusses the important aerobic and anaerobic adaptations to training and ways to achieve them including the role played by altitude training. Once again, there are many excellent and practical examples of training sets that will achieve the various results athletes seek. Finally, Jan provides several insightful suggestions concerning the training of age-group competitive swimmers.

I believe coaches and athletes at all stages of development can profit from the information in this book. Further, I believe the information presented will raise the standard of swimming performance throughout the world.

Dr. Ernest W. Maglischo
Successful coach and expert in physiology and biomechanics of competitive swimming

The ‘96 Olympic Games of Atlanta heralded a new era in the Dutch swimming. Kirsten Vlieghuis won 2 bronze medals (400 m and 800 m freestyle), Pieter van den Hoogenband (4th place) and Marcel Wouda (4th place) managed to join with the world’s top athletes and even our young relay teams unexpectedly reached the Olympic finals, something they had not succeeded in for many years.
There seemed thus to be a successful future in store for the Dutch swimming team.

We in the Netherlands, of course, knew about lactate but until then we had seldom monitored training by means of lactate measurements, or to put it another way, the lactate tests results and their interpretation missed any direct link or “translation” to the training practice.
What about the shift of the lactate curve? What is good? What is not?
What is the real message of the lactate readings for the coach?
What does a lactate value really tell us about the conditioning profile of the swimmer?
The sciences did not seem to be able to provide an adequate, effective and conclusive answer to all the questions I, as a coach, was confronted with in the swimming pool! I felt my own experience, my own perception and my own views on training matched better with the results in competition than the dry sometimes even implausible scientific measurements.

It was shortly after the Olympic Games that I got the opportunity to meet Jan Olbrecht, whom I, so far, only knew from his many (to me often unreadable) scientific contributions on lactate. Our first talk hit the mark; he provided the answers I had sought for so many years. Indeed, although I felt I was working in the right direction I could not always explain the swimmer’s adaptations and evolution. Now, Jan unraveled in a scientific but still simple and quite comprehensible way - which was very much in contrast with what I expected from a dry professor - these inexplicable “evolvements”. He corroborated my feelings as a coach and gave me the tools to understand what actually happened to my swimmers physiologically. Seen in this light, the lactate readings took on a completely new meaning. For the first time in my coaching career I had learned something new which was immediately workable and practicable in training.

I really must say that our collaboration for the past 3 years has been very productive and has played an important part in the success of our team. Indeed, in view of the lactate test results Jan determines the workload each swimmer can sustain and assesses for each swimmer individually the appropriate training load and the most adequate strategies and implementations in training. This new scientific training approach does not imply I have to throw away my own specific training methods but it enables me to individualize and structure the training planning and periodization in a systematic, purposive and scientific way. The discussion we have each time on the test results is always very useful and instructive and teaches me a lot about the metabolic impact of the different workouts.

I personally think this book should be considered as a manual for the modern coach who wants to know before planning and periodizing what kind of conditioning adaptations he may expect. It provides knowledge and shares experience but it is above all understandable and applicable to every training situation. It remains furthermore an enormous challenge to plan and apply the new training methods in the right way and at the right time. But do not think it will spare you work. Quite the contrary. It provides tools to systematically build up the appropriate training for each swimmer individually.

The title of the book is clear but is in no way exaggerated - the performances of the Dutch National Swimming Team in general and the Eindhoven team in particular speak for themselves. The part Jan, together with his wife and right-hand Pascale played in that success goes without saying for which I really want to express my gratitude.

Jacco Verhaeren
Coach PSV Eindhoven

I have known Jan Olbrecht for almost 20 years starting when we were both swimmers on the Belgian National Team. However, the story of our close collaboration as coach and scientist started in 1987.
I was at that time a young coach and the training methods I was familiar with and with which I apparently had success as a swimmer (endurance training until 5-6 weeks before the important competition, followed by 3 weeks specific anaerobic training and 2 weeks of tapering-off), were not always effective for me as a coach. I was thus on the look-out for a new approach, a more purposive way of coaching, “something” that would enable me to evaluate the planned as well as the completed training and to adapt or adjust it, where and when necessary.

My first introduction to lactate testing and Jan’s new training concepts indicated that this might be that new approach I had been seeking. Nevertheless, it took me a few months to master the test results and to put the subsequent training advice into practice. Every 5 to 6 weeks the swimmers were evaluated and so also was my planning. The lactate testings not only confronted me with the results of the completed training period but also indicated the changes or adjustments that needed to be made in the next mesocycle so the swimmer’s conditioning could evolve and their specific needs could be met.

Since 1988, I have had the great pleasure and the tremendous challenge of coaching some top level swimmers, finalists at the Olympic Games, World and European Championships (among others Stefaan Maene, Brigitte Becue, the Dutch National Swimming team).

Jan Olbrecht’s innovating scientific approach helps me structure the training individually and systematically. His way of combining science and practice surely helps me bring my swimmers to their top form at the right moment.

Stefaan Obreno
Head Coach National Dutch Swimming Team

INTRODUCTION

The swim coach is an artist molding intuition, feeling, creativity, scientific knowledge and experience into an efficient and well thought up training plan meant to help the swimmer bring top performances.

But, an athlete cannot win an Olympic gold medal nor set a World record just like that. The way to a top performance is long, complex and tricky and will undoubtedly raise thousands of questions such as, for example, how to design the endurance training of a sprinter? How to improve the anaerobic capacity? Which training exercises can or cannot be put together in one training unit? What is the most appropriate time for technique training? What is the super-compensation effect? What will determine the time needed to reach super-compensation? How to train the anaerobic power? How to ensure the most efficient way of training? Can heart rate or lactate tests be used to maximize training efficiency? How to plan and periodize age group training? How to avoid stagnation of the competitive performance? What is training planning? What is training periodization? How to prepare a long distance swimmer for 2 consecutive competition weekends?

This book is not only the result of more than 2 decades of research on planning and monitoring the training of competitive swimmers but it is above all the product of a very close collaboration with many national and international top coaches such as E. Maglischo, J. Verhaeren, St. Obreno, M. Lohberg, M. Pedroletti, M. Guizien, A. Paeck, H. Verbauwen, R. Gaastra, etc. They not only generously shared their information, experience, training concept and ideas but also confronted me with the sometimes huge gaps between theory and practice and challenged me to search for answers. This book does not claim to know it all or to provide all the answers. It is meant to stimulate or incite the coach to keep an open mind to any information he gets, to challenge him to create his own training philosophy, using his experience and vision in concert with the new scientific findings, to help him gain an understanding of the physiological effects the different types of training may induce and finally to assist him in maximizing training efficiency to achieve the results he is after.

There is of course not one single way of training, but there are a very few important principles that need to be respected; only a solid and well-conceived training plan with clear directions and based on systematic and superintended progressions can guarantee maximal benefit. *The key of success does not lie in training hard but in training purposively and carefully* . Failing to plan and follow a training and competition program will surely undermine your athlete's potential and inevitably result in frustration or, what is even worse, in overtraining or injuries.

How to use this book

This book has been written so the coach can use the information of each chapter independently as a foundation to create his own training system and philosophy.

It should be considered as a puzzle. The first two chapters of part I reveal the pieces:

* the most important definitions and principles of training
* the different types of training and exercises
* the training effects one can expect

Through chapter 3, which is a little bit more theoretical and complex, the coach will gain an understanding of the metabolic activity during swimming. This chapter could be skipped but many of the new training methods rest on metabolic research. Basic knowledge on metabolics in swimming will therefore surely help to take up the development of the new training methods to come. Chapter 4 provides tools and methods to define the swimmer's type and to determine individual training objectives and correct intensity zones. We chose to include the latter chapters in this part, as for an appropriate, well-balanced and thought-out planning and periodization (part II) it is of great benefit to first understand some important physiological processes induced by training. A better comprehension and a clear insight into the metabolic impact of training will enable the coach to give full vent to his fantasy and to *create his own winning training plan* .

Part II will guide and help the coach put all the puzzle pieces in place. After a short theoretical introduction (chapter 5, Definitions and Descriptions), you will find the basics of planning and periodizing (chapter 6 and 7): a detailed description of the successive stages the athlete has to go through to ensure a continuous performance improvement over several years. Finally in chapter 8 and 9 (templates) you will learn to frame and draw up step by step and methodically the training plan that will help you achieve your objectives.

In the end, when all the pieces of the puzzle are put in the right place, you will have a specific training plan which is just made out for your particular athlete and which will surely help him win the gold medal.

Part I TRAINING

Chapter 1 DEFINITIONS AND PRINCIPLES OF TRAINING

Training for athletics is scientifically defined as “the systematic and purposeful activation of the muscles with the objective to improve performance through morphological (structural) and functional adaptations” (Hollmann 1990).

Thus, it is considered training if there is an activation of the muscles. Although nutrition and eating habits may affect the success of training, the muscle development, and, consequently, the performance in competition, they are not considered training. Mental training, however, induces muscle activity. It fits therefore with the definition above and will be seen as a form of training.

Performance in competition depends on several determinant components that fall into 3 main classes (tab. 1):

- Technique
- Physical conditioning
- Psychology and mental strength

A coach who wishes to train an athlete for a high competitive performance must ensure that all these components are emphasized within the training plan.
In this book we will focus on the physical conditioning of swimmers, without losing sight of the fact that “technique” and “psychological capabilities” also strongly affect performance in competition. Indeed, despite possessing a high level of physical conditioning, a swimmer that is not able to control or cope with the competitive pressure will never perform at his best, nor will he exhaust his conditioning potential.

Factors affecting competitive performance

Technique	Physical Conditioning	Psychological Conditioning
Stroke technique, coordination Starts and turns …	Aerobic conditioning (endurance) Anaerobic conditioning Flexibility and strength …	Stress control Motivation …
Speed		

Tab. 1 Competitive performance depends primarily on the status of these components.

SUPER-COMPENSATION

Description

A healthy human body adjusts itself to stimuli provided by the environment. For example, if one travels to La Paz, Bolivia, which is located at an elevation of 12,000 feet above sea level, he certainly will, at first, have trouble breathing and experience difficulty even with light activity. After a few days however, his body will adjust to this extreme altitude and, after an adaptation period, he will even be able to resume nearly normal activity. Similarly if one travels to a "warmer" country, he will inevitably experience some discomfort for the first few days. After a few days, however, the heat will become tolerable and the level of discomfort will lessen because a healthy human body always adjusts itself to ongoing stress. Likewise, a body that is regularly exposed to a certain level of exercise will, after an adaptation period, be able to handle greater workloads and, consequently, be capable of performing better in competition.

The adaptation phenomenon is ruled by the **"principle of super-compensation"** which follows 4 distinctive phases (fig. 1):

- **Phase 1** : during this phase the swimmer completes a large volume of training; he becomes *tired* and his physical performance drops (curve decline)

- **Phase 2** : the so-called *recovery phase* induced by very low intensity training (regeneration training or active rest) and rest between training sessions. The physical performance will now return to the starting level (curve rise equals compensation for the hard work). This phase allows for several biological adaptations to take place:

  ✽ normalization of the cell environment: waste products are removed, the pH-values normalize and the cell structures recover

  ✽ recovery of neuromuscular stimulation processes: a tired muscle does not react optimally to stimuli from the nerves. This will be restored when the muscle recovers

  ✽ concentration and activity of enzymes and hormones will be restored

  ✽ energy sources are replenished. Glycogen and other fuel sources are restored

- **Phase 3** : the *super-compensation phase* : *physical performance increases above the initial level* . The swimmer can now handle the same load as before but with less strain or a more intense load with the same ease

- **Phase 4** : if training is not carried on, *the improvement in physical capacity will be progressively lost*

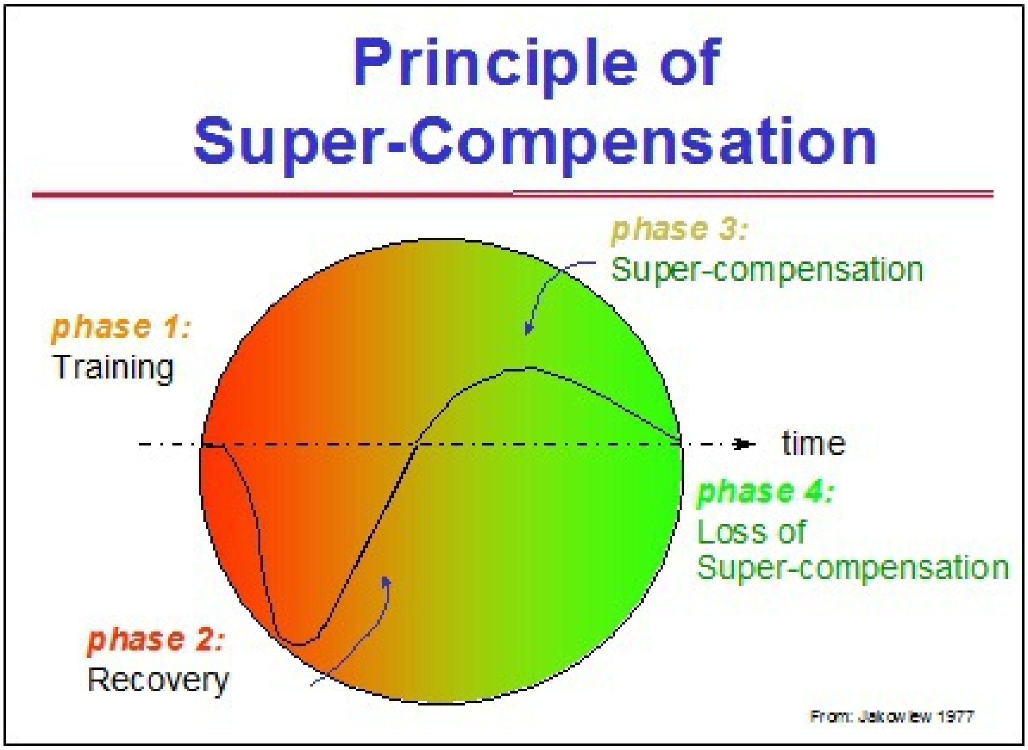


Fig. 1 Schematic representation of "the principle of super-compensation" (Jakowlew).

Requirements for a Successful Super-Compensation Process:

1. *A healthy body* : inflammation, overtraining, mental stress, etc. strongly reduce the possibility for super-compensation
2. *Adequate training intensity and volume* : This is probably the most delicate, even crucial aspect of successful training. Indeed, the training must be just long enough (volume) and just hard enough (intensity) to stimulate the body in such a way as to induce morphological (structural) and functional adaptations. When training is too hard and/or too long, it will break down the body too much and will actually impede the process of super-compensation. So, the real art is always adjusting intensity and volume to meet the purpose of the training as well as the conditioning and mental state of the athlete. We will come back to this later
3. *Enough rest (passive or active rest): rest or regenerative workouts will make up most of an athlete's training time. Insufficient rest or*

insufficient low intensity training (regeneration training) between important training sessions prevents the body from achieving super-compensation

This brings us to one of the most important and overriding principles in training:

"Rest or regeneration is the most essential part of training for inducing optimal biological adaptations"

Timing

With a healthy athlete the time necessary for super-compensation (fig. 2) depends on several factors:

* the type and duration of the workout
* the conditioning level of the athlete
* the recovery level of the athlete

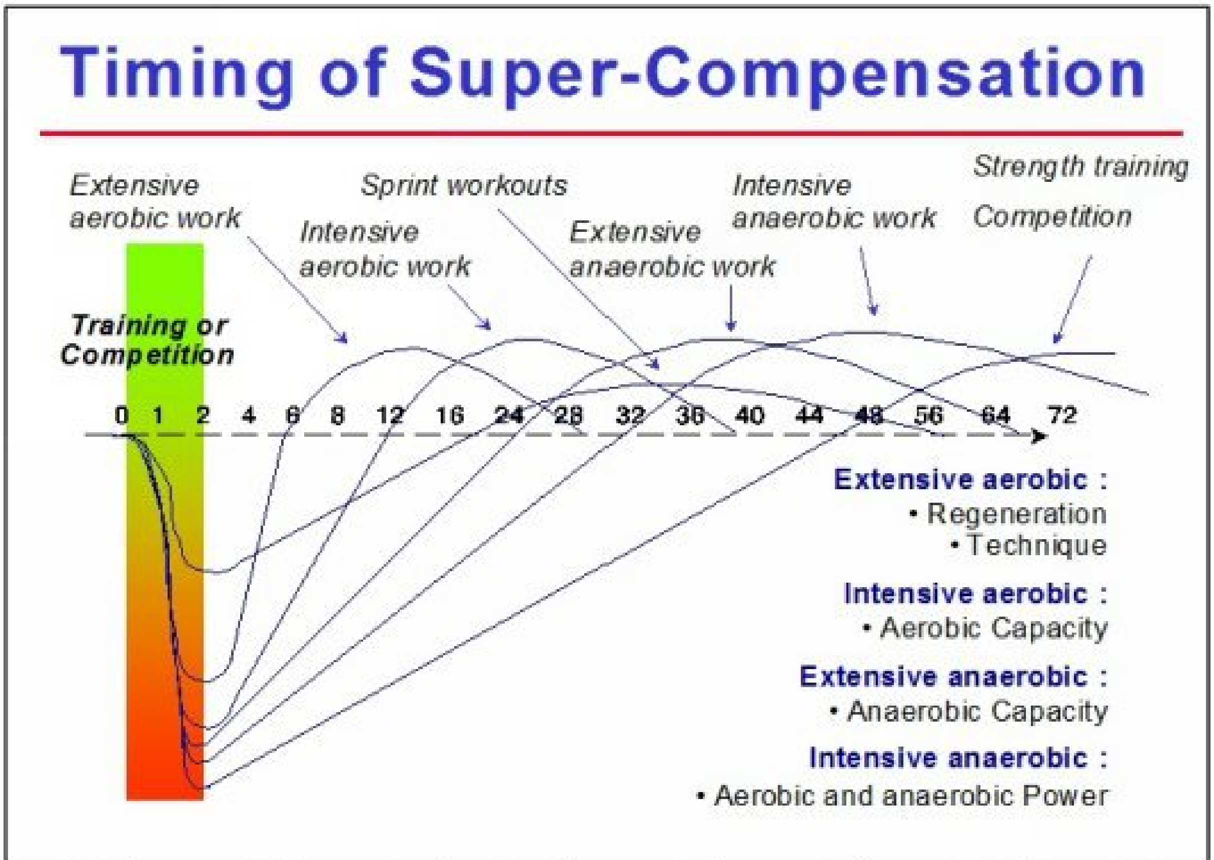
If the swimmer has a high conditioning level and/or is well rested, the time for super-compensation will be shorter. Mental stress and fatigue on the other hand, will slow down the process necessary to achieve super-compensation.

Furthermore, there should be a distinction between long and short sets in training. For example, an intensive anaerobic set of 600 m will require a longer time for super-compensation than a shorter set of the same type (e.g. set of 150 m).

* Long anaerobic set of 600 m - e.g. 3 x 200 m broken (4 x 50 m max. with 10 sec rest) and 10 minutes rest between each set
* Short anaerobic set of 150 m - e.g. 2 x 75 m max. (50 m + 25 m with 10 sec rest after the 50 m) and 10 minutes rest after the first set

The 150 m set will require only 2 days for super-compensation while the 600 m set will require a longer period (3 to 4 days).

These differences in timing for super-compensation are due to the duration of the various biological regeneration processes that take place during the recovery phase. While the replenishment of creatine phosphate will take only a few seconds to a couple of minutes to return to normal levels, the reloading of the muscle with glycogen may last up to 24 hours or, in some cases, even longer. The production of new enzymes or proteins may also take hours, sometimes even days to complete.



Training types	Extensive Endurance	Intensive Endurance	Sprints/ Short Sets	Extensive Anaerobic Training	Extensive Strength Training	Intensive Anaerobic Training	Intensive/ Strength Training/ Competition
From	8	24	30	36	40	40	48
To	12	30	40	48	60	60	72

Fig. 2 Time, expressed in hours, for different types of training to reach the maximal super-compensation (Olbrecht 1998 not published, used in oral presentations).

Long but extensive (low intensity) training sets cause little or no structural breakdown of the organism. In this case, only the expended energy supplies will have to be replenished, so the time necessary to recover from this type of effort will be quite short.

Intensive training units, on the other hand, swallow energy and are very destructive for the organism. They will, therefore, require a considerably longer regeneration period (recovery phase) because the energy supplies will have to be refilled and a number of proteins and enzymes will need to be reconstructed.

The principle of super-compensation rules every step of the training process. The planning, the periodization, even the type of training set and its volume and intensity must be continuously adjusted according to the individual super-compensation pattern

BIOLOGICAL ADAPTATIONS

The *adaptations of the human organism* follow a regular pattern. During the first 1 to 2 weeks of a new training cycle the body adapts quickly to the new stimulus. In the next few weeks the power of this same stimulus to provoke an adaptation will progressively fade to end up completely after 6 weeks (fig. 3). We therefore call week 1 and 2 of the adaptation process the *“fast adaptation phase”* and weeks 3 to 6 the *“stabilization phase”* .

After the body stops responding to the current training load (intensity and/or volume), it is necessary to increase it to induce further adaptation. This means that, after about 6 weeks, training should be modified by either:

1. increasing the training volume
2. increasing the training intensity
3. increasing the frequency per week of training units or intensive workouts
4. changing the proportion of workouts in each stroke (e.g. changing from freestyle to breaststroke while keeping volume and intensity the same)
5. making the training program more difficult by changing the training environment such as training at altitude or by reducing the regenerative training between intensive workouts (e.g. maintaining the same number of intensive workouts per week but with less extensive training units between them)
6. a combination of the changes in points 1-5

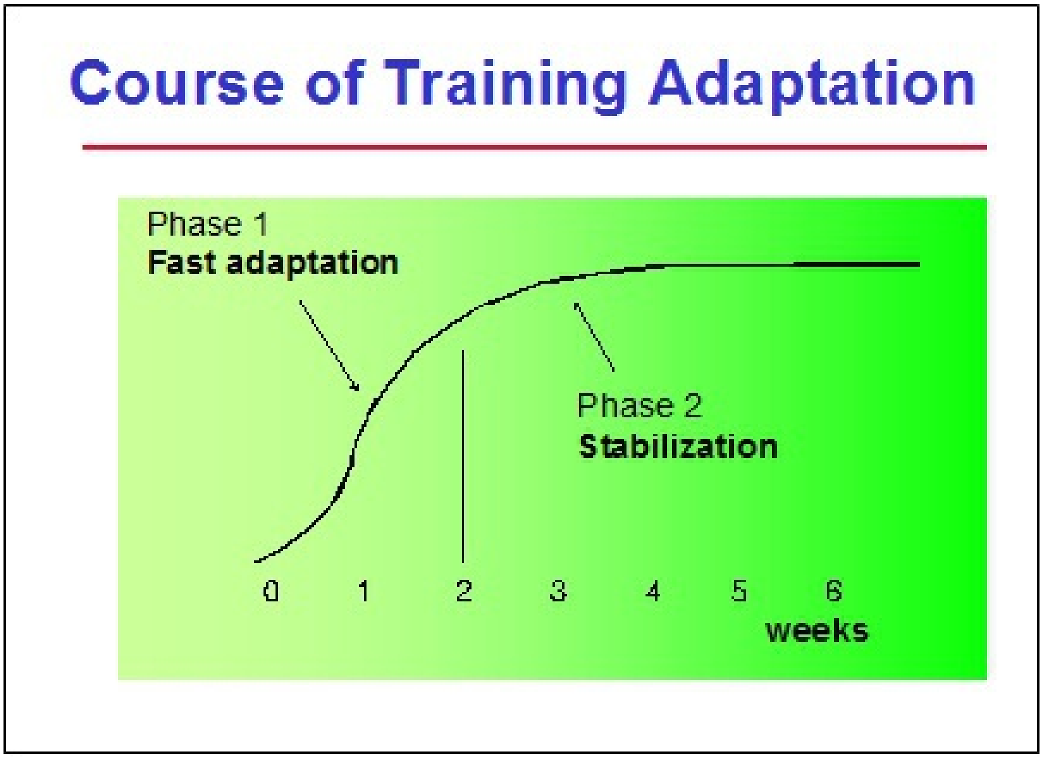


Fig. 3 Adaptation process in the organism as a result of a training stimulus.

Why wait 6 weeks before introducing a new stimulus or a new training load? Why not change the training plan after 2 weeks when the stimulus has induced its fast adaptation? The reason is that weeks 3 to 6 are necessary to stabilize the adaptations brought about in weeks 1 and 2.

Let's take an example of an adaptation in endurance:
For the improvement of the endurance capacity to take place, thousands of small cellular parts need to be rebuilt and/or newly produced. Some of them will be rebuilt quickly while others will require more time. If the training load is increased too soon, only the cellular structures that can adapt quickly will be able to follow the imposed training rhythm. All the others will fall behind; they will not be rebuilt or, at worst, be irrevocably lost. As a consequence, there will be no homogeneous development of the endurance capacity and, in due course, this may result in the swimmer breaking down and becoming overtrained.

Moreover, it is important to reduce both volume and intensity near the end of the stabilization phase in order to start the next training cycle in a fresh and relaxed condition.

In addition to the fact that the basic physiological adaptation pattern requires six weeks, there are two other equally important principles of training adaptation that must be considered. The first is called the **detraining process** while the second is referred to as the **declining return on training investment** .

DETRAINING PROCESS

Whenever the athlete stops training, a "detraining" process sets in, inducing a relatively rapid loss of the acquired adaptation. For example, half the increase in mitochondria from 5 weeks of training may be lost in 1 week of detraining (fig. 4). Since the number of mitochondria substantially determines the endurance capacity of the swimmer, it is obvious that the loss of 1 week of training will have a marked effect on the endurance.

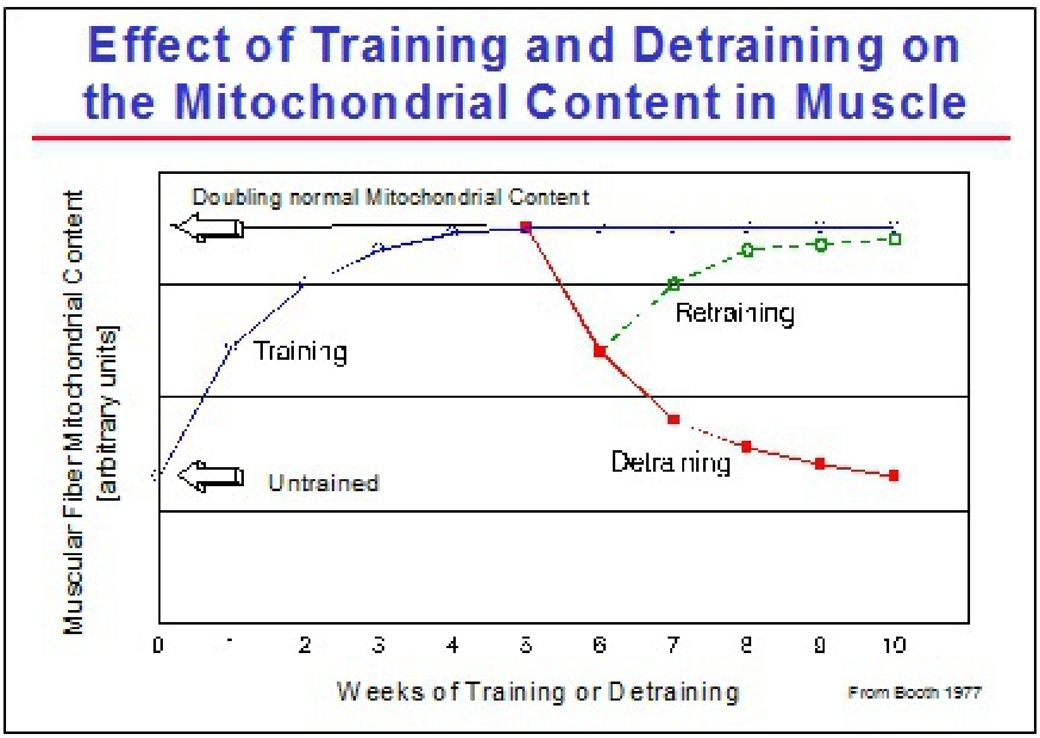


Fig. 4 The evolution of the number of mitochondria in the muscle cell as a function of the number of weeks of training, detraining and retraining.

It will require nearly 4 weeks of "retraining" for the swimmer to regain his condition after 1 week of detraining. Therefore, it is clear that only regularity and continuity in the training process will ensure progression of conditioning over several years.

Practical experience has taught us that the rate of detraining of the endurance capacity depends on the time that has been spent in building up this capacity. In other words, the more time the swimmer spends building up his endurance capacity, the longer this capacity will remain strong, even when he drastically reduces training. The following two examples illustrate this finding:

1. A top level swimmer, who was preparing for the World Championships, had to interrupt the last 7 months of her preparation quite frequently because of school commitments and injuries. Despite these breaks in training the swimmer improved her endurance capacity from 58 to 64 ml/kg per min within the last 8 weeks prior to the competition. However, a test 2 weeks after the World Championships revealed that her aerobic capacity had dwindled to 59 ml/kg per min: a loss of 5 ml/kg after only 2 weeks of rest. In contrast, her training mates, who got through the entire preparation period without interruption, managed to keep almost the same level of aerobic capacity they had before the competition despite 2 weeks of taper followed by 2 weeks of nearly no training (the transition period)
2. If a swimmer misses training because of an injury, providing it is not a serious injury or an infectious disease, the time he will need to regain his previous level of endurance capacity will be shorter if the build-up period has been long. Indeed, if a swimmer had a continuous build-up period of at least 4 months prior to his injury, then it would not be unusual for him to regain his previous aerobic capacity level after only 14 days of retraining. On the other hand, a swimmer who built up his aerobic capacity in only 5 weeks will undoubtedly lose more of his aerobic capacity after 2 weeks of forced inactivity and have to restart the build-up process again

These two examples tell us that:

- as long as the training period preceding a period of inactivity due to injury or illness - providing it is not serious - was **continuous** and **long enough** , there is no reason to panic. The swimmer will be able to regain his previous level of conditioning within a short time. Alternative training exercises such as strength training, gymnastics or even other sports like cycling (e.g. if shoulder is injured) will help to reduce the time needed to regain the previous level. However, recovering health and fitness remains, of course, the priority

- it is “vital” to grant the swimmer a week of rest after the last main competition before starting a new macrocycle
- it is physiologically possible to keep a swimmer in good form for several successive competitions for up to 4 weeks, provided the build-up period was long enough and the swimmer is highly motivated. The psychological and mental support will in this case prevail over conditioning

DECLINING RETURN ON TRAINING INVESTMENT

Studies based on scientific research have shown that the rate of improvement in conditioning is not linear to the increase of training load. As a matter of fact, the effect of an increase in training load (increase of intensity and/or volume) will be different depending upon the conditioning level of the swimmer. Indeed, the better the conditioning level, the greater the training load that will be needed to make even slight improvements. Consequently, the risks of injury and overtraining will be greater. Or, to put it differently, the better the swimmer, the less improvement a same increase in training load will produce. This phenomenon is called the **principle of declining return on training investment** (fig. 5).

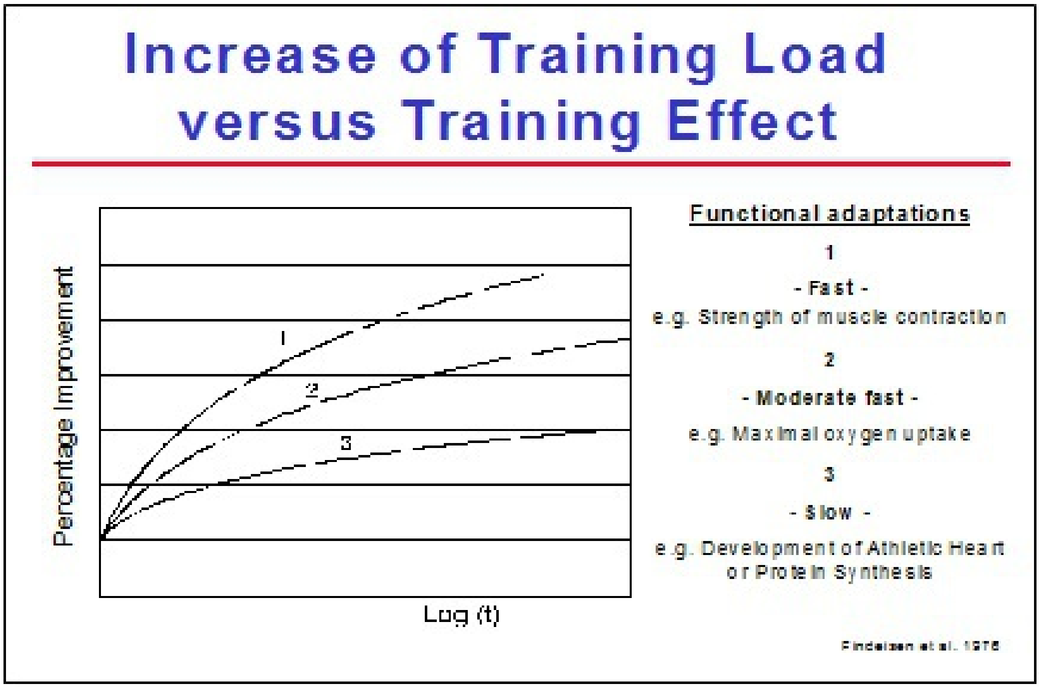


Fig. 5 Percentage of improvement related to the increase of training load.

The **principle of declining return on training investment** is applicable to all kinds of training adaptations:

- quick adaptation such as strength training
- fairly slow adaptation such as the improvement in VO_2 max
- slow adaptation such as the development of the cardiovascular system, joints and tendons, etc.

With this principle in mind it is clear that we should always look for the smallest increment of training load that will produce the greatest possible training effect. For example, as long as swimmers can improve their conditioning with one single training session a day, they need not train twice a day, except during training camps, of course.

Each change in the training load has to be introduced gradually and progressively over time. Increasing the training load of young, talented swimmers radically and extremely will result in successes in the short run, but stagnation will soon set in. When this happens, most athletes become discouraged and stop training.

The principle of declining return on training investment is a common rule in the training process, but its rate is very individual. Indeed, a given training load will not induce the same conditioning improvement for every swimmer. In other words, each swimmer adapts differently and at his own rhythm to a certain training load. So, patience is needed to reach the adaptation longed for. There is no point in trying to force it, particularly with less talented swimmers. High performance swimmers, on the other hand, nearly always have a high rate of improvement regardless of how strenuous the workload may be. The rate of improvement that results from a certain workload could therefore be a good standard to detect or identify new talent.

THE WAVE PRINCIPLE IN THE TRAINING PROCESS

To ensure a continuous improvement of conditioning, it is not only necessary to change the training load after approximately 6 weeks (see Biological Adaptations) but also to provide regular periods of regeneration training for the super-compensation to take place. The structure of an effective and successful training process will therefore be characterized by a continuous alternation of periods of increased training load (training units at a higher intensity and/or volume) and periods of reduced training (regeneration training).

This alternation of intensive or high volume training and regeneration sessions, often referred to as **the wave principle of training** , can easily be achieved by:

1. working with mesocycles (see The Mesocycle) of 2 to 7 weeks divided into two parts. The first part (the working phase) should encompass about 3/5 of the mesocycle and consist of high volume sets and some intensive workouts. The rest of the mesocycle (the recovery phase) should be a regeneration period consisting of only a few quality sets (workouts at high intensity) and a lot of regenerative training

The following figure (fig. 6) presents a 5-week mesocycle. During the first part of the cycle (week 1 to 3, the working phase), the training load is progressively increased by raising volume, intensity and frequency of the intensive workouts. For example, from week 1 to week 2 the total volume increases and the number of quality sets is doubled. In the third week, the frequency of the quality sets remains the same as in week 2 but the intensity and total volume of these sets increase during the second part of the mesocycle (week 4 and 5, the recovery phase) the training load should be reduced by decreasing volume as well as intensity (such as only 1 or 2 intensive workouts, which will be shorter than during the working phase)

2. alternating intensive and high volume sessions with regeneration training within each week of the mesocycle. It is absolutely essential to intersperse the intensive sets with a lot of very extensive work in order to enhance the effect of the progressively increased workload

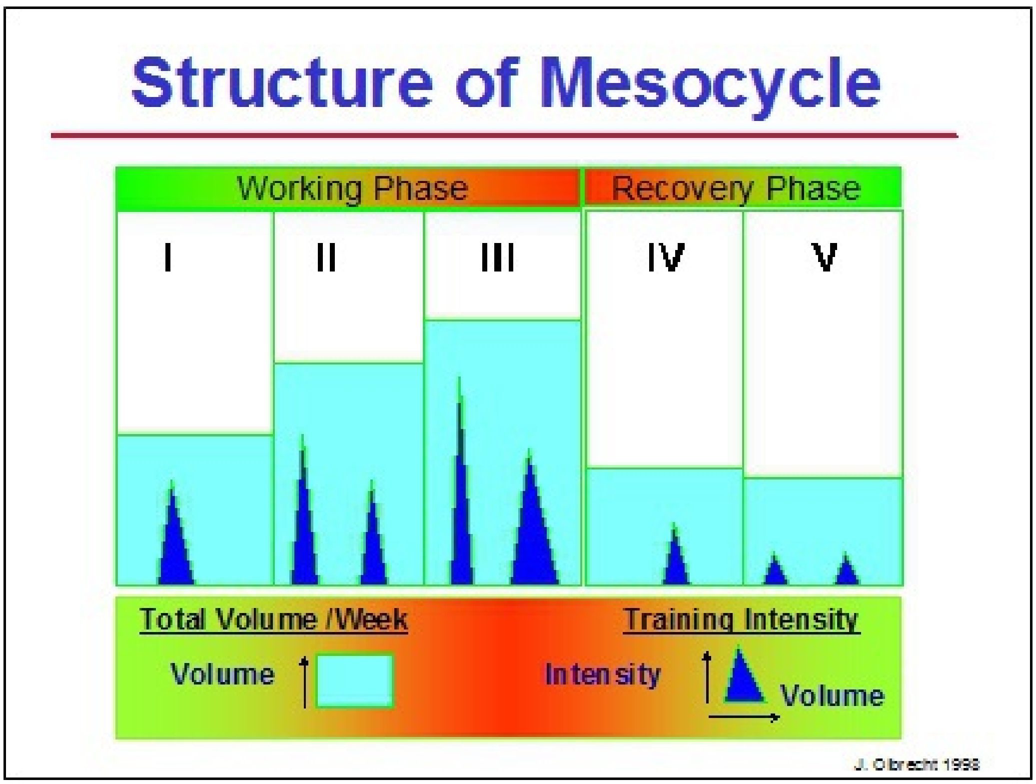


Fig. 6 The wave principle during a 5-week mesocycle (Olbrecht 1998 not published, used in oral presentations).

Variations on this mesocycle are always possible, sometimes even necessary, but the wave pattern must remain present, both in volume and in intensity. In chapter 6, Basics on Training Planning (see Mesocycle), we will focus on and discuss the pros and cons of the different types of mesocycles.

We can conclude that:

Considering the principle of super-compensation and the fact that the same type of training only effectively produces adaptations during approximately 6 weeks, it is extremely important, to use a wave pattern alternating intensity and/or volume with regeneration during the entire training process

Note: It is often claimed that “variation is necessary in training” to cheer up and motivate the swimmers. It must be clearly stated that variations in training are meant to meet training objectives and not to reduce the boredom of the swimmers. The coach will already have plenty of opportunities to vary training to motivate his swimmers as the training objectives change during the different steps of the training process.

Chapter 2 COMPOSITION OF TRAINING

TRAINING TERMS AND CONCEPT

Before describing the different types of workouts and the effects they may induce, it is necessary to define some training terms.

A training unit or training session (see The Training Unit) is one complete training period during the day that can be carried out in or out of the water. Elite swimmers, for example, often complete 2 training units in the water and 1 training unit on land each day. Beginners may do only 2 or 3 training units a week and all these are in the water.

A group of training units forms a specific training period or **training cycle** . These cycles are characterized by the time they last and will be repeated frequently during the swimming season.

The smallest cycle is called a **microcycle** (see The Microcycle). In swimming a microcycle generally covers 1 training week. A **mesocycle** (see The Mesocycle) is a succession of a few microcycles, lasts about 2 to 7 weeks and contains a phase of hard work followed by a phase of relatively easy training. The duration of this period will depend upon the training objectives during that training period and on the conditioning level of the swimmer. A **macrocycle** (see The Macrocycle) consists of several mesocycles. It stretches over several months and covers a specific season such as the short course season. It covers the base, competition and transition training periods.

There are usually 2 macrocycles per year, but the competition calendar sometimes forces the coach to plan for 3 to 4 macrocycles. This is only possible for the very top level swimmers, however. For age group swimmers, it is advisable to stick to 2 macrocycles per year since more macrocycles would hurt the long-term build-up of their conditioning. All this may sound complicated, but it will become clearer when planning and periodization are discussed in the second part of the book.

THE BASIC ELEMENTS OF A TRAINING EXERCISE - THE KEY TO SUCCESSFUL WORKOUTS

Within a single training session a coach designs various sets or repeats and drills. These repeats are not chosen at random, but according to the physiological adaptations he wants them to induce. Every workout, therefore, has an "*objective*" and has to be designed, planned and implemented in a way to produce the right adaptation at the right moment.

Let's give an example: a coach asks his swimmer to swim 5 x 400 m with 30 sec rest. This set or repeats can be swum at different speeds, but it is the adaptation wished for that will determine the speed of the set. If it is intended to improve the aerobic capacity (increase of VO_2 max), the coach may choose to split the first 400 m into 4 x 100 m - each 100 m with a negative split (each 25 m is faster with the last 25 m nearly at max.) and with about 90 sec rest between each 100 m -, followed then by 4 x 400 m at a very extensive (slow) pace with 30 sec rest. If, on the other hand, the second part of the set (4 x 400 m with 30 sec rest) is swum at a speed near to the lactate threshold (maximal lactate steady state) then, instead of improving his aerobic capacity, the swimmer will improve his aerobic power (definition of aerobic capacity and power, see Capacity and Power Training).

A set of repeats should not only be carried out correctly, but be prepared carefully as well. If during a training session the coach plans a set of sprints, it is obvious that he would not let his swimmer jump directly into the water and sprint. The swimmer will have to prepare for this exercise with a warm-up, in which he will accelerate to sprint speed a few times after easy swimming. In this way he will prepare his muscles for the tension that is to follow. If he fails to do so, the sprints will not only lose their effectiveness, but the swimmer will run a greater risk of injury or overload his muscles.

By manipulating the following 4 basic elements of a training exercise, the coach *selects the biological adaptations he wants the exercise to induce.* These features are:

1. The *set distance* (or *volume*) of the exercise (how many meters)
2. The *intensity* (how fast one is going to swim)

3. The *rest* (how long one waits before resuming the exercise)
4. The *repeat distance* or *interval* (length of each swim); an exercise is often divided into partial distances

Example**

6 x 200 m Back Rest = 30 sec Swimming time = 2:35

Volume = 1200 m
Intensity = 2:35
Interval or *repeat distance* = 200 m
Rest = 30 sec

What about Repetitions?

If we know the volume of the set and the distance of the repeats (intervals), we can easily deduce the number of repetitions. Moreover, experience teaches us that adaptations are primarily determined not by the number of repetitions of a workout but by its volume, intensity, rest and repeat distance. The effectiveness of the different training sets will therefore change by varying one or more of these 4 components.

The *type of adaptation* a training set will induce is determined by the *proper combination of the 4 basic elements of an exercise:*

volume
intensity
rest
repeat distance

CAPACITY AND POWER TRAINING

The various training exercises have 2 primary objectives:

1. the *"building-up"* of the various components a swimmer needs to improve his conditioning

2. the *"fine-tuning"* of these components to maximize their contribution in competition and to provide an optimal synergism between them. To

illustrate this, imagine 2 top level rowers in the same boat. If one of them rows with a slow frequency and the other with a rather high frequency or if the first one is a fast starter and the other a fast finisher, you can easily see that putting them together in the same boat will cause synchronization problems. Each of the rowers may be a champion individually, but together they will never perform well. A similar situation happens in team sports. You need a team to win and not a group of individual stars

The workouts that are principally meant to build up the swimmer's conditioning are labeled as ***capacity training exercises*** . The increase of maximal oxygen uptake (VO_2 max), the increased breakdown of carbohydrates (glycolysis) as well as the increase of maximal force fall under this category. The exercises used for the fine-tuning, on the other hand, are referred to as ***power training workouts*** .

The properties of the aerobic and anaerobic metabolism are the determining factors of an athlete's competitive performances. With respect to the building-up or fine-tuning of both conditioning components we will split aerobic (endurance) and anaerobic training, respectively, into capacity and power training and speak of:

- *aerobic and anaerobic capacity training* when we want to build up the maximal oxygen uptake and maximal lactate production rate respectively
- *aerobic and anaerobic power training* when we want to fine tune both capacities in order to maximize their utilization during competition. This fine-tuning will differ depending upon the distance of the event for which the swimmer is preparing

It is obvious that the *"building-up"* of capacities (capacity training) always will precede and last longer than the *"fine-tuning"* (power training).

Example: After several weeks focusing on building up the aerobic capacity of a long distance swimmer (increasing VO_2 max), it is not unusual that the swimmer's anaerobic capacity will increase too (increase of glycolysis). But, if this anaerobic capacity becomes too strong the percentage of the aerobic capacity at which the athlete competes will decrease which will prevent him from using his endurance capacity maximally during a 1500 m competition. The ideal situation would then be to have the 1500 m competition preceded by a period of 4-6 weeks during which the aerobic capacity is “kept in shape” and the anaerobic capacity reduced to an optimal level. This fine-tuning can be done either by considerably increasing the training volume or by inserting some endurance power training during the competition period (see Classification of Training Exercises).

Why not select training sets that make fine-tuning unnecessary?

There is no training set (even a workout at the maximum lactate steady state) that induces all of the biological adaptations needed to improve performance in competition. This becomes clear when we consider:

- that each of the biological adaptations not only requires a different type of training but also a different time to get induced
- that not all training sets are compatible. Some training sets work against the adaptations produced by other sets

If, for example, the objective is to improve the aerobic capacity, several things have to be taken into account:

a. An improvement in endurance capacity requires a lot of morphological (structural) adaptations such as building new enzymes, new capillaries, new cell particles, etc. The coach will therefore have to earmark much more time to improve the endurance capacity of his swimmer than to build up his anaerobic capacity (see Methodology for Training Planning)

b. As the improvements in aerobic capacity require various changes in the biological systems, it is clear that the timing of these changes will vary. Some of them will take days, others weeks; some of them will need low intensity workouts, others high intensity sets. It is thus very important, when trying to improve aerobic capacity, to schedule each training stimulus carefully

c. While building up the aerobic capacity during the base training period, it is likely that other conditioning components such as the anaerobic power of sprinters or the aerobic power of middle distance swimmers will deteriorate in quality. Therefore the coach has to consider whether or not to insert some cycles in this long base training period to adjust the loss in quality of the other conditioning properties

This example demonstrates why the conditioning cannot be improved by one single type of training and why it is a must for the coach to draw up, well in advance, a work plan for the available time and the different conditioning components to be trained. You cannot just make a mishmash of all the different types of exercises and expect all elements of the conditioning process to be covered.

So, the coach will have to draw up a *timetable or plan* before the start of the season. In this plan, he fixes the important competitions and broadly outlines "when" certain forms of training are important. He will foresee a period of *"building-up"* and once the basic conditioning properties have been developed, he will start the *"fine-tuning"* program to achieve the optimal contribution of all conditioning components and so the best possible performance in competition.

Through various training sets we aim at the *building-up* and the *fine-tuning* of the conditioning properties. In training science terms we speak of:

- ***capacity training* if we *build up* the conditioning components**
- ***power training* if we *fine tune* the conditioning components**

CLASSIFICATION OF TRAINING EXERCISES

Athletes, coaches, training advisers, even sports scientists often struggle with a variety of terms to identify the same type of exercise or even use a single term to refer to different types of exercises.

Interval training, for example, means different things to different people. Some coaches associate "interval training" with high intensity efforts while others extend the term to sets or repeats that are swum at threshold intensity or even below. The terms "extensive" and "intensive" may also refer to different characteristics of the training efforts depending upon whether they are used in Europe or in the US and Canada. European coaches use both terms to refer to the intensity of the training sets (respectively low intensity and high intensity) while the Americans and Canadians rather assign these terms to the volume of the exercise.

And, as if there was not already enough confusion, coaches often attribute different training objectives to the same type of exercise. “Threshold training” also called “VO_2 max-training”, for example, is, in the US, mostly used to increase VO_2 max, while in Europe this type of exercise is meant to improve the aerobic power, i.e. to maximize the use of the VO_2 max in competition. On the other hand, to improve VO_2 max, some coaches choose low intensity intervals and high mileage while others, for the same purpose, prefer high intensity repeats.

In an attempt to remove all sources of misunderstanding, we introduced a common coach’s language and a uniform classification of training exercises.

This classification, presented below, contains many of the terms coaches feel comfortable with. Each class groups the workouts with the same “class effect”, i.e. inducing the same major biological and functional adaptation. Designing a workout to induce just one specific biological adaptation is impossible. Most of the time there is a major effect (= class effect) coupled with minor effects whether desired or not.

The 4 classes with their specific class effect are:

	Class of workout	Class effect
1	Aerobic Capacity	Increase of maximal oxygen uptake[1] (VO_2max)
2	Anaerobic Capacity	Increase of the maximal glycolytic rate, also called the maximal lactate production capacity (PLamax = VLamax)
3	Aerobic Power	Maximize the use of the maximal oxygen uptake (%VO_2max) during competition efforts
4	Anaerobic Power	Maximize the use of the maximal glycolytic rate (%VLamax) during competition efforts

We selected these 4 workout classes because the major training effects (class effects) induced by each stand for:

* the main training objectives defined during the periodization process
* the key biological adaptations targeted in the conditioning program

These aspects will be of great help for:

1. the planning and periodization process. Indeed, it will allow the coach to create a clear overall plan well in advance (fig. 65, 66, 67, 68, 70 and 71) and to translate the objectives into correct workouts in a later stage of the training process. Once the swim season is split into mesocycles, the coach identifies the objectives he wants to achieve in each of them. According to these objectives, he will then determine the frequency per week of each of the 4 different classes of training workouts. Depending on the swimmer's conditioning profile the mix of the classes of exercises will vary

2. implementing "training adjustments" after training evaluation tests (see The Realization Phase). The records of the "times swum during the sets" will enable the coach to check if the "execution" of the training workouts meets the requirements of the class they were meant for. He will so have an overall but reliable picture of the content of each training period and be able to evaluate the training effectiveness by considering and weighing the training actually completed and the effects (the training, testing and competition progress) it induced. The training for the next period will be adjusted, i.e. the sequence, the frequency and volume of the different classes of training exercises will be adapted depending on the rate of success in achieving the desired training adaptations

Moreover, this classification system also enables the coach to create freely new training sets for each of the classes, knowing which main training effect to expect. The coach can "unleash" his imagination, fantasy and ingenuity to create whatever training workout he deems best. As long as the workouts meet the requirements of the class corresponding to the planned training objective, he can be assured that the exercise induces the training effect he wished for.

The criteria for the classification of the training workouts are based on a combination of scientific evidence and empirical expertise acquired after trial and error. We have been using this approach for over 10 years and have evaluated the outcome of different types of training workouts. We are very confident that this classification scheme is correct and that it works. As the book progresses we will discuss not only the physiology behind each objective but also illustrate our experience with various examples.

Table 2 delineates how each training objective/class effect can be achieved using the basic elements of a training exercise (volume, intensity, interval length and rest, see The Basic Elements of a Training Exercise).

Classification of Training Exercises

	Aerobic Capacity (Endurance Cap.=AEC)		Anaerobic Capacity (=ANC)		Aerobic Power (=AEP)		Anaerobic Power (=ANP)	
Type of swimmer	S	L	S	L	(S) M	L	S	M (L)
Volume*	Long	Very High	Moderate	Low**	110-90% Comp. distance		110-90% Comp. distance	
Interval	Short (100-300m)	Long (300-800m)	Very Short (25-75m)		Short progresses to Long (50-100m) => (100-300m)		Short (25-100m)	
Intensity*	Extensive alternated with intensive and short intervals in the same or next training session		Nearly all-out	All-out	Race Pace or somewhat faster		All-out	
Rest	Short (40-20s)	(20-10s)	Long (>= 2x swim time) (35s-1:30min)		Short progress. to Very Short (45-30s) => (10-20s)		Short (10-20s)	
	8x100m R=20s 1, 3 fast	6x500m R=20s 1, 2 (50fast/50slow)	6x(3x50m) R=1:20min P/3	3x(2x25m,1x50m) all-out after 25m R=30s after 50m R=90s	5x75m R=45s to 3x125m R=15s	12x100m R=30s to 5x300m R=20s	Brokens / Comp.Test 4x50m R=10s 25+50+25+50m R=5-10s	

*depends on conditioning level

Sprint and technique are not in this classification

** has been changed vs previous publication, see text

Adapted: J. Olbrecht: Schwimmen, Lernen und Optimieren 1994

Tab. 2 Classification of training exercises according to expected changes in aerobic and anaerobic capacity or power (S = sprinter; up to 100m, M = medium distance; 200 to 400m, L = long distance; 800m and longer).

Aerobic Capacity Training (Endurance Capacity)

Goal:	to increase the maximum uptake of oxygen per minute (physiological adaptation induced: rise of VO_2max).

This type of exercise lays the basis for the swimmer's conditioning and is therefore not only essential for the long distance swimmer but also for the sprinter. It will thus, take up the largest part of the base training period. During the competition training period these workouts will be performed less frequently and at a lower intensity as, during this period, their only purpose is to maintain the acquired aerobic capacity and to counterbalance the intensive training work (anaerobic capacity, anaerobic power or aerobic power workouts).

Checklist for the aerobic capacity exercises:

1. high volume (for sprinters, the intervals and the total distance of the workouts are generally shorter than for long distance swimmers)

2. low intensity during the major part of the exercise but "spiced" with some intensive bouts at the beginning of the workout

3. little rest between the intervals

An improvement in aerobic capacity depends to a large extent on the biological adaptations in the slow twitch fibers. This type of muscle fiber contains large numbers of mitochondria and oxidative enzymes. Mitochondria are the parts of the muscle cell that supply energy by oxygen consumption (aerobic energy). Increasing mitochondria size, density or both through training will result in an increase in the rate of aerobic energy supply and determine to a large extent the improvement of endurance.

Research has shown that these slow twitch muscle fibers are fully activated even at low and medium intensities. This corroborates the finding that the mitochondria, in these type I muscle fibers, are fully recruited to contribute to the energy supply even at a low to medium intensity (fig. 7). Higher training intensities will not recruit additional mitochondria in the slow twitch fibers. However, if we want to put training stress on the mitochondria of the oxidative fast twitch fibers, medium to high training intensity is needed. Thus, more intensive training will not lead to a larger training effect in the slow twitch muscle fibers but will transfer the training stimulus to another muscle fiber group, the oxidative fast twitch fibers (FTIIa).

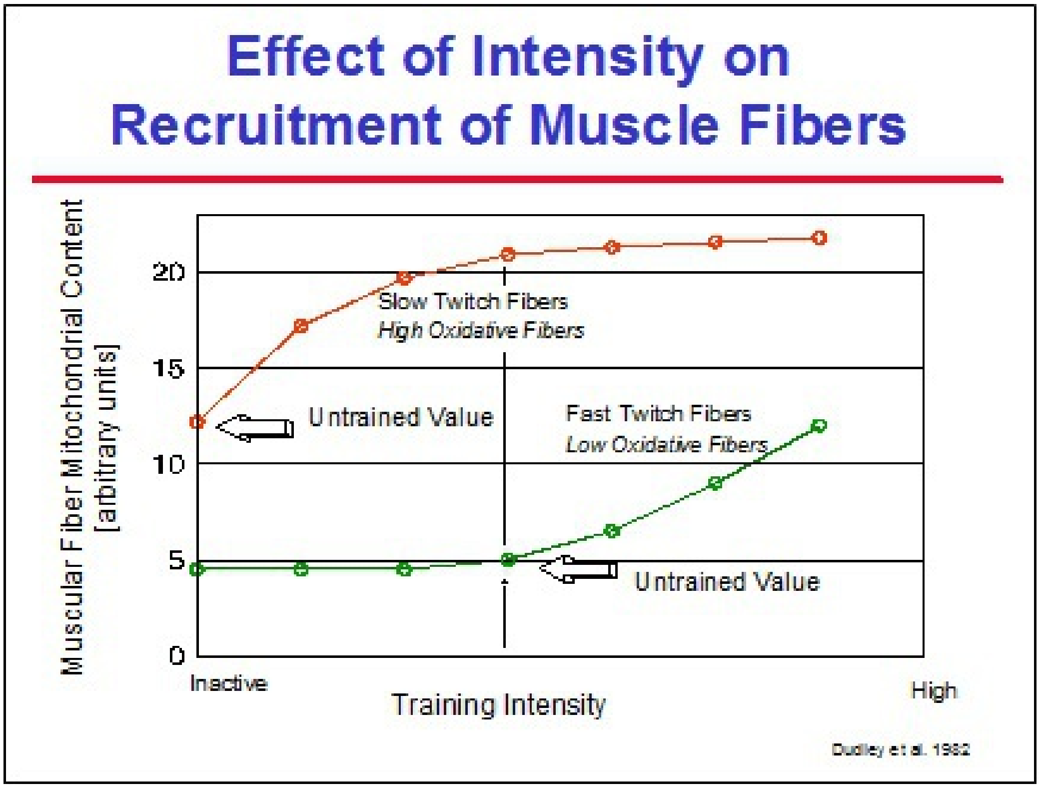


Fig. 7 Recruiting patterns and biological adaptations shown by the increase in mitochondria in the slow twitch muscle fibers and in the oxidative fast twitch muscle fibers (FTIIa) as a function of the training intensity.

Muscle biopsy research has underlined the importance of the intensity as well as the duration (and thus also volume or mileage) of the training exercises for increasing the number of mitochondria (fig. 8). It was found that the peak adaptation in mitochondria content seems to occur with shorter but highly intensive training exercises. This indicates that short, intense swims are very useful to raise the number of mitochondria. *Thus, not only long extensive (slow) workouts but also short, intense exercises are essential for the improvement of the aerobic energy supply process*. These short intensive swims seem to affect the recruitment of mitochondria and therefore focus the training stress on this part of the biological adaptation. Long extensive workouts on the other hand, are more suitable for improving the cardiovascular system, the homeostasis and the availability of substrates, and to induce other non muscle specific adaptations.

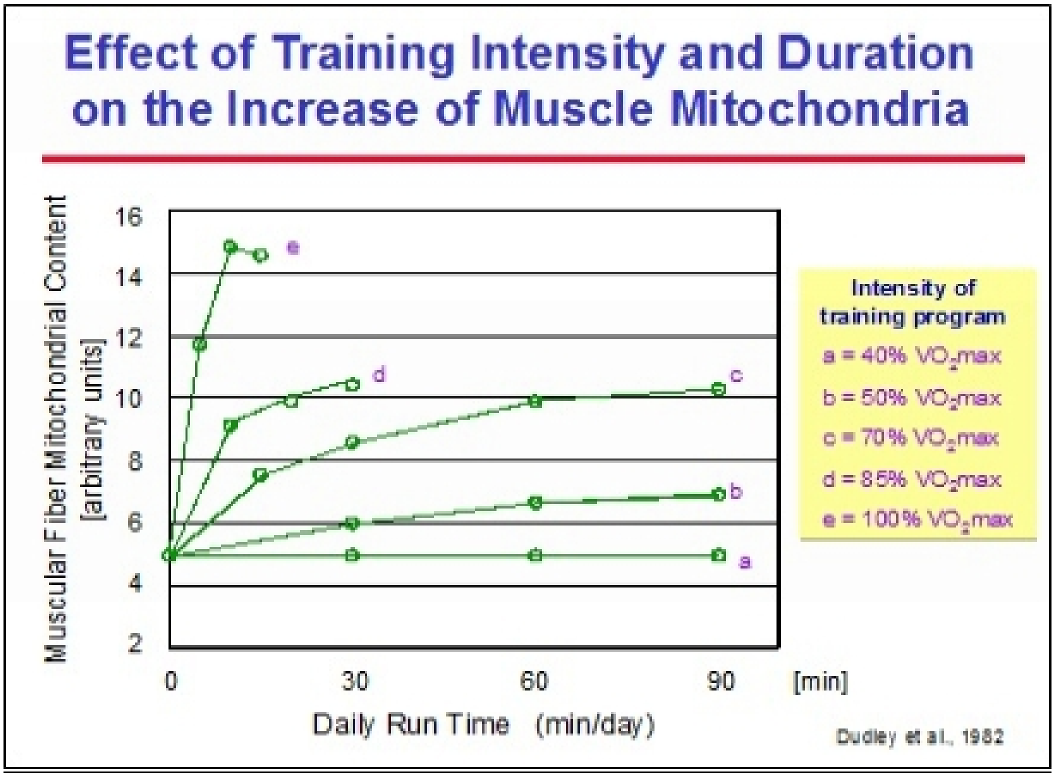


Fig. 8 Influence of the duration and the intensity of the training program on the increase of the number of mitochondria in the muscle fibers (adapted from Dudley et al. 1982).

How can we apply these scientific findings to training practice to improve the aerobic capacity?

1. Aerobic capacity training can only provide maximum benefit if long swims of low intensity are “spiced” with a few intensive and short stimuli. These “spices”:

   - may only be added in restricted quantity. For swimmers with mostly slow twitch muscle fibers the quantity is strongly restricted. For swimmers who have more fast twitch fibers, the quality or frequency per week of short intense bouts may be higher. The number of short intensive bouts necessary for aerobic capacity training depends primarily on the level of anaerobic capacity and muscle force (non-invasive indication = no biopsy), but will ultimately be determined by a systematic and continuous evaluation of training effects following different doses of short intensive swims

•

must be scheduled in the beginning of a workout and be followed by a long swim of low intensity

2. The intensity at which the long extensive swims have to be carried out can be low since even low intensity provides enough stress on the mitochondria to provoke adaptation. In other words, the swimmer does not have to swim very fast to recruit the slow twitch muscle fibers that are responsible for most of the aerobic capacity

Experience has taught us that we can combine short intensive repeats with long extensive workouts in 2 ways:

First (Type I), both the intensive and extensive work take place during the same training unit; the intensive bouts mostly in the beginning of the exercise and the extensive part at the end (= “extensive tail” at LA1 - see fig. 56 - or even slower). Drills, relaxation exercises or technique do not belong to the extensive tail. For respectively poor to well trained swimmers:

- ✻ the total volume of the set varies from 600 to 2800 m
- ✻ the “extensive tail” takes up from 2/3 to 5/6 of the whole set

Second (Type II), the extensive work happens in the training unit that follows the intensive one. Such an intensive session may contain a long anaerobic capacity workout, an aerobic or anaerobic power set or may be a competition

Example Type I*: both the intensive and extensive work take place during the same training unit

(2 x 100 m) + (2 x 150 m) + (2 x 250 m) + (1 x 400 m) total: 1400 m

- rest after 100 and 150 m = 20 sec; after 250 m = 45 sec
- 1st 100 m progressively faster each 25 m, 2nd 100 m at LA1
  (LA1 means that the pace will generate a level of 1 mmol/l lactate, see also fig. 56 ③)
- 1st 150 m 25 fast/25 easy
- The 250 m and 400 m are at a very slow pace - LA1

Example Type II:** short intensive and long extensive (slow) swims are distributed over 2 training units with the extensive work following the intensive work

Day 1: *main part 1* (anaerobic capacity exercise)

3 x (4 x 50 m) start every 1:15 total: 600 m

- 4 x 50 m odd repeats = as fast as possible and free stroke frequency (stroke rate)
  even repeats = as fast as possible but 3 strokes/50 m less than in odd repeats
- 1st set = Fly / 2nd set = best stroke / 3rd set = Freestyle

main part 2 (intensive aerobic set)

4 x 400 m Freestyle total: 1600 m

- rest = 30 sec
- Odd repetitions at LA4; even repetitions at LA2 (LA4 and LA2 means that the pace will generate a level of 4 mmol/l and 2 mmol/l lactate respectively)

Day 2: main part

4 x 800 m Freestyle total: 3200 m

- rest = 1 min
- each interval at LA1

The second example shows a way to improve the aerobic capacity through the sequence of an intensive training unit (Day 1: built up by an anaerobic capacity exercise and an intensive aerobic set) with only regenerative work during the next training unit (Day 2). In this way:

* the build-up of both the aerobic and anaerobic capacity can, to some extent, take place simultaneously
* even high volume regeneration work, performed at very a low intensity, plays an important part in the building-up of the aerobic capacity

Regeneration training strongly stimulates improvements in aerobic capacity and should therefore be classified within the category of endurance capacity (aerobic capacity) training. Regeneration training in one or more consecutive training units can follow an intensive unit such as an anaerobic capacity or power set, an aerobic power exercise or a competition. It is always very important to give the swimmer regeneration training the day after a competition. Besides, it is advisable to grant the swimmer a complete day of rest (a day with no training) each week, but never the day after a competition.

You will find more details about and guidelines for recording the different types of aerobic capacity training in the macro-, meso- and microcycles in chapter 9, Templates for Planning and Periodization.

Remarks:

* Aerobic capacity training loses its effectiveness quickly if the training stimulus is stopped or interrupted too often
* Too much variation in stroke or type of motion (e.g. alternating freestyle and backstroke or alternating freestyle pulling (arms alone) and then kicking) reduces the benefit of the aerobic capacity training. This is especially important for IM swimmers in the base training period. At that stage of the periodization it is recommended to improve the aerobic capacity in each stroke and to avoid medley sets. The coach should thus select one, or at a most two strokes for his main sets

Example*** Aerobic capacity for IM swimmers

Base Training Period		Competition Training Period
Not Appropriate	Appropriate	Appropriate
16 x 100 m IM 16 x 100 m Change every 50 m 16 x 100 m Change every 100 m	16 x 100 m Back followed next day with 8 x 200 m 4 x Breast, 4 x Freestyle 16 x 100 m Breast (1-6); Back (7-16)	4 x 100 m IM followed by 4 x 200 m IM 16 x 50 m Change every 50 m

* Focusing on the improvement of the aerobic capacity for too many weeks results:

- in a decrease of the aerobic power
- possibly in an increase of anaerobic capacity. This may be the reason for the decrease in aerobic power, however, such a decrease may also occur without an increase of anaerobic capacity

Anaerobic Capacity Training

Goal:	increasing the capacity to break down carbohydrates anaerobically (physiological adaptation induced: rise of glycolytic rate = VLamax[2]).

This type of training also plays an important part in the base training period and, to a smaller extent, in the competition training period where it is mainly used for maintenance purposes. Preferably, these exercises are inserted into the first half of the training session (see The Training Unit). This type of workout lays the basis for the tough anaerobic power work that will come up later and must therefore certainly precede the anaerobic power sets.

Checklist for the anaerobic capacity exercises:

1. short intervals (25-50 m)
2. each repetition must be swum somewhat slower than maximum pace for swimmers with a high anaerobic capacity, and at maximal speed for swimmers with a low anaerobic capacity
3. the rest must be at least as long as the exertion time span, preferably, even 2 times the exertion time
4. contrary to all other types of training, passive rest is better than active rest

A trick to control the intensity of the anaerobic capacity sets is to insert an all-out sprint early in the set:

- if the time of this sprint is more than 1.5 sec/50 m faster than the previous repetitions then the previous repeats were too slow and the remaining reps have to be swum faster
- if the time of this sprint is less than 1 sec/50 m faster than the previous repetitions then the previous swims were too fast and the remaining reps have to be swum slightly slower

Example*

2 x (4 x 25 m) total: 200 m

- rest = 20 sec
- 1st series best stroke / 2nd series Freestyle

Example** (for a 400 m IM swimmer)

4 x (5 x 50 m) start every 1:15 total: 1000 m

- series 1: Fly / 2: Back / 3: Breast / 4: Freestyle

It is generally held that the anaerobic capacity is, to a great extent, innately determined. *But* - and this is where our findings conflict with the common belief - *we found that the anaerobic capacity can be improved by training* , although it takes a long time (mostly 1 or 2 years or sometimes even more). The long period required for the improvement in anaerobic capacity to come about, may be the reason why most investigations missed it.

Moreover, not every swimmer, even with an appropriate training, manages to increase his anaerobic capacity. As the changes in anaerobic capacity - if they occurred - took place in swimmers older than 17 years of age, it can be assumed that they are not related to puberty but rather to the training process.

Thus, many of the anaerobic training exercises are planned and included in the training program with the understanding that some changes in innate anaerobic capacity will occur over the long- term rather than during the current macrocycle. It is, however, possible to observe changes in anaerobic capacity in the short-term, e.g. even during the course of a mesocycle, but these are then more temporary fluctuations rather than intrinsic changes (changes in the innate anaerobic capacity). We noticed, for example, that a high weekly mileage of extensive work temporarily reduces the anaerobic capacity. When training then shifts to less voluminous and more intensive work, it is common to see the level of the anaerobic capacity return to its innate level within 3 to 5 weeks. These findings on both the innate and temporary changes in anaerobic capacity are based on our training experience with top level swimmers but are not yet described in any specialized literature.

Anaerobic capacity workouts:

* are usually swum in the best stroke
* are compatible with frequency training (stroke rate training, see Special Types of Training)
* can be used as the “spice” for the endurance capacity training; we have indeed already seen in a previous section that the anaerobic workouts can be used to increase both the aerobic and anaerobic capacity

It may sound strange, but the anaerobic capacity is very important not only for sprinters but, to another extent of course, also for long distance swimmers. Indeed, distance swimmers mostly have an innate low anaerobic capacity, which enables them to swim fast at long distances in competition even with a moderate aerobic capacity. The problem for these swimmers arises when they also swim fast in training. Due to their low anaerobic capacity, the onset of lactate accumulation in a workout will occur at a much higher intensity than in swimmers with the same aerobic capacity but with a higher anaerobic capacity. The swimmers with the lower anaerobic capacity will come much closer to their maximal aerobic and anaerobic capacities at a subjectively easy effort and at a low measured lactate level. Moreover, these swimmers seem to have no warning system to suggest fatigue and will therefore “easily” swim too fast in training and break down their anaerobic capacity even further so that the above situation becomes exacerbated. If such swimmers, with a low anaerobic capacity, are not forced to slow down in training or if the break down of their anaerobic capacity is not counterbalanced with anaerobic capacity training, they are making a beeline for overtraining.

The frequency per week and volume per set of the anaerobic capacity training exercises is strongly determined by the quality of the individual athlete’s anaerobic capacity. If the swimmer has a poor anaerobic capacity, as is the case for most of the long distance swimmers, the coach has to “administer” this type of exercise in dribs and drabs. Let us, to illustrate this, compare a weak anaerobic capacity with thin ice. If you let a feather fall down on thin ice nothing will happen; the ice will remain intact. If, on the other hand, you try to walk on thin ice it will surely break. Thus, as long as the swimmer has a low anaerobic capacity, the frequency per week and the volume per set of this type of training has to be low (1 anaerobic capacity training every 3rd-4th training session and between 200 and 400 m per set). We also advise these swimmers to use mostly short fractions (25m) and to swim all-out; e.g. 3 x (2 x 25m, 50m) each 25m start every 45s and after each 50m 1 min rest. For swimmers with a high anaerobic capacity, which is the case for most sprinters, the exercise will rather be 12 x 50 m start every 1:30.

A good anaerobic capacity is thus important:

* for sprinters, to be able to swim fast on short distances
* for long distance swimmers, but to a lesser extent, to be able to accelerate at different times during long distance events and to better tolerate huge training loads. Lactate tests after long distance races revealed considerably higher lactate values for World class swimmers than for swimmers of a regional or even national level (fig. 20). This proves, for one thing, that anaerobic capacity is important even for long distance swimmers and also that this anaerobic capacity will in the end make the difference between winning and losing

Remarks:

* As the anaerobic capacity increases it will be increasingly difficult to build up a good aerobic power
* If a good anaerobic capacity is not coupled with a good aerobic capacity it will be difficult to improve the aerobic and anaerobic power

Aerobic Power Training

Goal:	maximizing the use of one's aerobic capacity (physiological adaptation induced: increasing the % of VO_2max that can be maintained during long distance exercises).

This type of exercise is necessary to complete the training work. It will, therefore, be principally planned during the pre-competition training period and it should be swum in the main stroke (medley swimmers will now swim medley and freestyle sets). From a physiological point of view, the importance of this type of workout increases in accordance with the length of the competition event for which the swimmer is preparing. For 100 and 200 m races, this type of training is not a must, but for races longer than 200 m it is an absolutely essential part of the pre-competition training period.

At first sight, the aerobic power training sets do not look so difficult, but experience has taught us, to our amazement, that elite swimmers, even the long distance swimmers who like them very much, can "digest" them only once or, at maximum, twice a week. This type of training seems to require a long recovery period and is therefore very "treacherous".

As results from this type of training should not be expected within 3 weeks, the coach is compelled to start this type of training at least 6 weeks before the important competition event.

Checklist for the aerobic power exercises:

1. stay around the competition distance (for a 400 m swimmer the workout may be 500 m long and be repeated 2 or 3 times)
2. the rest is always short (5 to 15 sec) and the repeat distance is adjusted to enable the swimmer to maintain competition speed or

just a little bit faster

3. as the swimmer gets closer to the competition, the coach will make these swim sets progressively harder by shortening the rest in proportion to the repeat distance. Example: in the beginning an aerobic power series consists mainly of 50 and 100 m intervals with 15 sec rest. In the following weeks the athlete swims longer intervals (100 and 200 m) but with the same or less rest

Example*

3 x (100 m + 50 m) total: 450 m

- rest = 15 sec after 100 m
- rest = 10 sec after the 50 m interval
- intensity = competition time (CT) for 400 m

Example**

week 1: 5 x (100 m + 2 x 50 m) total: 1000 m

- rest = 10 sec
- intensity = competition time (CT) for 800 m

week 2: 3 x (100 m + 200 m + 50 m) total: 1050 m

- rest = 10 sec; 5 sec after 50 m
- intensity = CT for 800 m

As mentioned previously, the coach may use an anaerobic capacity training exercise or a short aerobic power workout (maximum 600 m) to “spice” the aerobic capacity training. This is especially useful at the end of the base training period, just prior to the competition training period, to prepare for the longer aerobic power sets that will then be done. As an example, the coach can, near the end of the base training period, schedule a 6 x 100 m (with 10 sec rest) to be swum at 800/1500 m competition speed, followed by a 8 x 200 m at a very low intensity (LA1). A test competition of more than 200 m can also replace an aerobic power set in training, something that is often even preferred by the swimmers.

Remarks:

Aerobic power training tends to drive both the aerobic and anaerobic capacities lower if the majority of the remaining training swims are not slow enough. It is, therefore, very important to insert extra regeneration work in a training week in which aerobic power sets have been planned. The loss of anaerobic capacity, due to the aerobic power sets, is only a problem if it is already low. If, on the other hand, the anaerobic capacity is too high the aerobic power set will lower it to a more desirable level.

Anaerobic Power Training

Goal:	maximizing the use of one's anaerobic capacity (physiological adaptation induced: increasing the involved percentage of VLamax that can be maintained during a high level effort in training or in competition).

This type of exercise is also used in the competition training period. The anaerobic training at the end of the base training period is meant to increase the ability to produce more lactate (anaerobic capacity training). The anaerobic power training in the competition training period is a.o. used to toughen the swimmer against acidosis. This adaptation can be completed quite quickly (2 weeks).

While, during the pre-competition training period, the long distance swimmers will try to improve their aerobic power, the sprinters (50 - 100 m) and middle distance swimmers (200 - 400 m) will concentrate on increasing their anaerobic power. Compared to the aerobic power training, which may take up to 4 weeks to show any effect, anaerobic power will change much more quickly. Indeed, in swimmers with a moderate anaerobic capacity, the effects of this training will be clearly noticeable after about 10-17 days. In swimmers with a strong anaerobic capacity, on the other hand, the full effect will be reached after 4 weeks focused on the anaerobic power training followed by 2 to 3 weeks of tapering. It is advisable to split the 4 weeks of anaerobic power training into 2 x 2 weeks with 1 week of low intensity or regeneration work in between. Anaerobic training for more than 2 consecutive weeks may carry more risks than advantages.

Checklist for the anaerobic power exercises:

1. maximum swim speed (starting with the first repetition)

2. short repeat distances; these must allow a maximum swim speed
3. very little rest (5 to 10 sec depending on the interval used - 25 or 50 m)

The total distance of an anaerobic power set is very low (125 to 250 m). For top level swimmers, the volume can be increased to 600 m but should then be split in sets of 100 to 200 m with 10 to 20 min rest between each set.

Example*

2 x (4 x 20 m or pool width) total: 160 m

- the swimmer arrives and departs again in multiples of 5 sec
- all-out + breath control + push off from wall
- as fast as possible right from 1st repetition

Example**

2 x (4 x 50 m) total: 400 m

- rest = 10 sec (broken) / 10 to 20 min after 4 x 50 m
- all-out (as fast as possible from the 1st repetition)

Example***

2 x (25 m + 50 m + 75 m + 50 m) total: 400 m

- rest = 5, 10 and 15 sec / 10 to 20 min after first 200 m
- all-out

Training sets with repeat distances longer than 50 m and swum at maximum speed with a lot of rest (e.g. 4 x 100 m max., Rest = 10 min) can also be classified as anaerobic power training. These are very hard and the quality of execution often falls off. Therefore these anaerobic power sets should be:

- executed with shorter intervals (max. 75 m) and / or
- reserved only for very good swimmers (sprinters and Sr. national qualifiers)

An alternative for this type of exercise is to enter the swimmer in a local competition and let him/her swim 3 or 4 100 m events. This may be a greater motivation for the swimmer.

Remarks:

Similar to the aerobic power sets, this type of training tends to drive down both the aerobic and anaerobic capacities. So, for this class of exercise, it will also be necessary to plan, in conjunction with the anaerobic power workouts, very long and slow swims at a very low intensity to maintain the level of both capacities.

PARADOXICAL TRAINING EFFECTS

The 4 workout classes induce specific conditioning adaptations according to the principle of super-compensation. This super-compensation can only be reached if the workouts are completed in *the adequate volume* , with the *right intensity* and at *the right moment* . If this is not the case, unexpected consequences or undesirable adaptations of the conditioning will take place.

The conditioning properties that are most sensitive to these unexpected and undesirable adaptations are the anaerobic capacity and anaerobic power. An overdose of anaerobic work has the opposite effect of what is intended and will break down both the anaerobic capacity and the anaerobic power instead of building them up.

If this happened, a few weeks (generally 2 or 3 weeks but sometimes even more, depending on the size of the previous overdose) of very extensive (slow) or even regeneration training should be sufficient to see the anaerobic capacity and power improve.

This sounds paradoxical but we often see:

(a) that the exaggerated emphasis on anaerobic work in training results in the deterioration of both the anaerobic capacity and power

(b) subsequently a lot of extensive (low intensity) or regeneration training with only a small amount of anaerobic training will improve the anaerobic capacity and power

Swimmers who tend to overdo or to swim too fast in training often have unexpectedly good competition results in the beginning of the swim season or after being out for 2-3 weeks with an illness or an injury. However, they will not be able to sustain or improve these performances as the season progresses and will frequently swim as fast in training as in competition.

Similarly, the aerobic capacity often suffers from too much volume. A stagnation or even deterioration of the aerobic capacity due to too much mileage can quite easily be reversed by reducing the number of training sessions a week.

As the adequate proportion and timing of the different training exercises vary individually, the coach has to find, for each swimmer, the appropriate moment and the right balance between frequency, volume and intensity of the various types of workouts. Regular conditioning tests will help him to evaluate the effects of the training and plan for each swimmer the right training "regimen".

DOSING OF THE TRAINING EXERCISES

The following table provides you with:

1. a summary of the main purposes of the 4 classes of training workouts
2. the time needed to induce stable adaptations (time necessary for stable effects to show up)
3. the dosing scheme for each workout class
4. the secondary adaptations that may crop up

Considering the time necessary for the effects to show up, we, by analogy with other physiological literature, have observed "conditioning" improvements after a very short training period (1-2 weeks). These short-term changes are rather due to a better neuromuscular and vascular control (a more efficient metabolism, blood flow distribution and use of muscle fibers) than to changes in the muscle structure and will therefore be lost quickly with inactivity. Stable or longer lasting adaptations involve the remodeling of the muscle, such as the enhancement of the mitochondria content or an increase of capillaries, and require, therefore, much longer and continuous training periods.

The “dose” of the different workouts strongly depends on:

1. the quality of both capacities
2. the training period in the conditioning program

Table 3 gives an overview of the (main) adaptations induced by the 4 classes of exercises as well as the advisable “doses” of the different types of exercises according to the level of the aerobic (VO_2 max) and anaerobic capacities (VLamax) and the period in the training process. These are, of course, guidelines. Depending on the way swimmers respond to the various training structures, the dose of the different types of exercises may have to be adapted individually. In the pre-competition training period the coach has to pay attention to aerobic power and both capacities for a long distance swimmer and to the anaerobic power and both capacities for the preparation of a sprinter.

The frequencies per week are guidelines too, which means that they can be lowered or increased depending on the changes in both capacities and according to the training objectives. Take for example a long distance swimmer with a high aerobic and an anaerobic capacity that is too high; his coach could, as indicated in the table, easily plan 5 aerobic and 3 anaerobic capacity exercises per week during the base training period. But, as the swimmer’s anaerobic capacity proved to be far too high, it is preferable to replace one of the anaerobic capacity workouts with an extensive long swim to drive it down so the enhancement of an unfavorable conditioning component can be avoided.

Type of Exercise		Main Class Effect	Time for Effect	Undesired Effects	Advisable Dose Maximal Frequency* (max. distance per set) Base Training Period	 Competition Training Period
CAPACITY	Aerobic	⇧VO_2max	At least 8 weeks. Will be longer the better VO_2max	⇩Anaerobic Power ⇩Aerobic Power ⇧Anaerobic Capacity	Hi Type I > II ③④ VO_2 max: 3.5 (3000) 5 (2400) 1.5 (800) 2 (800) Low VLamax Hi	Hi Type II > I ③④ VO_2 max: 3 (2500) 3 (2000) 1 (1000) 2 (1000) Low VLamax Hi
	Anaerobic	⇧VLamax	At least 18 weeks. Will be longer the weaker VLamax	⇩Aerobic Power	Hi VO_2 max: 2 (400) 3 (800) 1 (250) 2 (600) Low VLamax Hi	Hi VO_2 max: 1 (400) 1 (800) 1 (250) 0.5 (500) Low VLamax Hi
POWER	Aerobic	⇧ %-use VO_2max ①	About 3 - 7 weeks. 3 for weak and 7 for strong VLamax	⇩Anaerobic Capacity ⇩Aerobic Capacity	Hi VO_2 max: 0.3 (900) 0.5 (1200) 0.2 (300) 0.3 (500) Low VLamax Hi	Hi VO_2 max: 1.5 (1200) 2 (1800) 1 (300) 0.8 (600) Low VLamax Hi
	Anaerobic	⇧ %-use VLamax ②	About 2- 6 weeks. 2 for weak and 6 for strong VLamax	⇩Anaerobic Capacity ⇩Aerobic Capacity	Hi VO_2 max: 0.2 (900) 0.2 (400) 0.2 (100) 0.2 (300) Low VLamax Hi	Hi VO_2 max: 2 (500) 3 (800) 1 (150) 1 (300) Low VLamax Hi

Tab. 3 Overview of the (main) adaptations induced by the 4 classes of workouts as well as their advisable doses depending on the level of the aerobic (VO_2max; Y-axis) and anaerobic capacities (VLamax; X-axis) and the period in the year plan. ① Important for long distance swimmers, ② important for sprinters, ③ frequency of aerobic capacity workout type I + type II. ④ Volume refers to the aerobic capacity workout type I. A frequency of 0.5 means once every 2 weeks.

* Swimmers with a low VO_2max are expected to complete 4 training units per week while for sprinters and long distance swimmers with a very high VO_2max the total number of training units per week may rise to respectively 10 and 12.

SPECIAL TYPES OF TRAINING

Although the exercises for technique, sprint, stroke frequency (rate), breath control, and strength training cannot be classified into the 4 exercise classes, their success rate is also a matter of proper timing and of the right balance between frequency, volume, intensity, rest and repeat distance.

Technique Training

For technique training there are 2 types of exercises:

- Type I: technique training swims to *teach* and *correct* the stroke pattern
- Type II: technique training swims (drills) to help *automate* the proper stroke pattern. In other words, the swimmer will acquire the stroke skills without thinking (= automation)

Type I: For this type of training the rules are fairly strict and should be followed exactly. To be able to learn the stroke patterns correctly the swimmer has to be fresh and fit; it is therefore important:

* to schedule these exercises at the beginning of the training session, just after the warm-up
* to ensure that the rest between the intervals is long (to allow full recovery) and will be used by the coach to make corrections or give feed-back to his swimmer
* to keep the volume and the repeat distance short and the intensity low

Type II: For the drills intended to automate the proper stroke pattern, volume and repeat distance should be made progressively longer, the rest shorter and the intensity higher. Their objective is to gradually achieve the correct form in the normal training rhythm.

Frequency (Stroke Rate) Training

Stroke frequency (the number of strokes per minute) is a very important component to achieve a good time in competition. Elite swimmers are

capable of reaching high frequencies as well as swimming fast at low frequencies.
Increasing the maximum arm stroke frequency (rate) is only useful if it makes the swimmer faster. Indeed, a swimmer will never get full benefit from increasing his stroke frequency (rate) if he starts slipping through the water. It is consequently important that, at any stroke frequency, he keeps catching the water properly.

There are therefore 2 types of training exercises:

- Type I: drills to teach the swimmer to swim faster while increasing his stroke frequency
- Type II: drills to help the swimmer increase his maximum stroke frequency (rate)

Type I: Requires short intervals to be swum at different frequencies. This will automatically induce variations in intensities. Specific tests can show at which frequencies the swimmer gets into problems.

As the intervals will be often repeated, the volume of the exercise will be quite high. The rest ranges between 30 sec and 1 min; the faster the swimming, the longer the rest.

If the exercise is carried out properly every increase in arm stroke frequency (rate) must result in a faster time. If this is not the case, the frequency (rate) should be lowered during a few repetitions. After a moment the swimmer will again be asked to increase the stroke frequency (rate) while swimming faster.

Example * * *

20 x 50 m Breast start every 1:15
- progressively from 1 to 4
- stroke frequency (rate): 34-38-42-46[3]

Remarks:

* This exercise requires an individual approach and is therefore time consuming

* These exercises are not advised for swimmers who still have to improve their technique

If the stroke frequency (rate) training is carried out correctly and swum fast enough, it can also be used as anaerobic capacity training

Type II: As this type of stroke frequency (rate) training is very similar to the sprint training (see below) the same rules concerning volume, intensity, rest and interval length required to improve the base speed will prevail here:

It is useless to start type II of the stroke frequency (rate) training if type I has not been successfully completed. Besides, the coach has to make sure the swimmer is strong enough to swim at higher frequencies (rates). Indeed, boys during their growth spurt and girls in general, often lack the strength for high frequencies (rates). They will need to improve their strength before they can start with type II of the stroke frequency (rate) training.

Another reason to defer the start of type II may be metabolic in nature. For example, already high lactate values at a moderate stroke rate may prevent a breaststroke swimmer from increasing his stroke frequency (rate) even more. The swimmer will not be able to mobilize the energy needed to properly carry out a faster stroke. Consequently, he will need to improve his aerobic capacity in breaststroke before starting with type II of the stroke frequency (rate) training.

Sprint Training

The requirements for sprint training are similar to those for technique training. It is important to distinguish between:

- Type I: sprint training exercises, in the true sense of the word, to increase the base speed
- Type II: sprint training exercises to train a swimmer to swim fast in a state of fatigue

Type I: Again for this type of training the rules are fairly strict and should be followed exactly. To be able to carry out these exercises at maximal speed and with a correct technique, the swimmer has to be fresh and fit; it is therefore important:

* to schedule these exercises at the beginning of the training session, just after the warm-up and after a few preparatory rhythm swims

* the rest between the intervals should be long (to allow full recovery) and may be active

* to keep the volume and the interval length short

Example*

2 x 25 m sprint start every 2 min

Example**

6 x 50 m (25 m fast with starting dive, 25 m easy)

- 1st, 3rd and 5th fast but fluent (with good technique)
- 2nd, 4th and 6th sprint

Type II: If we want the swimmer to maintain a good sprint performance in a state of fatigue, the above mentioned rules remain valid but the sprint exercises will be planned towards the end of the training session.

Breath Control

In this type of exercise we ask the swimmer to breathe once every x-number of strokes. Originally it was thought that this type of training, by analogy with altitude training, would restrict the availability of oxygen to be taken up by the red blood cells and in doing so, it would provide an extra stimulus for the improvement of the endurance capacity. This, however, has never been proven experimentally. Breathing a lot or a little has no effect on the swimmer's aerobic conditioning. It is therefore not appropriate to label this type of training as hypoxia training or to consider it as a surrogate to training at altitude. The term hypoxia training has been used erroneously for many years. It is now proper to refer to it as *breath control training* .

Breath control training has a favorable effect on the swimmer's movement in the water as well as on the technique. A swimmer's head is just like the rudder of a boat or the ailerons of an airplane. Every movement of the head will influence the swimming direction as well as the position of the body in the water. A stable position of the head during swimming reduces unnecessary positioning changes of the body and promotes water sensitivity and the proper execution of the stroke.

The objective of breath control training is therefore more the efficiency of the stroke than a conditioning adaptation.

Example *

5 x 100 m Freestyle total = 500 m

- rest = 20 sec
- breathe every 3 strokes the 1st 25 m
- breathe every 4 strokes the 2nd 25 m
- breathe every 2 strokes the 3rd 25 m
- breathe every 3 strokes the 4th 25 m

Another favorable effect of breath control training is that the swimmer learns to swim with fewer breaths, which helps at key points in the race. The “feeling of lack of air” leads to cramping and a stimulating urge to gasp for air. Breath control training will help to raise the threshold of the “feeling of lack of air” which means that the swimmer will not cramp nor gasp for air after the turns and during the final rush to the wall. The swimmer will also lose less velocity by maintaining a better technique.

The effect of the breath control training shows up after 8 to 14 days on a frequency of 1 breath control exercise every 3 days. This type of exercise is very beneficial during the last days of the pre-competition training period.

Example * *

2 x (6 x pool width)

- without breathing
- rest = maximum 5 sec

Strength Training

Strength training (ST) with weights for swimmers is a very complex and comprehensive subject. In this section we will confine our discussion to the classification and characterization of a few types of strength training. This will help the coach to coordinate strength training on land with water training (what strength training is compatible with what type of swim training) and to plan and periodize strength training correctly (when the different types of strength training are appropriate).

Types of Strength Training Strength training to increase maximum force (MFST) is considered to be capacity training while strength training to increase explosiveness (EXST) and strength training to maintain a high force involvement for a long period (ENST) are derived from maximum force training and are therefore labeled as power training. In table 4 you will find an overview of the characteristics of the different types of strength training exercises on land.

Types of Strength Training Exercises

	Capacity *Increase Maximal Force (MFST)*		Power *Increase Explosive Force (EXST)*		*Increase Endurance Force (ENST)*	
	S - M	(M) - L	S - M	(M) - L*	S - M*	(M) - L
Number of repetitions per set	2 - 4	3 - 6	15 - 30	20 - 40	40 - 70	40 - 100
Number of sets	1 - 3	1 - 2	1 - 3	1 - 2	2 - 3	2 - 3
Number of different exercises	5 - 12	5 - 8	5 - 10	4 - 6	4 - 5	4 - 8
Number of executions	10 - 50	15 - 60	75 - 250	80 - 300	300 - 500	300 - 800
Training units / week	1 - 3		1 - 4		1 - 4	

Remarks: Load: always maximal resistance with respect to the number of repetitions and mode of execution
Indicated ranges: dependent upon level of strength and years of experience with strength training
S: Sprinter (till 100 m) M: 200-400 m L: 800 m and longer
*: For this type of swimmers, this type of exercise is not a must

Tab. 4 Overview of the characteristics of the 3 types of strength training on land. All the exercises have to be carried out correctly to avoid injuries. The EXST exercises require an explosive (quick) execution.

Synchronization of Dry-Land Strength Training and Swim Training It is important to prepare the swimmers for strength training well in advance. We therefore advise at least 1 year of conditional gymnastics and 1 year of EXST with light weights before starting a real strength training program as described below. The first 2 years should, thus, be considered as an introduction to strength training, provide an adequate basis of general strength and teach swimmers how to work correctly with weights to avoid injuries once the "serious" work begins.

A swimmer who already has experience with strength training, will, during the base training period of the swimming season, primarily develop his maximal force. Indeed, maximal force must first be established before it can be "polished" into the form the swimmer needs in competition. In the base training period, maximal force training on land is compatible with the training objectives (less intensive work) in the water.

Once in the pre-competition training period, which on land starts 2 or 4 weeks earlier than in water, the sprinters should concentrate on explosive force (EXST) and to a lesser extent on endurance force strength training (ENST). The long distance swimmers, on the other hand, will in this training period focus on the endurance force strength training (ENST). The volume and intensity of the workouts with paddles should also progressively increase during the first 2 to 3 weeks of the land pre-competition training period.

Strength training can be continued until 2 weeks before the main competition, but it is advisable to reduce the load of the dry-land training toward the end of this training period to avoid negative interference with the intensive water work.

Composition of a Strength Training Unit Strength training (ST) starts, as does swim training, with a warm-up which is built up as follows:

1. some very extensive (low intensity) exercises to warm up the muscles
2. dynamic exercises at a low to moderate intensity (cardiovascular effect)
3. static muscle work to get the muscles ready for the high contraction tension in the main part of the strength training unit

The main part of the ST unit consists of exercises meant to achieve the training objectives, alternated with workouts that have a therapeutic purpose. Both types of exercises have to be balanced not only to avoid injuries, but, also, to achieve maximal effectiveness. In the exercises especially chosen to meet the training objectives it is important to alternate the active muscle groups. Moreover, “swimming propulsive” muscle groups always have to work dynamically (isotonic), while “body stabilization” muscle groups will work dynamically during the base training period and statically (isometric) during the competition training period.

The strength training should end with a cool-down that includes a lot of stretching exercises.

Training Load There are 2 different ways to fix the load for the different strength training exercises:

1. the loads used can be expressed in percentages of the highest weight (%Fmax) a swimmer can lift (press, push, ...) once. For each exercise of the ST program, the maximal weight has to be determined by means of a maximal test

Disadvantages:	• performing maximal tests is time consuming and exposes the swimmer to injuries
	• as the “strength conditioning” improves during the swim season, it is clear that the maximal weight a swimmer can lift once will also evolve and that, consequently, the maximal test will have to be repeated regularly. This recurrence of the maximal tests will certainly further increase the risk of injuries and may cause negative interactions with other training exercises or disturb the planning proceedings

	• The percentages of the highest weight (%Fmax) allocated to the ENST and EXST exercises may significantly differ from one swimmer to the other. Both types of ST require, therefore, an individual assessment of the workload. As this method uses the same %Fmax per type of ST exercise for all swimmers, the coach will never be sure whether he is working with the right load or not

2. As this first approach is too risky and time consuming, we give preference to a safer method whereby the maximal weight is assigned to the number of repetitions and the mode of execution

Example *12 x bench press - carried out quickly and explosively*

The swimmer has to carry out the 12 repetitions quickly and correctly with the highest maximal weight. If he manages to do so, the next time this exercise is on the program, he will have to increase the weight slightly. If, on the other hand, the weight proves to be too heavy, he will have to lower it and select a more appropriate load for the correct execution of the required 12 repetitions.

For the EXST exercises the correct execution of the repetitions will limit the maximal load. For the MFST and the ENST exercises, on the other hand, it is the number of repetitions that will restrict it.

In this approach, the coach first selects the number of repetitions and the mode of execution that meet the training objectives (tab. 4). He will then choose for every exercise the weight with which the swimmer can easily and correctly carry out the workouts. At the beginning of the strength training period the load is, thus, set rather low but will progressively be (from set to set) increased as the ST program progresses, providing the number of repetitions and the mode of execution are carried out correctly.

Advantage:	• this approach allows a progressive and individualized adaptation process. It is thus much safer and enhances the efficiency of strength training

These are, of course, general rules. The assessment of the swimmer's conditioning level may lead to prescribing variants of the strength training schedule described above.

Age Group Training

Safe and Appropriate Training In competitive swimming, it is not unusual for young children to train a lot. But, putting a maturing body through a strenuous or inappropriate training regime may be prejudicial:

- to his health
- to a continuous improvement of his performance in competition

In this chapter we will concentrate on a few important principles for safe and constructive age group training:

1. *Avoid too many repetitions of the same movement*

Studies have shown that young athletes have brittle joints, which make them very sensitive to overload. So, too many repetitions, even at a low intensity, may induce local complaints/injuries of the locomotor system. These complaints do not always appear in combination with symptoms of fatigue; it is for the coach, therefore, all the more reason to anticipate the local locomotor strain by providing variations in the movement patterns

The following table presents for age group swimmers the risks of injuries linked to some specific training exercises (tab. 5).

Load Type	Duration	Limiting factors	Risk of Injury	Usefulness for Age-Group Swimmers
Strength Training	5 sec	Contracting elements	Cartilage	0
Speed	15 sec	Energy-rich in phosphates	Muscle injuries	+
Anaerobic Power	1 min Anaerobic > 50% Aerobic > 50%	Glycolytic capacity and maximal strength	Catabolic	(+)
⇳	1-5 min Anaerobic (50-30%) Aerobic (30-70%)	Glycolytic capacity	Vegetative system	+
Aerobic Power	5-30 min Anaerobic (20-10%) Aerobic (80-90%)	Locomotor solidity	Tendons	+(+)
Coordination	-	Neuromuscular	-	++
Flexibility	-	Tendons and ligaments	Joints	++

0 Not Appropriate + unconditional limited usefulness (+) conditional limited usefulness (see text) ++ Appropriate

Tab. 5 The rate of usefulness of various types of training exercises for age group swimmers with respect to physiological and morphological limiting factors and the risks of injuries (adapted from Keul, 1982).

2. *The effects of the training exercises may vary according to the biological maturity of the swimmers*

The composition of the training program not only depends on the safety of the workouts but also on the biological age of the swimmers. Indeed, according to the biological maturity of the swimmer the training effect may vary quantitatively (same effect but to a different extent) as well as qualitatively (other effect)

- *Quantitative variation of the training effect*

For example, technique, reaction time and flexibility are well trainable at a very young age (5 to 8 years). Successful endurance capacity training is already possible from the age of 7. Studies revealed that 11 to 14 year old boys are most sensitive to endurance conditioning; the greatest difference in endurance between trained and untrained boys was namely observed just before and during puberty (fig. 9).

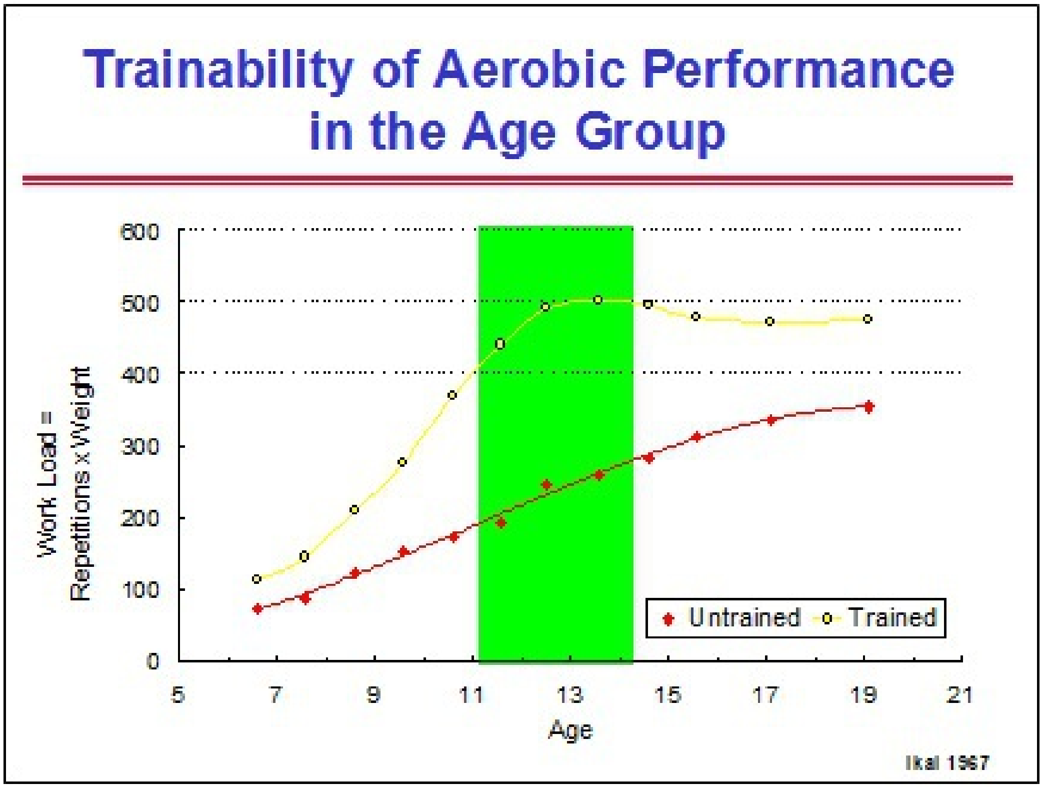


Fig. 9 Difference in endurance between trained and untrained boys just before and during puberty.

Other aspects of conditioning such as maximum force (MFST), endurance force (ENST) and anaerobic power, are only successfully trainable from the age of 16 for girls and 18 for boys (tab. 6).

Conditioning qualities		Age Group (f = female, m = male)							
		5-8	9-10	11-12	13-14	15-16	17-18	19-20	20+
Maximum Force	f				+	++	+++	+++	=
	m					+	++	+++	
Explosive Force	f			+	++	++	+++	=	=
	m			+	+	+++	+++		
Endurance Force	f				+	++	+++	+++	=
	m					+	++	+++	
Aerobic Conditioning	f	(+)	+	+	++	++	+++	=	=
	m	(+)	+	+	++	++	+++		
Anaerobic Conditioning	f			(+)	+	++	+++	+++	=
	m				(+)	+	++	+++	
Reaction Time	f		+	+	++	++	+++	=	=
	m		+	+	++	++	+++		
Speed	f			+	++	++	+++	=	=
	m				+	++	+++		
Coordination Technique	f	++	++	+++	+++	+++	++	+	+
	m	++	+++	+++	+++	++	+	+	+
Flexibility	f	++	++	++	+++	=	=	=	=
	m	++	++	++	+++				

\+ Somewhat trainable (+) Somewhat trainable under certain conditions (see text)
++ Well trainable +++ Very well trainable

Tab. 6 Overview of the trainability of different conditioning properties according to age.

- *Qualitative variation of the training effect*

Research has also shown that young athletes react differently than adults to certain types of training, such as endurance training. That is to say that the nature of the metabolic adaptations due to the endurance training will vary between young swimmers and adults. These qualitative differences are demonstrated in the following table (tab. 7).

Metabolic adaptations after aerobic training		
Adults	**Metabolic adaptations**	**Adolescents**
⇧⇧	Creatine kinase exhaustion	⇧
⇧⇧	Glycolysis	⇧
⇧⇧	Lactate production	(⇧)
⇧	Acidosis tolerance	Not Tested
⇧	Glucose consumption	Not Tested
⇧⇧	Oxidation capacity	⇧⇧
⇧⇧	Fat combustion	⇧⇧(⇧)
⇩	Aerobic-anaerobic transition	⇩
⇩	Adrenaline	⇩
⇩	Noradrenaline	⇩

Tab. 7 The metabolic adaptations in children and adults after aerobic training.

Preparation of Young Swimmers for Future Intense and Voluminous Training For a long-term and continuous improvement in conditioning the swimmer has to pass through the following 4 phases, each of them aiming at appropriate age-related training objectives.

Phase 1: initiation and acquisition of a proper stroke technique (start at 6 - 8 years)

The main purpose in this phase is the acquisition of a proper basic stroke mechanism and the improvement of flexibility, reaction speed and different coordination abilities. The aerobic capacity training will gradually be introduced. As children of this age not only enjoy intermittent work and variation in physical activity, but are also susceptible to local locomotor complaints, a lot of the aerobic capacity work is performed by means of games. Besides the development of the aerobic capacity, these games also sustain the improvement of the coordination abilities, reaction time and flexibility.

In this phase the swimmer should also spend a lot of time on “non swimming” exercises (fig. 10). This non-specific training is extremely important and is meant to strengthen the swimmer’s physique to help him withstand the voluminous and more intensive build-up (phase 3) and top level training (phase 4) with the least possible risk of injuries.

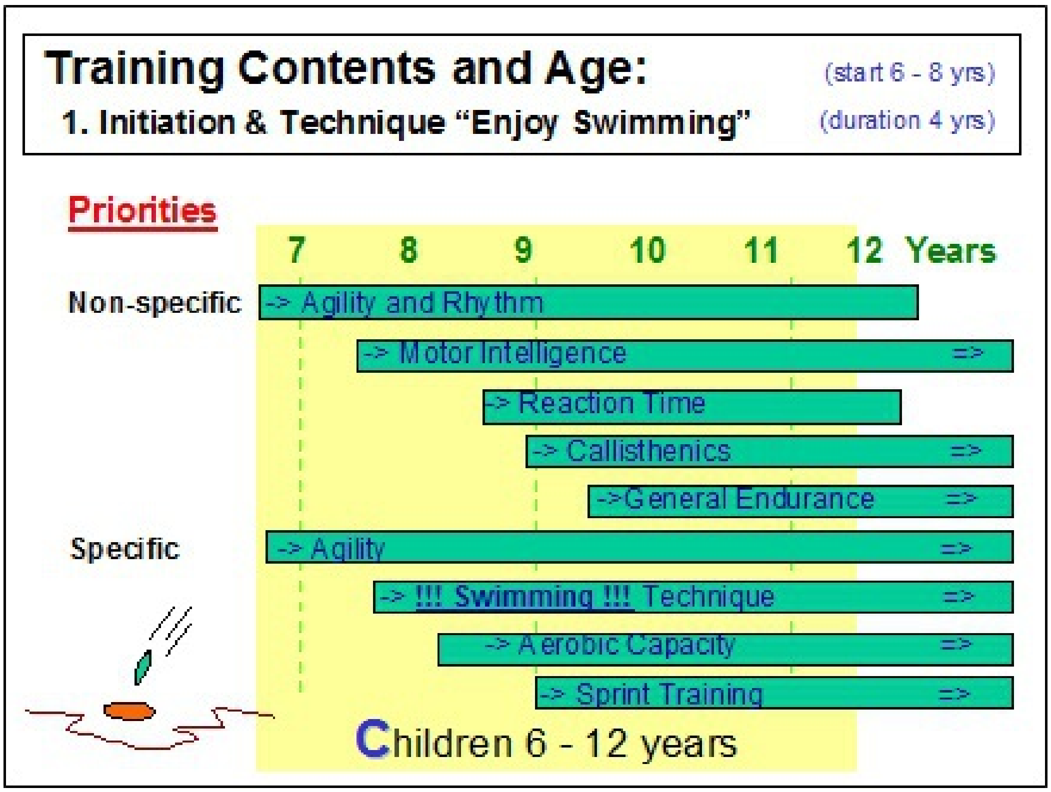


Fig. 10 Overview of the suitable types of training for the 6 to 12 year old age group swimmers.

Phase 2: *basic training (start at 10 - 12 years)*

The swimmer will spend progressively more time on his water training (specific training) without however neglecting the non-specific training forms. It is also time now to introduce new types of specific workouts (fig. 11).

Phase 3: *build-up training (start at 14 - 16 years)*

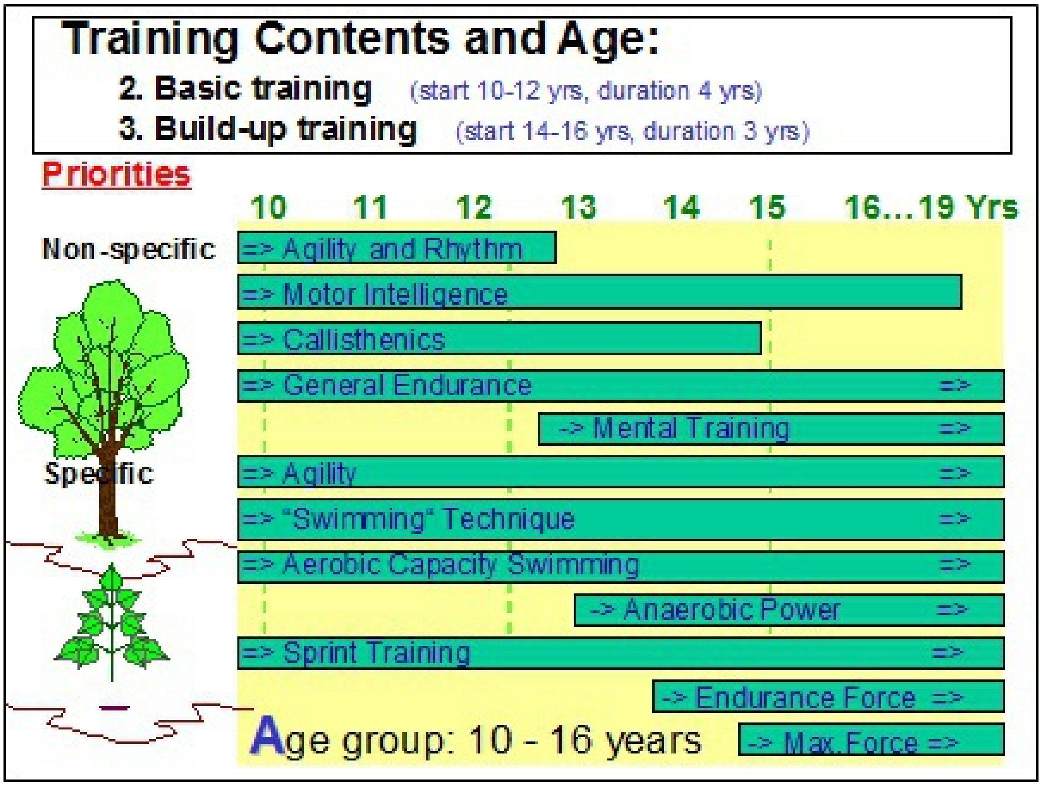


Fig. 11 Overview of the suitable types of training for the 10 to 16 year old age group swimmers.

Phase 4: *top level training (start at 17-19 years)*

Both phases 3 (fig. 11) and 4 (fig. 12) contain all aspects of training. It remains important, however, to supplement the water training with non-specific training forms and to progressively increase the training load over the years.

For more detailed information we can advise interesting specific literature on this subject, such as E. W. Maglischo's *Swimming Even Faster* and K. Wilke and O. Madsen's *Coaching the Young Swimmer* .

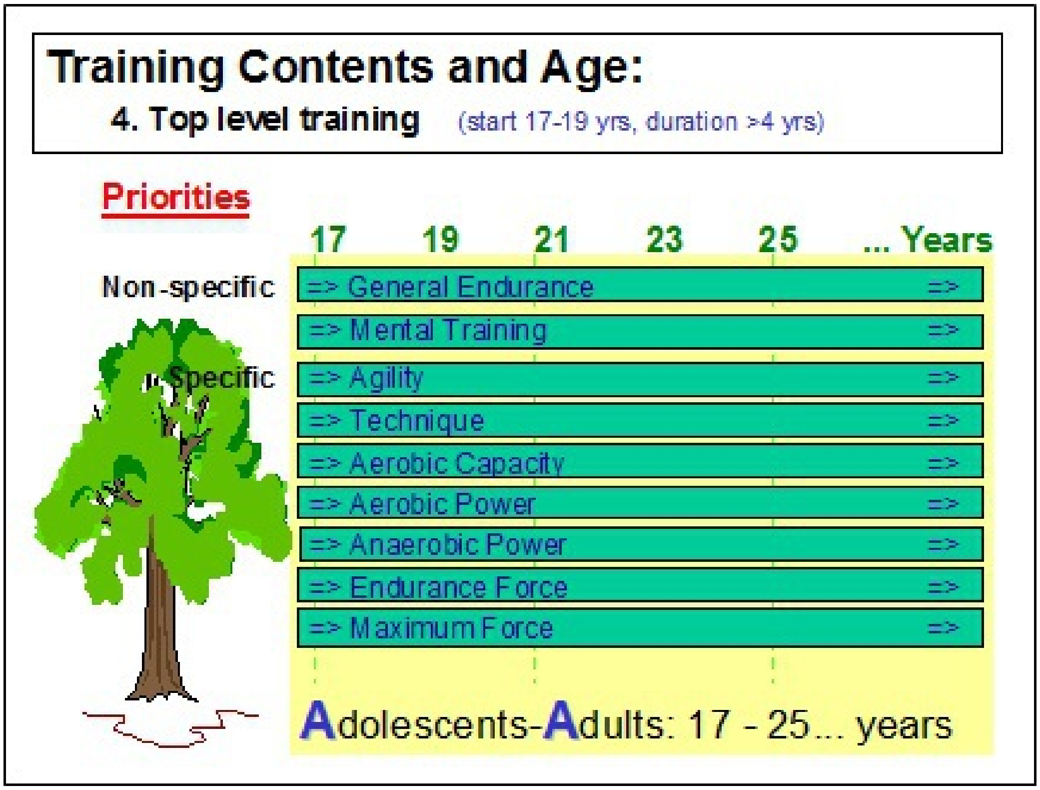


Fig. 12 Overview of the suitable types of training for swimmers aged 17 and older.

The next table shows the number of training units per week of the different types of exercises according to the swimmer's age (tab. 8).

Instructions for the use of table 8:

1. For age group swimmers with a high level of conditioning the number of training units per week can be higher than indicated in table 8. In this case we will rather increase the number of endurance capacity sessions and only occasionally add an anaerobic capacity training unit (see following point)

2. The physiological profile of the swimmer rather than his performance in competition determines the volume and frequency of the different types of training per week. Indeed, to guarantee long- term progression a coach should not be misled by temporary good meet results but gear the types of training and training volume to the swimmer's physiological profile. Swimmers who are aerobically and anaerobically strong, can, of course, sustain more training sessions per

week or train for a longer period than members of their peer group who are not as strong. Nevertheless, too much training at a young age may, in virtue of the principle of declining training return (see Declining Return on Training Investment), leave little room for improvement as the swimmer gets older

Types of Training	Age Group							
	5-8	9-10	11-12	13-14	15-16	17-18	19-20	20+
Training units per week	2-3	3-4	4-6	5-8	5-8	5-10	6-11	6-11
(non-specific units/week)	(1-2)	(2)	(2)	(2)	(2-1)	(2-1)	(1)	(1)
In water								
Endurance (Type I)	1	1-2	1-2	2	2	2-3	2-3	2-3
Capacity (Type II)	-	-	1	1-2	1-3	1-3	1-3	1-4
Anaerobic Capacity			1 (+)	1	1-2	1-3	1-3	1-4
Aerobic Power					1(+)	1	1-2	1-2
Anaerobic Power				1 (+)	1-2	1-2	1-3	1-3
Sprint Training	1	2	2-3	2-3	2-3	2-4	3-5	3-5
Technique Training	2-3	3	2-3	2-3	2-3	1-3	1-3	1-3
Dry-land								
Maximal Force	1g	2g	2g+ 1(+)	2g+ 1(+)	1g+2	2	2-3	2-3
Endurance Force (g*)								
Explosive Force								

Tab. 8 Number of training units per week of the different types of exercises according to the swimmer's age. (g*) For young swimmers we advise gymnastics to strengthen their constitution. (+) Trainable under certain conditions; see text.

3. Experiments have also shown that it is not the weekly volume of anaerobic work but rather the number of training units per week (frequency) in which such exercises occur, that determines the improvement of the anaerobic conditioning. Significant anaerobic adaptations take place even with a very low volume of anaerobic work.

This means that anaerobic training can be programmed at a young age, but only on the condition that:

* its volume per training session remains very low (total 100 - 200 m)
* the extensive work (95% of weekly volume) is really done at a very low intensity. Young swimmers are very motivated to swim fast or to compete with each other during nearly every training session. If the coach does not manage this attitude, many unexpected adaptations may occur which could reduce the potential for improvement at a later age and even have some adverse effects on the young swimmer's health swimmer's health

Older sources in the literature advise us to start with anaerobic training at a later age (see + in tab. 6) but, if both of the above mentioned conditions are met, there is, in fact, no scientific/clinical reason for delaying it (see (+) in tab. 6 and 8 above)

4. The table is based on the chronological age of the swimmer. However, in children, the chronological age can differ from the biological age (maturity). It is common to see a 6 foot 12 year old boy standing on the starting blocks next to another boy of the same age who is only 4 1/2 feet tall. Therefore, it is the measured or estimated biological age rather than the chronological age that will determine the selection of the different types of exercises

5. The frequency of the different types of training per week also depends on the objectives of the different periods of the year plan. Both types of strength training (ENST and EXST) will not be programmed during the base training period but only during the competition training period, which, as mentioned before, starts 2 or 4 weeks earlier on land than in water, even if the swimmer is physiologically ready to handle the workouts

Chapter 3 METABOLIC ACTIVITY DURING SWIMMING

AEROBIC AND ANAEROBIC METABOLISM: DESCRIPTION AND SIGNIFICANCE FOR SWIM TRAINING

The muscular system's performance capacity is determined principally by the amount of energy made available for muscular activity. The more energy that is made available per second, the harder (more intensely) the muscle can operate. If, for any reason less energy is made available, the intensity of muscular activity will decrease, forcing the swimmer to slow down.

The production of this energy occurs in the muscle cell and is transmitted to the working muscle elements via ATP (adenosinetri phosphate). The working muscle captures the energy carried by ATP, converting ATP at that point into energy-poor ADP (adenosinedi phosphate) or AMP (adenosinemono phosphate).

ATP is present in the muscle in only very small quantities (5.5 mmol/kg of muscle measured while wet; Dawson 1983) and must, therefore, be supplemented during exercise. If this does not happen, the muscle will very quickly experience an energy deficiency, forcing the swimmer to interrupt or slow down the exercise until sufficient energy is again available.

Three different biochemical processes can recharge energy-poor ADP to energy-rich ATP (rephosphorylation of ADP, fig. 13). Two of these processes can occur without requiring oxygen and are therefore called *anaerobic metabolic systems;* one is the anaerobic alactic energy production system and the other is called the anaerobic lactic energy production system. The third reaction requires oxygen to generate energy-rich ATP. We call this the *aerobic metabolic system* .

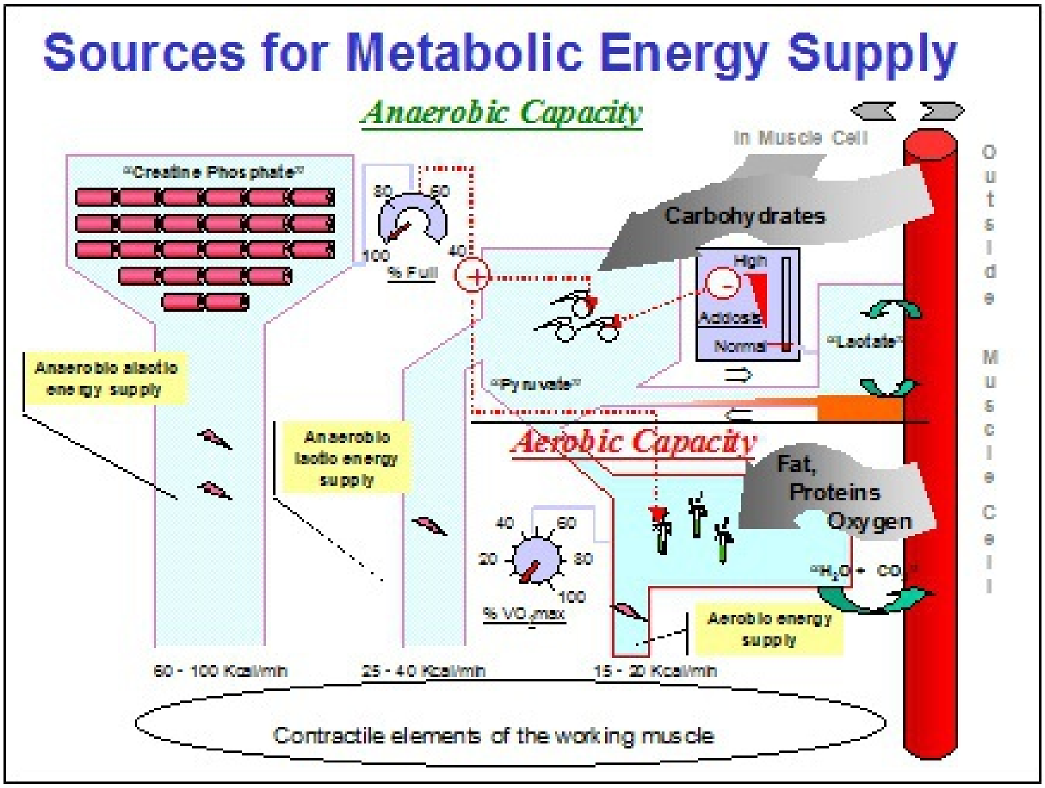


Fig. 13 Schematic representation of the 3 energy producing systems that provide for the manufacture of energy-rich ATP.

From a metabolic point of view (assuming no interaction of technique and body stature), swimming a best time is proportional to the total energy (Etot) supplied by the 3 energy delivery systems.

$$E_{tot} = E_{AL} + E_{LA} + E_{O_2}$$

E_{AL} - anaerobic alactic energy

E_{LA} - anaerobic lactic energy

E_{O_2} - aerobic energy

This means that the more energy that can be delivered per second by the 3 energy systems, the better the performance in competition. It is important to state that the highest total delivery of energy (Etot) is not automatically obtained by maximizing the energy producing capacity of each of the 3 systems. Indeed, to achieve the highest total energy (Etot), each system has to be trained or improved in such a way that it does not inhibit the maximal energy delivery of the other systems.

We must also take into account that a swimmer with a better technique and body stature (such as body length) will achieve a faster time from an equal production of energy. Experimental research showed, for example, that the taller the swimmer, the less aerobic energy is needed for a given speed (fig. 14).

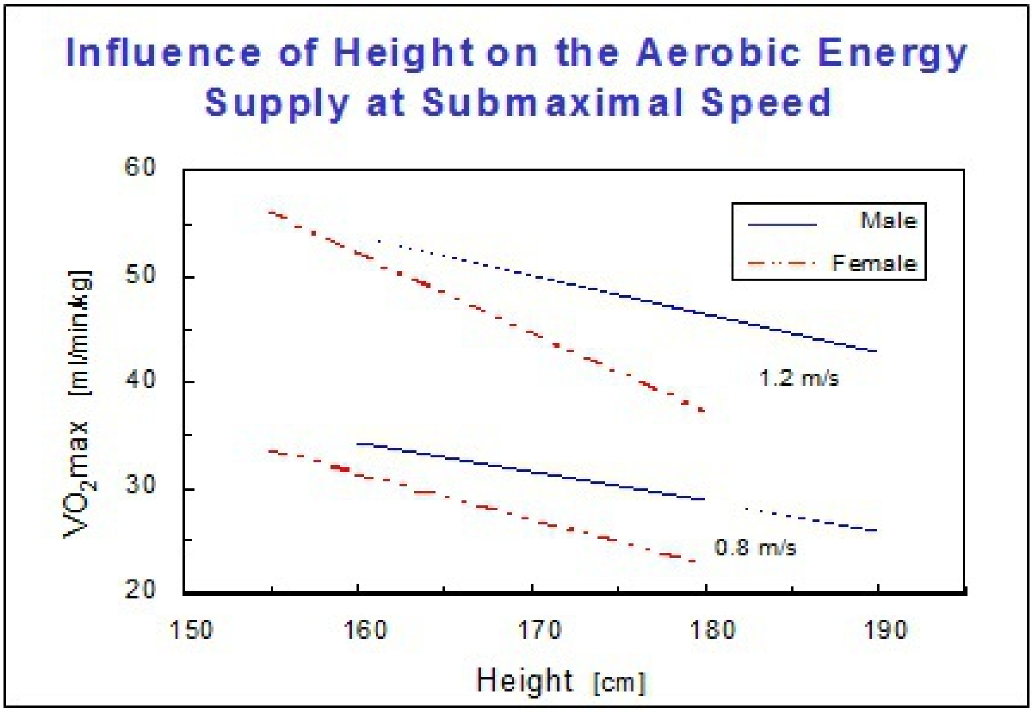


Fig. 14 Influence of body stature on the relation between speed and aerobic energy supply (based on data from Madsen 1982).

We must not lose sight of the fact that with every swimming exertion, the 3 energy delivery systems are activated. Even short sprints of 15 - 20 m, which are often regarded as pure anaerobic alactic exertions, will stimulate the other 2 systems. The contribution made by each of the 3 energy delivery systems will vary from exercise to exercise depending on:

* the duration of the effort
* the intensity of the effort
* the rest period between successive efforts

For training to be effective, it will be very important to improve each of the 3 energy supply systems with respect to:

* their importance/involvement in the competition event
* the swimmer's individual trainability of each of the 3 systems

That means that "the more the better" is not valid. For example, if a long distance swimmer trains his anaerobic capacity (breakdown of glycogen to pyruvate) too strongly in comparison to his aerobic capacity, or if he responds quickly to anaerobic training, the contribution from his aerobic capacity will be reduced during long distance competition events. Consequently, this swimmer will never be able to reach the best possible long distance competition performance of which he is capable according to the level of his aerobic capacity.

It is therefore important to balance the acquired capacities of the aerobic and anaerobic energy delivery systems in order to achieve the best possible competition performance. This will be discussed further as fine-tuning and is an especially important issue during the pre-competition training period.

The following will describe in detail the meaning, the importance and the trainability of the 3 energy supply systems. It is important to remember that, as stated in the very beginning of the book (tab. 1), it is not ONLY the conditioning that will determine the performance in competition. Indeed, despite a high total energy supply (Etot) the competition performance will always be restrained if the other determinant components such as technique, mental strength, etc. are not adequately developed. A good transfer from "internal energy" (= Etot) into "external energy", i.e. the mechanical energy that provides swimming speed (= function of the propulsion forces that overcome the water resistance), is very important and will, for top level swimmers, determine whether they win or lose. As a consequence, technique and conditioning must both be conscientiously trained.

ANAEROBIC ALACTIC ENERGY PRODUCTION (CREATINE PHOSPHATE SYSTEM)

Description

As mentioned above, ATP can be generated in various ways. One method of generating energy-rich ATP consists of transferring 1 phosphate (P) from creatine phosphate (CP) directly to the energy-poor ADP. Energy-rich ATP is thus obtained again and only creatine (C) remains as residue. No lactate is produced during this process and no oxygen is required. This is why this system is called the *anaerobic alactic energy system.*

$$ADP + CP \Rightarrow ATP + C$$

Compared to the other 2 energy production systems the anaerobic alactic system or creatine phosphate system:

* delivers energy immediately
* produces the most energy per second (60 - 100 Kcal/min or 250 - 420 Kj/min; R. Guillet 1980)
* can work at maximum level for only a few seconds

This form of energy supply is analogous to energy that is delivered by batteries. As soon as it is switched on, the battery provides energy. The more energy that is required from the battery, the sooner the energy supply will be drained. The residue, creatine or the empty battery in the example, is rechargeable after the exercise.

Significance for Swimming

This production system is very suitable for explosive exertions, such as short sprints or starts, which require an explosive energy supply. Since the creatine phosphate content in the muscle is limited, it can only provide energy for a few seconds and it will be necessary to switch fairly quickly to another form of energy production. On average, human beings have at their disposal between 11 and 23 mmol of creatine phosphate per kilogram of muscle weight (Dawson 1983, Di Prampero 1981), which provides just enough energy to sustain a maximum effort for about 4 seconds.

However, a poor sprinting speed cannot usually be traced back to a shortage of creatine phosphate. In most cases, it should be attributed to a deficiency in strength, coordination and technique.

Trainability - (low)

The supply of creatine phosphate depends on the bulk of muscle and the type of muscle fiber. The fast twitch muscle fibers contain more creatine phosphate than the slow twitch muscle fibers (fast twitch: 22.0 $\pm$ 2.0 and slow twitch: 13.2 $\pm$ 1.3 mmol/kg measured while wet, Meyer 1982). A large quantity of fast twitch fibers per kg of muscle thus implies a greater amount of creatine phosphate per kg of the total muscle weight.

To what extent training can change the proportion between fast twitch and slow twitch muscle fibers still remains a heated point of debate. It is known that the type of muscle fiber is determined by the maximum frequency of stimuli discharged to the muscle by the motor-neuron during an exertion (Freund 1983, Staron 1997, Stephens 1977). Fast twitch fibers have motor-neurons that can stimulate the muscle with very high frequencies while the motor-neurons of the slow twitch muscle fibers maintain rather low maximum frequencies.

Sending high frequencies of nerve impulses to the muscle via electrostimulation can, over a period of several weeks, make slow twitch fibers evolve into fast twitch fibers (Bacou 1996, Barjot 1998, Buchthal 1980). However, it should be noted that, when the new fast twitch muscle fibers return to the control of their proper motor-neurons, which are only able to send muscle stimulation impulses at a low frequency, they revert once again to slow twitch fibers. Furthermore, because of the fact that most motor-neurons can barely be urged to vary their frequency pattern, the “new fast twitch fibers” are not going to use their newly acquired capacities (explosiveness, etc.) and are, in time, going to revert back to slow twitch fibers.

There exists only a small percentage of motor-neurons which can be trained to alter their maximum stimulation frequency. If we wish to increase the reserve of creatine phosphate we will have to rely upon a subgroup of the fast twitch muscle fibers (the fast twitch oxidative fibers or fast twitch muscle fibers that utilize oxygen) which can be stimulated by this small percentage of motor-neurons.

Another possibility to increase the CP reserve is to increase muscle mass. However, as long as the proportion between slow twitch and fast twitch muscle fibers remains the same, the quantity of creatine phosphate per kg of muscle also remains unchanged and, as a result, we can expect only a slight return. An improvement in performance via an increase in muscle bulk can often be traced back to a gain in strength, which in the water can be converted into speed. This, however, can be expected only as long as the disadvantages of an increase in muscle bulk do not offset the gain in strength.

ANAEROBIC LACTIC ENERGY PRODUCTION

Description

Once ADP can no longer be recharged sufficiently by creatine phosphate into ATP, more and more energy-poor ADPs will be available. These ADP molecules will activate the 2 remaining systems for delivering energy.

We will start to discuss the system that as first and without using oxygen (anaerobic) is going to convert energy-poor ADP into energy-rich ATP.

In the *initial phase* glycogen (= carbohydrates which are stored in the liver and in the muscle) is broken down into pyruvate. For each pyruvate that is liberated in this process, a very small quantity of energy (used to form ATP) is released. Since this process occurs very quickly and in many locations simultaneously, the total quantity of energy production per second is high (lower than that of the anaerobic alactic system but much higher than that of the aerobic energy supply system). Glycogen is the only fuel that can be used for this process, which is, therefore, called glycolysis (from the Greek "lysis" which means to break down).

In a *second phase* , pyruvate will accumulate if not enough oxygen is available to eliminate it. This accumulation indirectly slows the activity of the glycolysis (the initial phase). In order to prevent this inhibition, pyruvate is converted to lactate. Because lactate is produced, this anaerobic system is called the anaerobic lactic system.

<Phase 1> - **Glycogen + ADP => ATP + pyruvate** (= glycolysis)

<Phase 2> - **pyruvate** <=> **lactate**

Formulas only describe the biochemical reactions qualitatively and do not reflect the quantity of substrates

Considering both phases as the anaerobic lactic energy supply, it:

- produces energy almost immediately

- produces lactate in addition to energy at higher activity levels
- produces less energy per second than the creatine phosphate system, but more energy per second than the aerobic system (about 2-3 times faster than the aerobic system)
- may weaken during long lasting efforts or high volume training if the glycogen supply becomes low
- is provisioned only by glycogen/carbohydrates

In contrast with the creatine phosphate energy delivery system, which does not exert any negative influence on the other 2 systems, the production of lactate is going to lead to acidification, which, if it becomes too strong, will significantly hinder further energy production. The acidosis will block the activity of important enzymes that enable the glycolysis to take place and, as a result, reduce the energy provided by the glycolysis. The athlete will then no longer have enough energy at his disposal to continue his exertion with the same intensity and he will have to slow down.

The acidosis depends strongly on the rate (= quantity per second) at which pyruvate is converted into lactate. At a low rate of conversion, the acidosis can easily be neutralized by the buffer capacity of the muscle cell. At a high rate of conversion, the buffer capacity of the muscle cell will no longer be able to counter that acidosis and the intra-muscular pH will decrease. At this stage, the acidosis becomes so pronounced that it will reduce further glycolysis activity and, consequently, the energy production by this system.

Considering figure 13, it becomes obvious that there are 2 possible reasons for an excess of pyruvate that causes a high rate of conversion to lactate:

1. pyruvate is not sufficiently eliminated by oxidation because the aerobic energy supply system is insufficiently developed
2. the production of pyruvate is very high

Moreover, this figure also makes clear that a very high capacity to produce pyruvate (= the anaerobic capacity) is of practically no benefit if the capacity of the aerobic energy supply system to use pyruvate (= the aerobic capacity) is too low. In such a situation, one will very quickly, and even at low levels of glycolytic activity, achieve a high accumulation of pyruvate, which will, even at a low swimming speed, inevitably induce a high conversion of pyruvate to lactate and a high acidosis.

Once again, to achieve the best possible performance in competition, the aerobic and anaerobic capacity must both be well-balanced. The idea “the more the better” is not valid with respect to the anaerobic capacity.

Significance for Swimming

The anaerobic lactic energy delivery system could be thought of as an auxiliary system; this system takes care of an energy shortage when the body has insufficient energy at its disposal to continue the intensity of the exercise. The percent of energy contributed by the anaerobic lactic system is largely determined by the duration of the maximal effort.

The longer the competition distance, the more careful the swimmer must be not to induce an acidosis that will slow down the further release of energy. The swimmer is thus forced to use his anaerobic lactic energy supply system at a low extent in order to avoid a reduction of the energy supply/swimming speed during the race. In a short competition event, a “collapse” of the energy supply/swimming speed can occur sooner - provided it does not happen before the end of the race -. Therefore a sprinter can use his anaerobic lactic energy supply system much more without the risk of a break down during the race.

For a 100 m race, a fast swimmer will have no problems with the onset of acidosis curtailing further energy supply if it occurs after 50 seconds. But if the same swimmer’s lactic system is utilized at the same intensity during a 200 m event, the “collapse” of his energy supply/swimming speed will happen too early and will cause him great difficulty in the second half of the race; the swimmer will almost inevitably “die” before he gets to the wall.

The real art in competition is choosing the swimming speed that will involve the anaerobic lactic system as much as possible without inducing the early (before the end of the race) onset of acidosis that will slow the release of energy.

The longer the distance, the less one can use the anaerobic lactic system. Therefore, the longer the distance in competition for which a swimmer is preparing, the less this system needs to be developed. This does not mean that long distance swimmers do not need to develop their lactic system. Experimental data (fig. 20) show that top distance swimmers utilize the lactic energy supply system better than less talented distance swimmers.

The shorter the race, however, the more important it is for a swimmer to develop a strong anaerobic lactic performance capacity and the more time has to be spent on building up this capacity.

Trainability (medium)

The most crucial step for increasing the anaerobic lactic energy supply is to improve the capacity to produce pyruvate. This capacity is referred to as the “anaerobic capacity”. It may be somewhat confusing that we do not include the anaerobic alactic energy supply into the anaerobic capacity. However, since the trainability of the anaerobic alactic system is very limited, and since its contribution to the competition performance ranges from low to very low (perhaps with the exception of a 50 m sprint), it is for practical reasons that we use the name “anaerobic capacity” to label the athlete’s capacity to produce pyruvate.

Since the conversion from pyruvate into lactate and vice versa occurs very quickly, the ratio between the muscle concentrations of lactate and pyruvate is described as being 1/1. You may therefore find in literature on the subject the label “production of lactate” for what we refer to as “the production of pyruvate”. You may further find that the units of the anaerobic capacity are expressed in “mmol of lactate per liter and per second” instead of “mmol of pyruvate per liter and per second”. This may sound confusing but is in fact quite simple since they are counterparts.

In this book, the label “anaerobic capacity” reflects the capacity to produce pyruvate and will, based on the 1/1 ratio between lactate and pyruvate, be expressed accordingly in mmol of lactate per liter and per sec.

As previously mentioned, the most crucial step in increasing the anaerobic lactic energy supply, is improving the anaerobic capacity. This capacity becomes even more important if one’s aerobic capacity increases. An increase in aerobic capacity will eliminate more pyruvate and therefore a higher production of pyruvate (anaerobic capacity) will be required to achieve a certain blood lactate concentration after exercises. We observed that, to attain 4 mmol/l of lactate in the blood, well endurance trained World class swimmers produced about 20 - 25% more pyruvate than regional level swimmers (fig. 53).

The anaerobic capacity varies greatly from athlete to athlete. The share of the different types of muscle fibers in athletes strongly determines their anaerobic capacity. Fast twitch muscle fibers show higher glycolysis activity and can produce far more lactic energy than the slow twitch muscle fibers. An athlete with proportionally more fast twitch muscle fibers is thus able to achieve a higher anaerobic capacity and to produce more lactate/pyruvate than an athlete possessing predominantly slow twitch muscle fibers.

Because training can exert only limited influence on the original proportion of fast twitch and slow twitch muscle fibers, it has been believed until recently that:

- the anaerobic capacity can only be improved to a small extent
- the only benefit of the anaerobic workouts is the achievement of high blood lactate values in competition (anaerobic power training)

However, new data show that over years some swimmers made some amazing improvements in their anaerobic capacities. These swimmers were 17 years of age or older when they were first tested and were followed for at least 4 years. During that time, their training programs were continually adjusted to stimulate the enhancement of their anaerobic capacity.

Although it takes years to increase anaerobic capacity, it can easily be reduced in a short time. A high volume training program that consists almost exclusively of long extensive (low intensity) endurance exercises over the course of 1 or 2 weeks will suppress the swimmer's anaerobic capacity. The swimmer will lose explosiveness and the capacity to accelerate during races of more than 50 m. Performance in short sprints (25 m or less) normally remains the same. If these short sprints become slower, it is usually not due to a weakened anaerobic lactic capacity, but rather to a decline in technical skill, muscle strength or neuromuscular fitness. The anaerobic capacity can be brought back to innate levels through anaerobic capacity training.

After a long and extremely extensive training period, swimmers will often experience difficulties in completing and tolerating short intensive swims. Lactate tests do indeed reveal an increase in swimming speed at low concentrations of lactate, but the lactate readings after a race drop off considerably. Competition performances over short distances are, therefore, generally disappointing. Through simulation it can be shown that faster swimming at low concentrations of lactate is not the result of improved endurance, but rather of the weakened capacity to produce lactate (anaerobic capacity).

A similar drop in anaerobic capacity can also be found when one trains anaerobically too often and too long (volume of set). This is more a result of not getting enough rest to reach super-compensation rather than of an incorrect choice of intensity. In this situation, the swimmer often also loses aerobic capacity, but after a rest period of 1 or 2 weeks both his aerobic and anaerobic capacities will return to their normal level.

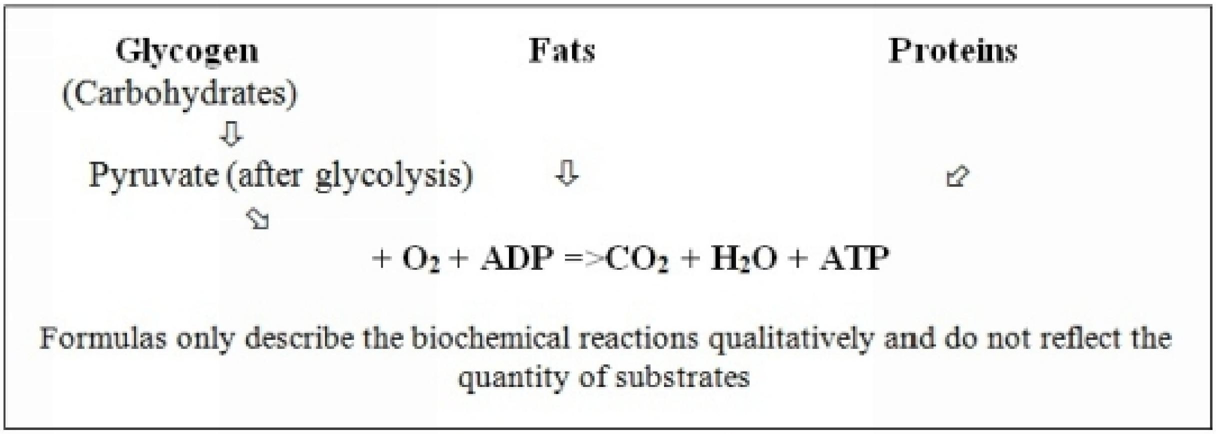
AEROBIC ENERGY PRODUCTION

Description

When an increase in energy-poor ADPs activates the anaerobic system to produce energy, it also activates the aerobic system at the same time. As the word "aerobic" indicates, oxygen (O_2) is involved with this system.

The maximum amount of energy provided by this aerobic energy delivery system is proportional to the maximum uptake of oxygen (or in other words, to the maximum use of oxygen = VO_2 max). The swimmer's maximal capacity to provide energy in this way is called the aerobic capacity.

Fats and proteins as well as carbohydrates (namely pyruvate from glycolysis) are used as fuel for the production of new energy that converts ADP to ATP. The waste products of this process are carbon dioxide (CO_2) and water (H_2O).



The specific characteristics of this system are:

* a limited production of energy per second compared with the other two sources of energy (15 - 20 Kcal/min or 63 - 84 Kj/min, R. Guillet 1980)
* a very prolonged release of energy
* its waste products have virtually no negative influence on the other two energy processes
* sluggish run-in time before the system contributes to maximum delivery of energy during exercise (about 50 - 90 seconds depending on the intensity of the exercise)

All of the factors that have an influence on the transport of oxygen are also going to influence the capacity of the aerobic system to produce energy. Therefore, the aerobic capacity will be positively influenced by:

* a very extensive network of blood capillaries (the network of the smallest blood vessels) in and around the muscle
* an increased level of hemoglobin (i.e. an increased number of oxygen carriers in the bloodstream; standard values are 14.5 mg% for women and 15.5 mg% for men)
* an increased cardiac output (i.e. the product of heart rate and stroke volume)
* a more efficient intake of oxygen in the muscle through myoglobin

Significance for Swimming

As a result of the above characteristics this energy delivery system is very suitable for prolonged exertions (delivers energy for an extended period of time, with no negative waste products, etc.). The better the aerobic system is developed, the more oxygen can be used and the faster one can swim during prolonged exercises. The aerobic capacity is therefore the major factor determining performance in long distance events.

Nevertheless, the amount of energy that is released by the aerobic system per second will never be as great as the amount of energy that can be produced by both anaerobic energy delivery systems. Since the contribution of both anaerobic energy systems decreases during long efforts, it is never possible to swim as fast during prolonged efforts as during short exertions.

Does this mean that a sprinter does not need a well-developed aerobic system?

It is a common belief that sprinters do not need the same maximum aerobic capacity as long distance swimmers. But nothing is further from the truth (except perhaps for a 50 m sprint). Sprinters must possess a strong aerobic capacity, even as strong as that of long distance swimmers. This is confirmed by:

* experimental research comparing top level sprinters with top class long distance swimmers
* the simulation program that indicates the underlying processes of aerobic and anaerobic capacity that are necessary to realize a top performance

It can now be stated that sprinters (100 m-swimmers) are more complete athletes than long distance swimmers. World class sprinters must combine a superb aerobic capacity with a superb anaerobic capacity. In contrast, top long distance swimmers are able to attain World class performances with a strong aerobic capacity in combination with a lower anaerobic capacity. It can be concluded that the better the sprinter’s aerobic capacity, the better the results he can expect in competition.

There is also another reason for swimmers in all events to achieve a strong aerobic capacity. Indeed, a good level of aerobic capacity speeds up the regeneration process between intensive workouts.

As a consequence for training, the better one regenerates from an intensive workout, the more frequently he can perform specific intensive exercises in a training cycle without any risk of overtraining. With respect to competition, faster regeneration will allow the swimmer to achieve several consecutive top performances within a few days. This becomes more and more important with the introduction of semi-finals between preliminary and final heats in competition.

Furthermore, a high aerobic capacity speeds up:

* the reconstruction of creatine phosphate from creatine (this process takes place immediately after the exertion)
* the elimination of lactate

Swimmers with a well-developed aerobic capacity are going to eliminate lactate more quickly both during and after the exertion.

We have pointed out that a swimmer with a higher aerobic capacity will eliminate lactate more quickly during an exertion. Therefore, identical lactate readings reflect a greater share of anaerobic energy delivery in swimmers with a high aerobic capacity than in swimmers with a lower endurance capacity

This finding is of great importance for handling and interpreting the lactate readings and for determining training intensities, training goals and training adjustments (see Blood Lactate).

Nevertheless, the return of the concentration of lactate to resting levels cannot be equated with a complete recovery. It so happens that many other recovery processes are required before we can speak of a complete recuperation. Thus, stiffness of the muscles, for example, days after serious exertion must be attributed, not to lactate remaining in the muscle, but rather to an incomplete recovery of the biological structures.

The aerobic system is important for the delivery of energy and thus for the *competition performance* in events of more than 50 m. But, more importantly, it also promotes the *trainability* and the *recuperative capacity* of the athlete. A swimmer with a good endurance capacity recuperates much faster and is able to work through longer and more intense training units per week without any symptoms of overtraining. A good endurance capacity also provides the ability to recover faster between competitions that are scheduled in close succession

Trainability (high)

The aerobic system is the most trainable of the 3 energy delivery processes. The biological adaptations that go hand in hand with an improvement in aerobic capacity (increase in the maximum intake of oxygen) are very complex and take place slowly. In planning, therefore, it is necessary to allow sufficient time for building up the aerobic capacity (at least 8 weeks).

Building up the aerobic capacity requires quite a lot of extensive work (see Classification of Training Exercises). Since the anaerobic capacity is a very important provider of fuel for the aerobic energy supply system, it is important to keep this anaerobic capacity at an appropriate level. Too much volume and too extensive (low intensity) workouts may decrease the anaerobic capacity to a level that does not provide enough fuel from carbohydrates to the aerobic energy supply. In such a situation, other sources of fuels, such as fat or proteins, will be broken down in order to provide fuel for the aerobic energy supply system.

For long distance swimmers who have to compete in races of more than 1.5 hours, using fuels other than carbohydrates may be of great benefit. For the classic long distance swimming events (up to 1500 m), there is no real benefit to train the athlete for switching from carbohydrates to fat or proteins. This means that, for these classic events, we will always need a sufficiently well developed anaerobic capacity in addition to a well-developed aerobic capacity.

We will, therefore always, even in the base training period of a season, combine short intensive workouts and long extensive swims in order to improve the aerobic capacity.

Experience has taught us that aerobic capacity training can very easily be combined with sprint training and with strength training that is focused on the improvement of maximal muscular force (see Strength Training). The negative interaction of extensive (slow) endurance training on sprint speed and explosiveness can then be minimized and the stroke frequency (rate) as well as the sprinting speed can be largely maintained.

In order to monitor the level of the anaerobic capacity, lactate readings will be used to feed a simulation program that calculates the status of the anaerobic capacity. Other observations such as a decrease of the swimmer’s sprinting speed (max. 25 m) and stroke frequency (especially important in breaststroke) despite sufficient regeneration from the voluminous work, may indicate that the anaerobic capacity is decreasing.

METABOLIC ACTIVITY DURING REST

In a state of rest the 3 energy delivery systems provide the energy necessary to live (fig. 15). When human beings breathe, they use little oxygen (about 200 - 300 ml/min). The low lactate readings (between 1 and 2 mmol/l) at rest indicate that only a small amount of energy is produced by the anaerobic lactic metabolism.

The total amount of energy produced at rest is needed for the cardiac and lung work but also for repairing and restructuring cell compounds in order to guarantee the good functioning of the human organism. The metabolic activity providing this energy is called the “rest or basal metabolism”.

The creatine phosphate system is fully loaded and ready to supply energy for any onset of activity.

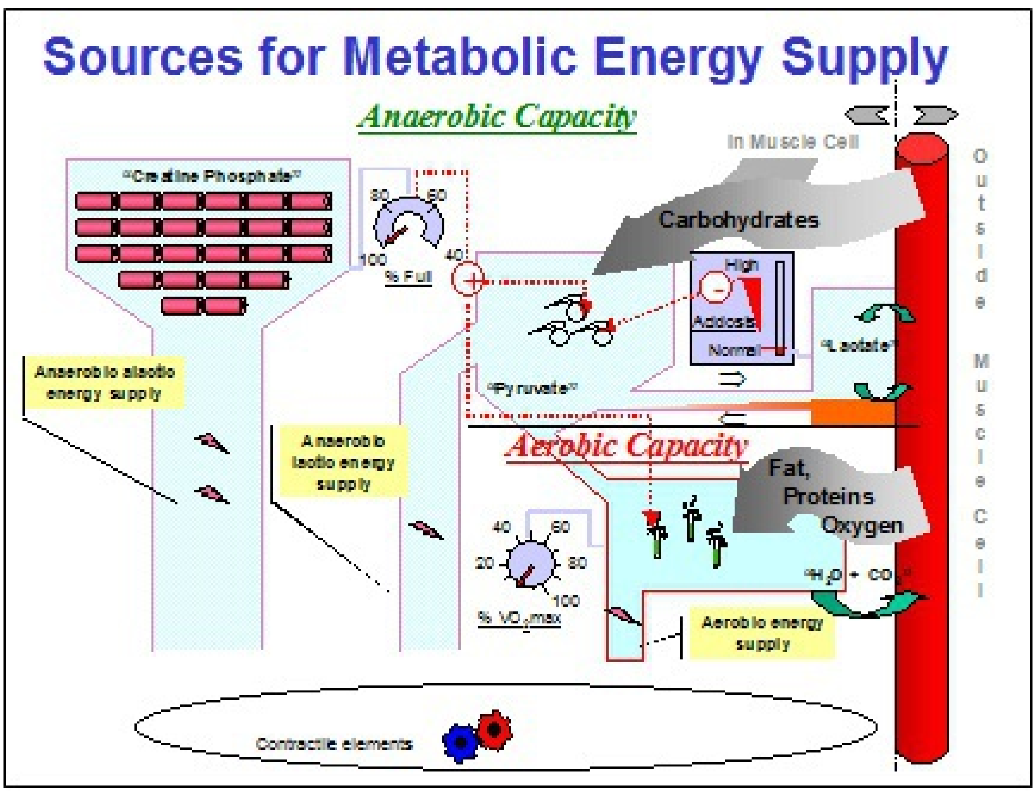


Fig. 15 In a state of rest the 3 energy delivery systems provide as much energy as human beings need to live. The low organic activity is

represented by the small cogwheels while the low energy production of the aerobic and anaerobic lactic systems is represent

METABOLIC ACTIVITY DURING COMPETITION

At the beginning of a race, the delivery of energy is immediately covered by the anaerobic alactic metabolism (the creatine phosphate system = CP). Even before this energy system is exhausted (40 - 60% of CP energy is still available), the aerobic metabolism is activated (fig. 16). But because this aerobic energy delivering system is a slow starter it cannot provide the energy necessary to sustain a maximum effort. This means that the third system, the anaerobic lactic metabolism, will come into operation and, in time, lactate will accumulate in the muscle. Some of this lactate will be burned up by the aerobic system and deliver energy. The remaining lactate stays in the muscle or flows gradually into the bloodstream, where the concentration of lactate increases.

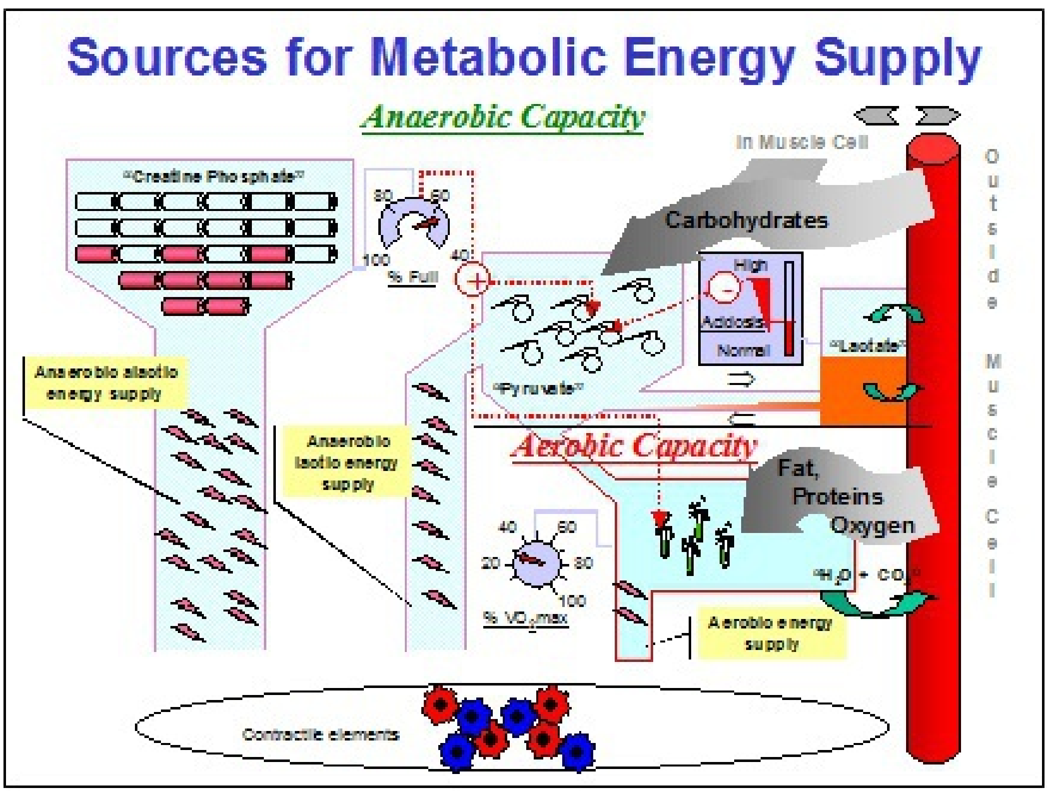


Fig. 16 Schematic representation of the activation of the 3 energy supply systems at the beginning of a maximal effort (see text).

The intensive action of the lactic system causes acidosis in the muscle. In order to avoid the damaging results of excessive acidosis, the activity of the anaerobic lactic system will decrease as the acidosis increases (fig. 17), reducing the delivery of energy and the swimming speed.

To limit excessive acidosis and thus to minimize the sharp reduction in speed, it is important for the swimmer to spread out and balance his efforts across the entire distance of the race. The duration of the effort will, therefore, also strongly determine the contribution of the anaerobic lactic system.

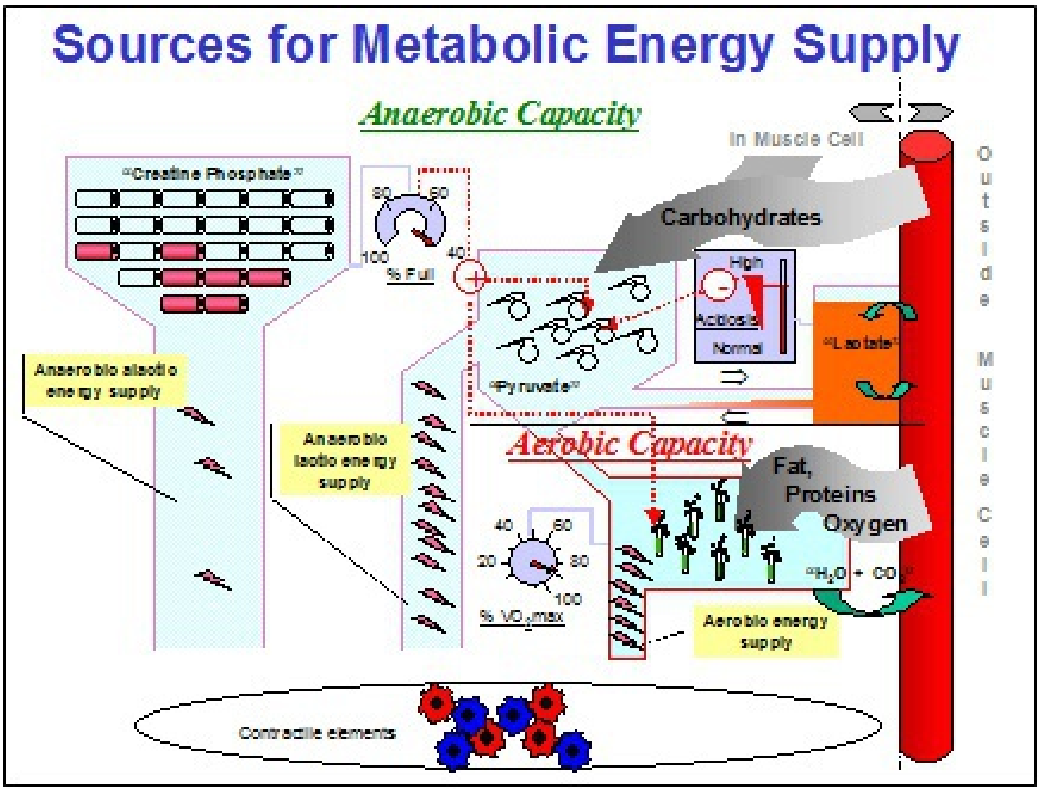


Fig. 17 Schematic representation of the activation of the 3 energy supply systems at the end of a maximal effort.

For sprinters, it is estimated that about 50% of the total delivery of energy during a 100 m competition event comes from the anaerobic lactic system (fig. 18). The remaining 50% consists of creatine phosphate and aerobic energy.

For long distance swimmers, on the other hand, the anaerobic lactic contribution is lower and the amount of energy contributed by their aerobic and creatine phosphate systems may increase to about 65%. Since the creatine phosphate share for efforts over 50 seconds is small, the shift from 50% to 65% comes about principally through a greater contribution by the aerobic delivery of energy.

The type of the swimmer (sprinter or long distance swimmer), thus, has an influence on the percentage of aerobic and anaerobic energy delivery.

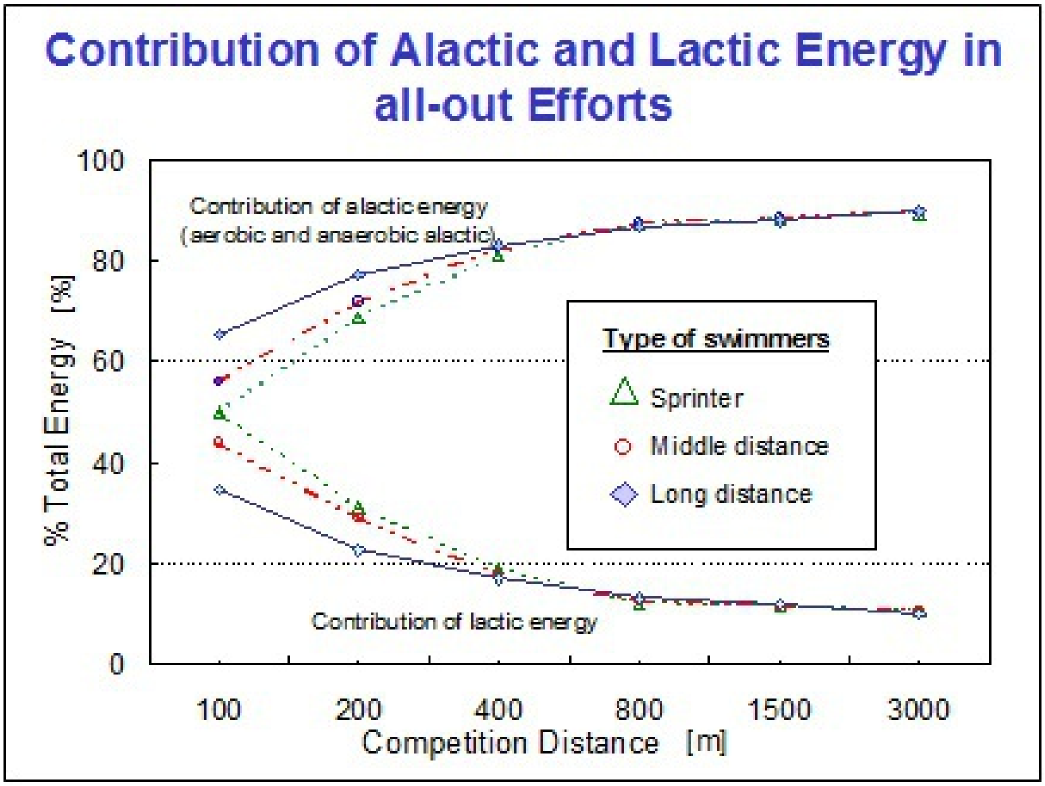


Fig. 18 Contribution of the anaerobic lactic (bottom) and the alactic (aerobic + anaerobic alactic) energy delivery during competition efforts over 50 to 1500 m for sprinters, middle distance and long distance swimmers (3000 m is a maximal test in training).

The swimmer’s performance level (international as opposed to regional level), on the other hand, has virtually no influence on the percentage of alactic (aerobic + CP energy) versus lactic energy delivery during competition (fig. 19).

As the competition distance increases, the share of aerobic energy contributed also increases, while the percentage of energy delivered by means of lactate production decreases. From a competition distance of 800 m upwards, the shares of lactic and alactic energy used are practically the same for sprinters and long distance swimmers.

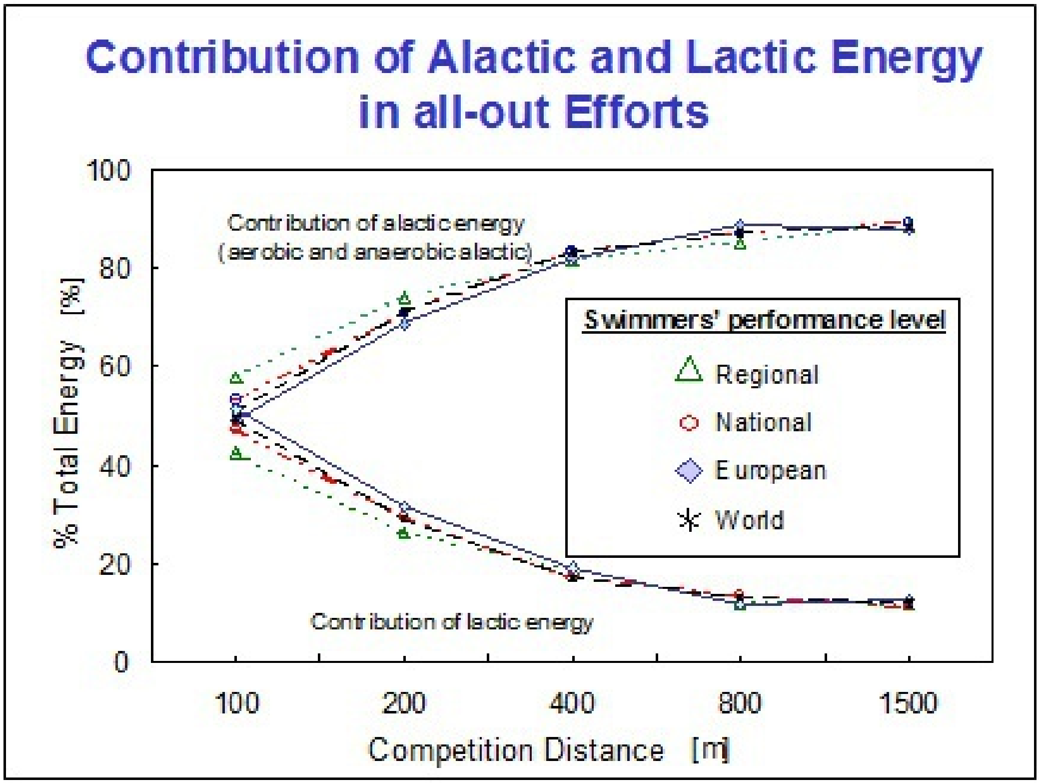


Fig. 19 Contribution of the anaerobic lactic (bottom) and the alactic (aerobic + anaerobic alactic) energy delivery during competition efforts over 50 to 1500 m for swimmers of regional, national, European or World class level.

As the variation in the alactic (aerobic + anaerobic alactic) and the lactic energy supply in an 800 m race is very small, an extra contribution of lactic energy will often determine who wins and who loses. This is why the best swimmers (European and World class swimmers) reach the highest lactate readings even over long distances (fig. 20).

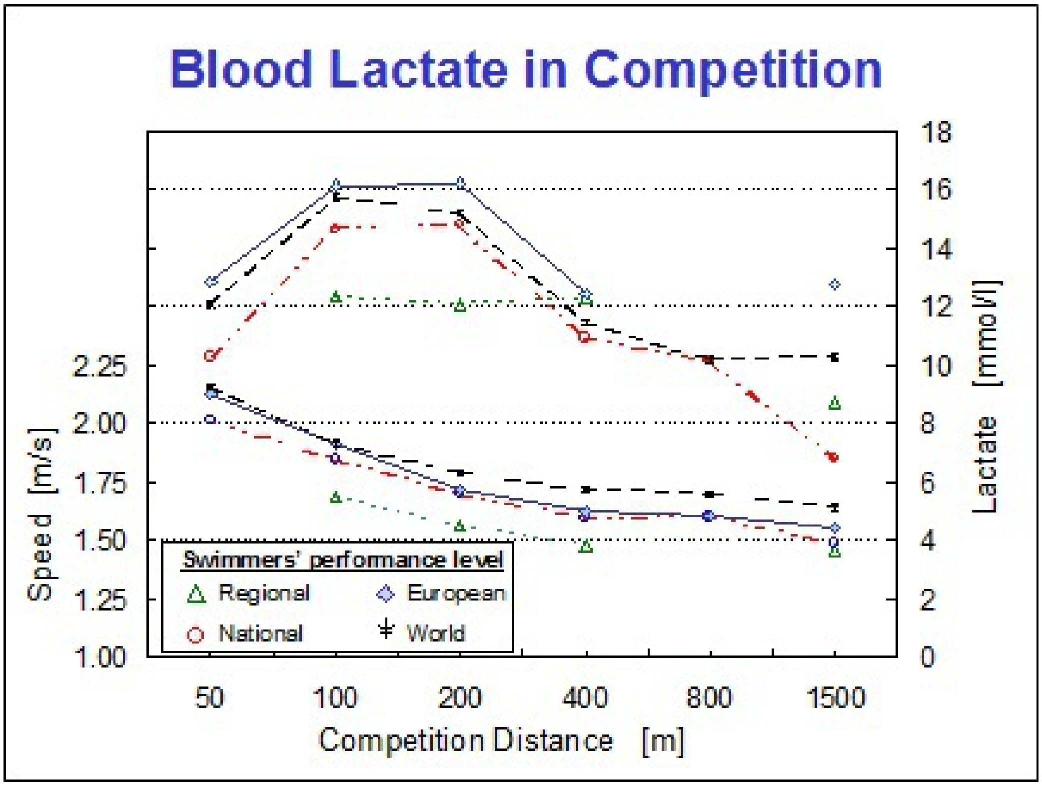


Fig. 20 Maximal concentrations of lactate measured at races from 50 to 1500 m for which swimmers of World class, European, national and regional level were specifically prepared

Furthermore, sprinters reach higher lactate readings than the middle and long distance swimmers in shorter events (fig. 21). When sprinters participate in competitions over longer distances, they are no longer able to activate their energy systems in a balanced way. Indeed, due to their high anaerobic capacity, lactate will quickly accumulate in the muscle and lead to an excessive acidosis even at a low level of their maximal oxygen uptake (VO_2 max = aerobic capacity). So, even at a low share of their aerobic capacity, they will be forced to slow down. The production of pyruvate (anaerobic capacity) will decrease and in spite of having the capacity to produce a large amount of lactate, they will, at the end of the race, have lower blood lactate readings than the long distance swimmers. Of course, these sprinters could start slower to avoid an excessive muscular acidosis and, as a result, reach the same lactate values as the long distance swimmers, but they would then have to swim at a submaximal level of both their aerobic and anaerobic capacities, which is certainly not conducive to good performances

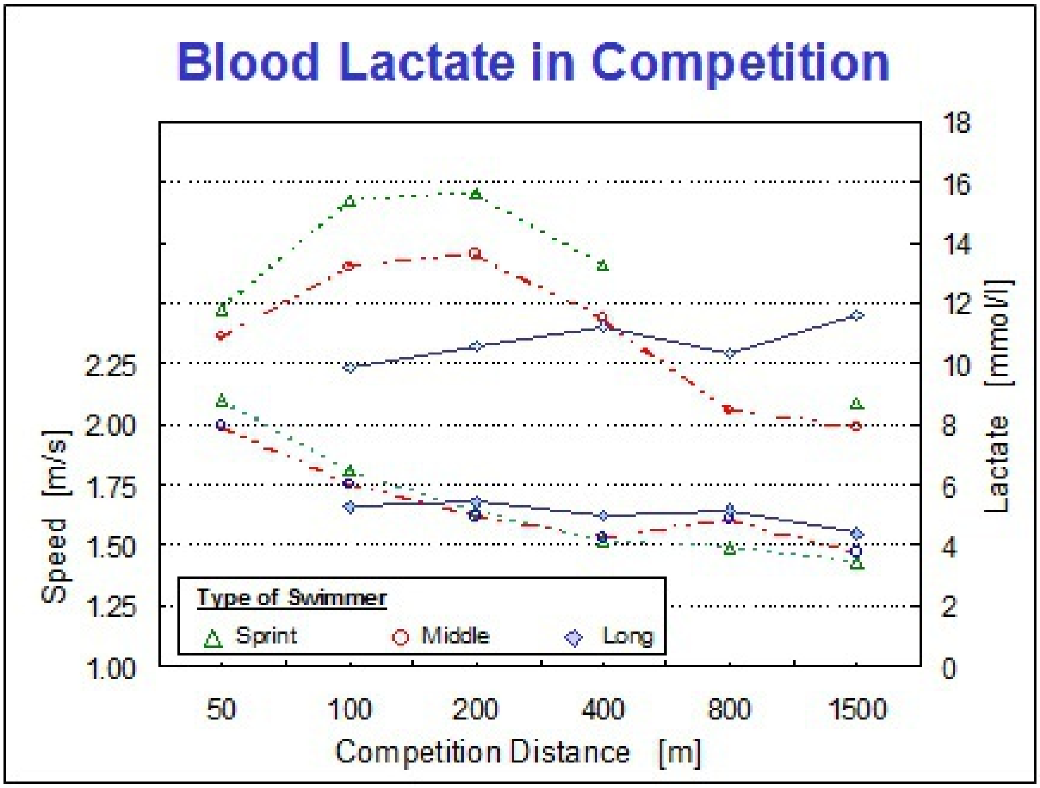


Fig. 21 Maximal lactate concentrations measured at races from 50 to 1500 m for sprinters, middle distance and long distance swimmers.

Since the maximal lactate concentration is physiologically limited, it is extremely important that swimmers be able to swim fast even at low concentrations of lactate. Referring to the metabolic scheme (fig. 13), this implies that more pyruvate must be absorbed by the aerobic energy supply system instead of being converted to lactate. We have actually observed that swimmers who achieve the best results in competition also swim fastest at low concentrations of lactate. This principle is valid for both short (fig. 22) and longer (fig. 23) swimming distances.

This points to the fact that, in swimming, endurance (= having a strong aerobic capacity - VO_2 max) has a very strong effect on the results in competition. For the distance swimmers this seems of course obvious, but the same principle is also valid for the sprinters.

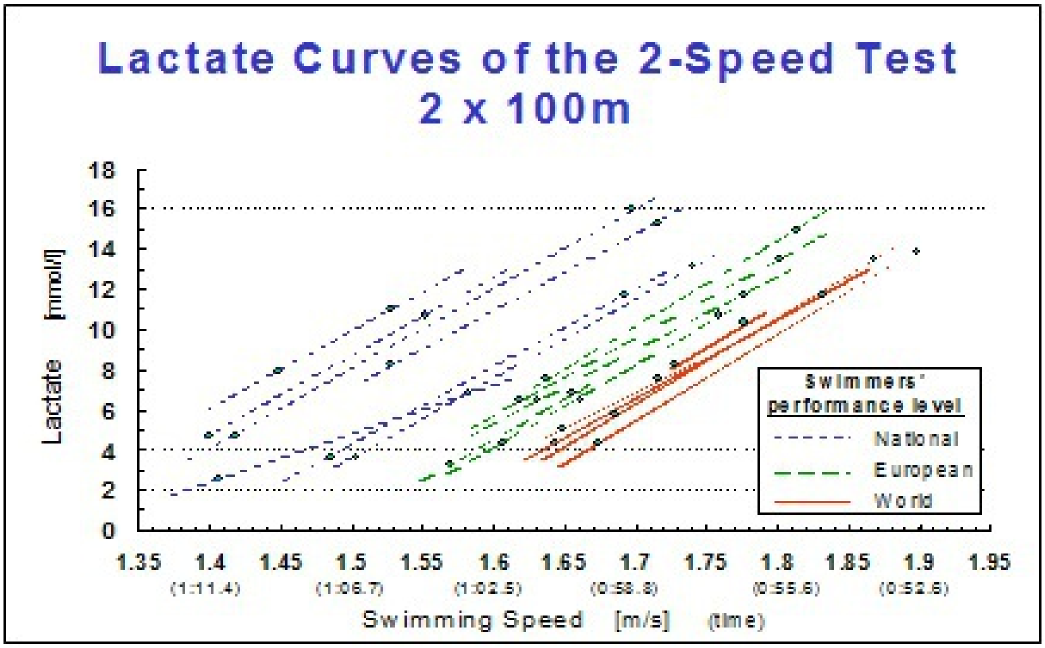


Fig. 22 The swimmers with the best competition results over 100 m achieve the fastest times at low lactate concentrations (2-Speed test, see Standard Lactate Test).

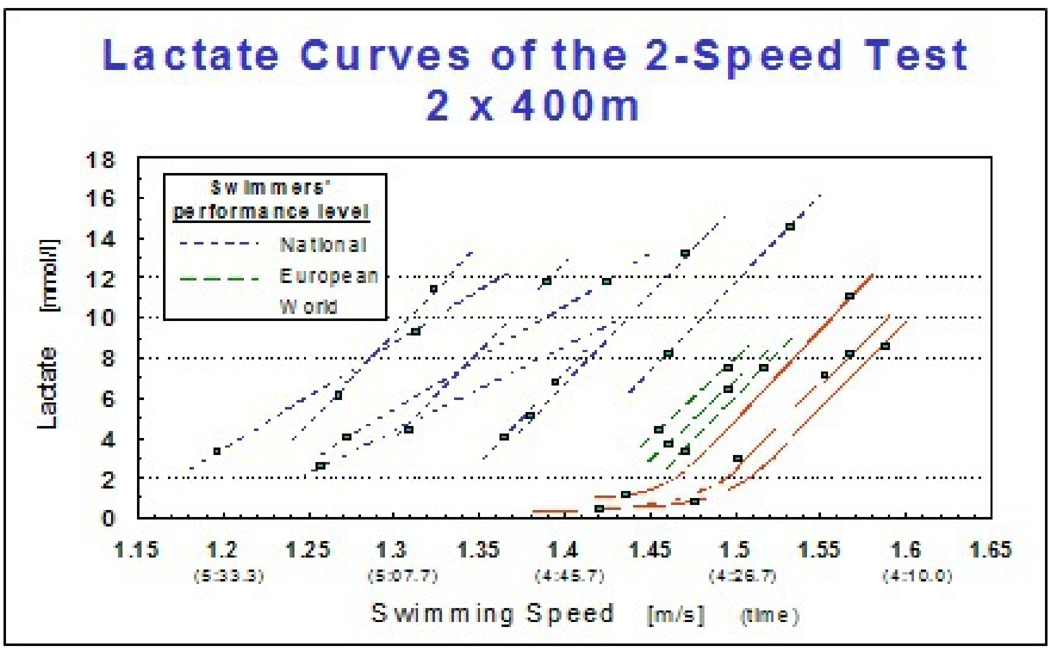


Fig. 23 The swimmers with the best competition results over 400 m achieve the fastest times at low concentrations of lactate (2-Speed test, see Standard Lactate Test).

Exceptions to this principle are possible if the swimmer can compensate for his lack of endurance by reaching higher levels of lactate (fig. 24). This compensation, however, is limited.

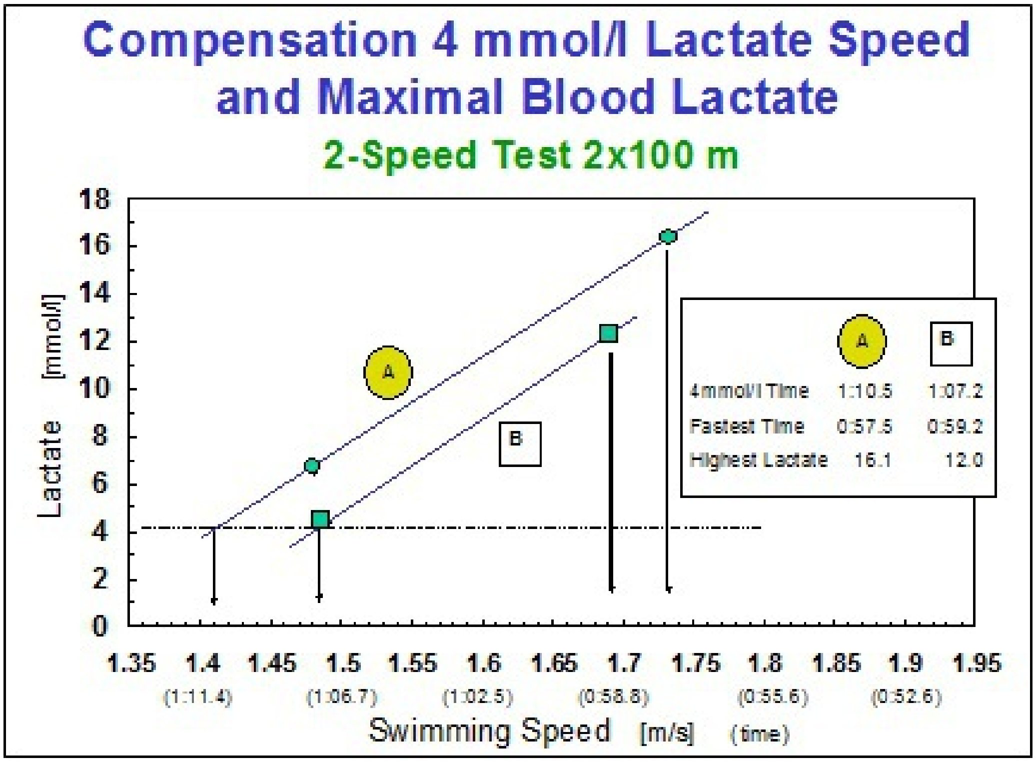


Fig. 24 Swimmer A achieves a faster time in competition than swimmer B because swimmer A compensates for a slower swimming speed at low lactate concentrations by reaching higher maximum lactate levels in competition (2-Speed test, see Standard Lactate Test).

> To summarize these findings we can confidently assert that top performances at World level are possible only on the condition that the swimmer has <u>both</u>:
>
> - *an excellent aerobic (endurance) capacity* (being able to swim fast with little lactate and reach a high VO_2max) and
> - *a very strong anaerobic capacity* (being able to produce a large amount of pyruvate/lactate)

However, it would be too simplistic to assume that the coach's job consists only of improving both the swimmer's aerobic and anaerobic capacities. Indeed, the aerobic and anaerobic capacities need to be developed in exactly the right proportion to each other in order to achieve the best performance in competition.

Examples:

- A distance swimmer with an anaerobic capacity that is too high cannot activate his aerobic (endurance) capacity to its highest level.

Result: he has a weak aerobic power and, despite a good aerobic capacity, he will register poor performances in long distance competitions

- A sprinter with an aerobic capacity that is too low will go into acidosis more quickly and will therefore not be able to activate his anaerobic capacity to its highest level.

Result: He will have a weak anaerobic power. A better aerobic capacity would enable him to burn more pyruvate instead of converting it to lactate and thus delay the acidosis in the muscle cell. With the same anaerobic, but a higher aerobic capacity, he would be able to better utilize his anaerobic capacity (higher anaerobic power) and, as a result, perform better in short events

Adjusting both the aerobic and anaerobic capacities to one another (fine-tuning) is one of the main objectives of the pre-competition training period. However, in order to achieve a maximal training effect the coach should also consider:

- the quality of the swimming technique
- the swimmer's level of endurance and explosive force

Example:

If during the pre-competition training period a coach observes, a breaststroke swimmer experiencing a loss of movement efficiency (technical problem) at higher stroke frequencies, it is too late to adjust technique. But, because the quality of the anaerobic capacity supports high stroke frequencies, it may then be of great benefit to "sacrifice" some anaerobic capacity and power training in favor of aerobic capacity and frequency exercises. For a "strong butterfly swimmer" on the other hand, such a fine-tuning would be fatal as he needs a very strong anaerobic power to benefit from his strength. For him we would rather sacrifice some aerobic capacity and power training to strive for maximal anaerobic power.

During the preparation of the swimmer, both the aerobic and anaerobic capacities must be developed *sufficiently* but also in *a mutually balanced proportion*

METABOLIC ACTIVITY DURING TRAINING EXERCISES

Continuous Swimming

This type of exercise consists of at least 15 min continuous swimming and is especially useful in combination with short intensive workouts to improve the aerobic capacity.

The activation mechanism of the energy supply systems is the same as in competition:

* first the anaerobic alactic metabolism is activated
* the aerobic metabolism starts about half-way through the energy reserve of the anaerobic alactic system
* if the intensity (swimming speed) requires more energy than both systems can supply, the anaerobic lactic metabolism will contribute to make up the shortfall

As pointed out in the description of the anaerobic lactic metabolism, the maximal involvement of this system is strongly determined by the duration of the effort. The input of the anaerobic lactic system will therefore be limited during this type of long duration exercise. Indeed, according to the swimming speed maintained during the exercise, about 87 to 97% of the energy is provided by the aerobic metabolism.

The highest continuous swimming speed that can be maintained during this effort also represents an estimation of the swimmer's endurance and can therefore be used to fix the training intensity of endurance sets for every swimmer individually (see The 30-Minute Test).

Lactate Steady State Nearly all types of continuous swimming sets lead to a "steady state" in the activity of the 3 metabolic processes:

* the oxygen uptake and the activity of the 2 anaerobic processes remain constant
* once this steady state of the 3 systems has set in it can be maintained during the entire exercise

There will be a slight increase in the lactate concentration in the beginning of the exercise at very low submaximal swimming speeds. Within the first 5 minutes, however, the oxygen uptake will increase, steadily reducing the supply of energy by the anaerobic lactic system causing the lactate concentration in the blood to drop. At this point, a balance between the production and elimination of lactate sets in, causing the blood lactate concentration to remain constant during the course of the exercise and so to enter a steady state (fig. 25).

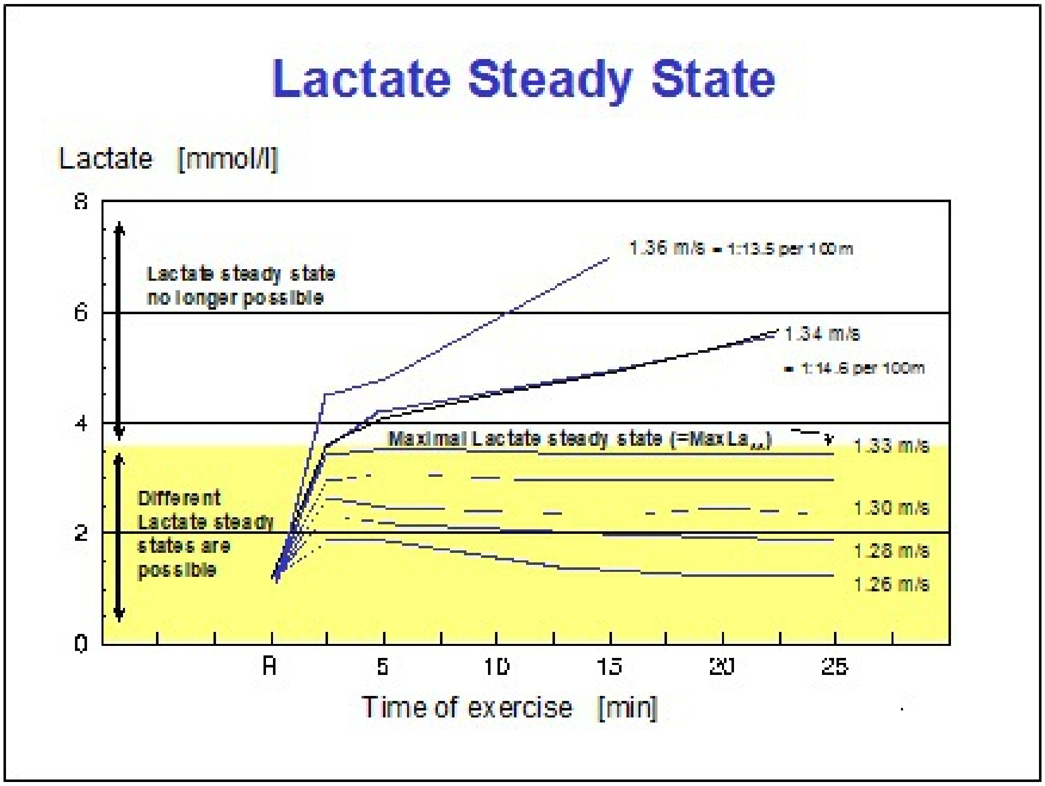


Fig. 25 As long as the swimming speed leads to a balance between lactate production and lactate elimination, it is possible to reach a lactate steady state (see text).

When the swimmer swims slightly faster, the lactate will also increase in the beginning of the exercise, but after about 5 to 7 minutes it will again settle into a lactate steady state. The lactate concentration during this exertion, however, will be higher than during the slower exercise. Thus, depending on the swimming speed, a lactate steady state can be attained at different lactate concentrations.

At even faster speeds the swimmer will reach a point where the lactate concentration will not remain constant, but instead will increase steadily. At this swimming speed, there is no longer a balance between lactate production and lactate elimination, the lactate production being higher than the lactate elimination. This causes the lactate to accumulate progressively and no longer leads to a lactate steady state. The highest lactate value that remains constant is called the *maximum lactate steady state* ($MaxLa_{ss}$) or lactate threshold. We have found lactate steady states up to 6 mmol/l, although the maximum level differs greatly from one athlete to another. Long distance swimmers have, on average, a lower maximum lactate steady state than sprinters. S.P., a World class long distance swimmer from Germany, for example, had a $MaxLa_{ss}$ lactate concentration of 2.5 mmol/l, while we once registered a $MaxLa_{ss}$ value of 9.3 mmol/l in a sprinter.

Not only in the world of swimming, but also in many other sports the hypothesis has, for quite some time, been put forth that training at the maximal lactate steady state was the magic and optimal method to improve the athlete’s endurance (aerobic capacity/power). However, nothing is further from the truth. The components that make up an athlete’s endurance are very complex and it is fantasy to think that just one single intensity is capable of developing all of the different components that are necessary for its improvement. Thus, the aerobic metabolic performance of an athlete has to be built up by means of different types of exercises and at different swimming speeds. Continuous swimming is one of those types of workouts that can be used to build up an athlete’s endurance.

Interval Swimming

In this type of exercise the total distance is interrupted by periods of rest. Interval sets can be used:

* to improve the aerobic and anaerobic capacity and power
* as sprint training
* as frequency training (stroke rate training; see Frequency Training)

Interval sets will be composed differently depending upon whether their objective is to affect the aerobic or anaerobic processes. The balance between rest, volume and intensity of the workout will determine to what extent both metabolic systems will be activated. By changing the balance between those different training elements (rest, volume and intensity) you can gradually move through the 4 types of exercises (fig. 26).

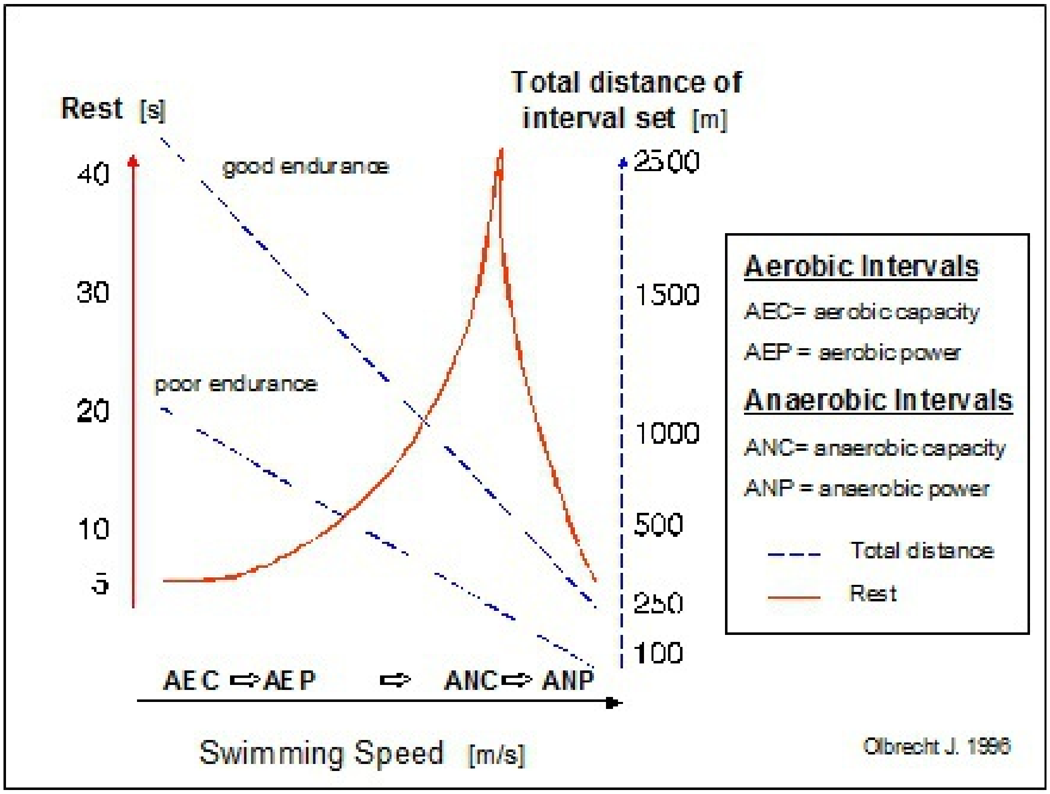


Fig. 26 Schematic representation of the transition to different types of swimming sets according to swimming speed (intensity), rest and total distance (volume). AEC = aerobic capacity, ANC = anaerobic capacity, AEP = aerobic power, ANP = anaerobic power.

You can enhance the anaerobic character of an exercise by:

* increasing the swimming speed (intensity) and/or
* increasing the rest periods (the rest is passive) and shortening the repeat distances (intervals)

The last method strongly involves anaerobic metabolism at the beginning of the exercise with only a low involvement of the aerobic system. It is therefore very suitable for swimmers with a low anaerobic capacity, such as the majority of the long distance swimmers. Indeed, these swimmers often run the risk of overloading their aerobic capacity system by swimming too fast too often in training. Moreover, the total volume of the workout will decrease as the intensity rises.

Aerobic and anaerobic power training sets are always characterized by short rest periods (5 to 15 sec) but differ in total volume. The aerobic power exercises require a high total volume whereas the anaerobic power workouts are already very effective at a low total distance (maximum 600 m per training unit).

The rest period of the exercise will not change according to the level of the swimmer. The swimming speed (intensity) and the total volume, on the other hand, will depend strongly on the swimmer's aerobic capacity: the better the aerobic capacity, the faster and the longer the repeat set can be.

Aerobic Intervals Building up the endurance (aerobic capacity and power) requires a sustained training stimulus on the same muscle group, which automatically implies high volume, submaximal intensities, little rest and little variation.

The aerobic training exercises serve principally to activate the aerobic energy supply system and, to a much lesser extent, the anaerobic lactic metabolism. If the swimming speed of the aerobic training exercises is too high, the swimmer runs the risk of ruining his super-compensation, of inducing unwanted adaptations, and even possibly of impeding the effect of other training swims. Moreover, if endurance training is swum too fast, the swimmer will not be able to continue increasing the intensity when it becomes really necessary to do so.

Depending on the type of swimmer, we can expect lactate values for the aerobic capacity and power training between 1 and 2.5 mmol/l, and 2 and 7 mmol/l respectively. For swimmers with a good aerobic and/or a weak anaerobic capacity, it is recommended to remain at low lactate values. For these swimmers, lactate values of 2 mmol/l can already represent the same metabolic stress as 5 mmol/l for swimmers with a weak aerobic and a strong anaerobic capacity.

Calculations revealed that, in order to achieve 3 mmol/l lactate values during different aerobic interval exercises (fig. 27), a well endurance trained swimmer (speed at 4 mmol/l lactate on 400 m = 1.5 m/s, i.e. 4:26.7) has to provide 25% more anaerobic lactic energy than a less endurance trained athlete (speed at 4 mmol/l lactate on 400 m = 1.333 m/s, i.e. 5:00.0). Moreover, the rate of the anaerobic lactic energy supply increases as the intervals of the sets lengthen but hardly ever changes as the rest period increases from 10 to 30 sec, even if the swimmer has to swim faster to reach 3 mmol/l. The percentage of the total energy provided by the anaerobic lactic system remains nearly the same for all sets swum at 3 mmol/l whereas the percentage of the aerobic and anaerobic alactic system respectively decreases and increases as the interval of the sets shortens.

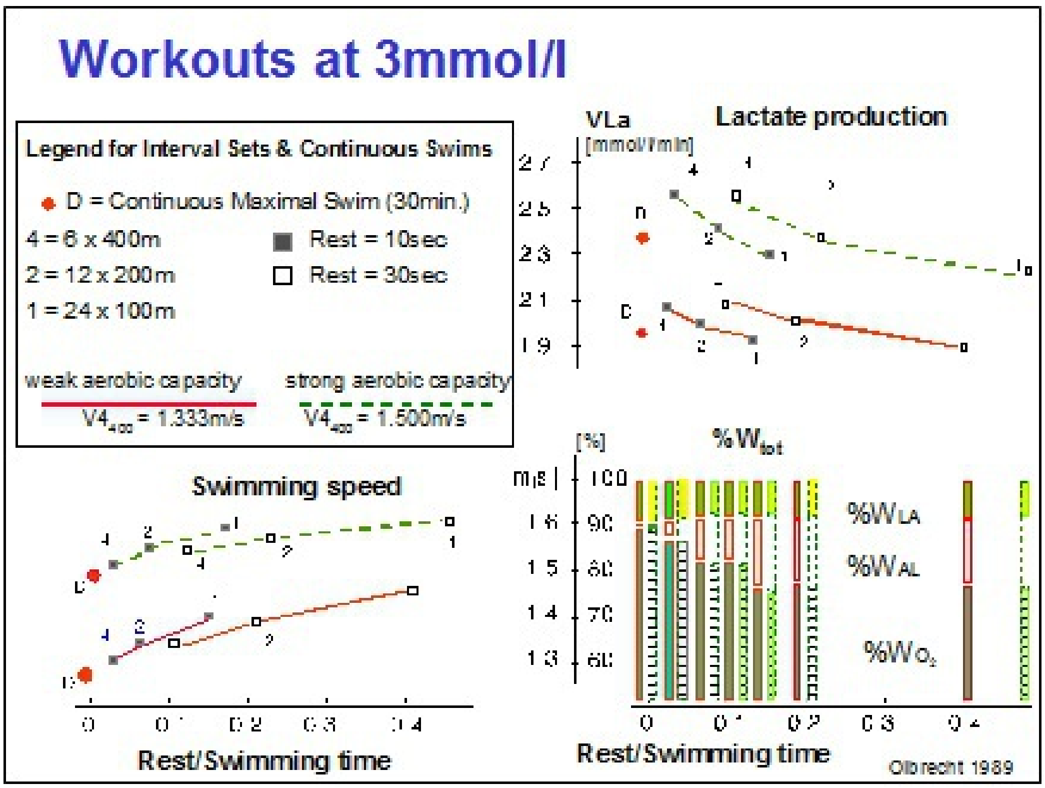


Fig. 27 Swimming speed (intensity; m/s), rate of lactate production (mmol/l/min) and partition of the total energy ($\%W_{tot}$) in anaerobic lactic ($\%W_{LA}$), anaerobic alactic ($\%W_{AL}$) and aerobic energy ($\%W_{O2}$) in different aerobic interval sets leading to a lactate value of 3 mmol/l for a well endurance trained (speed at 4 mmol/l over 400 m = $V4_{400}$ = 1.5 m/s) and less endurance trained ($V4_{400}$ = 1.333 m/s) swimmer.

During the aerobic interval sets the blood lactate concentration measured after each repetition increases within the first 5 min. During the next repeats lactate will decrease until a lactate steady state is reached (fig. 28; 6 x 200 m R 30” and fig. 44; 6 x 400 m R 30”).

It is thus not unusual to measure, for a constant swimming speed, higher lactate values after the first repetition of a set than at the end of the workout. This can be attributed to the increasing contribution of the aerobic metabolism as the exercise progresses.

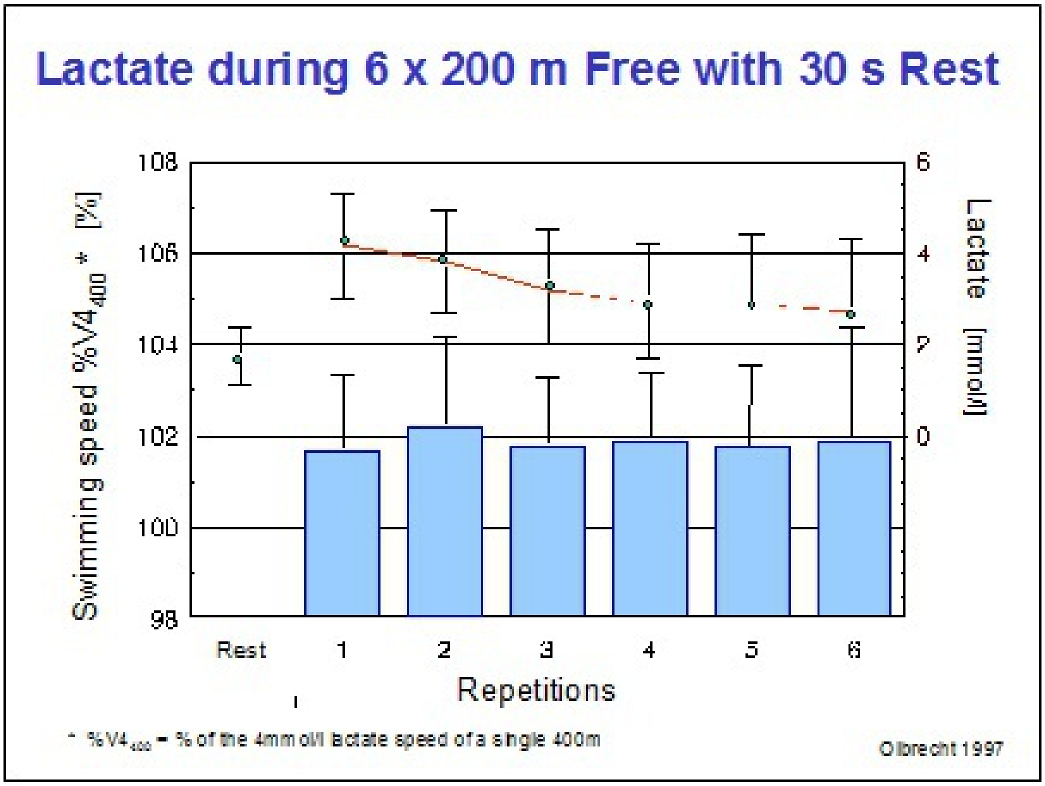


Fig. 28 Evolution of the blood lactate concentration during an aerobic interval training session (6 x 200 m with 30 sec rest).

Aerobic power workouts require a high total volume, short rest and a higher swimming speed resulting in a progressive increase of the lactate concentration over the repetitions. The exercises will be swum faster than at the maximal lactate steady state speed, which is also the case in long distance races. It is, therefore, important to plan aerobic power sets during the pre-competition training period to enable the swimmer to mobilize the maximum amount of aerobic and anaerobic energy during the race.

We have been able to determine experimentally the influence of rest and repeat length on the relation between lactate and swimming speed (fig. 29). This allows us to estimate from the lactate-speed relationship of a 400 m swim, the swimming times to be swum in training sets according to desired lactate ranges (see fig. 54).

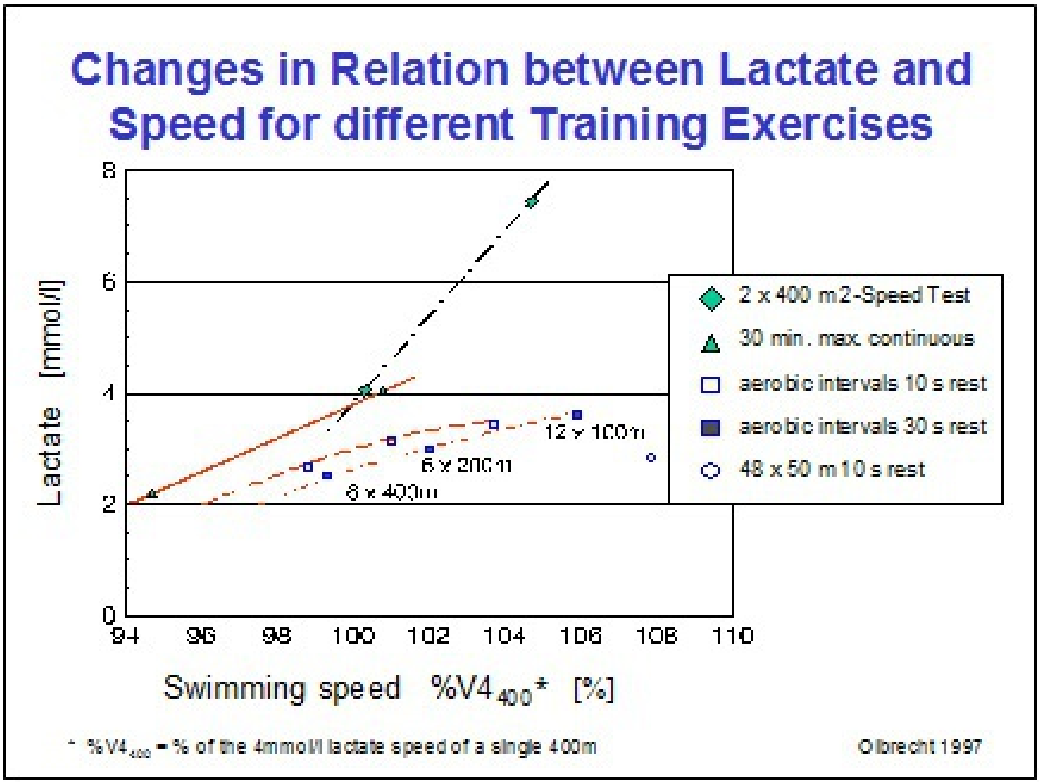


Fig. 29 Influence of rest and interval length on the relationship between lactate and swimming speed.

Anaerobic Intervals As the swimming speed increases, the anaerobic lactic metabolism will get progressively more involved, reducing the possibility of the athlete's attaining a lactate steady state. The sets will be shorter than the aerobic intervals and be swum either progressively faster or at a constantly high speed.

Another way to stress the anaerobic metabolism is to lengthen the rest periods (the rest must be passive) and to shorten the intervals (25 to 75 m maximum). This strongly involves the anaerobic metabolism at the beginning of the exercise and is the gentlest way to increase the proportion of anaerobic work in training. It is therefore very suitable for swimmers with a low anaerobic capacity, such as the majority of the long distance swimmers.

Whether the training objective is to improve the anaerobic capacity or the anaerobic power, the coach will choose one or even a combination of the

above mentioned procedures.
If it is the aim of the training to raise the anaerobic capacity, the repeats will, of course, be swum fairly fast and alternated with long passive rest periods. It is, however, important to limit the acidosis, which acts as a brake on the glycolysis. The blood lactate concentrations that can be expected after the anaerobic capacity training sets can range from 5 (for long distance swimmers) to 10 mmol/l (for sprinters).

For the anaerobic power training, the rest periods should be very short and each repeat should be swum at maximal speed (all-out) right from the start even if the swimmer is afraid of “dying” (he feels incapable of maintaining the pace to the wall). The total distance of the exercise will consequently be limited (175 to 250 m). The set may be repeated during the same training unit but only after a regenerative swim. However, sets of 500 m or more at maximal speed during 1 training session are risky for overtraining and should only be attempted by top level swimmers.

In anaerobic power sets the 3 metabolic processes are driven to their limits, inducing a high level of acidosis. As a result, lactate values, from 9 to far above 15 mmol/l can be expected. For a good execution and efficiency of the exercise the swimmer needs to be mentally prepared and fully motivated. It is therefore obvious that it is best to restrict anaerobic power workouts in training to broken sets (3 to 5 intervals of 25 to 75 m swum all-out with only 5 to 10 sec rest). For longer repeat distances it is more appropriate to plan 3 or 4 short races in a test competition. The swimmer is not expected to swim best times in these competition events, but rather to consider the races as anaerobic power training and as an opportunity to try out strategies such as going out fast, etc.

Sprint Intervals The repeat distances of the sprint exercises are short (max. 15 sec) and are swum at maximal speed with enough rest to allow full recovery. The aim of this exercise is to raise the swimmer’s basic speed (different than swimming fast in a tired state).

During these exercises the majority of the energy will be supplied by the anaerobic alactic metabolism. In contrast to what is reported in the specific literature, we observed an increase of the blood lactate concentration after several sprints with long rest breaks. This indicates that beside the anaerobic alactic system, the other energy supply systems are also activated.

But the lactate values registered after these exercises are not so high that they impede adaptations necessary for the improvement of speed.

Chapter 4 DETERMINING TRAINING INTENSITY AND CONTENT

The effectiveness of a training session depends on:

1. the correct *choice* and *sequence* of the various types of exercises within a training period (see part II, Planning and Periodization of Training)
2. the correct *execution* of the exercises to induce the desired adaptations. For the coach this comes down to carefully choosing and respecting the appropriate:

   - intensity (swimming speed)
   - volume
   - rest
   - repeat distance (interval length)

Why is the correct execution of the exercises so important?

The combination of intensity (swimming speed), volume, interval length and rest between the intervals (repeats) determines when and how the various metabolic processes are activated. The way these processes contribute to the energy supply will induce specific biological adaptations. Therefore, to bring about certain improvement in conditioning, the intensity, volume, rest and interval length of the workout have to be well-balanced to meet the objectives of the exercise.

Many years of observation, testing and evaluation have revealed that the most frequent error in training is that the intensity of the exercise is not matched with its objective. As a result of this:

1. the biological adaptations actually induced in the body differ from those desired
2. the effectiveness of the subsequent exercises is affected

For example, aerobic exercises swum too fast will not only impede the desired improvement in aerobic endurance, but will also fatigue the

swimmer unnecessarily preventing her/him from executing subsequent training exercises correctly, i.e. swimming fast when required.
Apart from training with the wrong intensity (swimming speed) swimmers often swim too much (too high volume) according to their capabilities.

Rest breaks and repeat distances (interval lengths) are quite closely related to the training objective and are practically the same whatever the level of the swimmer may be (World class or regional level; see tab. 2). Intensity and volume, on the other hand, depend very much on the swimmers' conditioning and need thus to be carefully and individually chosen.

In the following section we will consider some techniques to ensure the appropriate training intensity and volume:

1. the swimmer's subjective perception of intensity
2. blood lactate
3. heart rate
4. percentage of best times
5. 30-minute test or T30

THE SWIMMER'S SUBJECTIVE PERCEPTION OF INTENSITY

The easiest way to assess the training intensity is to ask the swimmer about his perception of the difficulty of the exercise (was the set easygoing? Was it hard? etc.). In general we can state that:

- aerobic and anaerobic capacity training are subjectively perceived as easy to moderate intensive
- aerobic and anaerobic power training, both metabolically very exhaustive, are respectively perceived as highly intensive and exhaustive

Experience, however, taught us that the swimmer's subjective perception of training intensity very much depends on the speed and volume he is accustomed to in training. Indeed, the same subjective sensation may, for example, correspond to different intensities with different swimmers or the same intensity may arouse different subjective sensations. A swimmer who often swims too fast in training will therefore, subjectively feel good during intensive training sessions, while a swimmer who rarely swims intensively will, with the same intensity, feel totally exhausted.

During a training session we measured lactate after an aerobic interval set and asked the swimmers to estimate subjectively the intensity on a scale coded from 1 to 20. This scale, also called the Borg scale, expresses the feeling of intensity in numbers from 1, which is very easy, to 20 which is extremely difficult (Borg 1982; the new version is coded from 1 to 10, also see Maglischo, Swimming Even Faster, 1993). In doing so it allows swimmers to quantify their subjective feeling of intensity.

There was only a very slight correlation between the subjective feeling of the training intensity and the actual (objective) metabolic workload measured by means of lactate (fig. 30). Indeed, we found, for example, for the level 12 on the Borg scale (more intensive) variations in measured blood lactate concentrations from 2.9 to 7.8 mmol/l. For the same concentration of lactate at, for example, 4 mmol/l the feeling of intensity varied from intense (10 on the Borg scale) to very highly intense (16 on the Borg scale).

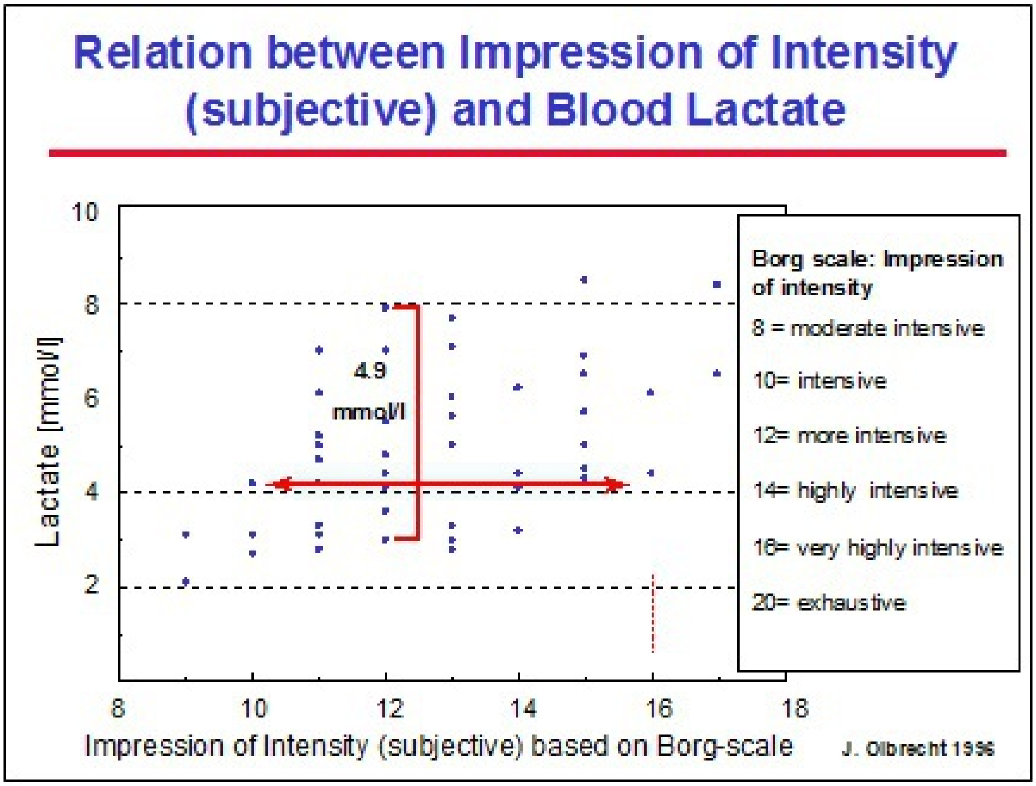


Fig. 30 Relation between the subjective feeling of intensity during a training set (using the Borg scale) and the objective metabolic workload (blood lactate concentrations).

This means, in training terms, that one and the same subjective impression of intensity easily can represent a totally different activity of the 3 metabolic processes and therefore induce different training effects. A coach who relies exclusively on the subjective impression of his swimmers to determine the intensity of the different types of training exercises could, thus, have some nasty surprises. Indeed, the planning and periodization may be correct and the swimmers' subjective experience of intensity may also match totally what the coach expects. However, if the feelings of the swimmers misrepresent the actual metabolic intensity, the training adaptations that occur may be undesirable.

Determining the real training intensity, i.e. the workload on the swimmer's organism, is thus a *conditio sine qua non* for reaching the desired training effect. The coach must therefore have at his disposal an *objective* measure to precisely determine the training intensity.

This does not mean that the swimmer’s “subjective feeling of intensity” is unimportant. Of course not, but it needs to be “calibrated” with objective measurements such as lactate. If swimmers are regularly tested they will progressively be able to match their subjective feeling of intensity with the lactate readings. They will learn to estimate quite exactly the intensity of the training exercises, and, therefore, provide the coach with much more reliable and meaningful feedback.

Conclusion:

The subjective perception of intensity:

- is very easy to use
- may differ from the real training intensity
- provides the coach with useful feedback on the completed training
- should, if possible, be complemented with an objective measure of intensity (preferably lactate)

BLOOD LACTATE

In contrast to the subjective perception of intensity, blood lactate concentrations allow to assess the real workload of an exercise on the muscles. Lactate is therefore the best reflection of the real training intensity.

Although lactate has already been used to optimize training since the 1970’s, there was little scientific evidence on the optimal test protocol or on the interpretation and implementation of the test results. Very often test results were conflicting with observations in training and/or competition or with the coach’s experience. As a result, lactate tests gradually fell into discredit. However, a lot of top coaches, who learned to understand the real meaning of lactate, consider now lactate tests as an indispensable even essential tool for maximizing the effectiveness of their training.

It is thus very important for the coach to acquire a good knowledge on the basic principles of lactate production, distribution and elimination in order to properly use lactate in training. So, before getting on with the description of the test protocols and the use of lactate in training, let’s first briefly focus on the basics of lactate.

Why using lactate?

Lactate is an intermediary product in the process of delivering energy (fig. 13) and is released through the anaerobic lactic metabolism when the energy delivered by the anaerobic alactic and aerobic system becomes insufficient. This may happen:

- in the beginning of an effort when creatine phosphate (CP) is exhausted and the aerobic system does not yet deliver enough energy due to its slow start or
- during an effort when the workload (swimming speed) is so high that the aerobic energy supply approaches its maximal capacity. In this case it is clear that lactate tests reflect the performance of both, the anaerobic lactic and aerobic energy supply systems.

The blood lactate readings, therefore, provide us with information on:

1. the activation and contribution of the aerobic and anaerobic systems
2. the performance of both systems, when combined with the swimming speed

Lactate as a Measure for Training Intensity

The use of lactate to determine the intensity of the training exercises enables us to link these intensities to the activity of the 3 different energy delivering systems in the muscle. Because these lactate based training intensities provide information on how the exercises trigger the metabolic system, it is possible to relate the induced training adaptations to the metabolic characteristics of the muscular activity during a certain training period. In time, lactate tests will provide for each individual swimmer, an insight into the biological adaptations that can be expected from the training exercises.

Relevance and Reliability of the Lactate Readings

If you want the lactate tests to provide relevant and reliable information, make sure:

- the tests are performed in the *specific sport* , even in the *specific stroke* you wish the information to be used for
- the *highest post-exercise lactate concentration* is used for the evaluation

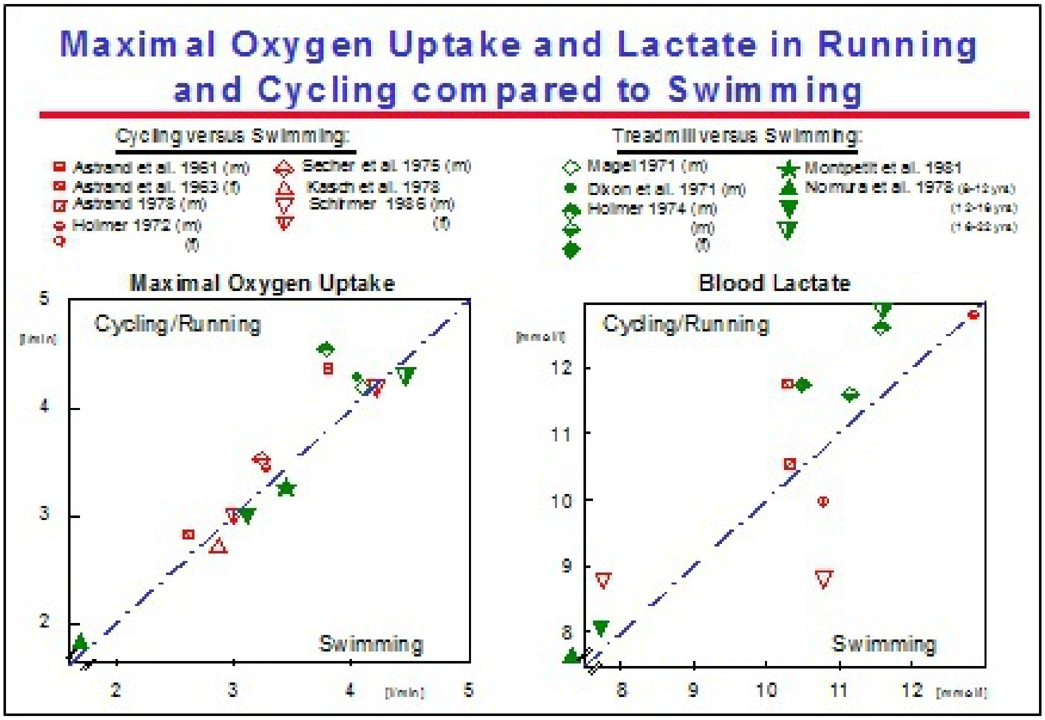


Fig. 31 Maximal oxygen uptake and blood lactate in running and cycling as compared to swimming. Plots represent mean values from different studies.

Sport Specific Testing a swimmer, for example, on a bicycle ergometer or treadmill will virtually produce no reliable information on the swim specific conditioning of that swimmer (fig. 31, 32 and 77) and will therefore provide no useful advice for swim training. Indeed, figure 32 clearly shows that there is no scientific evidence that the swimmer who has a higher 4 mmol/l workload (watt/kg) on the bicycle ergometer also swims faster at 4 mmol/l. The 4 mmol/l swim speed is an important parameter for estimating the training intensity as well as the swim performance in competition. It is quite obvious, however, that the 4 mmol/l workload on the bicycle will not provide any valid indication for competition performance or training intensity in swimming.

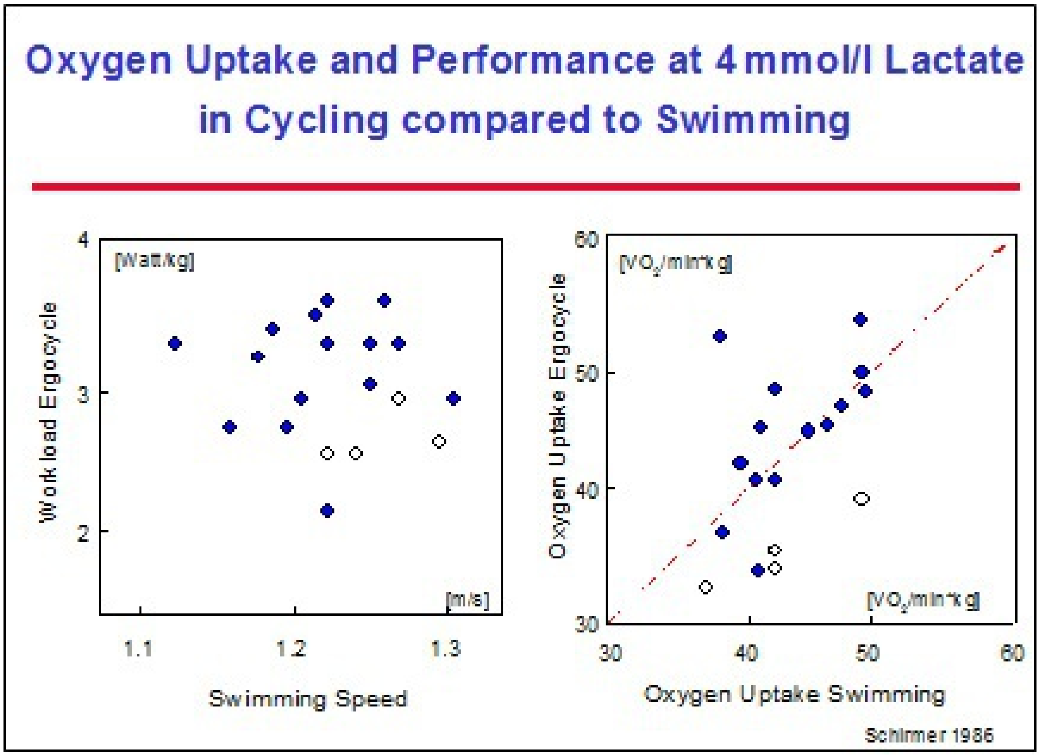


Fig. 32 Oxygen uptake and performance at 4 mmol/l lactate in cycling as compared to swimming. Plots represent individual values.

Stroke Specific Likewise back-, breaststroke and good fly swimmers will have to be tested in their personal best stroke if the coach wants to use the lactate test readings for optimizing respectively back- breaststroke or fly training. Indeed, although changes in 4 mmol/l speed (V4) in freestyle are roughly correlated with the changes in V4 in back-, breaststroke and fly, they cannot always be transferred to nor observed in the other strokes (fig. 33). The correlation is too weak to reliably assess the aerobic endurance (V4) in back-, breaststroke or fly based on the measured V4 in freestyle.

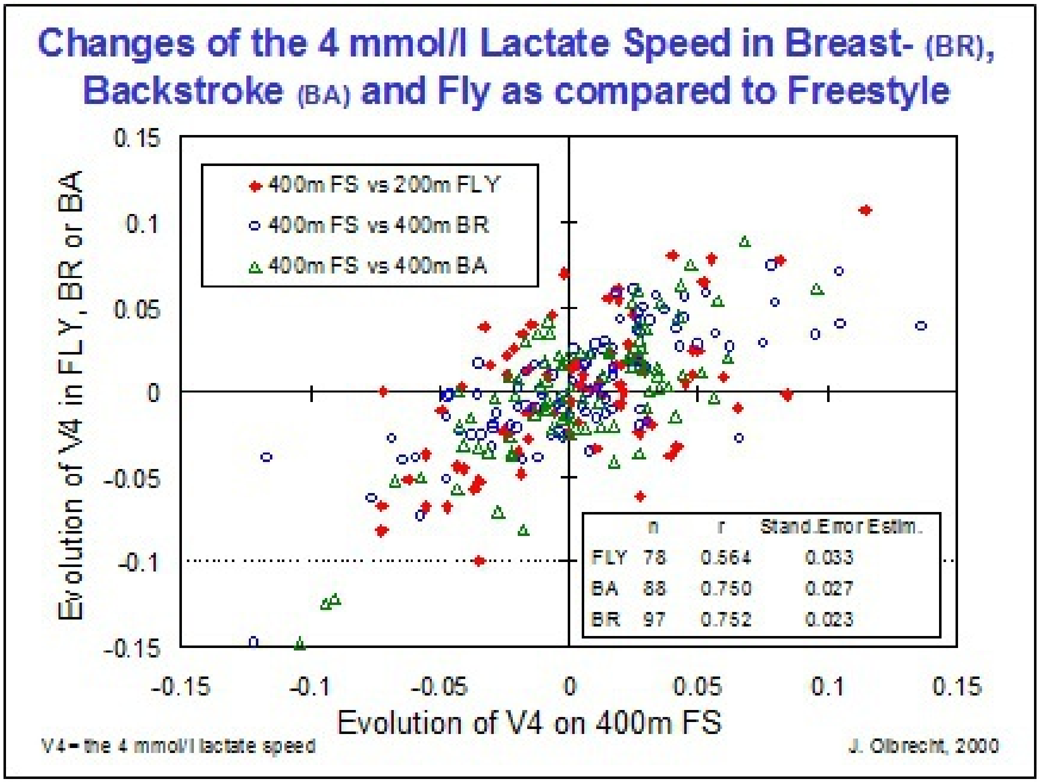


Fig. 33 Changes in 4 mmol/l speed (V4) in freestyle are only roughly correlated with the changes in V4 in back-, breaststroke and fly. Indeed, changes in V4 in freestyle cannot always be transferred to nor observed in the other strokes. The correlation is too weak to reliably assess the aerobic endurance (V4) in breast-, backstroke or fly based on the measured V4 in freestyle.

Blood Sampling For a reliable lactate test it is also important to take arterial blood (from the ear lobe, preferably, or from the finger tip) and to catch the highest blood lactate concentration (= the highest post-exercise lactate reading) as only this value tells something about the lactate production in the muscle. As lactate moves from the muscle into the bloodstream, where it is finally measured, there will always be a delay between the time the peak muscle lactate concentration is reached and the time the highest blood lactate value occurs (fig. 34). To be sure to catch the highest blood lactate concentration after an exercise, at least 2 blood samples should be taken.

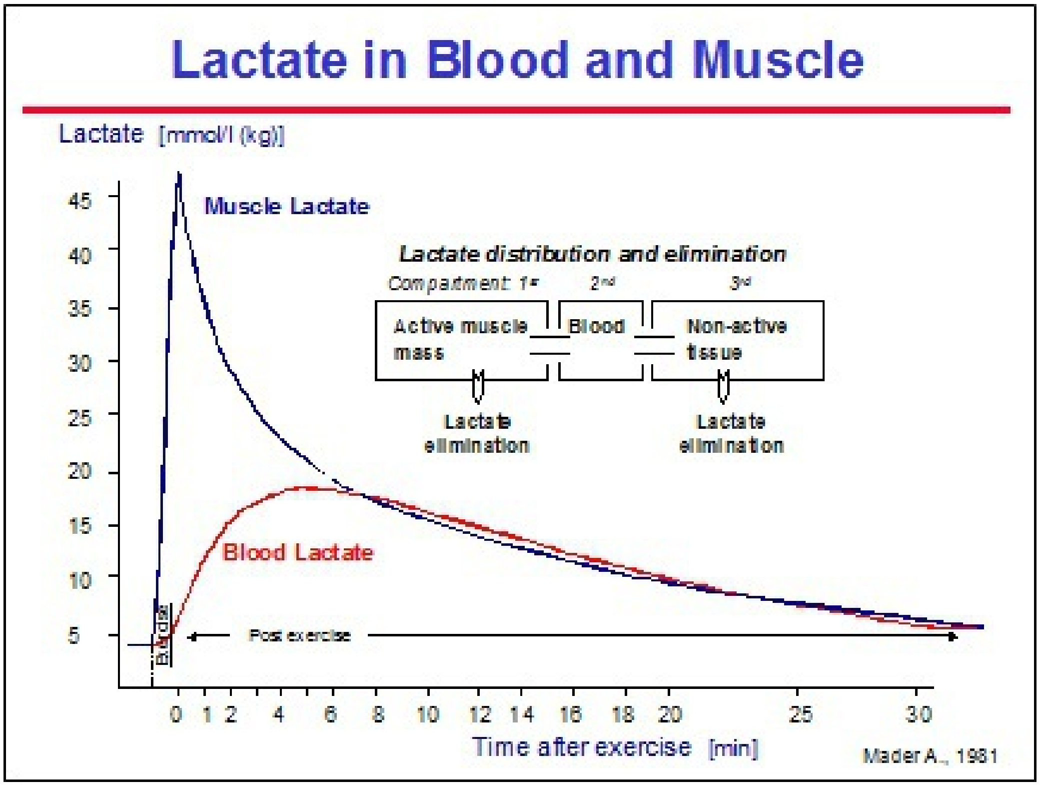


Fig. 34 Evolution of the lactate concentration in the muscle and in blood during and after an exercise (Mader et al. 1981).

After a submaximal swim of more than 100 m, 2 blood samples will usually do; the first sample in the 1st minute after the effort, and the second sample in the 3rd minute. If the second reading is higher than the first one, it will be necessary to take a third blood sample in the 5th minute, and, perhaps even several additional samples until the blood lactate concentration declines.

The more intensive and/or the shorter the effort (100 m and less), the later the peak blood lactate concentration appears and the more blood samples will have to be taken. Therefore, after a maximal effort we start taking blood samples in the 3rd minute, and then every 2 minutes until blood lactate declines. With an instant analyzer it is easy to observe whether lactate is decreasing or not. But, if no direct lactate measurement is available, you will have to take blood samples in the 3rd, 5th, 7th, 10th and even 12th min after the short all-out effort to make sure you do not miss the highest blood lactate concentration.

Basics for a Reliable Interpretation of Lactate Readings

The interpretation of the lactate tests is not as simple as it seems. Indeed, different training exercises may, for the same swimmer, produce the same blood lactate concentration. Nevertheless it may correspond to different training efforts/loads/intensities and thus induce different biological adaptations (training effects). Moreover the same lactate concentration measured after a certain training set may, for different swimmers, represent a different metabolic activity of the working muscle and therefore have a different meaning for each of them.

The new findings in lactate research provide an answer to the numerous paradoxes of the classic interpretation of the lactate readings. In the following section we will deal with a new scientific theory which enables the coach to have a better understanding of lactate and to interpret the lactate values in a reliable way.

It is only possible to get at the real meaning of a lactate reading if the following 3 aspects are taken into account:

- the absolute value of the reading
- the proportion of the swimmer’s aerobic and anaerobic capacity that is involved in the effort
- the characteristics of the exercise (volume, intensity, rest and interval)

The Lactate Curve For as long as lactate has been used in sports to optimize training efficiency, the lactate curve has been considered as the starting point for describing and defining the athlete’s conditioning profile. We will, therefore, first discuss the origin and meaning of the lactate curve.

When, for example, a swimmer is asked to perform a set of progressively faster swims, lactate will be low after the first and very extensive repeat, but will rise clearly from a certain speed on (= lactate onset). The faster the subsequent swims, the higher the lactate reading will be. Plotting the swimming speed against the respective lactate values in a lactate-speed diagram and connecting all these points with a line (curve) shows the lactate curve (fig. 35).

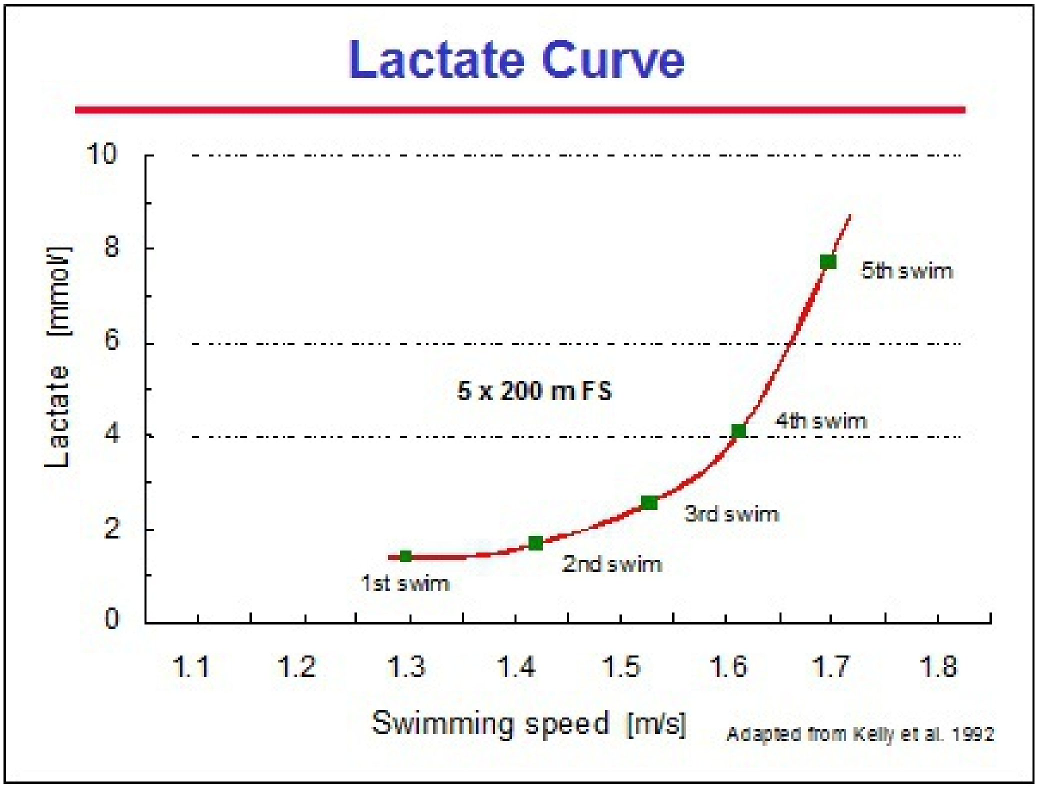


Fig. 35 The lactate curve shows the relation between lactate and speed for a specific exercise (adapted from Kelly et al. 1992).

This curve can now be used to estimate the speed for a given lactate value or vice versa, to estimate the blood lactate concentration for a given speed. Figure 36 shows the estimation of the swimming speed generating 4 mmol/l blood lactate as well as the expected blood lactate concentration corresponding to the swimming speed of 1.6 m/s (4:10 min) for a 400 m swim.

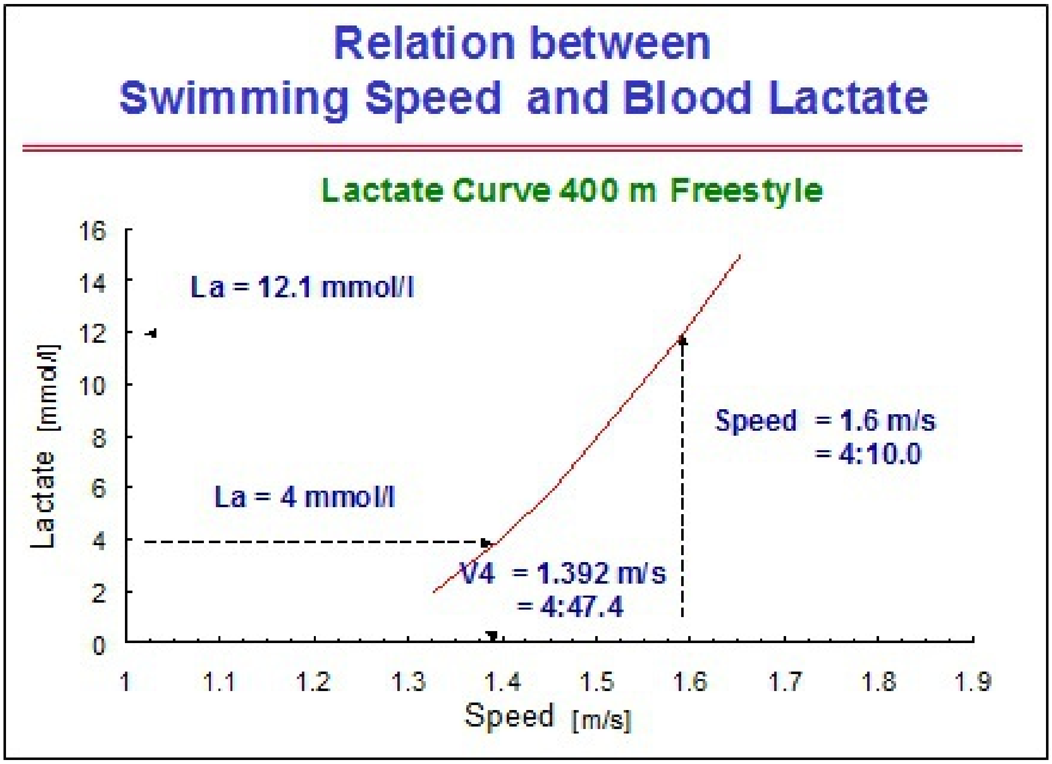


Fig. 36 The lactate curve shows the blood lactate concentration that can be expected at different swimming speeds, or vice versa, the swimming speed that can be expected at different blood lactate concentrations. In this example 4 mmol/l lactate corresponds to 1.392 m/s (4:47.4) and 1.6 m/s (4:10) corresponds to 12.1 mmol/l lactate.

The lactate curve reflects thus the relation between speed and lactate for the specific exercise the swimmer performed. Indeed, depending on both the distance of each swim and the rest between each repetition, the shape and/or position of the lactate curve will change:

* the longer the distance of the interval, the earlier (at lower swimming speed) the lactate onset (fig. 37)
* the longer the interval test distance, the higher the lactate concentration for any swimming speed faster than the velocity at the lactate onset (fig. 37)
* a short rest between the swims will produce an exponential curve while a complete rest will lead to a nearly linear curve (fig. 38)
* in comparison with a complete rest between the swim intervals, a short rest will lead to a later lactate onset (at a higher speed) but to

higher lactate concentrations at moderate to intensive swimming speeds (fig.38)

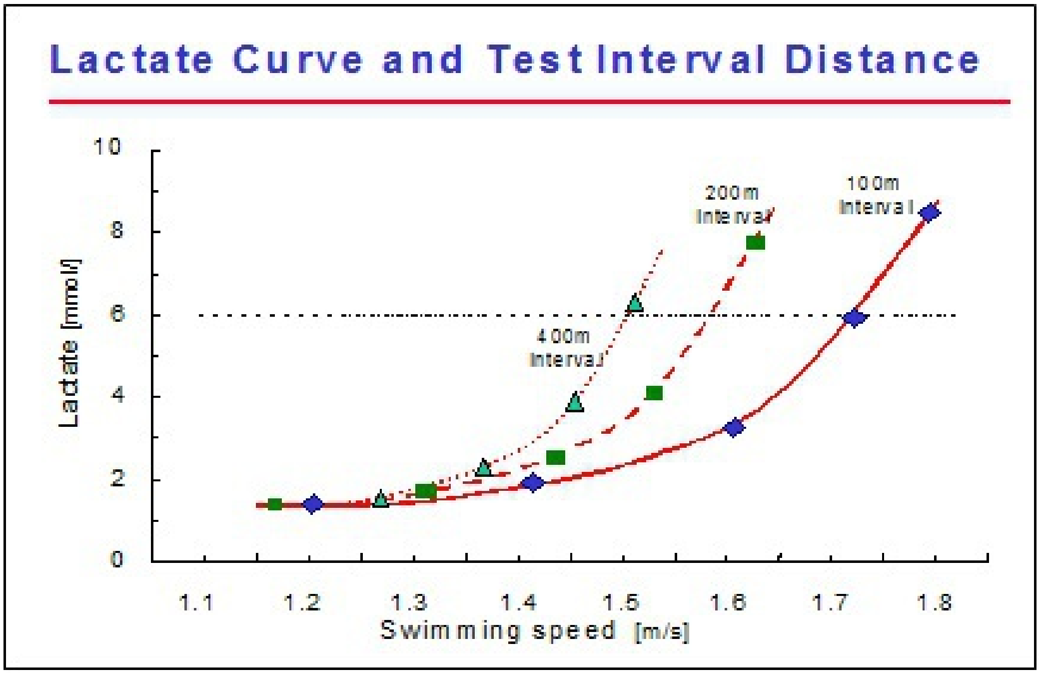


Fig. 37 The longer the test distance, the earlier (at lower swimming speed) the onset of blood lactate.

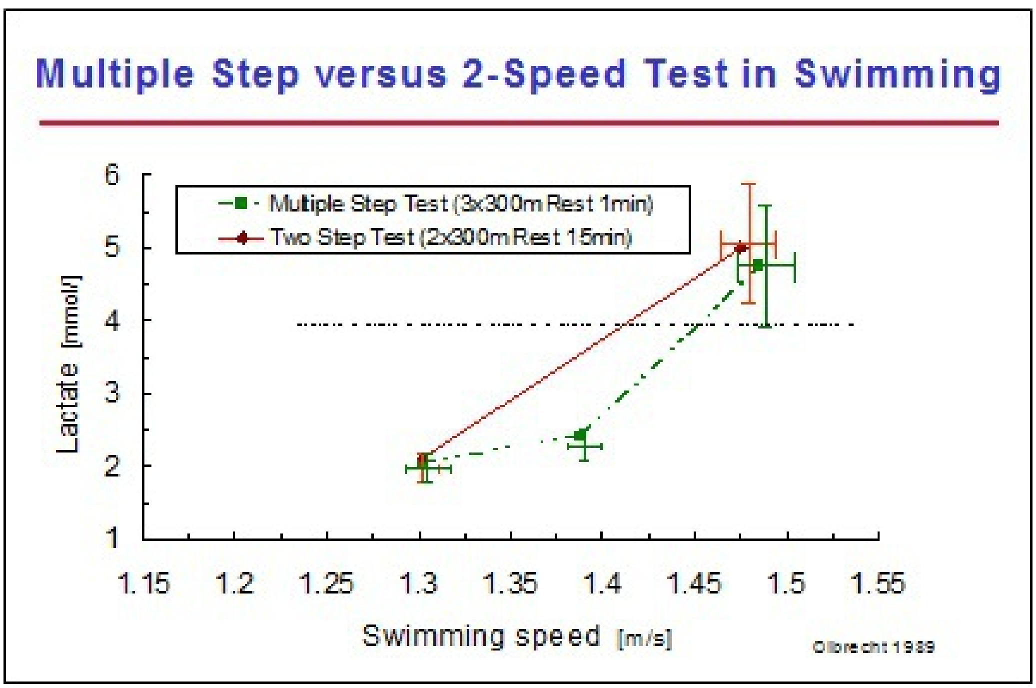


Fig. 38 Influence of test protocol on the lactate curve. All swimmers performed both types of test protocols.

Shift of the Lactate Curve: Interaction of the Aerobic and Anaerobic System on the Lactate Curve For exercises longer than 2 minutes, lactate in the bloodstream increases sharply (= lactate onset) when the swimmer approaches the limit of his aerobic capacity (= maximum oxygen uptake or VO_2 max). The higher the swimmer's VO_2 max (aerobic capacity), the faster he will be able to swim before the onset of the blood lactate occurs. In the lactate-speed diagram the increase of the aerobic capacity will, thus, pull the lactate curve to the right. As a result, the swimming speed corresponding to 4 mmol/l lactate (V4) will increase. This reasoning has contributed to the belief of many years that a faster V4 is unconditionally linked with an improved VO_2 max.

This approach was simple and became widely used but there quickly appeared some unexplainable problems with lactate testing. Many coaches and sports scientists noticed that, as the season progressed, the V4 or other measure of endurance often did not change or even got worse as the swimmers were getting faster in competition. Some questioned whether there was any further improvement in aerobic endurance or that the lactate tests were not a good measure of aerobic endurance. Since we believed that the theory was good and the aerobic capacity was increasing there must be something missing from the interpretation of the lactate tests.

Indeed, we observed that the onset of lactate as well as V4 did not occur at the same percentage of VO_2 max for every swimmer (fig. 39). For example, swimmer A has a sharp increase of lactate in the blood at about 60% of his VO_2 max, while this occurs at about 70% of swimmer B's aerobic capacity. This difference in aerobic contribution can be explained by the swimmers' level of anaerobic capacity (the maximal capacity to produce lactate/pyruvate per second = glycolysis). The higher the anaerobic capacity (VLamax), the sooner the anaerobic system will be involved in the energy supply and the sooner the lactate onset may occur. As opposed to an increase of the aerobic capacity, which will pull the lactate curve to the right, the increase of the anaerobic capacity will pull the lactate curve to the left and consequently slow down V4. Meanwhile an increase of the anaerobic capacity will also lower the percentage of VO_2 max at the lactate onset and at V4. As a result: if V4 (= the 4 mmol/l lactate speed) remains the same but the anaerobic capacity increases, V4 will occur at a lower percentage of VO_2 max and thus be related to a higher VO_2 max.

As an improvement of the anaerobic capacity generally occurs as the season progresses and the anaerobic training is increasingly emphasized, V4 may remain the same or even get worse despite an unchanged or even improved VO_2 max.

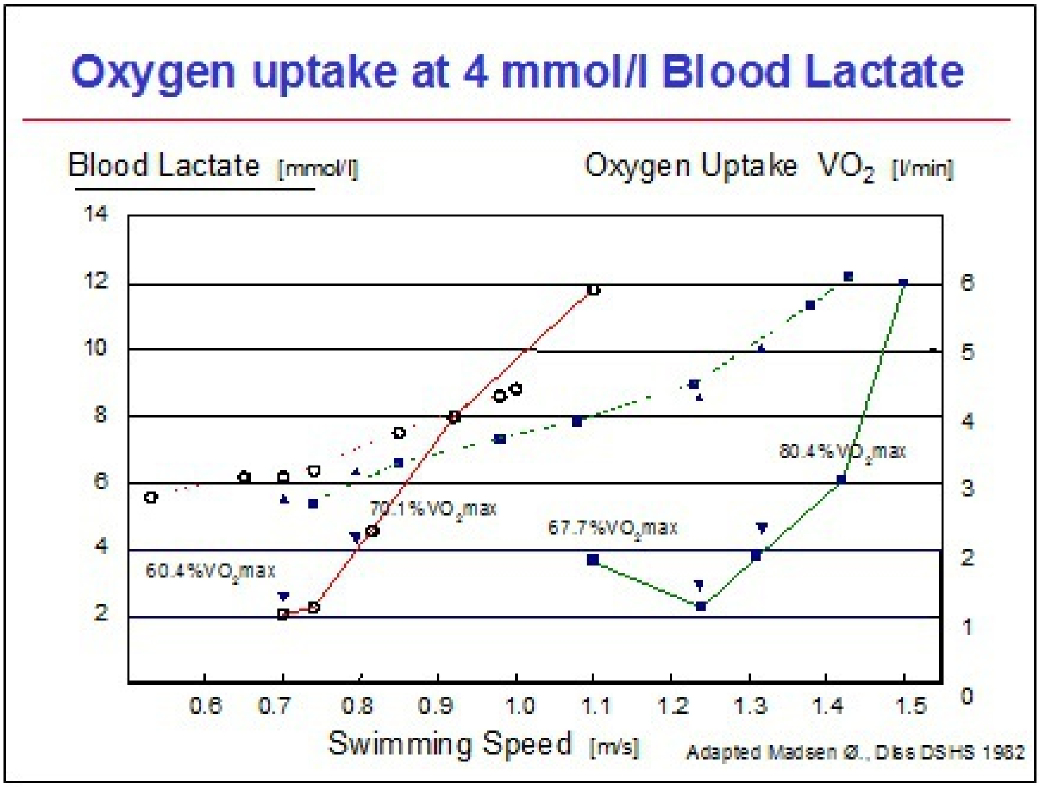


Fig. 39 Oxygen uptake (- - -) and lactate concentration (——) during a set of progressively faster swims. The $\%VO_2$ max reached at the lactate onset and at the 4 mmol/l blood lactate is indicated for a student (o) and an elite swimmer (■).

Conclusion:

The position of the lactate curve is not only determined by the aerobic capacity as has been stated for years, but also by the anaerobic capacity. Simplified we can say that:

- an increase in aerobic capacity will cause the lactate curve to move towards the right
- an increase in anaerobic capacity will force the curve to shift to the left
- a decrease in aerobic capacity will move the curve to the left
- a decrease in anaerobic capacity will shift the curve to the right

The final position of the lactate curve will depend on the balance between the aerobic and anaerobic capacities

This can graphically be illustrated by a pulley system on each side of the lactate curve (fig. 40). On the right side you have the aerobic capacity (VO_2 max) pulley. The higher VO_2 max, the higher the force that pulls the lactate curve to the right. On the left side you have the anaerobic capacity (VLamax) pulley. The higher VLamax, the higher the force to pull the curve to the left. The final position of the lactate curve will be where both forces are balanced.

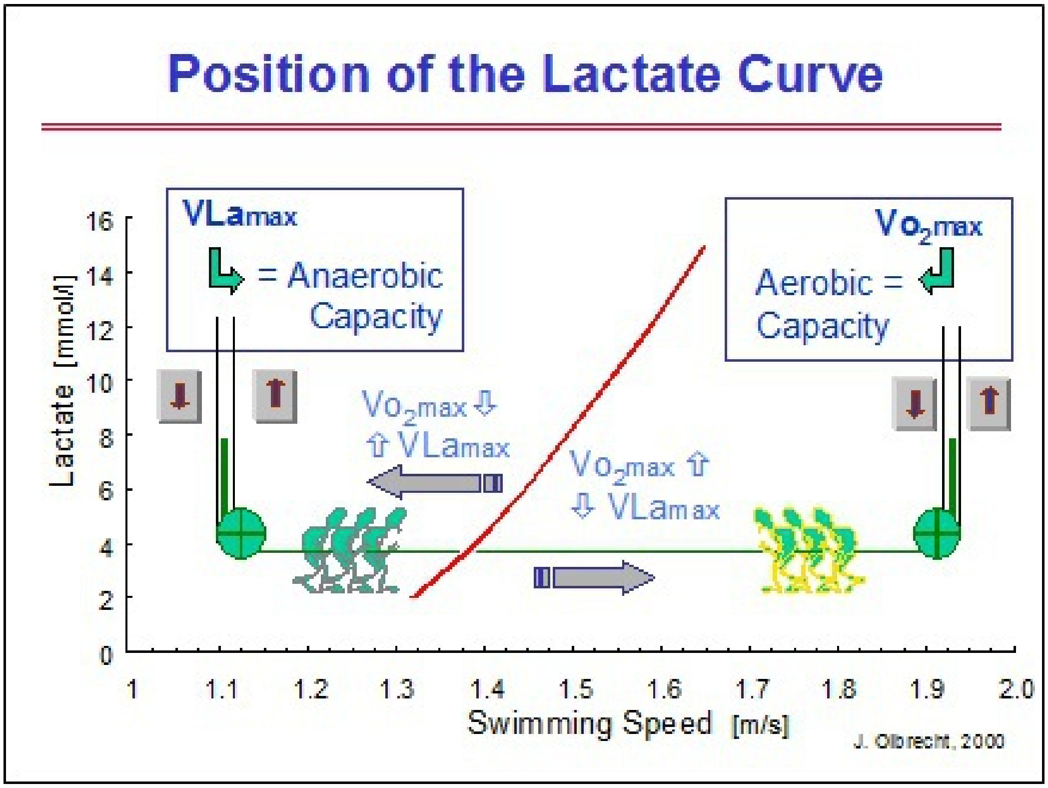


Fig. 40 The final position of the lactate curve depends on the balance between the aerobic and anaerobic capacities.

With this in mind, it is obvious that a sprinter with a high VLamax and also a “World class” aerobic capacity will never have a lactate curve positioned as far to the right as a top level long distance swimmer’s who has the same aerobic capacity but a lower VLamax. The example given in figure 41 clearly shows that the lactate curve of the World class sprinter is located to the left of the World class long distance swimmer’s curve. This is not the result of a difference in aerobic capacity between the World class sprinter and long distance swimmer, but of a different balance between their respective aerobic and anaerobic capacities. Indeed, as their aerobic capacities are nearly identical, it is the sprinter’s higher anaerobic capacity that will hold the curve on the left.

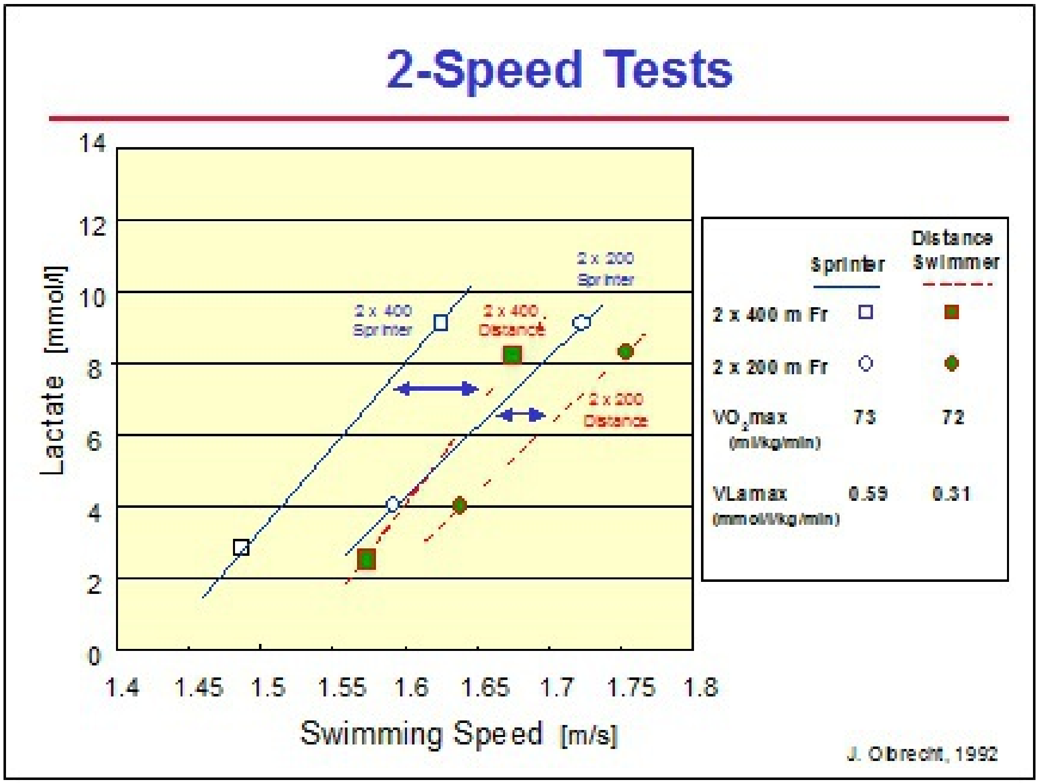


Fig. 41 The position of the distance swimmer's lactate curve to the right of the sprinter's curve does not always reflect a better aerobic capacity, but can just as easily come about through a weaker anaerobic capacity (see text).

Experimental research, where maximal oxygen uptake was measured during swimming, confirms this finding and shows a nearly identical aerobic capacity (VO_2 max) in World class sprinters and distance swimmers. The difference between their lactate curves is therefore not produced by a different aerobic capacity but is due to a different level of anaerobic capacity (the sprinter's much higher anaerobic capacity).

Interpretation of the Lactate Curve For years the interpretation of the lactate curve and readings was thought to be quite simple:

The more the curve to the right = the better the aerobic capacity and therefore
- a shift to the right = a better aerobic capacity
- a shift to the left = a worse aerobic capacity
- no shift at all = aerobic capacity unchanged

This classic interpretation, however, resulted in a lot of misleading evaluations of the conditioning and incorrect advice on training. Research has now shown that, as both the aerobic and anaerobic capacities define the position of the lactate curve:

* it is essential to consider both capacities together before evaluating and interpreting the shifts of the lactate curve
* there are not 3 but 13 ways to explain the shifts of the lactate curve, making the interpretation much more intricate, complex, and delicate but actually more accurate and reliable (fig. 42)

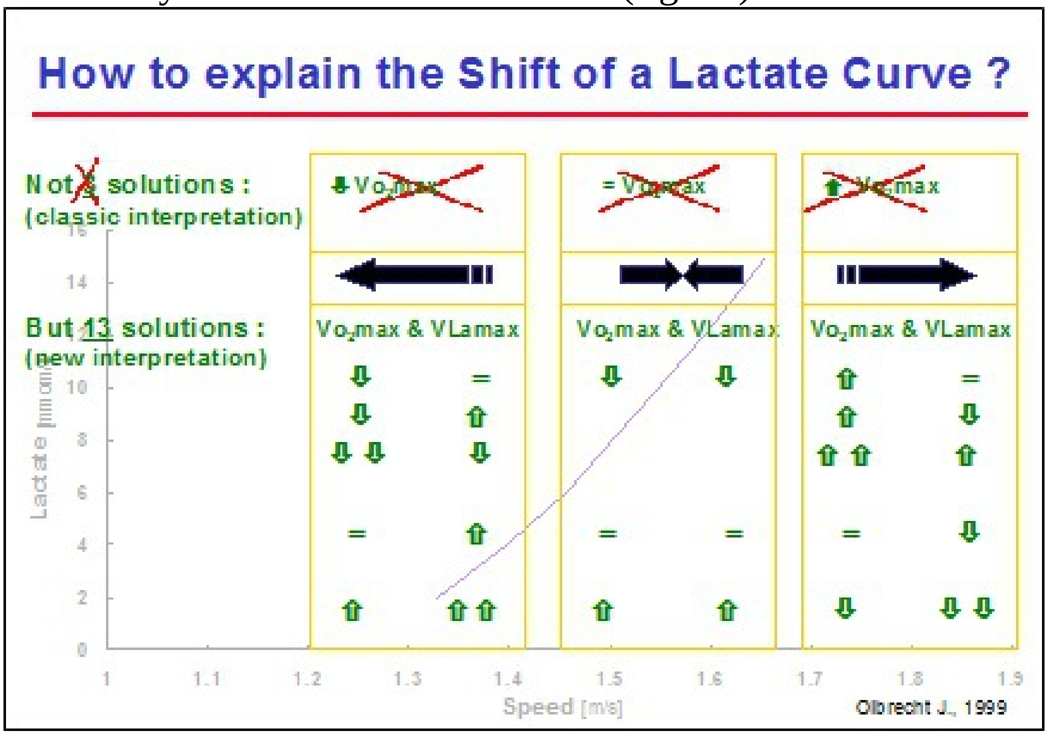


Fig. 42 Not 3 (classical interpretations) but actually 13 different reasons for changes in the position of the lactate curve.

For example, the movement of the lactate curve to the right does not automatically reflect an improvement of the aerobic capacity. Indeed, as the anaerobic capacity decreases, the force to pull the curve to the left weakens causing a shift to the right without any change in aerobic capacity. The lactate curve moves thus to the right not because of an improvement in aerobic capacity but of a decrease in anaerobic capacity. The opposite is also possible. A shift of the lactate curve to the left is not unconditionally a reflection of a lower aerobic capacity, but can also come about through an improvement in anaerobic capacity; the pulling force towards the left increases while the pulling force to the right remains the same, causing the lactate curve to move to the left (fig. 43). The classical interpretation of the lactate curves in figure 43 points to a deterioration in aerobic capacity, however, the simulation model explains that the shift to the left is the result of an improved anaerobic capacity and an unchanged aerobic capacity. This explanation proved to be much more in line with the swimmer's progress and achievements in training and in competition.

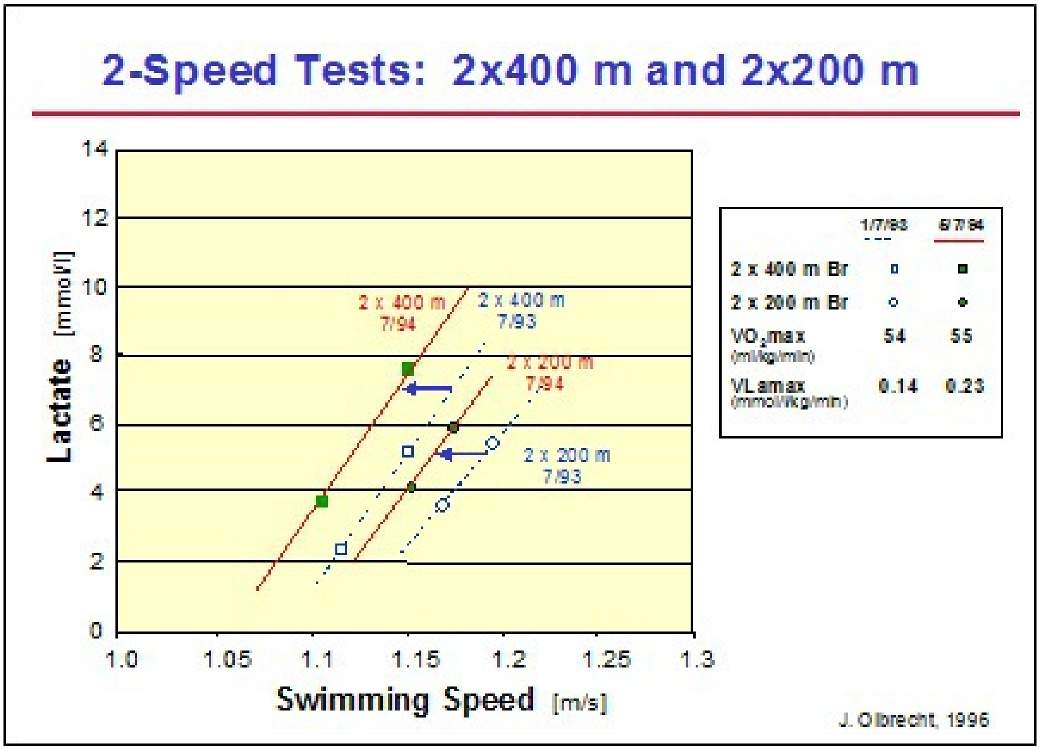


Fig. 43 Shift of the lactate-speed curve to the left despite an unchanged aerobic capacity.

And the whole process behind the move of the lactate curve can become even more complex. For example, a deterioration of the aerobic capacity can also cause the lactate curve to shift to the right if the anaerobic capacity decreases even more. Thus, the pulling force to the left drops off more than the pulling force to the right, resulting in a move of the lactate curve to the right.

Lactate Readings in Training The characteristics of the exercises also strongly influence the meaning of the lactate reading. Indeed, a 3 mmol/l lactate reached for example after a single 400 m or after a set of 6 x 400 m with 30 seconds rest needs to be interpreted differently as it has to be ascribed to:

* a different use of the metabolic processes

* different swimming speeds

To prove this we asked a group of swimmers to swim a series of 6 x 400 m with 30 seconds rest at a constant speed corresponding to 3.5 mmol/l blood lactate in a single 400 m free swim (fig. 44). After the first repetition a mean blood lactate of 3.5 mmol/l was actually reached. After the 2nd and 3rd repetition likewise swum at the same constant speed, we detected a decrease in the concentration of lactate to 2.7 mmol/l on average, which then hardly changed during the remainder of the exercise.

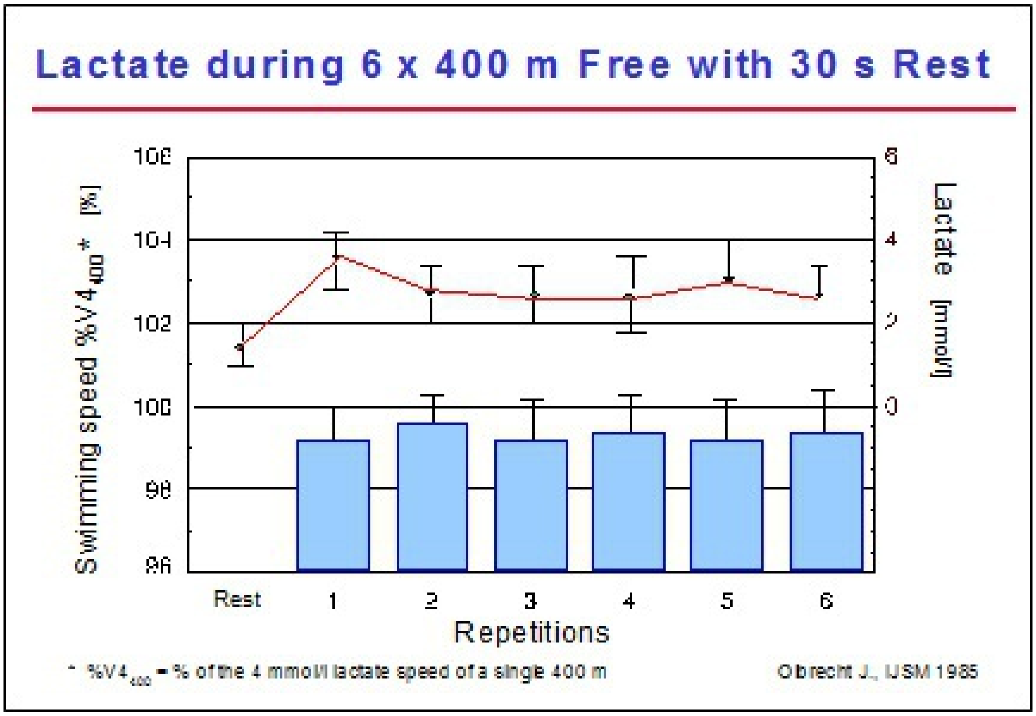


Fig. 44 Evolution of the blood lactate concentrations during a series of 6 x 400 m with 30 seconds rest, swum at a constant speed corresponding to 3.5 mmol/l in a single 400 m free swim.

What has actually happened?

Over the first 3 repetitions the contribution of the aerobic metabolism, which requires a certain lead time before being able to deliver the desired amount of energy, progressively increases, gradually taking over the work of the anaerobic lactic system. This results not only in a reduction of the lactate/pyruvate production but also in an increase of the lactate/pyruvate elimination, as lactate/pyruvate is used as fuel. Consequently at the end of the workout less lactate reaches the bloodstream. The concentration of lactate measured in the blood will thus decline.

If, nevertheless, the coach wants to have this exercise ended at 3.5 mmol/l he will have to ask his swimmer to swim progressively faster during the course of the series. The swimming speed that in the series of 6 x 400 m corresponds to 3.5 mmol/l is thus clearly higher than the speed that leads to 3.5 mmol/l after a single 400 m swim.

Interpretation of Lactate Readings in Training In training as well as during a test, the balance between the aerobic and anaerobic capacities influences the meaning of the lactate value. If a coach asks 2 swimmers to complete the same set at the same lactate concentration, he often assumes that both swimmers train at the same intensity. This however is hardly ever the case.

For example, 2 swimmers, whether or not with the same V4, complete the same training set at 3 mmol/l. The one with the higher aerobic capacity will use more lactate/pyruvate and therefore be forced to produce more of it than his training mate with the lower aerobic capacity (fig. 27). As a result, the anaerobic contribution will be higher for the well endurance trained athlete despite the same 3 mmol/l lactate value after the same workout.

This can be demonstrated by the simulation of a 6 x 400 m set with 30 sec rest, for 3 swimmers with different aerobic and anaerobic capacities, at 4 different paces. The paces were chosen so that each swimmer would reach blood lactate concentrations of 1, 2, 3 and 4 mmol/l at the end of the workout (fig. 45). Swimmer A has both a higher aerobic and anaerobic capacity than swimmer B (see right upper part of the graph). Nevertheless, the capacities of both swimmers result in the same lactate curve (see lower right) and in the same swimming speeds that produce 1, 2, 3 and 4 mmol/l lactate after the simulated workout (see lower left). Despite the same lactate curve, the same swimming speed and the same lactate concentration during the 6 x 400 m interval set, the aerobic and anaerobic capacities of swimmer A are definitely less charged than those of swimmer B; indeed from 1 till 4 mmol/l swimmer A uses his aerobic and anaerobic capacities respectively for 74 till 78% of VO_2 max (aerobic capacity) and for 6 to 10% of VLamax (anaerobic capacity) while swimmer B, for the same range of lactate values, uses his aerobic capacity for 83 to 86% and his anaerobic capacity for 15 to

22% (see upper left). This training set will, thus, affect both swimmers differently and induce different training adaptations, i.e. an improvement of aerobic capacity in swimmer A and an improvement of aerobic power in swimmer B. If swimmers A and B perform the same training program in the same way for several weeks, swimmer B will run a higher risk for overtraining because he will be stressing both his aerobic and anaerobic metabolism more intensely.

Swimmer C is a classic example of a good long distance swimmer with a good aerobic and a weak anaerobic capacity. For this swimmer, the 6 x 400 m at 1, 2, 3 and 4 mmol/l lactate will lead to a much higher training load on his metabolic system than for swimmers A and B. In order to decrease the training load to a normal level both swimmers C and B will have to slow down on the 6 x 400 m and train at lower lactate levels than swimmer A.

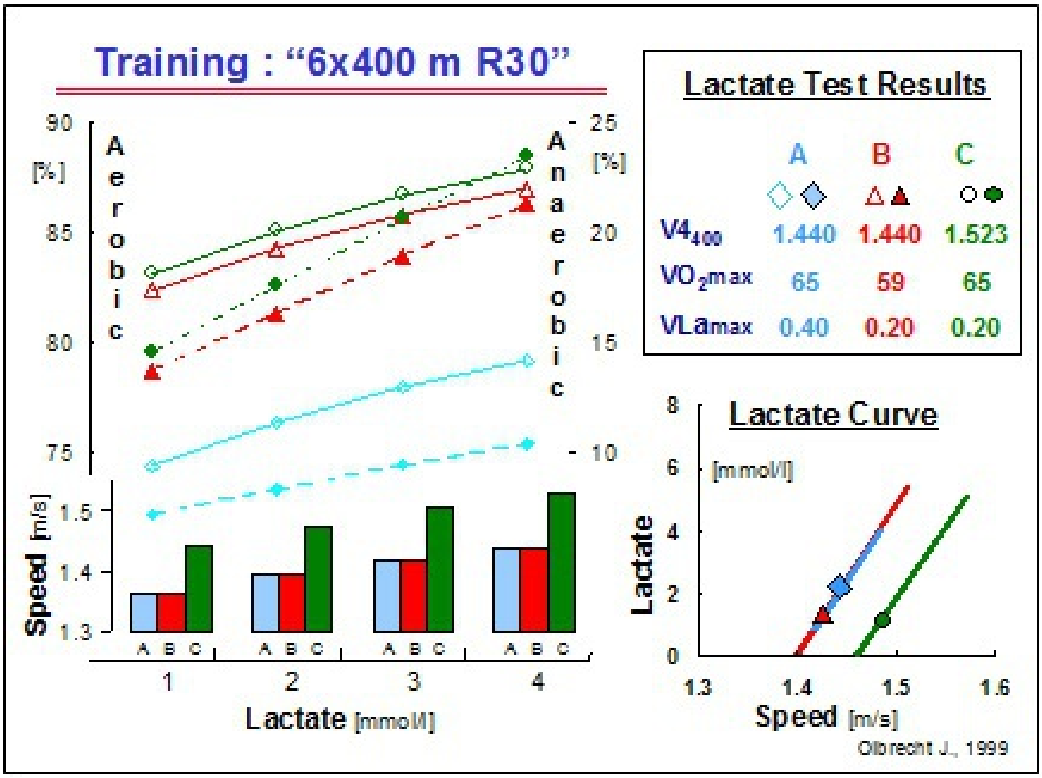


Fig. 45 Different metabolic impact of a 6 x 400 m with 30 sec rest leading to 1, 2, 3 and 4 mmol/l at the end of the set for 3 swimmers (A, B and C) with different aerobic and anaerobic capacities. These results were obtained by simulation (see text).

Thus, if a swimmer combines a well developed aerobic capacity with a low anaerobic capacity - which is usually the case with long distance swimmers

- it is obvious that:

    * a higher contribution of aerobic and anaerobic work will be necessary for him to reach, for example, 3 mmol/l lactate. He will be forced to use more of his aerobic and anaerobic reserves than his training mates with a higher anaerobic capacity
    * he will also have difficulty accumulating lactate during long continuous swims or long interval sets. It is, indeed, not so unusual for him to swim flat out in a long training set and "founder" exhausted on 2 or 3 mmol/l lactate

On the other hand, swimmers with a modest anaerobic and a good aerobic capacity will easily achieve high but accumulated lactate values in long (distance) exercises or sets. In fact, it often happens (see fig. 21) that good long distance swimmers reach higher levels of lactate than sprinters after a long distance event even though their anaerobic capacity is lower. Indeed, the high anaerobic capacity of the sprinters will quickly lead to the acidification of the muscle cells protecting them by inhibiting the energy processes and, resulting in a slow down of the effort and virtually no further accumulation of lactate since its production has been dimmed. Moreover, this quick and substantial activation of the anaerobic system will impede a high contribution of the aerobic system during efforts lasting more than 2 minutes.

Conclusion:

The balance between the aerobic and anaerobic capacities determines:

- to what extent a training set at a given lactate concentration taxes the metabolic system and thus also
- the lactate concentration which corresponds to a given training load

What about Lactate Elimination? The lactate elimination process occurs during as well as after the effort and affects the position of the lactate curve. Basic research has shown that the lactate elimination rate mainly depends on the oxygen uptake during the exercise and on the lactate concentration available in the muscle. Based on these findings the simulation program (see Advanced Lactate Test) calculates the maximal elimination rate as well as the elimination of lactate during the effort to assess its contribution to the position of the lactate curve.

What about the Lactate Thresholds? Lactate tests and the aerobic and anaerobic threshold, whether or not the individual anaerobic threshold or the lactate steady state, are often mentioned in the same breath. Until now, we have spent very little time discussing the lactate thresholds in swimming for several reasons:

- There are a lot of different definitions of thresholds, with respect to the aerobic and anaerobic energy supply, each with a different physiological meaning. The only metabolic threshold that can be considered as a reliable reference for the endurance performance of an athlete is the maximal lactate steady state (MaxLa $_{ss}$ in Europe and lactate threshold in the US) which is defined as the highest swimming speed that can be maintained for more than 20 min at a constant lactate concentration. At this point (MaxLa $_{ss}$) lactate elimination equals lactate production. The speed that can be reached at MaxLa $_{ss}$ determines the aerobic power of the athlete. A high lactate elimination capacity will allow the anaerobic energy supply to contribute to a greater extent during a long effort without an increase of the lactate concentration
- The experimental determination of MaxLa $_{ss}$ is very work and time intensive; time and efforts that are lost for training. For elite swimmers, however, we avoid losing time for training by calculating the MaxLa $_{ss}$ with a simulation program that uses lactate readings from different tests
- The different methodologies to determine the metabolic thresholds based on multiple step tests are not only very time consuming but are also subject to intensive discussions on their validity and reproducibility. It is very easy to determine a fixed aerobic (at 2 mmol/l lactate) and a fixed anaerobic threshold

(at 4 mmol/l lactate); the methodology to determine an individual anaerobic threshold (varying from 1 to 6 mmol/l), on the other hand, is much more complex and all the more controversial as far as the reliability and validity of the method are concerned

In the early period of the history of lactate testing it was often suggested that the anaerobic threshold was the optimum training intensity for improving aerobic endurance. Later studies as well as practical experience have shown that this is not the case. Indeed, from the perspective of the physiology and the training science there is very little scientific evidence about the anaerobic threshold being the unique and best intensity for endurance training. The biological adaptations, which are necessary to achieve an improvement in aerobic endurance, are very complex and cannot be generated through one single training intensity. Only a combination of different types of exercises, each of them inducing specific training effects, will improve the swimmer's aerobic conditioning and in the end ensure success in competition.

So, as there is neither scientific nor anecdotal evidence that a training workout at the aerobic or anaerobic threshold has any advantage over a good balanced program of different types of training, the cost in time needed to determine a reliable anaerobic threshold (= the lactate threshold or $MaxLa_{ss}$) is not offset. The information gained by the determination of the lactate threshold will not be of any additional help to improve training efficiency. Therefore the determination of an aerobic or anaerobic threshold should be reserved for academic study purposes only.

Moreover, all speculations as well as the lack of a clear statement regarding thresholds provoke with coaches and athletes more confusion than advantages. For this reason we will use the terms fixed aerobic or anaerobic threshold, the individual anaerobic threshold or the lactate steady state as little as possible to avoid the misleading link to the "ideal training intensity".

Lactate Testing - Design and Execution

We can state that lactate measurements reflect the activity of both the aerobic and anaerobic metabolism. In combination with swimming speed they allow to:

1. determine the aerobic and anaerobic capacity of the swimmer
2. detect the strong and weak aspects in conditioning with regard to both capacities
3. determine, for the next training period, the training objectives, the dosage and configuration of the various types of training exercises
4. estimate the training intensity and volume appropriate to the various training assignments
5. evaluate the changes in metabolic performance and so to assess the effectiveness of training and to eliminate as much useless training exercises as possible

Age Lactate tests can be carried out regardless of the age of the swimmer. However, converting the test results into practical training advice will become easier if the athlete can produce a great deal of lactate. Since young swimmers are usually not yet capable of producing much of it, using lactate tests is often more difficult with them.
But, exceptionally, some young swimmers even as young as 8 can produce more lactate than an adult swimmer. It would therefore be wrong to determine for whom lactate tests might be meaningful on the basis of age alone. Knowing whether a young swimmer is capable of producing a lot or only a little lactate is extremely helpful in putting together a correct and appropriate age group training program.

Frequency The usefulness of lactate testing and its frequency per year can best be determined by the performance level of the swimmer. For a young swimmer with a rather low performance level, one simple lactate test per year is sufficient to characterize in broad outline his physiological profile and to pass on to the coach the most important directives for an appropriate training framework. The adjustments to the training intensity during the year can then be made in accordance with a series of alternative tests, such as, for example, the 30-minute test, the percentage of the swimmer’s best times, the heart rate, etc. (see later). The swimming technique of these swimmers is often quite poor so that the training process should be focused on this aspect of performance (technique) more so than on the improvement of the conditioning.

Criteria for a Reliable Test Protocol The better the swimmer's performance, the more accurate the training must be, the more useful the lactate tests become and the more often they should be scheduled (at most every 6 weeks) regardless the swimmer's age. The testing procedure will also become more complex, vary according to the specialty of the swimmer and be supplemented with lactate measurements during training (spot lactate tests) and competition.

To provide relevant information about the swimmer's metabolic performance level, the lactate test should meet following conditions:

- the test has to be a *swimming effort* (sport specific) carried out in the stroke (stroke specific) for which the information on conditioning is wished (see Relevance and Reliability of the Lactate Readings)
- enough blood samples need to be taken to ensure the assessment of the highest post-exercise blood lactate concentration (see Relevance and Reliability of the Lactate Readings)
- preference should be given to a *single effort* or to several efforts separated by total recovery rather than to a step test in order to avoid interactions of short rest breaks and the metabolic response of previous intervals on the lactate readings. These interactions would only make the interpretation of the lactate values much more difficult
- strength training and very exhaustive workouts (with high volume and/or at high intensities) should be avoided within 24 hours before the lactate test, as these efforts strongly affect the reliability of the test interpretation

Indeed we have learned from practice that the lactate response may be disturbed for 24 hours after strength training. Lactate accumulation at a submaximal workload seems to occur at an earlier stage and the lactate concentration after an all-out effort is less than expected. Taking such a disturbed lactate response as a reference for determining the actual physical conditioning profile, for evaluating the training efficiency and for deducing training guidelines for the coming training period will certainly be “misleading”

A similar interaction of less magnitude, happens if an intense water training session precedes a lactate test within 24 hours

- the test situation should always be comparable (pool length, morning or afternoon, comparable timing in periodization, diet, ...)

We have observed that in the afternoon one could swim about 1 sec/100 m faster for the same submaximal lactate concentration as compared to the morning session (fig. 46)

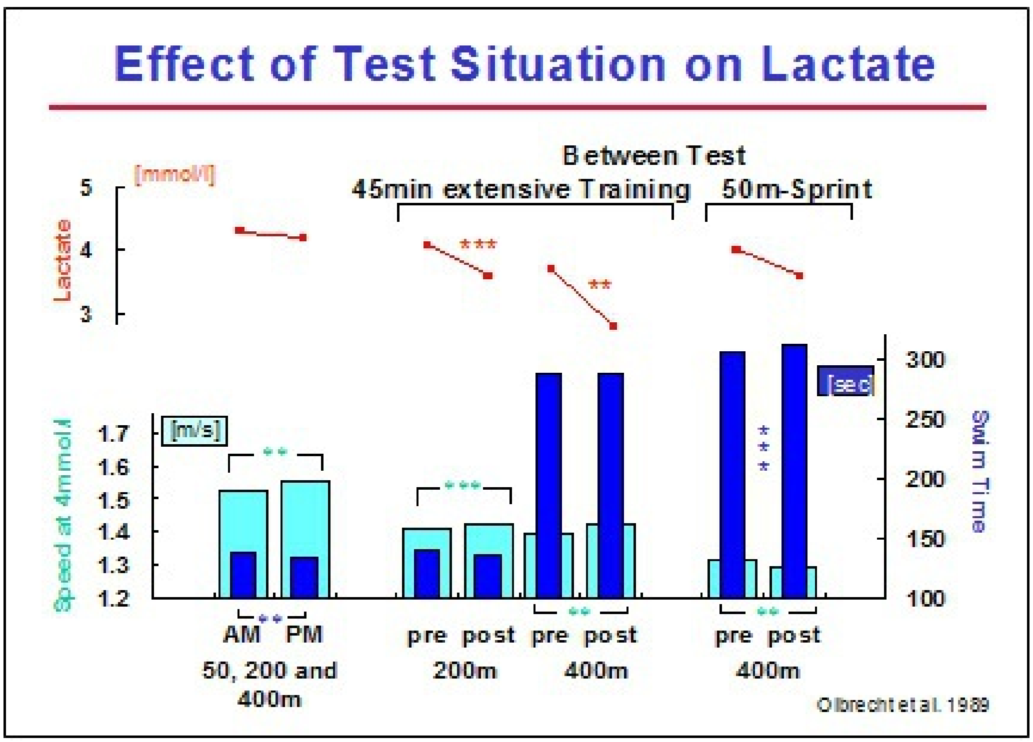


Fig. 46 Influence of test situation on the lactate readings.

This means that a lactate test should be performed at approximately the same time of the day, particularly when working with top swimmers. With these swimmers the room of improvement is restricted because they are already strong performers and it is important to ensure that small differences are really due to conditioning changes and not the consequence of a different testing time

Besides the time of the day, the coach has to ensure that “surrounding workouts” do not influence the lactate readings (fig. 46). Indeed, if for example, a 45-minute extensive workout (around 1.5 mmol/l) is inserted between 2 submaximal swims (200 or 400 m), the second effort will be swum statistically significantly faster but with a statistically significantly lower post-exercise lactate concentration as compared to the first effort which was swum immediately after the warm-up (1100 m). A 50 m sprint between two 400 m submaximal swims, on the other hand, has the opposite effect on the lactate-speed relation. In this test design (400 m submax., 15 min regeneration swim, 50 m sprint, 15 min regeneration swim and then the second 400 m submax. swim) swimmers will show no significant different lactate concentrations after the first and second 400 m swim, but they will therefore swim significantly slower during the second 400 m

Moreover, it is also essential to carry out the lactate tests at comparable points of time in the training periodization. Indeed, if a swimmer has just finished a long period (more than 7 days) of high volume or intensive training, the glycogen load of his muscles may be decreased and therefore attenuate the glycolytic activity which is responsible for producing lactate. In such a situation, the lactate production will be decreased and will result in a later onset of blood lactate accumulation (fig. 47) and a lower lactate concentration at exhaustion

This may give the impression that the swimmer improved in endurance, but in fact it is not a higher oxygen uptake (= real endurance improvement) but a lower lactate production capacity that causes the shift of the lactate curve to the right. It is therefore advisable to avoid lactate tests immediately after an extended high volume or intensive training period such as a swim camp, and to allow at least 3 days of regeneration training before testing

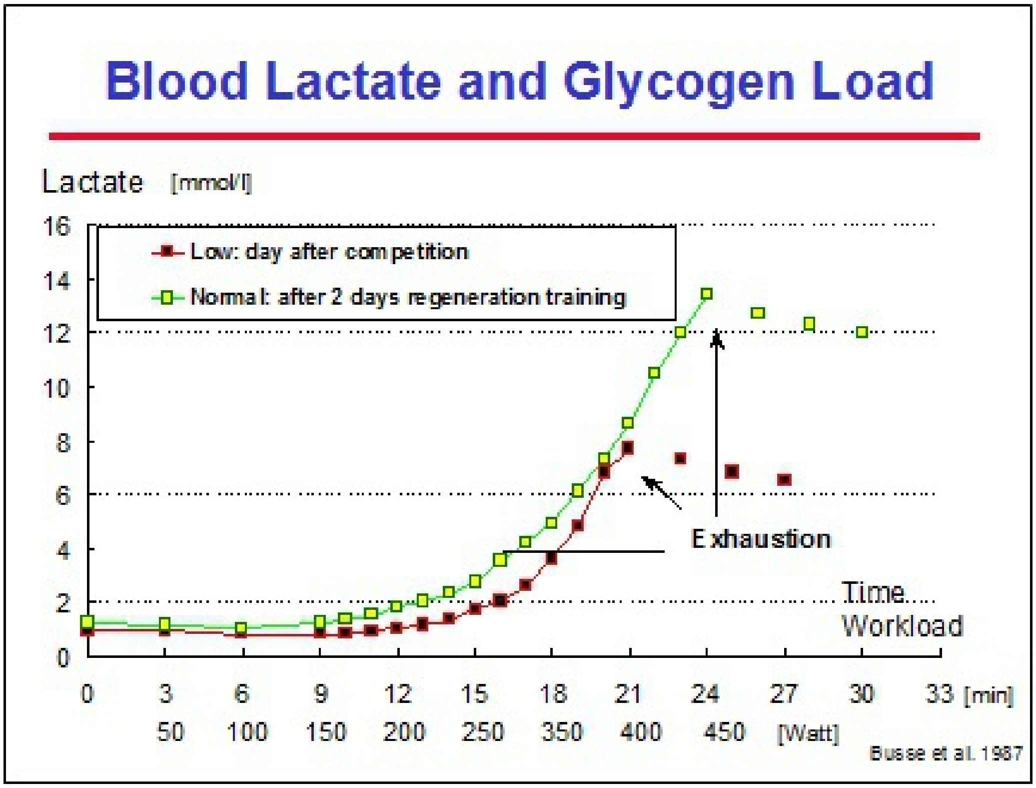


Fig. 47 Influence of the muscle glycogen content on the lactate curve.

Similarly, lactate tests after an altitude training camp should, from a scientific point of view, not be carried out within 14 days after return to sea level. This, however, makes little sense in practice. Indeed, the purpose of altitude training camps is usually to prepare athletes for major competitions that take place between 20 and 28 days after returning to sea level. Therefore, it is not appropriate for the coach to wait 14 days before getting any training information for the pre-competition fine-tuning. We therefore often advise to check for changes in the conditioning profile within 3 to 5 days after return to sea level, knowing however, that these results do not include nor represent all the effects of the altitude training

Not only the training (volume and/or intensity) completed but also the diet may affect the muscle glycogen load and consequently the glycolysis and the lactate results. We recommend, therefore, in order to avoid any interaction of the diet on the lactate test, to keep it as similar as possible during the days preceding the lactate test

To summarize we can state that, to avoid as much as possible all these interaction factors, lactate tests have to be planned well in advance (while periodizing the macrocycle) and preferably at the end of a recovery week or possibly in the first week of a mesocycle. If any of the interactions mentioned above cannot be avoided - training always has the highest priority and should never yield to a test -, they should at least be taken into account when interpreting the test results

- the test should always allow the evaluation of both the *aerobic* and *anaerobic metabolism* . This can be done by:

✽ combining both a submaximal aerobic and an all-out anaerobic effort (see Standard Lactate Test). To assess the *aerobic endurance* , the swimming speed at 4 mmol/l lactate (V4) during a submaximal effort lasting at least 2 minutes (optimal is more than 5 minutes so the oxygen uptake and the blood lactate can approach a steady state) can be used. The faster the V4 in such a submaximal effort, the better the aerobic endurance. For an estimation of the *anaerobic capacity* the exercise has to be all-out and should not last longer than 2 minutes (preferably 45 seconds to 1:30 min); a 50 or 100 m competition would also do, especially to avoid as much as possible a lack of motivation. We can assume that the higher the lactate value after the short all-out effort, the better the anaerobic capacity. However, this is only a rough approach. It does not enable the coach to quantify the anaerobic capacity but certainly gives an idea on how to interpret the obtained relation between lactate and speed. The use of table 10 will confirm the impact of the anaerobic capacity on the lactate values after the submaximal effort and so help to interpret the observed lactate-speed results

✽ using the simulation program (see Advanced Lactate Test) which quantifies both the aerobic and anaerobic capacities and reveals their respective contribution to the measured lactate and speed (Olbrecht 1989)

Standard Lactate Test: the 2- and 1-Speed Test One of the first lactate test protocols which met the criteria mentioned above and allowed the evaluation of both the aerobic and anaerobic metabolism was Mader's 2-Speed test (Mader 1980).

Test Protocol

It consists in swimming twice the same distance, in the same stroke but at different speeds. The first repetition is swum at submaximal speed while the second is all-out. Before each repetition, 5 minutes of complete rest is given to the swimmer and between both repetitions there must be at least 15 minutes rest, preferably active. Blood samples, to determine the highest post-exercise lactate concentrations, are taken in the 1st, 3rd and possibly in the 5th minute after the submaximal effort and in the 3rd, 5th, 7th, 10th and, if necessary, also in the 12th minute after the maximal effort (fig. 48).

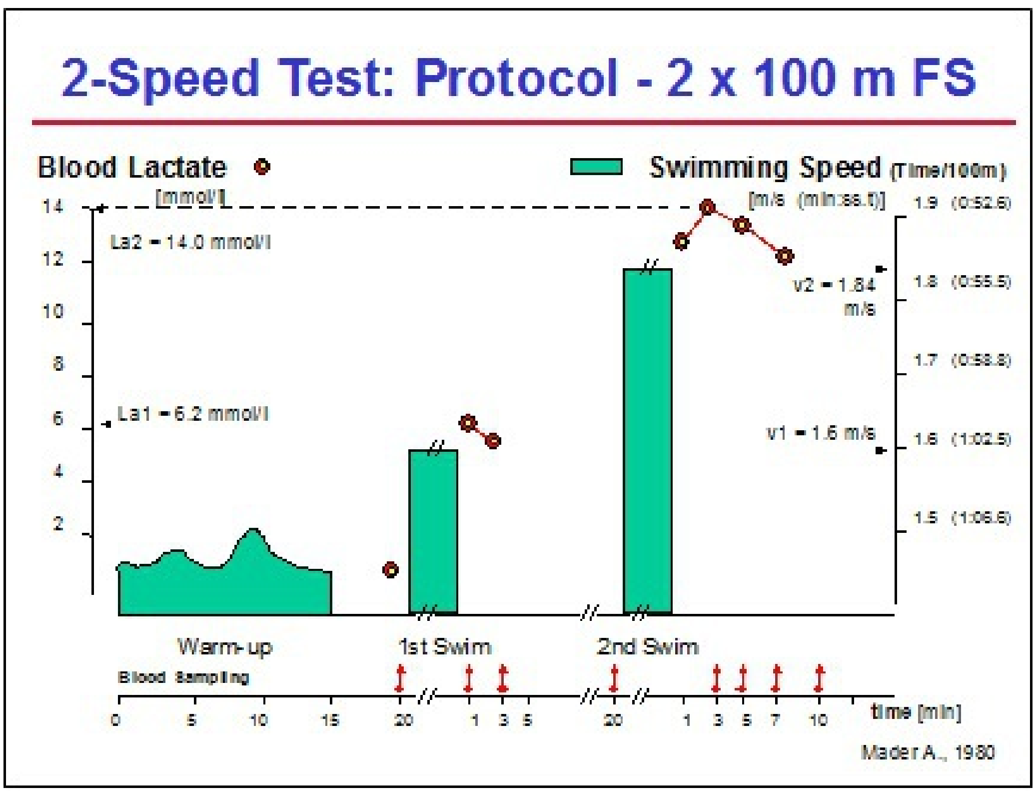


Fig. 48 A swimmer swims twice the same distance; the first repetition at submaximal speed, the second all-out. After the submaximal swim, lactate is measured in the 1st and 3rd minute while after the maximal effort blood samples are taken in the 3rd, 5th, 7th and 10th minute till a decrease of lactate sets in (2-Speed test protocol, Mader 1980).

It could be shown for each test distance and for each stroke that, once above 2.5 mmol/l, the increase in lactate for a given increase in speed (= the slope of the curve) was nearly the same for all male or female swimmers (Mader 1980, Olbrecht 1989). This allows us to plot both swimming speeds against their corresponding highest post-exercise lactate concentration into a lactate-speed diagram, and to connect both points by a straight line in order to obtain the relation between speed and lactate, i.e. the lactate curve (fig. 49). This curve reflects the actual aerobic and anaerobic performance of the tested swimmer, and is only valid for the tested exercise (interval distance and rest). Other test distances or another test protocol will of course generate other lactate curves (see Lactate Curve).

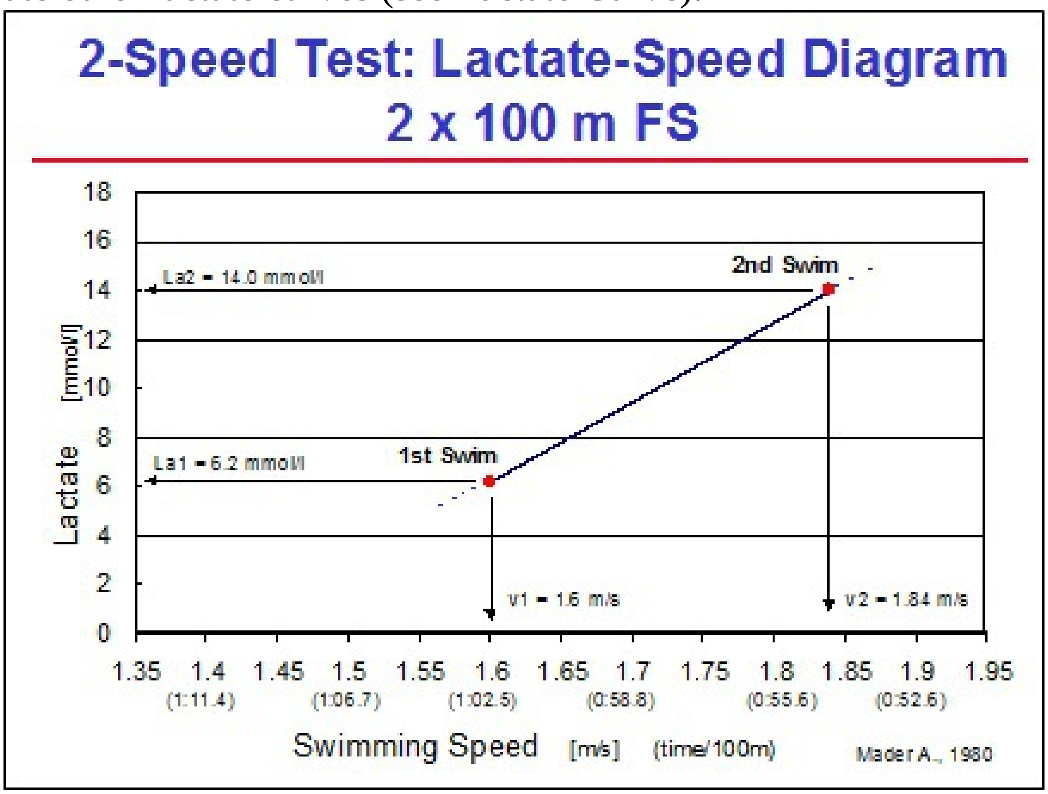


Fig. 49 The lactate curve, i.e. the relation between speed and lactate, can be designed by plotting the swimming speeds of a 2-Speed test against the corresponding highest post-exercise lactate concentrations in a lactate-speed diagram and by connecting both points by a straight line.

Based on the nearly constant slope per stroke, distance and gender, we simplified the 2-Speed test and developed the 1-Speed test (Olbrecht 1989). The test distance is swum only once and 1 plotted point in the lactate-speed diagram defines where to draw the lactate curve (fig. 50 (a)).

There are however some limitations to the 1-Speed test. Indeed, since small variations in the nearly constant slope are possible, the 1-Speed test can only be used to draw/estimate the lactate curve for a limited range around the measured lactate values, i.e. the lactate reading ± 4 mmol/l and not lower than 2.5 mmol/l. A submaximal 1-Speed test will therefore provide reliable information on the lactate-speed relation only for a range of submaximal intensities (fig. 50 (b)) while an all-out 1-Speed test will guarantee the same reliability for near maximal to maximal efforts (fig. 50 (c)). A submaximal 1-Speed test is thus not suitable for accurate competition performance predictions in swimming.

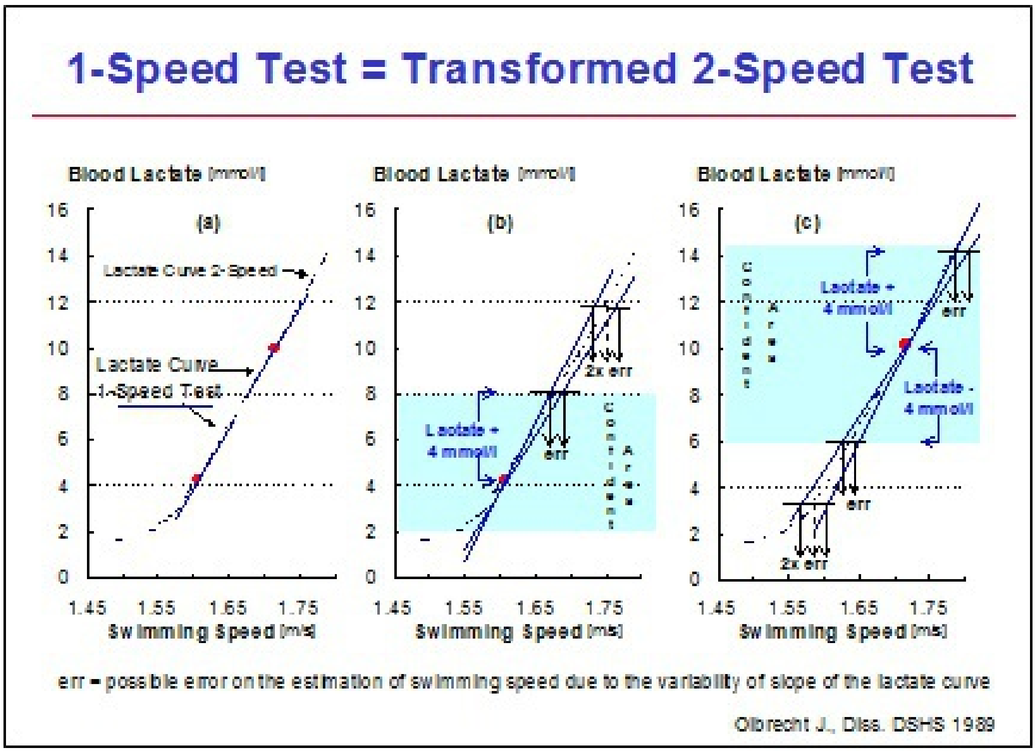


Fig. 50 Based on the nearly constant slope of the lactate curve per stroke, distance and gender, the test distance may be swum only once. One plotted point in the lactate-speed diagram will then define where to draw the lactate curve (a). Limitation of reliability of the lactate curve drawn up after a 1-Speed test: due to small variations in the slope of the lactate curve, the probability of a significant estimation error on speed for a desired lactate value (= err) or vice versa becomes more important the more the wanted lactate value is removed from the measured concentration (b) and (c). It is therefore important to limit the interpretation of the lactate-speed relation

within a lactate range of ☐ 4 mmol/l around the measured lactate value but above 2.5 mmol/l.

The combination of a submaximal 1-Speed test in freestyle with an all-out 1-Speed test in the swimmer’s specialty stroke may provide, for the same time and cost, more information than a 2-Speed test. Indeed, it gives:

* a very good indication on the swimmer’s aerobic endurance and subsequently on the appropriate training intensities for freestyle
* an idea of the swimmer’s aerobic endurance and of the appropriate training intensities in the specialty stroke when the all-out test is performed over at least 200 m
* an indication on the swimmer’s anaerobic capacity in his specialty stroke

Depending on the swimmers’ specialty, we designed following combinations of 1-Speed tests (tab. 9).

	Test protocols using the 1-Speed test		
Personal Best Stroke	Endurance Test “submaximal” - aerobic		Short All-Out Test “maximal” - anaerobic
Freestyle	400 m FS		100 m FS
Fly	400 m FS		100 m FLY
*alternative**	*400 m FS*	*200 m FLY*	*100 m FLY*
Backstroke	400 m FS		200 m BA
alternative	*400 m FS*	*200 m BA*	*100 m BA*
Breaststroke	400 m FS		200 m BR
alternative	*400 m FS*	*200 m BR*	*100 m BR*

* only for top level fly swimmers

Tab. 9 Depending on the swimmers’ specialty, we designed following combinations of 1-Speed tests.

For freestyle and butterfly swimmers the test procedure consists of one 400 m free at submaximal speed (about 20 seconds above 400 m best time) and one all-out 100 m swim in respectively freestyle or fly.

Breaststroke, backstroke and top butterfly swimmers should theoretically be tested over a 400 m in freestyle at submaximal speed followed by a submaximal 400 m and a 100 m all-out effort in their respective discipline. For financial reasons this protocol can be reduced to a submaximal 400 m freestyle and a 200 m all-out effort in the swimmer's specialty. The 400 m freestyle because, even for these swimmers, the greatest part of the training is carried out in freestyle, making the assessment of the training intensity in this stroke necessary. The 200 m distance because it provides fairly reliable information on both the aerobic and anaerobic metabolic performance in the swimmers' specialty stroke.

If the swimmers' performance level becomes better and some more money can be invested, you can add a 200 m submaximal 1-Speed test in their best stroke between the 400 m submaximal swim and the maximal effort. The all-out test will then be a 100 m max. This variation of the test protocol is useful for breaststroke, backstroke and top level fly swimmers.

Individual medley swimmers will be subjected to a submaximal 400 m freestyle and a 200 m medley swum at maximal speed. The all-out test can sometimes vary according to the objective of the next training period. If, for example, during the base training period, breaststroke has to be improved, the 200 m medley will be replaced by a 200 m breaststroke all-out or a 200 m breaststroke 1-Speed test at submaximal speed will be added to the test program in-between the submaximal 400 m freestyle and the 200 m medley all-out.

The choice of test stroke and test protocol depends thus very much on the information the coach wishes to gain about the conditioning of his swimmer and on the swimmers' performance level.

Evaluation of the Conditioning Profile

Aerobic Endurance - The 400 m freestyle, and to a smaller extent the 200 m, swum at submaximal speed are very important tests since they enable to determine the swimmer's aerobic endurance. A simple reference of the lactate curve to quantify the aerobic endurance is the swimming speed reached at 4 mmol/l lactate on the 400 m ($V4_{400}$) or 200 m ($V4_{200}$) (male, fig. 51; female fig. 52).

With some caution (see Shift of the Lactate Curve and fig. 58) we can state that the faster the speed at 4 mmol/l lactate over 400 and 200 m, the better the aerobic endurance. A 100 m lactate test, on the other hand, is too short to give a reliable picture of the aerobic endurance.

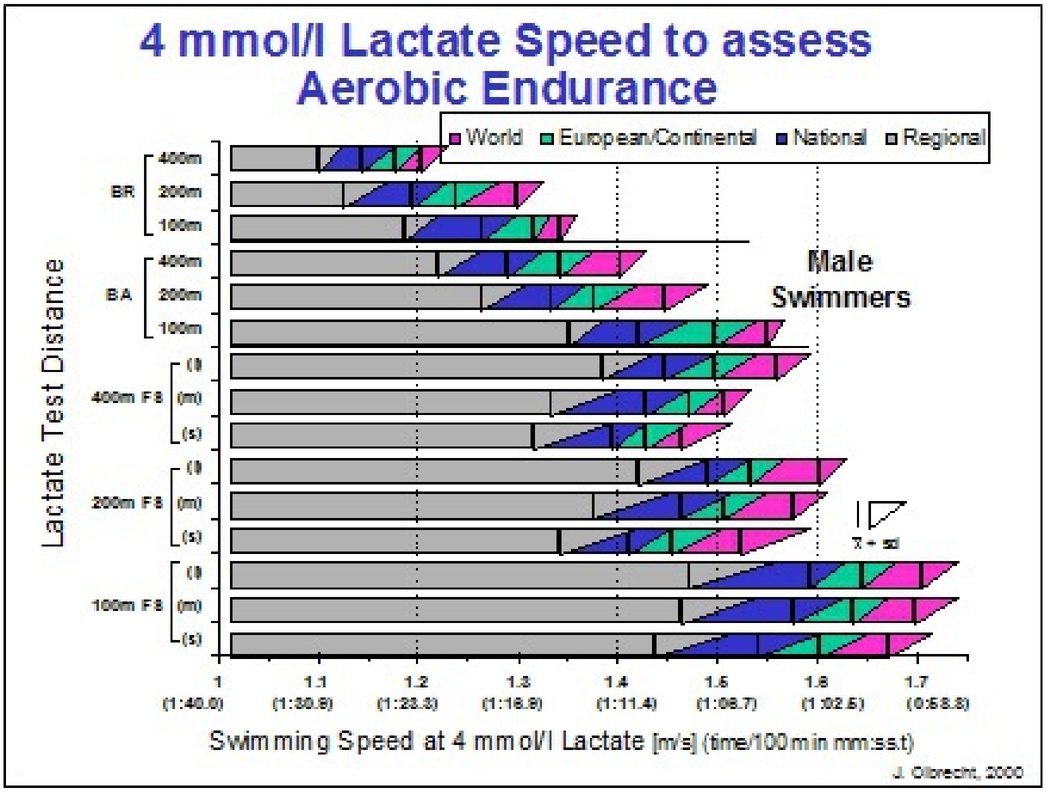


Fig. 51 Mean 4 mmol/l lactate speeds (V4) on 100, 200 and 400 m for male swimmers according to their performance level in competition (World Class, European, national and regional level). The V4 on 400 and 200 m can be used to assess the swimmers' aerobic endurance. sd = standard deviation.

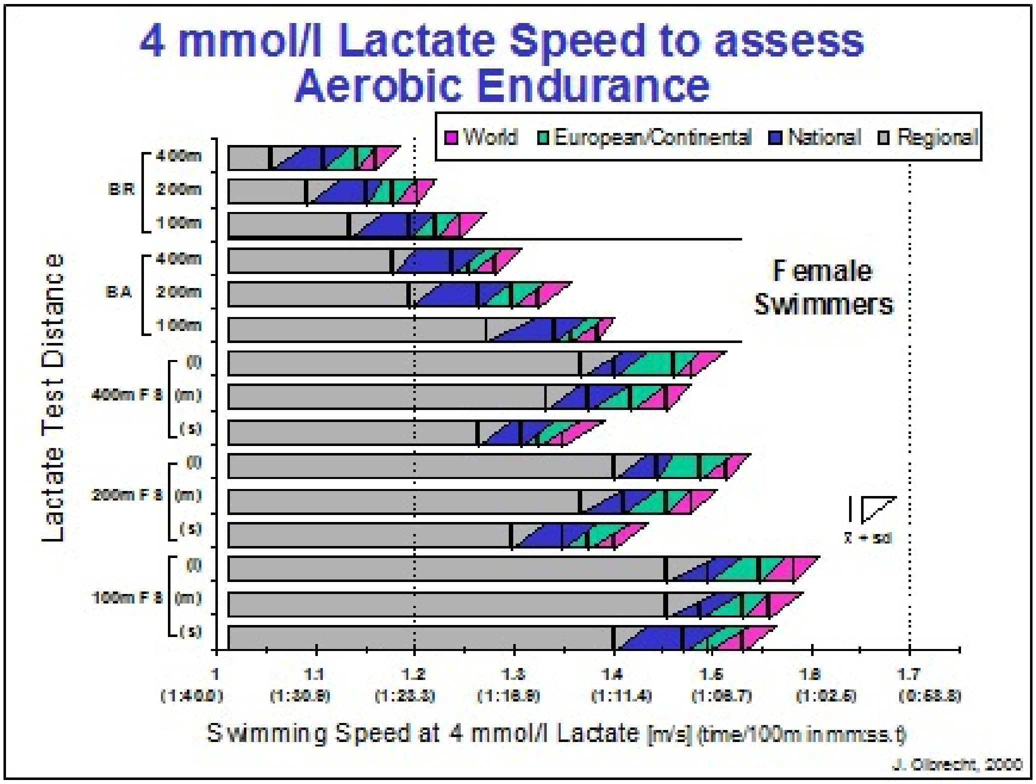


Fig. 52 Mean 4 mmol/l lactate speeds (V4) on 100, 200 and 400 m for female swimmers according to their performance level in competition (World Class, European, national and regional level). The V4 on 400 and 200 m can be used to assess the swimmers' aerobic endurance. sd = standard deviation.

It must however be noted that the 4 mmol/l lactate speed:

* should not be considered as the anaerobic threshold and not as the unique and best training intensity for endurance training
* may have a different meaning for different swimmers

It has been demonstrated that the stronger the swimmer's aerobic capacity, the greater also the contribution of the anaerobic lactic energy system to reach a certain lactate value (Olbrecht 1989). With respect to the 4 mmol/l lactate concentration at the end of a single 200 and/or 400 m swim, we found that the well endurance trained World class swimmer produces up to 25% more lactate per minute than the swimmer of a regional level with a weaker aerobic capacity (fig. 53).

As a higher uptake of oxygen leads to a greater elimination of lactate/pyruvate during exercise, the well endurance trained swimmer will have to produce more pyruvate/lactate per minute and to deliver more anaerobic work to reach the same concentration of lactate at the end of the exercise. The percentage of the total energy supply contributed by the aerobic and anaerobic metabolism to achieve 4 mmol/l lactate does change clearly when varying the test distance but is only slightly affected by the performance level of the swimmer.

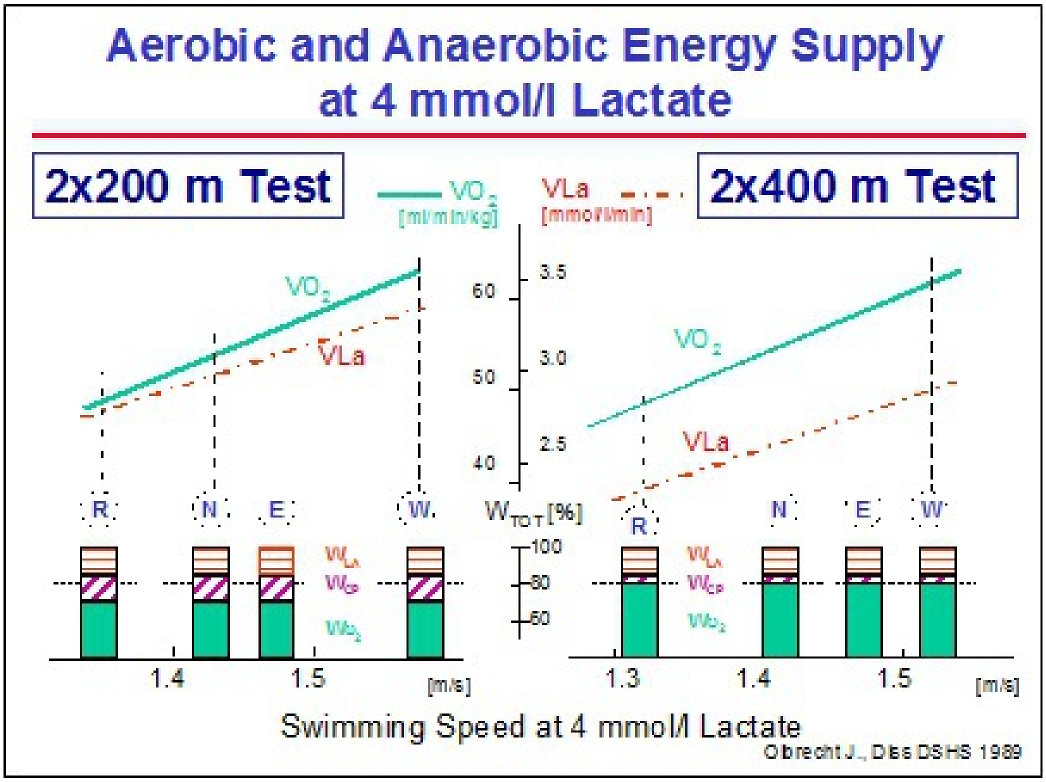


Fig. 53 Oxygen uptake (VO_2) and production of lactate (VLa) as well as the share of the total energy supply (%W_{tot}) provided by the anaerobic lactic (%W_{LA}), anaerobic alactic (%W_{AL}) and aerobic metabolism (%W_{O2}) at various 4 mmol/l speeds over a single 200 ($V4_{200}$) and 400 m ($V4_{400}$). R, N, E and W stand for the swimmer’s performance level: regional, national, European or World class.

Experimental studies (a.o. Madsen 1982) have corroborated that a swimmer with a low anaerobic capacity (cannot produce much lactate per minute) has to invest a higher percentage of his aerobic capacity to attain the same lactate reading as his swim mate with a higher anaerobic capacity. Elite long distance swimmers with a strong aerobic but mostly a low anaerobic capacity, invest some 90% of their measured maximum oxygen uptake at the 4 mmol/l lactate speed. Elite sprinters, on the other hand, with both a strong aerobic and anaerobic capacity, reach 85% of their maximum oxygen uptake at this 4 mmol/l lactate speed (less aerobic capacity, less lactate/pyruvate elimination, earlier lactate accumulation, lower %VO_2 max at 4 mmol/l). Less endurance trained student swimmers attain at 4 mmol/l about 78% of their measured maximum oxygen uptake (fig. 54).

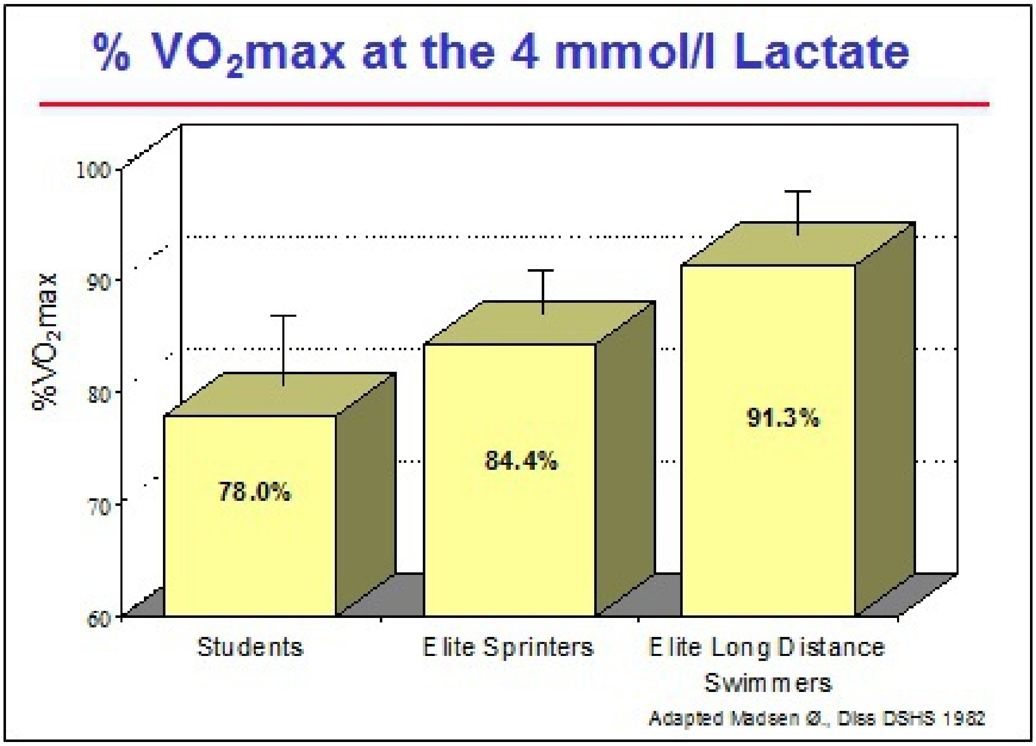


Fig. 54 Influence of a low and high aerobic and anaerobic capacity on the percentage of maximum aerobic capacity (%VO_2 max) that is reached at 4 mmol/l (see text).

Anaerobic Performance - The short all-out effort (200 or 100 m 1-Speed test or a 50/100 m competition lactate concentration) is useful to estimate the swimmers' anaerobic capacity. With some caution (see fig. 59) we can say that the higher the post all-out maximal lactate value, the better the anaerobic capacity. This maximal lactate concentration, however, is only a rough estimation of the anaerobic capacity (VLamax) and will never be as reliable as the VLamax calculated by the simulation program (see Advanced Lactate Test).

Because the simulation approach is not available to everyone yet, we set up a chart of criteria to characterize and help confirm the swimmers' levels of aerobic and anaerobic capacities (tab. 10, Capacities’ Level Chart). By means of this table the coach can typify and place his swimmers in one of the 4 categories below:

1. swimmers with a **high aerobic** and a **low anaerobic** capacity
2. swimmers with a **high aerobic** and a **high anaerobic** capacity
3. swimmers with a **low aerobic** and a **low anaerobic** capacity
4. swimmers with a **low aerobic** and a **high anaerobic** capacity

This typification is very important to reduce the risks of misinterpreting test results and of providing wrong training advice from the classic, less sophisticated standard lactate tests.

Let’s take an example:
A lactate test (e.g. a 400 m freestyle submaximal 1-Speed test followed by 15 min active rest and a 100 m freestyle 1-Speed test all-out) revealed a good aerobic endurance (e.g. a high 4 mmol/l lactate speed on 400 m) and a low maximal lactate level after the short all-out effort (the 100 m freestyle max.). Based on the classic interpretation method these test results indicate that the swimmer has a good aerobic but a low anaerobic capacity. If this diagnosis fits with the swimmer’s typification by means of the Capacities’ Level Chart (tab. 10), the coach may be quite confident about the interpretation of the lactate readings. If, on the other hand, the typification refutes the lactate results, i.e. a high anaerobic capacity deduced from the chart versus a low anaerobic capacity derived from the lactate results, caution is recommended. This contradiction may indicate that the low maximal lactate value after the 100 m freestyle all-out is not due to a weak anaerobic capacity but rather to a lack of anaerobic power (low use of anaerobic capacity). The analysis of the completed training period will shed light and provide evidence to corroborate this hypothesis.

Whether the training for the next period should be adapted or not as a result of this conditioning profile (high aerobic and anaerobic capacity and low anaerobic power) will actually depend on the current training period:

- In the base training period there is no need for special training adjustments except to avoid training exercises such as voluminous extensive workouts which could further break down the aerobic power

- In the pre-competition training period, on the other hand, the anaerobic power exercises have to receive the highest priority

	Low **Anaerobic Capacity**	*High*	
	• No better competition performances after intense or extensive and voluminous work	• Performs very good even when not tapered	
High	• No real feeling of exhaustion after races and impression of being able to swim faster	• Several best times during successive days of competitions	*High*
	• Swimmer performs best in competition shortly (4-6 days) after voluminous training	• Very fast recovery from training and competition	
A	• Swimmer does not like short interval workouts or fartlek exercises	• Reaches high lactates after short and long events	**A**
e	• No clear improvement of competition results if taper lasts longer than 1 week	• Is fast on short and long events	**e**
r	• No high lactates after maximal short as well as long distance events		**r**
o	• Competition best times on long distances are relatively better than on short events		**o**
b	• Best times in long and short course pools are nearly the same		**b**
i			**i**
c		• Reaches high lactates after short and rather low values after long events	**c**
	• Develops overuse-injuries easily	• Swimmer "dies" in last part of event	
	• No high lactates after maximal short events, but high values possible after long events	• Only 1 (or 2) good events in competitions of more than 1 day	
Low	• Bad results on long events and moderate performances in short competitions	• Bad results on long events but very good in short races	*Low*
	• Slow recovery from training and competition	• Best performances after long rest	
	Low **Anaerobic Capacity**	*High*	

Tab. 10 Capacities' Level Chart: chart of criteria to characterize the swimmers' level (high and low) of aerobic and anaerobic capacity.

The example just provided shows that it is extremely important to "confront" the classic interpretation of the lactate readings with the Capacities' Level Chart in order to validate the "lactate based" training advice and training objectives for the next period. In the case of a contradictory diagnosis (classic interpretation of a standard lactate test versus Capacities' Level Chart) always choose for the least risky training advice. If your judgment results in training that is too "soft", it is always possible to catch up later, however, once you have gone too far, i.e. trained too hard, it may take up to 6 weeks to recover.

Determination of the Training Objectives

Lactate tests also permit a prognosis of the competition results. This, of course, should not be a goal in itself but it helps to "track" shortcomings in the swimmer's conditioning. The close relationship between the speed at 4 mmol/l lactate and the speed in competition allows for example, to check whether the swimmer's aerobic endurance is strong enough to achieve the competition time he hopes for (fig. 55).

Suppose a swimmer wishes to swim a time of 4:11 min in a 400 m freestyle competition. According to figure 55, he will have to get a 4 mmol/l speed of 1.457 m/s (4:35 min) over 400 m freestyle. If the 400 m freestyle lactate test reveals a 4 mmol/l time slower than 4:35 min, a shortfall in aerobic endurance will be diagnosed. Depending on the remaining time before the main competition the swimmer is preparing for, the coach may have to spend more time on improving the deficient aerobic capacity and to delay the pre-competition fine-tuning.

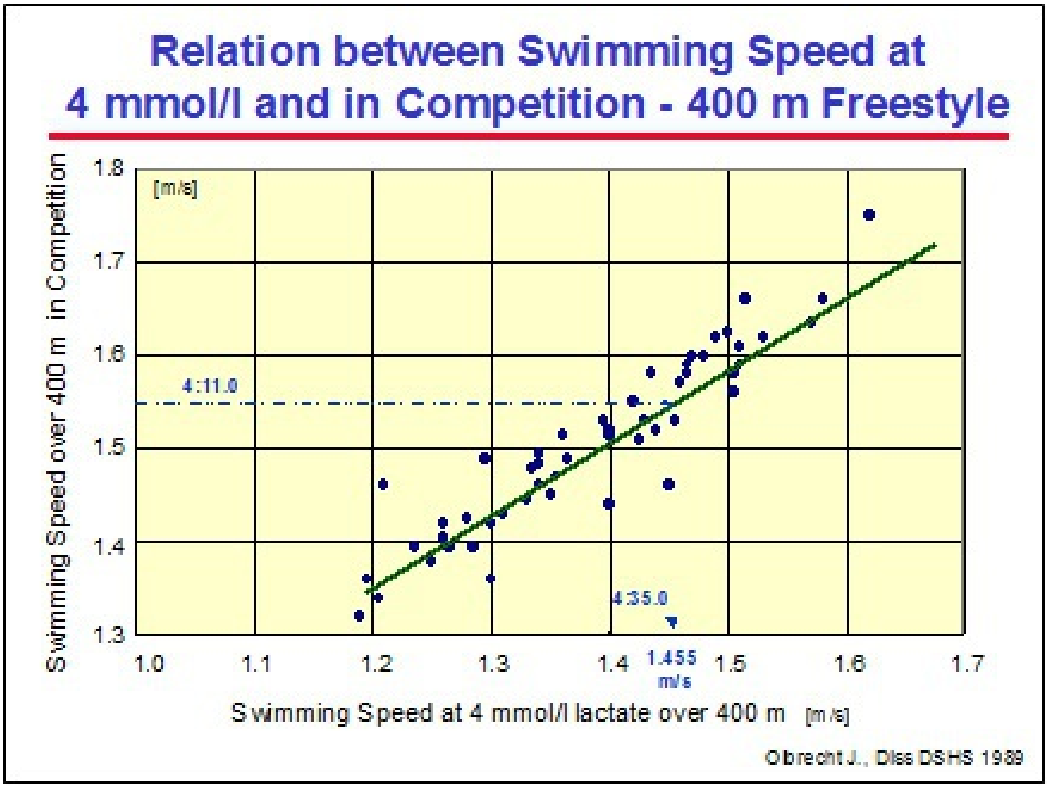


Fig. 55 Relation between the 4 mmol/l lactate speed over 400 m freestyle ($V4_{400}$) and the 400 m freestyle competition speed.

If the established 4 mmol/l lactate speed is only *slightly* slower than 4:35 min and, if the swimmer can reach high lactate readings in competition, the shortfall in endurance can be compensated for with a strong anaerobic capacity (high lactate readings) or anaerobic power. In this case, if it fits with the periodization, the coach can start the pre-competition fine-tuning as scheduled.

If the swimmer reaches a 4 mmol/l lactate speed, which is faster than 4:35 min, he is theoretically capable of realizing his competition goal (4:11 min). If he is, nevertheless, unsuccessful in swimming this goal time in competition, his failing can be blamed on a shortfall in either anaerobic or aerobic power (utilization of respectively the anaerobic and aerobic capacity). A comparison of the actual training structure and lactate readings with those of previous mesocycles will then help in figuring out the reason(s) for the current shortfall (anaerobic or aerobic power) and in selecting the objectives for the next training period.

The importance of the anaerobic capacity increases for short competitions and therefore must also be reckoned with when the lactate results predict competition performances that are actually beyond the swimmer's grasp.

A shortfall in anaerobic capacity can be deduced from the lactate readings registered after a short maximal test. If these readings are high, it is the aerobic and to a smaller extent the anaerobic power that need to be improved. If, on the other hand, they are low, the anaerobic capacity will be polished up first and only then the anaerobic power.

Assessment of Training Intensity and Volume

Intensity - Experiments have enabled us to statistically establish changes in the relationship between lactate and swimming speed according to the different types of exercises used in training (Olbrecht 1985). We can estimate the blood lactate concentrations for different swimming speeds on series of 100, 200 and 400 m intervals with 10 and 30 seconds rest as well as blood lactates during long distance swims lasting at least 20 minutes from a single 400 m swim (fig. 56 for freestyle). This estimation is different for males and females and is based on group averages. It is therefore advisable to check by lactate measurements during training (spot lactate tests) whether these calculations need to be individually adjusted.

This calculation model enables the coach to individualize the intensities (swimming speed) of the different workouts according to the swimmer's conditioning profile in order to make sure he reaches the required blood lactate concentrations corresponding to the training objectives by the end of the set.

Freddie Fast — Test date: 12/31/96 — Pool Length = 50m

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TEST: 1-DISTANCE TEST 1x400 m FS ①

		1st SWIM
Time	(s):	4:42.50
Velocity	(m/s):	1.416
Max.Lactate	(mM):	3.6

DIAGNOSIS ABOUT THE TESTED DISTANCE ②

Distance: 400m

	Linear		Exponential		
Lactate	Speed	Time	Speed	Time	Lactate
2	1.386	4:48.49	1.368	4:52.22	2
4	1.423	4:41.00	1.426	4:40.41	4
6	1.460	4:33.88	1.472	4:31.70	6
8	1.497	4:27.12	1.511	4:24.56	8

GUIDELINES FOR ENDURANCE TRAINING ③

Intensity and swimming times scale for Endurance Training Exercises

Extensive -->-->-->-->-->-->-->-->-->-->-->-->-->--> Intensive

		LA 1		LA 2	LA 3	LA 4	LA 5
Training Type	:	REG EXE	MIE		INE	EAN	
- Set ----- Rest			< Θ>				
100m	10s :	1:13.7	1:11.3	1:09.0	1:08.3	1:06.5	1:05.0
	30s :	1:12.3	1:09.8	1:07.5	1:06.6	1:04.8	1:03.2
200m	10s :	2:30.9	2:25.9	2:21.3	2:19.1	2:16.4	2:13.1
	30s :	2:29.0	2:24.1	2:19.5	2:17.4	2:13.4	2:11.3
400m	10s :	5:05.8	4:55.9	4:46.7	4:43.8	4:38.2	4:32.8
	30s :	5:04.6	4:54.7	4:45.5	4:39.5	4:34.6	4:29.4
continuous swimming			average 100 m time				
20 - 45 min	:	1:18.8	1:16.3	1:14.0	1:12.1	1:10.2	1:08.5

<Θ> = Main Training Intensity

TREND OVER TIME ④

		BEST	LAST	ACTUAL	GOAL	COMP*
V4/400m FS	Date :	30.04.88	17.07.95	31.12.95		
	Time :	4:21.82	4:36.31	4:41.00	4:35.29	4:10.0
V4/200m FS	Date :	23.09.89	12.06.95	31.12.95		
	Time :	2:19.35	2:23.84	2:24.51	2:20.5	2:01.0

(* Goal for competition in the future)

Fig. 56Computer print-out of the results of a 400 m freestyle 1-Speed test swum at submaximal speed. ☐ Measured time and lactate, ☐ calculated relationship between swimming speed/time and lactate for a single swim of 400 m, ☐ statistical prediction of swimming times for series of 100, 200 and 400 m repeats with 10 and 30 sec rest breaks as well as for a continuous swimming lasting more than 20 minutes, ☐ overview of the most important test results. REG = regeneration, EXE = extensive endurance, MIE = middle intensive endurance, INE = intensive endurance, EAN = extensive anaerobic training.

Thus, according to figure 56 Freddie Fast should be able to swim a series of 200 m freestyle with 10 seconds rest in 2:21 min if the coach wants him to train around 2 mmol/l lactate. If the coach wants his swimmer to complete a 400 m repeat set with 30 seconds rest at a blood lactate concentration of 4 mmol/l, the latter will have to cover each 400 m in about 4:35 min. A continuous long distance swim at 3 mmol/l will have to be completed at 1:12 min per 100 m.

The computer print-out (fig. 56) is essentially an aid for the coach. It enables the coach to link, for every tested swimmer, the swimming speed during training to an objective measure of intensity. Whether the swimmer has to train at 1, 2, 3, 4 or above 4 mmol/l lactate, how often he is going to do that per week and to what extent, will depend on the characteristics of his current conditioning profile as determined by the results of the lactate tests.

Since the conditioning profiles can vary a great deal from swimmer to swimmer, training intensity and volume may also vary individually despite the same training objectives, the same training period and the same competition event to prepare.

For example distance swimmers will normally have to swim endurance training sets at lower lactate concentrations than sprinters to avoid the overload of their aerobic and anaerobic systems. This finding, supported by scientific evidence (see Interpretation of Lactate Readings in Training), actually concurred with the empirical observations of coaches working with lactate in training. They noted that, during submaximal workouts, sprinters reach higher blood lactate values than long distance swimmers, are less fatigued and regenerate faster. But, although sprinters, in comparison to long distance swimmers, mostly have a higher anaerobic capacity, this high anaerobic capacity is relative as it can greatly vary within the same type of swimmers.

So, when possible, the coach should estimate the swimmer’s anaerobic capacity by means of lactate values in short all-out races (50 - 100 m), by lactate after a short maximal effort in training (100 m), by the Capacities' Level Chart or by the simulation program to assess the physiological impact of a workout at a given lactate concentration.

As a matter of fact, if high concentrations of lactate are registered during short distance testing in training or in competition, the lactate readings in training may also be somewhat higher. On the other hand, when the maximum readings are low, training will need to be completed with great caution, i.e. at low lactate concentrations. Furthermore, swimmers who reach low maximum lactate readings during these short distance testings in training or in competition usually have a distorted feeling of intensity (hard exercise is felt as light), as a result of which they often swim too fast in training. These findings certainly point to the importance of using a lactate test that includes both a submaximal and maximal swim.

Volume - The aerobic endurance (V4 on 400 m) can also be used as an indicator for the training volume a swimmer can sustain (tab. 11). Indeed, the better his V4, the more training units per week and/or the more distance per training he is able to complete without over-loading. The training frequencies and mileage per training session presented in table 11 prevail during the largest part of the training season. It is clear that during training camps, the coach will take the opportunity to serve up more training units per week and higher volumes per session. During taper or regenerative periods, on the other hand, where the recovery of the swimmer is of primary importance (after illness or overtraining for example), training frequencies and volume per session will remain lower than indicated in the table.

Training Units/Week (mileage/training unit)	Time at 4 mmol/l lactate on 400 m FS				
	slower than 6:00 min	5:30 min	5:00 min	4:40 min	faster than 4:20 min
< 13 years	2 - 3 (1200 - 2200)	4 - 5 (1800 - 2800)	5 - 6* (2000 - 3500)		
13 - 15 years	3 (1200 - 2800)	4 -6 (2000 - 3500)	5 - 8* (3000 - 4500)	6 - 9* (3500 - 5300)	
15 - 18 years	3 - 4 (1400 - 3500)	4 - 7 (2500 - 5000)	5 - 9 (3700 - 6500)	6 - 10 (4000 - 7500)	7 - 10 (4800 - 8300)
> 18 years	4 (1500 - 3500)	5 - 8 (3000 - 5500)	6 - 9 (4200-7000)	8 - 10 (4300 - 8000)	8 - 12* (4700 - 9000)

Tab. 11 Number of training units per week and mileage per unit according to the swimmer's aerobic endurance measured by the 4 mmol/l lactate speed over 400 m. * = Only applicable for 4 to 6 weeks a year, swim camps not included.

To summarize:

- On the basis of a 400 m freestyle at submaximal speed we can easily estimate for various training exercises the swim times corresponding to the desired lactate concentrations. These lactate values provide an objective standard for the intensity at which the swimmer is training

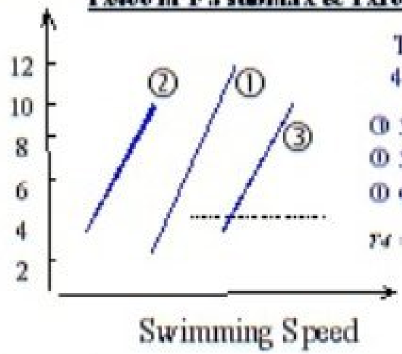
- The lactate value during training, which is in accord with a specific training objective, is not the same for every swimmer. Indeed, even when the training objective is the same, the lactate value during training will have to be adjusted according to the swimmer's aerobic and anaerobic capacities. Moreover, the weekly volume and frequency of training exercises at different lactate concentrations also depend on each individual swimmer's level of these 2 capacities

- This may lead to a different training approach for each swimmer despite the fact that they are all in the same point of the season and are preparing for the same competitive event

Example:

1-Speed Test:	test in 50 m pool	La-Spot Test:

Lactate



1x400 m FS submax & 1x100 m FS max

T4 on 400 m	Lmax on 100 m
① 5:13.7	7.5 mmol/l
② 5:25.3	8.9 mmol/l
③ 4:50.5	10.9 mmol/l

T4 = time at 4 mmol/l lactate

(ad test ③)
6x100 m FS start 3min

mean split: 1:04
La= 7.3 mmol/l

PROGNOSIS FOR ENDURANCE TRAINING (ad test ③)
Intensity and swimming times scale for Endurance Training Exercises

Extensive -->-->-->-->-->-->-->-->-->--> Intensive

	LA 1		LA 2	LA 3	LA 4	LA 5
Training Set :	REG EXE		MIE	INE	EAN	
- Set		◎				
100m :	1:23.3	1:18.7	1:16.2	1:15.5	1:13.4	1:11.8
200m :	2:34.7	2:40.3	2:35.9	2:33.5	2:30.6	2:27.0
400m :	5:35.6	5:25.2	5:16.4	5:13.6	5:07.2	5:01.0
contin swim - average 100 m times						
20 - 45 min :	1:27.2	1:24.3	1:21.4	1:19.3	1:17.2	1:15.8

Background:

Swimmer

- *Boy, 15 years old*
- *Good experience (6yrs) / Max 5 training units per week*
- *Objective: realizing 2:05 to be selected for Age Group National Championships*
- *Aerobic endurance (T4 on 400 m) to achieve 2:05 on 200 m FS should be at least 5:00*

(ad test ①)
TRAINING ANALYSIS – MAIN SWIMMING SETS

Sets	*Swim times*	*Estimated LA*
100 m	*Slowest: 1:12*	*5 mmol/l*
	Fastest: 1:08	*8 mmol/l*
200 m	*Slowest: 2:23*	*4 mmol/l*
	Fastest: 2:20	*8 mmol/l*
400 m	*Slowest: 4:50*	*6 mmol/l*
	Fastest: 4:42	*9 mmol/l*

TEST ①

Key Results: This swimmer has a good aerobic endurance which nevertheless is insufficient (5:14) to achieve the selection time of 2:05 on 200 m FS. The anaerobic capacity, however, is too low for a middle distance swimmer, even for a boy aged 15.

Training Advice: The aerobic endurance should be improved. Based on his current conditioning profile, the swimmer gets an intensity table enabling him to perform the endurance workouts at the appropriate swimming speeds (80% of aerobic interval sets at LA1). The coach is also advised to increase the number of anaerobic capacity sets per week.

TEST ②

Key Results: After the first lactate test we aimed at the improvement of the aerobic endurance. The second lactate test 2 months later, however, showed a decrease of the swimmer's aerobic endurance (5:25). The analysis of the intensity of his swimming during the last training period (see left upper box) revealed that most of the aerobic endurance sets were completed at speeds which produced blood lactate concentrations between 5 and 8 mmol/l. This was much too high for the aerobic endurance training to be effective. Moreover, the coach thought he could compensate for the lack of training time by swimming the sets somewhat faster.

The anaerobic capacity improved which was one of the training effects hoped for.

Training Advice: Swimming less but faster induces different training effects than swimming more and slower. You cannot compensate for less swimming by swimming faster sets. So the coach was advised to control speed in training more rigorously and to keep the anaerobic capacity training unchanged since it improved progressively.

TEST ③

Key Results: After a further 2 months a 3rd lactate test was performed. A strong decrease of the training intensity had induced a strong improvement of the swimmer's aerobic endurance within the last training period. His aerobic endurance is now strong enough to realize the 2:05 min selection time for the National Age Group Championships. Moreover, the anaerobic capacity had further improved.

Training Advice: There being 6 weeks left before the selection meeting and since the swimmer's aerobic endurance was strong enough, the coach was advised to replace 1 anaerobic capacity set with an anaerobic power exercise during each of the next 2 weeks. During the 3rd week the training volume should be increased and the intensity clearly lowered in order to refresh the aerobic endurance. The 4th week should start with 2-3 regeneration days before carrying on with 7 very intense training days (1 anaerobic capacity set and 2 anaerobic power workouts). The taper should begin after this microcycle.

Advanced Lactate Test: Simulation of the Aerobic and Anaerobic Capacities The most important physiological parameters to describe the conditioning profile in swimming are the aerobic (VO_2 max) and anaerobic (VLamax) capacities. These parameters play a decisive part not only in determining the swimmer's maximal competition performance but also in the way the aerobic and anaerobic systems contribute in the metabolic energy supply during exercise.

With the advanced evaluation model for the interpretation of lactate tests (Olbrecht 1989) by means of a simulation program that was patterned after Prof. Mader's - German University of Sports Sciences of Cologne - theoretical model of metabolism, we can now determine both the aerobic and anaerobic capacities of technically well skilled competitive swimmers. Unlike the classical method of interpretation, where the assessment of the conditioning is directly based on the relation between lactate and speed, i.e. the position of the lactate curve in the lactate-speed diagram, this new evaluation technique uses the lactate values to determine the 2 decisive factors - the aerobic (VO_2 max) and anaerobic capacities (VLamax) - that generate the lactate-speed relation.

These capacities:	• are thus the key factors in the determination of the individual metabolic profile
	• can be used to simulate on computer the individual metabolic response that can be expected on different types of training exercises. According to the results of the simulation we then decide whether the metabolic reaction induced by a workout serves the purpose of the training objectives or not

A comparison between the measured and simulated oxygen uptake (VO_2) and lactate during different submaximal running efforts has shown a very close similarity (fig. 57).

This approach is thus quite different from most other lactate testing procedures which, to describe the conditioning profile of the swimmer, use:

* a single speed such as the speed at the lactate or some other anaerobic threshold (e.g. the speed at 4 mmol/l) or,
* the slope and/or shape of the lactate curve

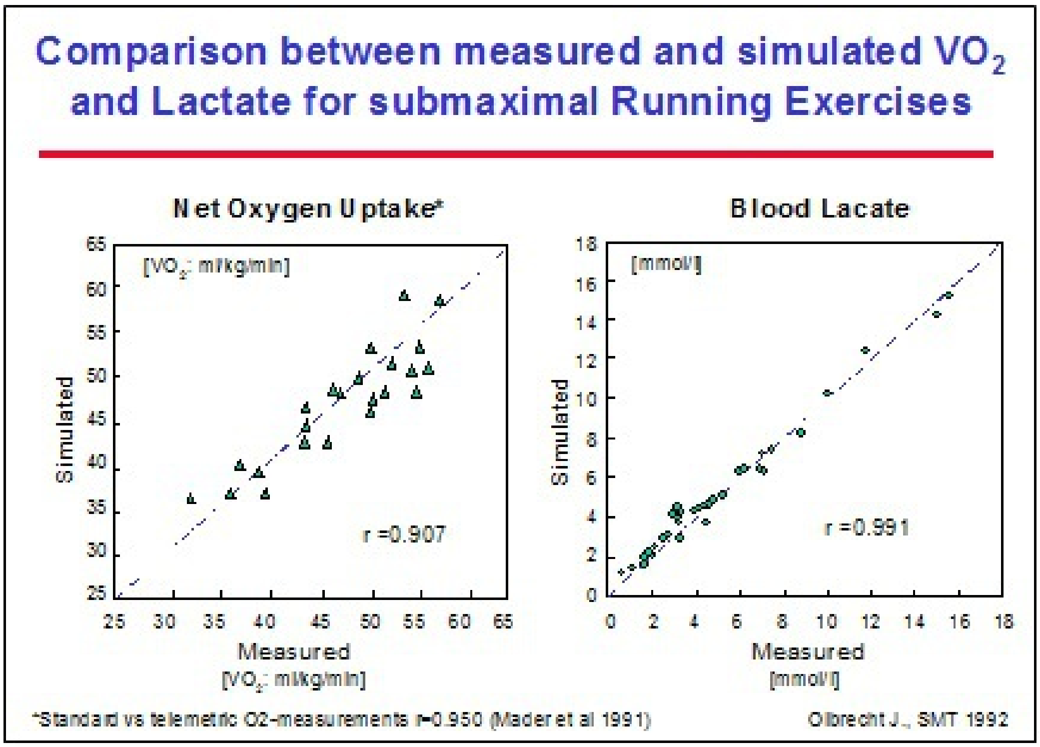


Fig. 57 Measured and simulated oxygen uptake (VO_2 , left) and lactate (right) at different submaximal running efforts have proven to be quite similar (Olbrecht 1992).

Indeed, this new model discloses the metabolic process behind blood lactate values and provides insight into the often paradoxical evaluations of the conditioning profile as well as the often contradictory and discrepant training advice emanating from the classical method of interpretation.

A comparison between the classic interpretation of the shift of the lactate curve (move to the right and the left is explained by respectively an increase and decrease of VO_2 max; see Interpretation of the Lactate Curve) and the interpretation based on the use of the simulation program for determining the aerobic and anaerobic capacities, reveals that:

* only about 60% of the classic interpretations of the aerobic capacity fits with the more sophisticated valuation (fig. 58). This means that out of 5 lactate tests classically evaluated, on average 2 interpretations are incorrect and will inevitably lead to a wrong training advice
* an improvement or deterioration in a 100 m all-out test can only be explained by respectively an improved or deteriorated anaerobic capacity in 50% of the cases (fig. 59)

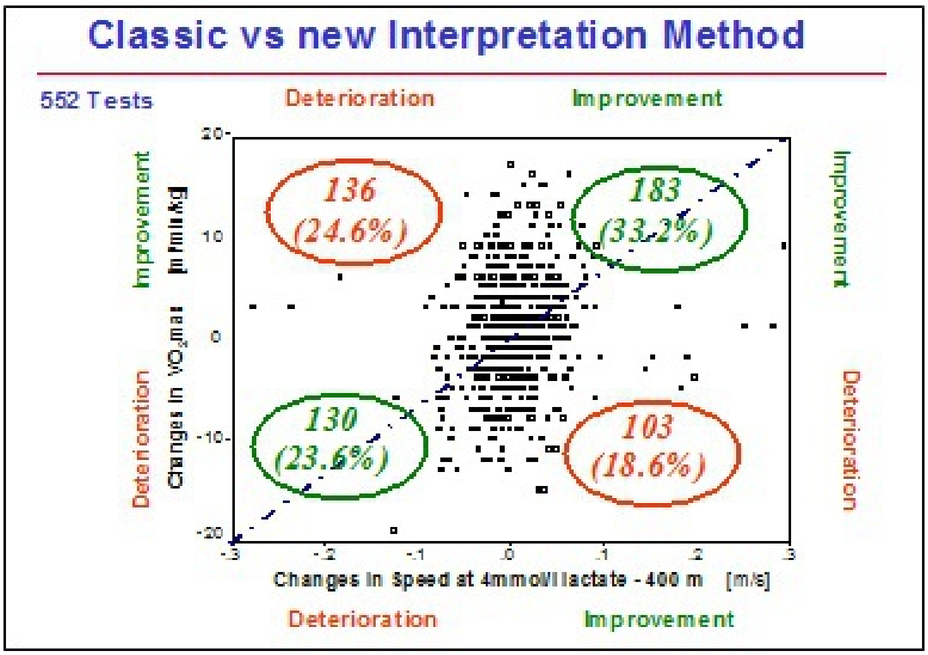


Fig. 58 About 60% of the classic interpretations of the aerobic capacity (VO_2 max) fits in with the more sophisticated valuation.

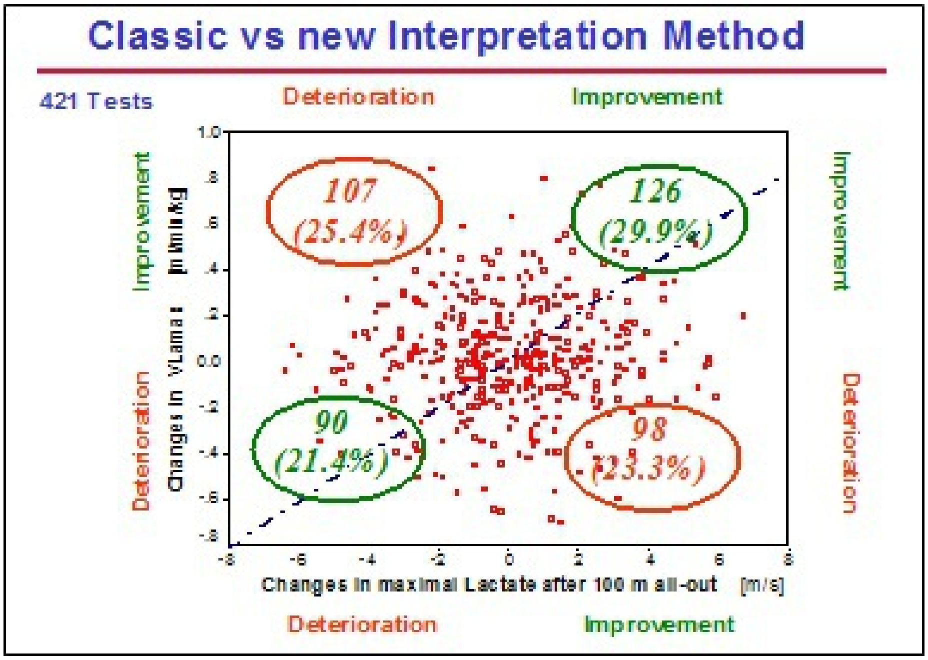


Fig. 59 An improvement or deterioration in a 100 m all-out test (Lamax) can only be explained by respectively an improved or deteriorated anaerobic capacity (VLamax) in 50% of the cases.

Why simulate VO $_2$ max when it is possible to measure it?

It is indeed possible to measure the maximal oxygen uptake but the results are hardly ever correct, not to mention representative. As a matter of fact to be somewhat representative, the VO_2 max of a swimmer has to be measured by free swimming in the swimming pool (possibly also in a swimming flume, although not every flume is suitable) and not on a cycle ergometer or a treadmill (see Relevance and Reliability of the Lactate Readings). This, however, is time consuming and requires considerable expertise.

Besides, the equipment for oxygen uptake measurements is expensive, quite delicate and sensitive to error. Let's give an example: an error of 100 ml/min ($\pm$ 50 ml/min) is quite a good standard in VO_2 max measurements. For a swimmer of 70 kg this represents an error of 1.43 ml/kg/min. This very same error in oxygen uptake results in a difference in time of about 1.5 sec per 100 m for swimmers. It is, thus, not possible by means of VO_2 max measurements to make a reliable distinction between 2 swimmers if their best times do not differ by more than 1.5 sec per 100 m. In other words, the accuracy of the oxygen uptake values is insufficient to distinguish between 2 swimmers who cover a 400 m freestyle in, respectively, 4:44 and 4:50.

The accuracy of the lactate readings on the other hand allows us to discern between 2 swimmers whose 400 m freestyle best time differs by only 1 second. It is, thus, clear that the lactate test enables a much more accurate analysis of the metabolic profile.

Moreover, to measure VO_2 max, the swimmer has to wear a mask, which considerably increases the water resistance, seriously hinders and impedes him and, last but not least, undermines his motivation.

We can conclude that measuring oxygen uptake:

* brings about a lot of practical and financial problems
* lacks the power to determine with enough accuracy the metabolic profile of elite swimmers
* limits the swim specific relevance of the test results when compared with lactate readings

Due to the specific dynamics of oxygen uptake no test protocol will ever be able to assess directly the real VO_2 max. The measured VO_2 max will always be underestimated. By means of our simulation program, however, we are able to calculate VO_2 max and this calculated maximal oxygen uptake always turns out to be higher than the measured one.

Test Protocol

The input for the simulation program are the times, distances and post-exercise lactate readings of different submaximal 1-Speed tests. The swims are separated by a rest period to get the swimmer back to "resting" lactate and oxygen uptake levels. It is important to start each interval from a resting lactate level to avoid any interaction with the post-exercise reading.

For example, a swimmer performs a 400 m, 200 m, 150 m and 100 m backstroke in one test session. Each interval will be swum at a constant and submaximal speed but will be progressively faster as the distance becomes shorter.

Lactate readings will be determined in the 1st, 3rd and if necessary also in the 5th minute after each swim. The rest period between the swims should be of approximately 10-20 min; 5-15 min of active rest during which the swimmer is asked to swim at regeneration speed in order to get back to a “resting” lactate and at least 5 min of passive rest to get back to “resting” oxygen uptake values.

After this rest period, he can then start the next interval. Breaststroke and freestyle swimmers follow the same test protocol but in their respective best stroke. Butterfly swimmers will swim the freestyle program. Individual medley swimmers will perform the test in the stroke the coach wants to stress in training.

The objective of our evaluation algorithm is to parse the post-exercise lactate readings into the contribution and peak value of both the aerobic and anaerobic metabolic processes. The traditional “step test” is therefore only partially appropriate for this type of evaluation method. The simulation program calculates:

* VO_2 max and VLamax (respectively the aerobic and anaerobic capacities)
* the contribution of aerobic and anaerobic energy supply for different type of exercises at different levels of lactate
* the lactate elimination rates during different exercises as well as the maximal elimination rate
* the maximal lactate steady state ($MaxLa_{ss}$, lactate elimination = production)
* the anaerobic power

This simulation model has been validated with more than 800 tests in swimmers ranging from age group swimmers to World champions but should, as it is quite elaborate and expensive to use, be currently reserved for athletes of at least national level.

Evaluation of the Conditioning Profile

We have evaluated over 200 elite swimmers in the last 6 years using this simulation approach. Figure 60 presents the mean aerobic and anaerobic capacity results observed in female and male sprinters, middle and long distance swimmers. In table 12 you will find the results of the top 3 swimmers for each stroke (freestyle, backstroke, butterfly (test in freestyle) and breaststroke).

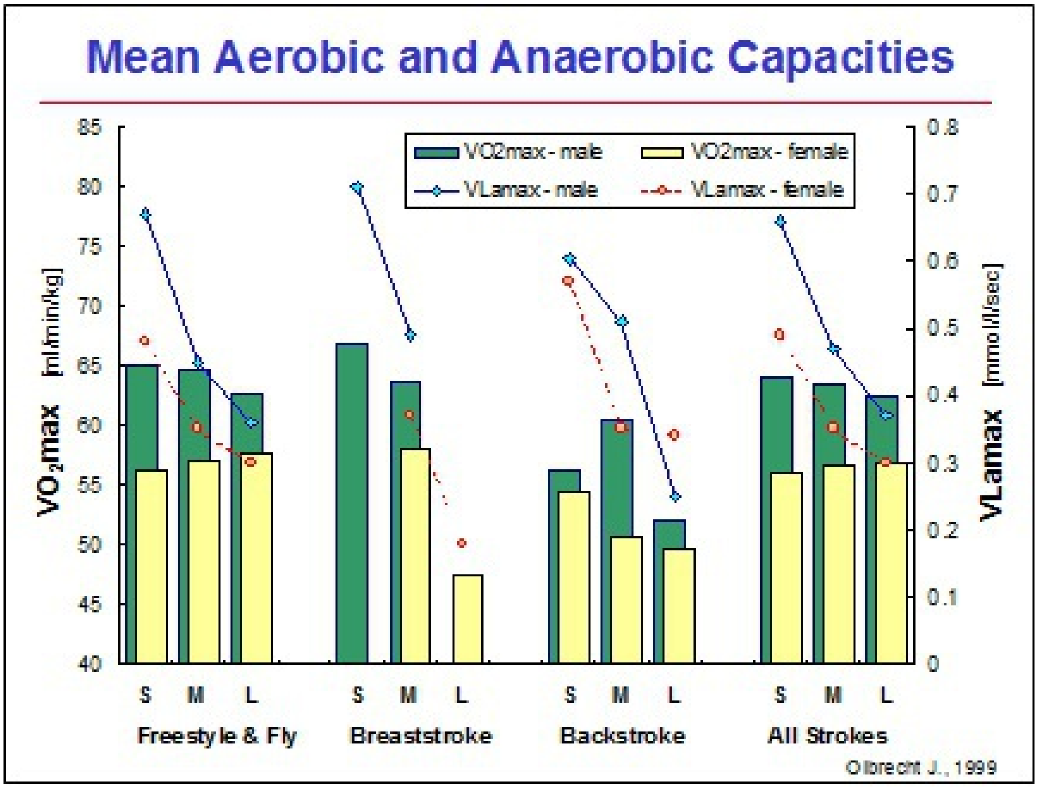


Fig. 60 Mean aerobic and anaerobic capacity calculated for female and male sprinters, middle and long distance swimmers. Due to the specific dynamics of the oxygen uptake, the measured VO_2 max will always be underestimated.

WOMEN	Top 3 level Range of Best Times (50 m pool) (times)	(initials)	Range of Capacities Aerobic	Anaerobic
SPRINT 50m	24.88 - 25.91 - 26.04	IDB - GC - TB	64 - 64 - 64	0.55 - 0.74 - 0.95
FREE 100-200m	54.89 - 56.33 - 56.90	IDB - TB - MVR	64 - 64 - 63	0.55 - 0.95 - 0.76
400m	4:10.52 - 4:11.71 - 4:12.10	KV - IA - CG	67 - 68 - 63	0.37 - 0.60 - 0.4
800m	8:31.60 - 8:34.56 - 8:36.51	KV - IA -CG	67 - 68 - 63	0.37 - 0.60 - 0.40
BACK 100-200m	2:16.53 - 1:03.54 - 1:05.01	SW - BS - SVD	59 - 57 - 56	0.46 - 0.63 - 0.61
BREAST 100-200m	2:27.66 - 1:10.17 - 1:12.52	BB - IK - HB	66 - 73 - 60	0.74 - 1.04 - 0.54
FLY* 100-200m	58.43 - 1:01.11 - 1:01.44	IDB - GC - FD	64 - 64 - 68	0.55 - 0.74 - 0.61

MEN	Top 3 level Range of Best Times (50 m pool) (times)	(initials)	Range of Capacities Aerobic	Anaerobic
SPRINT 50m	22.06 - 22.54 - 22.87	PVH - MV - JK	84 - 72 - 77	1.00 - 1.25 - 1.03
FREE 100-200m	1:46.41 - 1:50.49 - 49.87	PVH - JK - MV	86 - 77 - 72	0.97 - 1.03 - 1.25
400m	3:54.46 - 3:55.56 - 4:02.95	MW - PVH - BR	80 - 80 - 73	0.88 - 0.79 - 0.66
1500m	15:30.61 - 15:47.55 - 16:30.55	MW - SR - SM	80 - 79 - 77	0.88 - 0.81 - 0.88
BACK 100-200m	1:59.64 - 57.10 - 2:01.51	SM - KEZ - RR	79 - 72 - 74	0.75 - 0.86 - 0.72
BREAST 100-200m	1:00.80 - 1:02.50 - 1:02.80	FDB - MW - MF	73 - 76 - 77	0.99 - 0.80 - 0.89
FLY* 100-200m	53.68 - 54.08 - 1:59.20	SA - KJ - FS	77 - 78 - 73	0.68 - 1.09 - 0.94

Tab. 12 Range of the best aerobic and anaerobic capacities of the top 3 swimmers together with a range of their long course best times. * Fly swimmers are tested in freestyle.

Aerobic Capacity - It is quite easy to evaluate the aerobic capacity as it can never be too high. Globally we can state that, whatever the swimmers' type (sprinter or long distance swimmer), the higher his aerobic capacity the better:

* his chances of a good performance in competition

* his ability to handle and assimilate more training volume, higher training intensities and longer mesocycles

Anaerobic Capacity - The evaluation of the anaerobic capacity, on the other hand, is much more difficult. First, it is the event the swimmer is preparing for, that determines what level of anaerobic capacity is expedient. A 50 m sprinter, for example, should have a very high anaerobic capacity (>0.9 mmol/l/kg/min) as a high level provides him with a lot of energy in a very short period of time and allows him to maintain his maximum sprint speed till near the end of the 50 m. This huge delivery of energy will, of course, provoke a sudden and strong acidosis in the muscle, but since the event only lasts 22 to 35 sec, the onset of the inhibition of the anaerobic lactic system due to this acidosis will only occur in the very last part of the race.

The longer the distance of the competition event, the lower the anaerobic capacity has to be in order to:

* avoid a premature inhibition of the anaerobic system due to the acidosis it induces at its maximal contribution during the race (= anaerobic power)
* enable the aerobic capacity to be involved as much as possible (= maximizing the anaerobic power)

Secondly, the evaluation of the swimmer's anaerobic capacity also depends on his level of aerobic capacity. Indeed, a better aerobic capacity allows a higher contribution of the anaerobic capacity for the same onset of acidosis. This means extra benefit in terms of energy delivery, as more energy will be supplied not only by the aerobic but also by the anaerobic chain before an inhibition of the energy production due to the acidosis sets in. The simulation program can estimate the optimal anaerobic capacity for a given aerobic capacity and competition event (stroke and distance).

Determination of the Training Objectives

The huge advantage of the simulation program is that it opens a window on the metabolic processes behind the "numbers (= lactate readings and speed)". Training objectives are now fixed according to the strengths and weaknesses of the aerobic and anaerobic metabolic systems with respect to the competition event the swimmer is preparing for.

The periodization is the other essential factor that influences the determination of the training objectives. During the base training period, priority is given to the improvement of the aerobic and, if necessary, also of the anaerobic capacity. If it is, indeed, the anaerobic capacity that needs to be improved, the coach will not increase the volume of the anaerobic capacity sets but instead raise their frequency per week by either adding an extra anaerobic capacity training session in some weeks or by programming an anaerobic capacity set as the intensive part of an aerobic capacity workout. Moreover, the effect of the anaerobic capacity sets should be screened off from any negative interference with surrounding workouts such as too voluminous exercises. Long distance swimmers may, during this training period, have an anaerobic capacity that is somewhat too high since it decreases the risk of overloading the aerobic energy system and so protects them from overtraining.

If, on the other hand, the anaerobic capacity is strong enough, the coach can increase the training volume and design the intensive part of the aerobic capacity workouts with whatever intense exercise, even an aerobic power like mini-set.

During the pre-competition training period (i.e. the peak form inducing part of the competition period) the objective normally is for the long distance swimmers to improve their aerobic power and for the sprinters to raise their anaerobic power. But, since the potential to build up aerobic or anaerobic power strongly depends on the level of the respective capacities, the coach may decide to break the rule. That is to adapt, to a certain extent, the principal objective of this training period and to first focus on minor adjustments of the anaerobic capacity in order to establish a better balance between both capacities before building up the required power. Ideally the balance between both capacities should already be established at the end of the base training period, but practice shows that this is only rarely the case.

The length of the pre-competition training period and the frequency of the power workouts will also be adjusted according to the acquired level of both capacities. Sprinters with both a high anaerobic and aerobic capacity will not only need a longer pre-competition training period but will also need more anaerobic power training sets than those with a lower anaerobic capacity. Long distance swimmers with a very low anaerobic capacity will, whatever the level of their aerobic capacity, clearly need only a few aerobic power workouts to reach peak form. On the other hand, long distance swimmers combining a high aerobic with a good anaerobic capacity will need a longer pre-competition training period and more sets of aerobic power training than the distance swimmers with a lower anaerobic capacity.

Assessment of Training Intensity and Volume

With the classic 1- or 2-Speed test we are able to determine for each tested swimmer the swimming speed for sets of 100, 200 and 400 m fractions (interval distances) or for a continuous swim that will lead to the desired blood lactate concentration at the end of the workout (see fig. 56). This test, however, does not provide any evidence whether this desired blood lactate concentration meets the intensity fitting in with the objective of the exercise.

Since we are now able to reproduce swim efforts on computer, we can check, prior to the execution of the training set, which blood lactate concentration fits with the metabolic activity we want to induce in the muscle (see fig. 45). This makes the characterization of the training stimuli much more accurate. The proactive check together with the findings from the analysis of the previous training periods, that show how each swimmer responds to the different types of training exercises, determine the allocation of the correct intensity to the objective of the training set. This is particularly important for aerobic endurance sets, since this group of workouts is most sensitive to an inappropriate intensity (mostly too high).

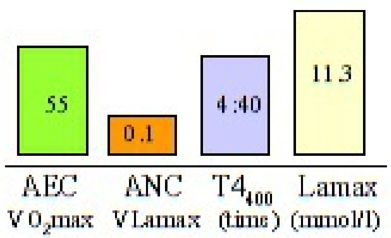
Conclusion:

The classic interpretation of lactate tests can give a distorted picture of the development in conditioning and can mislead coaches and athletes where the determination of the training objectives and intensities is concerned. Thus, for top athletes, where the slightest mistake in training can decide on success or failure, it is absolutely necessary to reject the classical method of interpretation and to use the lactate readings only to determine the characteristics of the athlete's metabolic processes. It is important that the recommendations for training be based on these characteristics and definitely not be deduced directly from the lactate-speed relation, i.e. the lactate curve

Example:

Test: 08 Oct. 1994

Results



Test Interpretation

- Very low anaerobic capacity (ANC), even for a good 400 m swimmer
- Good aerobic capacity (AEC)
- Very high anaerobic power (ANP)

($T4_{400}$ = time at 4 mmol/l lactate over 400m FS)
(Lamax = highest blood lactate after 100m all-out)

Background:

Swimmer

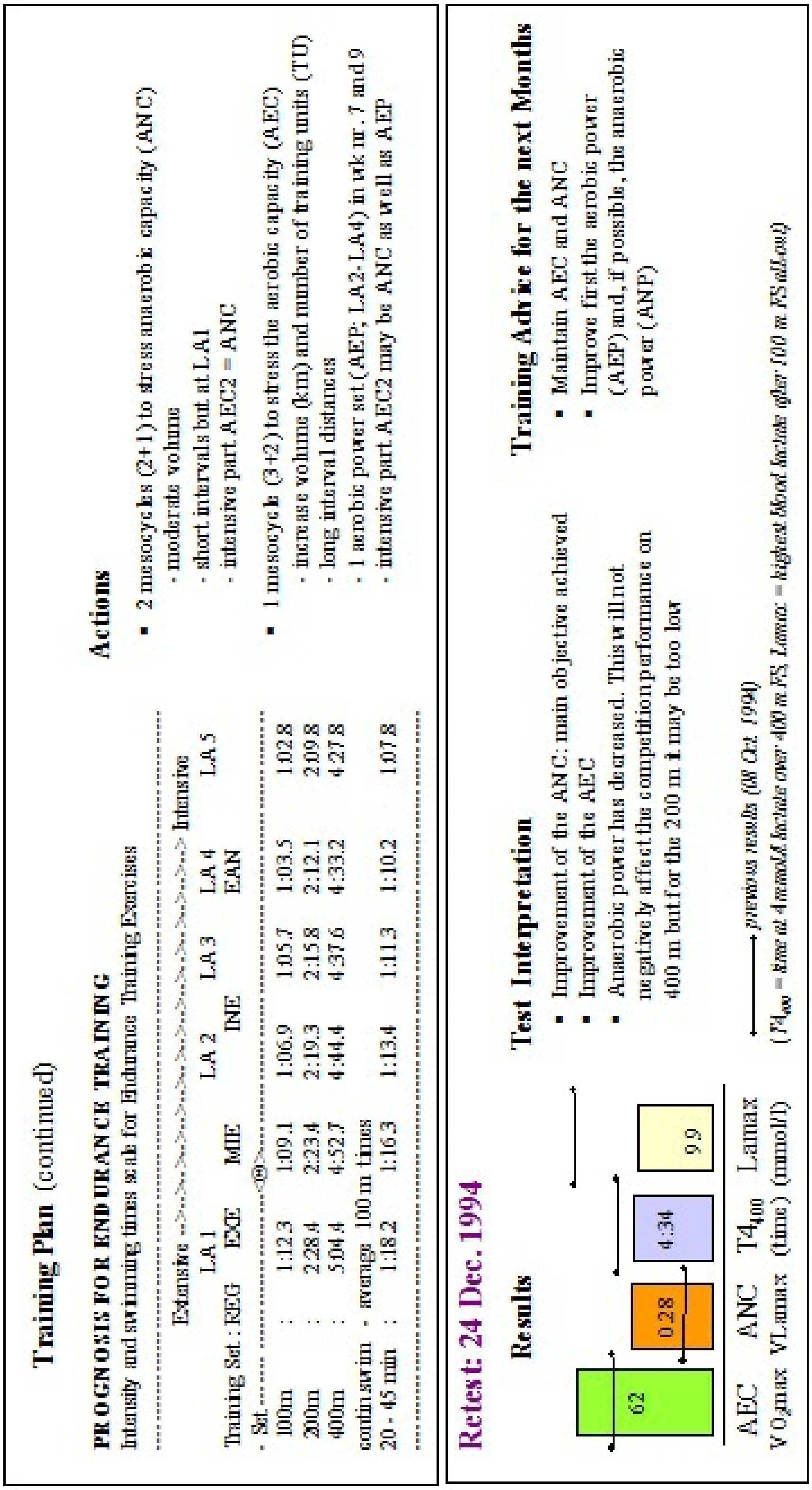
- *Boy, 16 years old*
- *Personal best stroke 200 and 400 m Freestyle*
- *Good experience (5 yrs) / Max 8 training units per week*
- *Objective: Top competition beginning of February 1995*

Training Plan

Wk Nr.	1	2	3	4	5	6	7	8	9	10	11
Km	35	40	25	35	45	28	40	45	55	35	30
TU	8	8	7	8	8	7	8	9	10	8	7
REG	3	3	6	3	4	6	4	6	6	7	6
AEC1(2)	1(2)	1(2)	-	1(2)	1(2)	-	2(2)	2(3)	1(2)	1	-
ANC	2	2	1	1	3	1	1	1	2	-	1
AEP	-	-	-	-	-	-	1	-	1	-	-
ANP	-	-	-	-	-	-	-	-	-	-	-
COMP	-	-	x	-	-	x	-	x	-	-	x
	mesocycle 3			mesocycle 4			mesocycle 5				

Training Advice

- Highest priority: increase of ANC
- Avoid moderate intensity, remain at LA1 or even slower for extensive endurance work
- Intensive work through ANC sets and competitions
- Maintain/increase AEC



Training Plan (continued)

PROGNOSIS FOR ENDURANCE TRAINING
Intensity and swimming times scale for Endurance Training Exercises

Extensive -->-->-->-->-->-->-->-->-->-->-->-->--> Intensive

Training Set :		LA 1 REG	LA 1 EXE	MIE	LA 2 INE	LA 3	LA 4 EAN	LA 5
- Set				<®>				
100m	:		1:12.3	1:09.1	1:06.9	1:05.7	1:03.5	1:02.8
200m	:		2:28.4	2:23.4	2:19.3	2:15.8	2:12.1	2:09.8
400m	:		5:04.4	4:52.7	4:44.4	4:37.6	4:33.2	4:27.8
contin swim - average 100 m times								
20 - 45 min	:		1:18.2	1:16.3	1:13.4	1:11.3	1:10.2	1:07.8

Actions

- 2 mesocycles (2+1) to stress anaerobic capacity (ANC)
  - moderate volume
  - short intervals but at LA1
  - intensive part AEC2 = ANC
- 1 mesocycle (3+2) to stress the aerobic capacity (AEC)
  - increase volume (km) and number of training units (TU)
  - long interval distances
  - 1 aerobic power set (AEP; LA2-LA4) in wk nr. 7 and 9
  - intensive part AEC2 may be ANC as well as AEP

Retest: 24 Dec. 1994

Results

Test Interpretation

- Improvement of the ANC: main objective achieved
- Improvement of the AEC
- Anaerobic power has decreased. This will not negatively affect the competition performance on 400 m but for the 200 m it may be too low

Training Advice for the next Months

- Maintain AEC and ANC
- Improve first the aerobic power (AEP) and, if possible, the anaerobic power (ANP)

⟶ *previous results (08 Oct. 1994)*

($T4_{400}$ = time at 4 mmol/l lactate over 400 m FS, Lamax = highest blood lactate after 100 m FS all-out)

Heart Rate

The heart rate is a frequently used, but nevertheless coarse and inaccurate, parameter for determining training intensity. There are several reasons for this:

1. the heart rate reflects the heart's reaction to numerous internal and external stimuli. The physical stress during training is certainly an important internal stimulus, but is not the only factor that influences the pulse rate. Indeed, at a same training intensity the swimmer's heart rate may vary depending on his mental stress, his degree of tiredness and the condition of his health, but also according to external environmental factors, such as the temperature, the wind and the length of the set. The heart rate, therefore, reflects more than just the degree of metabolic activity during exertion
2. slight differences in heart rate during training can, furthermore, be coupled with great shifts in the nature of the exercise. Training with the same heart rate can, therefore, mean a different type of training from day to day. If you also take into account the influence of the external environmental factors mentioned above, then, despite accurate measurements, the heart rate cannot be regarded as a reliable means of checking training intensity (tab. 13) or defining the type of exercise

Example**

A swimmer completes the following sequence where heart rate (Polar®) and lactate are measured:

4 x (2 x 100 m breaststroke) 45 sec rest and progressively per 2

Repeat a-b	Time repeat a (heart rate)	Time repeat b (heart rate)	Lactate after b (mmol/l)
1 - 2	1:28.0 (155)	1:27.9 (157)	1.70
3 - 4	1:22.5 (160)	1:22.7 (161)	2.50
5 - 6	1:19.2 (163)	1:19.4 (165)	3.80
7 - 8	1:18.2 (167)	1:17.9 (167)	7.50

Tab. 13 Evolution of the heart rate in function of the intensity (lactate) and the swim times (a and b).

What will the appropriate pulse rate be if we want to train at 3 mmol/l lactate?

According to the table above the appropriate pulse rate for this swimmer will be 162-163. But, we know that the pulse rate can easily swing 2 beats per minute either way depending on mental stress, degree of fatigue, health condition and lots of external environmental factors. This means that you can never be sure whether this pulse rate of 162 beats per minute today corresponds to 160 or 164 beats per minute. The preceding table, furthermore, teaches us that a discrepancy of 4 heart beats per minute will quickly be accompanied by a variation in lactate of about 2 mmol/l. How is it then possible to advise a precise intensity for this swimmer via the heart rate even if we still grant that an electronic heart rate monitor gives an accurate pulse rate measurement? If you determine the heart rate by counting the number of heart beats over ten seconds, then your level of accuracy will decline by at least 6 beats per minute either way. Would it then not be more accurate to ask for a swimming time of 1:21 - 1:22?

3. the heart rate is also very dependent on the training an athlete is accustomed to doing. The swimmer who often trains intensely will, after a time, tend more quickly towards slower heart rates during intense training exercises; on the other hand a swimmer who normally takes it easy will quite quickly reach faster heart rates at the slightest increase of training intensity
4. finally, it is impossible to determine the contribution of aerobic and anaerobic work by reading the heart rate. One may reach 140 bpm in training but what does that 140 bpm tell us about how good or how weak the anaerobic capacity is or about the percentage of anaerobic work that was done? The heart rate does not tell us anything about what happens in the muscle

The preceding argument does not mean, however, that the heart rate is a useless control parameter.

1. It is of great importance to convince swimmers to vary training intensities. Coupling training tasks to different heart rates can “encourage” the swimmer not to train at the same tempo repeat after repeat and set after set. Nevertheless, the heart rate does not

unequivocally typify the nature and degree of metabolic activity during exertion (high or low uptake of oxygen, production and elimination of lactate, ratio of aerobic and anaerobic work). As a result, neither the nature of the training exercise nor its effect can be determined accurately on the basis of the heart rate

Example: a heart rate of 140 beats per minute may not be the optimum intensity for endurance training for some swimmers, but the coupling of this heart rate to the training task will, nevertheless, prevent many swimmers from completing the exercise too quickly

The heart rate is *not an accurate means for controlling intensity and typifying training exercises*, but it can certainly be used to vary the intensity in training. However, in this case neither the exact nature of the training exercise nor its effect can be accurately determined

In this context we must also be critical of the value of the “Conconi Test (heart rate deflection point)”. Training at different percentages of the heart rate deflection point may certainly induce meaningful variations in intensity, but nevertheless this will not give us insight into the metabolic reactions that lie behind the heart frequencies (fig. 61). Missing this insight may not be so dramatic for recreational athletes but makes a scientific monitoring of training with elite swimmers impossible since you can never evaluate what training stimulus evoked what metabolic reaction

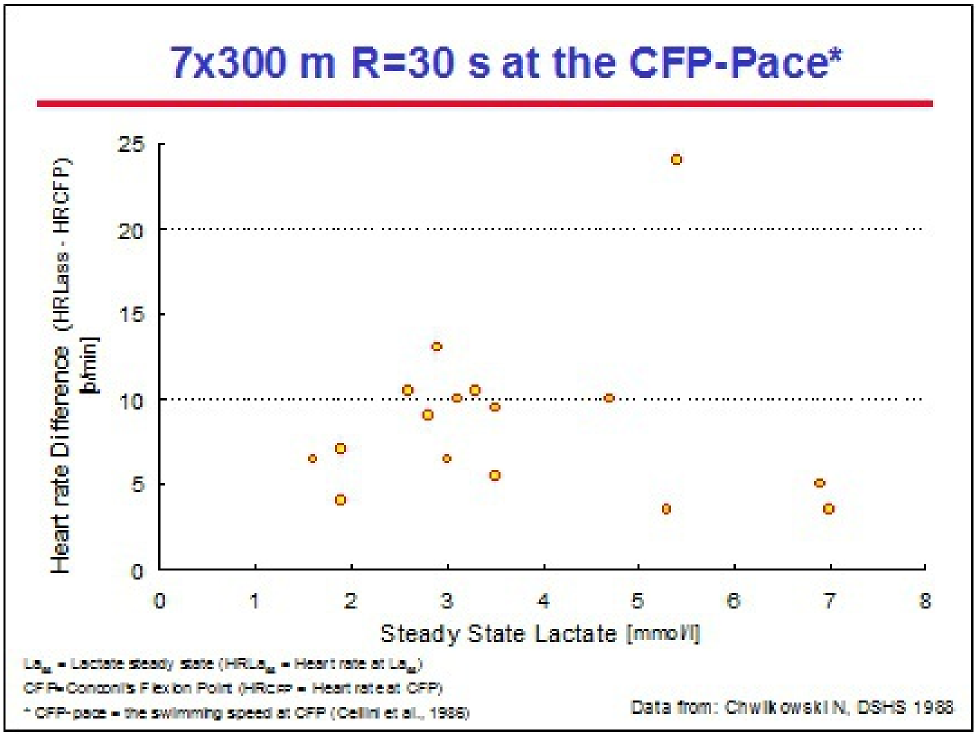


Fig. 61 Higher heart rates after a lactate steady state exercise (7 x 300 m, Rest: 30 sec at the swimming speed corresponding to Conconi's flexion point) are not related to post-exercise lactate values.

2. The fact that the heart rate reflects more than the metabolic reaction to an exertion stimulus certainly causes the heart rate to be a *valuable means for evaluating* the general condition of the whole organism. Heart rates can tell us how well the body handles training in combination with other factors such as strength training, acclimatization to altitude (fig. 62), health, tiredness and so on. In this respect we can appeal to the morning heart rate which is measured by counting the heartbeats during one whole minute while still lying. A higher than usual morning heart rate for more than 3 days is a clear sign that there is something wrong. This problem may be related to training but may have other origins too; nevertheless, it is a signal requiring a thorough analysis of the training and the environment

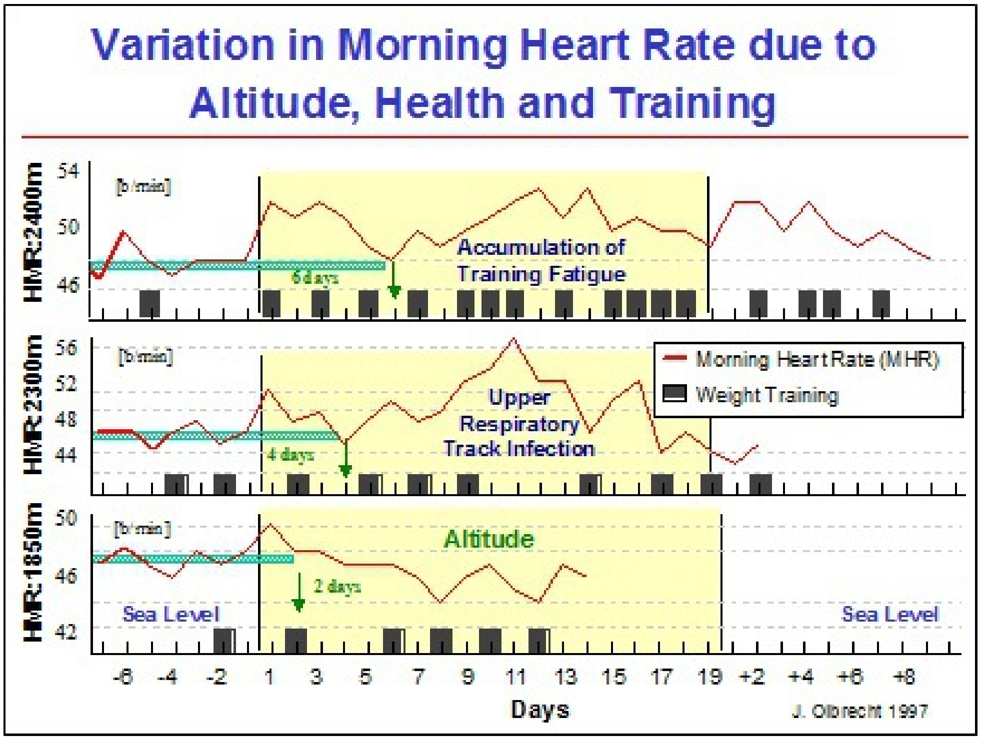


Fig. 62 Observance of fluctuations in morning pulse before, during and after quite a long training period in altitude and under the influence of strength training (middle graph) and health problems (top graph).

Also a higher than normal heart rate during frequently recurring sets at submaximal swimming intensities in combination with lower than normal maximal heart rates during all-out sets clearly indicate fatigue, emerging health problems or excessive training. These symptoms are, it is true, not serious in themselves, but get in the way of training adaptations and are an invitation to a careful and global (training and environment) analysis/evaluation of the last 14 days of training

As long as the heart rate is assessed at its true value, with all its possibilities and limitations, it is a valuable parameter. However, this is also true for parameters such as best time percentage, lactate, etc.

Percentage of Best Times

Studies show that the training intensity can be calculated more precisely based on a percentage of the best times than on the heart rate (Olbrecht 1997). Table 14 contains the percentages of best times at 100, 200 and 400 m for male and female swimmers, which can be used for calculating the training intensity of the 100, 200 and 400 m series respectively, each with at least 30 sec rest.

Female Lactate Level	n x 100 m AS	n x 100 m AW	n x 200 m AS	n x 200 m AW	n x 400 m AS	n x 400 m AW
2	69	78	73	82	83	87
4	74	82	80	86	85	89
6	77	86	84	89	88	92

Male Lactate Level	n x 100 m AS	n x 100 m AW	n x 200 m AS	n x 200 m AW	n x 400 m AS	n x 400 m AW
2	65	74	72	82	80	86
4	72	81	79	84	84	89
6	74	84	82	87	88	92

Tab. 14 Percentage of best times in freestyle and the corresponding lactate values for anaerobically strong (AS) and anaerobically weak (AW) swimmers for interval sets with at least 30 sec rest (e.g. 80% of 1:10 = 1:10 + 20% of 70 sec = 1:24).

To be able to use the percentage of best time exactly, it is important to know if the swimmer is anaerobically strong (AS) or anaerobically weak (AW). It is generally assumed that sprinters are anaerobically strong and thus are classified in the AS category while the long distance swimmers are classified as anaerobically weak (AW). However, a lactate value after a competition as well as the Capacities' Level Chart (tab. 10) allow a more precise categorization of AS or AW. The respective percentage of best time should also be lowered between 1 and 2% if the rest between the intervals is short (10 - 20 sec).

If the coach does not know whether the anaerobic capacity is weak or strong, these percentages may lead to the prescription of erroneous training intensities. For example, table 14 shows that a 5 x 400 m freestyle series at 87% of the personal freestyle best time corresponds to an endurance training (lactate = 2) for an anaerobically weak female swimmer (AW) and to a mixed aerobic/anaerobic training session (lactate = 6) for an anaerobically strong swimmer. These 2 swimmers completing the same series (5 x 400 m freestyle with 30 sec rest) at the same percentage of their best times (87% in this example) are experiencing very different training intensities.

Anaerobically strong (AS) and anaerobically weak (AW) swimmers activate the 3 energy supply systems in a different way for identical percentages of their best times. Let's take the example of 2 swimmers, each having the same 400 m freestyle best time (4:30.0); supposing the AS swimmer registers 15.5 mmol/l lactate in a 400 m freestyle competition, while the AW swimmer reaches 10.2 mmol/l lactate. If both get the same swimming assignment, for example 5 x 400 m freestyle at 87% of their best time, i.e. 5:05.0, they will swim each 400 m some 35 seconds slower than their best time. Since in a 400 m freestyle 1 mmol/l lactate corresponds to a time difference of about 4.2 sec, we can roughly estimate that the lactate concentration will be about 8.3 mmol/l (35 sec/4.2 sec) lower than in competition. The calculated 87% of the personal best time thus means a moderately intense aerobic/anaerobic training session (15.5 - 8.3 = 7.2 mmol/l) for the anaerobically strong swimmer (AS), while the same 87% represents endurance training (10.2 - 8.3 = 1.9 mmol/l) for the anaerobically weak swimmer (AW).

The determination of the training intensity based on a percentage of the best time is not "foolproof", but this method is more precise than specifying workout intensities based exclusively on the swimmer's subjective feeling of intensity or his heart rate.

The 30-Minute Test or T30

As we have seen in a previous section, both the swimmer's aerobic and anaerobic capacities are essential components in the determination of the training objectives and training intensities and can be assessed quite well by lactate tests. But, not every swimming club has the financial means to have these tests done. Therefore, we have sought out other less expensive forms of testing. The 30-minute test (T30) provides a fairly accurate estimation of the swimmer's endurance and offers an acceptable alternative for the 400 m lactate test.

The evaluation of the aerobic capacity by the 30-minute test is of course not as accurate as using lactate tests, but nevertheless it enables the coach to:

* measure the swimmer’s aerobic endurance
* purposefully build up the training
* calculate the training intensity for the endurance exercises

This test requires the swimmer to swim as far as possible in 30 min at a steady speed. The first time the swimmer uses this type of test he can aim for times that are 1 to 2 sec/100 m slower than his 1500 m or 800 m split times.

This test can also be conceived as a training exercise. The swimmers start by the minute and the coach stops each swimmer individually between 29 min 40 sec and 30 min 20 sec always at the end of a lap. He then registers the final time and the total distance each swimmer completed. The distance covered should then be converted to the distance that would have been covered at the same steady swimming speed in exactly 30 min. Therefore we use following formula:

$$\text{distance in 30 min} = \frac{1800 \times (\text{distance swum})}{\text{time swum}}$$

in which the time swum is expressed in seconds (min x 60 + seconds)

Example:
A swimmer covers 2150 m in 29 min 39 sec

Time swum = (60 x 29) + 39 = 1779 sec
Distance in exactly 30 min = 1800 x 2150 / 1779 = 2175 m

Research has shown that the average swimming speed recorded during the 30-minute test (total distance/time in seconds) corresponds closely to the aerobic capacity determined by lactate tests (fig. 63). It was established that generally the longer the distance swum in 30 min, the better the aerobic endurance. The average swimming speed of the swimmers with a very strong aerobic endurance is lower than the 4 mmol/l lactate speed of the 400 m submaximal lactate test. Swimmers with a weaker aerobic endurance, on the other hand, are capable of completing the 30-minute test at a higher velocity than the 4 mmol/l lactate speed of 400 m submaximal lactate test.

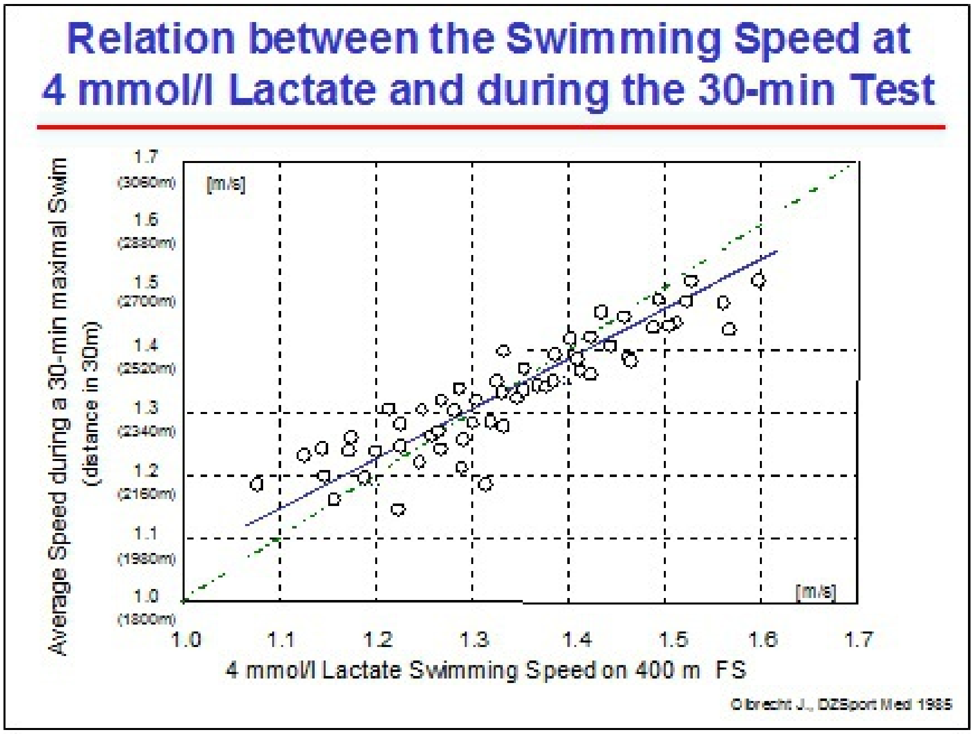


Fig. 63 Relation between the 4 mmol/l swimming speed and the swimming speed (total distance) during a 30 min maximal test.

This form of testing allows not only to estimate the aerobic capacity based on the distance swum in about 30 min, but also to derive swim times for a few classic extensive and intensive endurance training sets (table 15 and 16 for resp. female and male swimmers aged 14 and up).

Distance swum in 30 Min	Females 100 m Series Ext	 Int	200 m Series Ext	 Int	400 m Series Ext	 Int
1825 m	1:38	1:33	3.22	3:10	6:50	6:27
1850 m	1:37	1:32	3:20	3:07	6:45	6:22
1875 m	1:36	1:30	3:17	3:05	6:41	6:17
1900 m	1:35	1:29	3:15	3:03	6:36	6:12
1925 m	1:34	1:28	3:13	3:01	6:32	6:08
1950 m	1:33	1:27	3:10	2:59	6:28	6:04
1975 m	1:32	1:26	3:08	2:57	6:23	6:00
2000 m	1:31	1:25	3:06	2:55	6:19	5:56
2025 m	1:30	1:24	3:04	2:53	6:15	5:52
2050 m	1:29	1:23	3:02	2:51	6:11	5:48
2075 m	1:28	1:22	3:00	2:49	6:07	5:44
2100 m	1:27	1:21	2:58	2:47	6:03	5:40
2125 m	1:26	1:20	2:57	2:45	5:59	5:36
2150 m	1:25	1:20	2:55	2:43	5:56	5:33
2175 m	1:24	1:19	2:53	2:41	5:52	5:29
2200 m	1:23	1:18	2:52	2:40	5:48	5:26
2225 m	1:22	1:17	2:50	2:38	5:45	5:22
2250 m	1:21	1:16	2:48	2:37	5:42	5:19
2275 m	1:21	1:16	2:47	2:35	5:39	5:16
2300 m	1:20	1:15	2:45	2:33	5:35	5:12
2325 m	1:19	1:14	2:43	2:32	5:32	5:09
2350 m	1:19	1:13	2:42	2:30	5:29	5:06
2375 m	1:18	1:13	2:40	2:29	5:26	5:03
2400 m	1:17	1:12	2:39	2:27	5:23	5:00
2425 m	1:16	1:11	2:37	2:26	5:20	4:57
2450 m	1:16	1:10	2:36	2:24	5:17	4:54
2475 m	1:15	1:10	2:35	2:23	5:14	4:51
2500 m	1:14	1:09	2:33	2:22	5:12	4:48
2525 m	1:14	1:08	2:32	2:20	5:09	4:46
2550 m	1:13	1:08	2:31	2:19	5:06	4:44
2575 m	1:12	1:07	2:29	2:18	5:04	4:41
2600 m	1:12	1:07	2:28	2:16	5:01	4:38
2625 m	1:11	1:06	2:27	2:15	4:58	4:35

Tab. 15 Times for extensive (EXT) and intensive (INT) endurance series of 100, 200 and 400 m with 10 to 30 sec rest for females (if the time actually swum was more or less than 30 min, use the formula in the text to estimate the "30 min distance").

Distance swum in 30 Min	Males					
	100 m Series		200 m Series		400 m Series	
	Ext	Int	Ext	Int	Ext	Int
1850 m	1:44	1:37	3.41	3:28	7:25	7:03
1875 m	1:42	1:36	3:36	3:23	7:16	6:54
1900 m	1:41	1:35	3:31	3:18	7:07	6:45
1925 m	1:39	1:33	3:26	3:14	6:58	6:36
1950 m	1:38	1:32	3:23	3:11	6:54	6:30
1975 m	1:36	1:30	3:19	3:07	6:46	6:22
2000 m	1:35	1:29	3:16	3:04	6:38	6:14
2025 m	1:33	1:27	3:12	3:00	6:32	6:08
2050 m	1:32	1:26	3:09	2:57	6:26	6:02
2075 m	1:31	1:25	3:07	2:55	6:19	5:56
2100 m	1:30	1:24	3:04	2:52	6:14	5:51
2125 m	1:29	1:23	3:02	2:50	6:10	5:47
2150 m	1:28	1:22	3:00	2:48	6:06	5:43
2175 m	1:27	1:21	2:58	2:46	6:02	5:39
2200 m	1:26	1:20	2:56	2:44	5:58	5:35
2225 m	1:26	1:20	2:55	2:43	5:55	5:31
2250 m	1:25	1:19	2:54	2:41	5:51	5:27
2275 m	1:23	1:18	2:52	2:39	5:47	5:23
2300 m	1:22	1:17	2:49	2:38	5:43	5:19
2325 m	1:21	1:17	2:47	2:36	5:39	5:15
2350 m	1:20	1:15	2:45	2:33	5:35	5:11
2375 m	1:19	1:14	2:43	2:31	5:31	5:08
2400 m	1:18	1:13	2:41	2:29	5:27	5:04
2425 m	1:17	1:12	2:39	2:27	5:23	5:00
2450 m	1:16	1:11	2:37	2:25	5:19	4:56
2475 m	1:15	1:10	2:35	2:23	5:15	4:52
2500 m	1:14	1:09	2:33	2:22	5:12	4:49
2525 m	1:13	1:08	2:31	2:20	5:08	4:45
2550 m	1:13	1:07	2:29	2:18	5:05	4:42
2575 m	1:12	1:06	2:28	2:16	5:01	4:38
2600 m	1:11	1:06	2:26	2:15	4:58	4:35
2625 m	1:10	1:05	2:24	2:13	4:54	4:31
2650 m	1:09	1:04	2:23	2:11	4:51	4:28
2675 m	1:09	1:03	2:21	2:10	4:48	4:25
2700 m	1:08	1:02	2:20	2:08	4:45	4:22

Tab. 16 Times for extensive (EXT) and intensive (INT) endurance series of 100, 200 and 400 m with 10 to 30 sec rest for males (if the time actually swum was more or less than 30 min, use the formula in the text to estimate the "30 min distance").

Caution about the 30-minute test:

The 30-minute test cannot replace the lactate test as it does not estimate the swimmer's anaerobic capacity. Moreover, for the calculation of the aerobic capacity we have to rely on the average contribution of anaerobic energy at maximal effort during 30 min of swimming. If, then a swimmer with an anaerobic capacity that is above average puts more anaerobic energy into the 30-minute test and swims a few dozen meters more because of his anaerobic input, the total distance swum will overestimate the true aerobic capacity. A swimmer with a weak anaerobic capacity, on the other hand, will use only little anaerobic energy in the 30-minute test, causing his aerobic capacity to be somewhat underestimated.

By way of example:
Swimmers A and B cover the same distance (e.g. 2075 m) during the 30 min maximal continuous swimming test. If swimmer B has a better anaerobic capacity and uses more anaerobic energy during the test, the contribution of his aerobic energy supply system will, despite an equal performance, be smaller than that of swimmer A. If in this case, lactate tests are not available, the swimmer’s individual use of anaerobic energy cannot be determined. The coach runs then the risk, when evaluating the results, of overestimating the aerobic capacity and the “trainability” of swimmer B, and of underestimating both with swimmer A.

The reliability of the 30-minute test also depends very much on the swimmer’s motivation to perform maximally. Therefore it is important for the coach to fit this test into the training process in such a way that the swimmers can be motivated to do the best they can.

Other Tests

There are of course lots of other testing forms, mostly variations of the above tests and with about the same pros and cons. Some are based on lactate tests, others on heart rates, percentages of best times in competition or percentages of swimming speeds during continuous exercises, etc. For the interpretation and/or comments on these types of tests we refer to the respective sections in this chapter.

The purpose of this chapter is to provide the tools for a critical approach of the different types of testing and to stimulate the creation of a reliable measuring system for intensity and volume.

Part II PLANNING AND PERIODIZATION OF TRAINING

Chapter 5 DEFINITIONS AND DESCRIPTIONS

TRAINING PLANNING

> **The planning of training denotes the preparatory work of the coach and consists of structuring the training process in a systematic way according to the training objectives and the swimmer's conditioning level. It is based on the coach's experience and on his knowledge of sports sciences**

A training plan is *a table of contents* which lists, for swimmers of every performance level, all the elements needed to help them improve their competitive performance. It usually covers more than one year but the time allocated to the different elements is still quite rough.

The key elements recorded in a training plan are, in order of importance:

* the main competitions (e.g. participation in school competitions, in international events,...)
* the main conditioning / mental / technical improvements to be achieved
* the expected evolution in performance (the key objectives for each year, e.g. make top 3 at NJO's in 2 years, break 1 minute in 100 m backstroke,...)
* a general description of the swim and dry-land training (e.g. other sports activities to complete the swim training, possibly altitude training,...)

* the number of training units (sessions) and the total mileage per week (e.g. when to start with training twice a day,...)
* the tests to support and control the effectiveness of the training (e.g. video analysis, lactate tests,...)
* non sport related commitments (sponsor or school commitments,...)

As the evolution of the swimmer's conditioning is not a hundred percent predictable, training plans will be subject to modifications during the season according to successive evaluations of the training, competition and test results. External factors too, such as school commitments, health and injuries, obligations to sponsors or federations, may force the coach to adapt his "plans".

Why planning?

* To provide insight and overview about the training process
* To boost the swimmer's motivation
* To control the timing of the changes in conditioning by means of tests, competitions and evaluations of the training results
* To achieve a long-term continuous improvement in competition performance

A training plan should be set up early in the swimmer's career. For beginning competitive swimmers, establishing a training plan is fairly simple as, for this group, training is mainly directed towards fundamentals such as technique, flexibility and endurance, with only a limited amount of different training types.

However, once the coach is dealing with elite swimmers the training plan becomes much more complex not only because of the variety of types of training which have to be framed in the right sequence, but also because of the many sport and non sport related factors that have to be taken into consideration (tab. 17).

Level of the swimmer	Sport and non sport related factors
Beginning competitive swimmers	Training groups / clubs School or other commitments
Advanced swimmers	High school / university commitments National meets (NJO / Sr. nationals)
Elite swimmers	Sponsor commitments Olympic trials International commitments

Tab. 17 Sport and non sport related factors that affect the planning.

For example, if the athlete is in college, he or she may have commitments based on the academic schedule. In designing a training plan for elite swimmers, the coach will also have to spend more time on the analysis of the training completed in the previous years as well as on the evaluation of technique and conditioning.

TRAINING PERIODIZATION

The first step in the training process is the training planning which consists of gathering all the necessary ingredients (competitions, evaluation tests, types of training, all kinds of other sport or non sport activities,...) to improve competition performance. The second step, called the training periodization, arranges, organizes, schedules and fixes *the timing as well as the duration* of each type of training, testing, etc. *for one year* according to the fixed objectives.

The training periodization is the break down of a training year into different sequential and mutually dependent training periods (cycles) to bring the swimmer into peak form at the right moment

The **peak form** is a temporary but stable condition, which enables the swimmer to realize a top performance in his specific discipline. Reaching peak form is the culmination of 3 distinctive development periods:

- the *base training period* which consist in building up the basic conditioning components

- the *competition training period* split in:

  ✽ a pre-competition training period or peak form inducing training period. During this period all efforts are focused on the fine-tuning and optimization of the acquired basic conditioning components (e.g. aerobic and anaerobic capacities, mental preparation, stress tolerance) in order to realize a top performance

  ✽ an in-between competition training period or peak form maintaining training period. The main objective here is the stabilization of the peak form

- the *transition period* : a period of relative rest with a temporary regression of conditioning after the peak form was reached

It is important for the coach to comply with these 3 main training periods as well as with their sequence because:

- many biological and psycho-physiological adaptations to the training effort require a steady rhythmic pattern that involves:

  1. the increase in intensity of the activity
  2. the stabilization of the activity
  3. the reduction of the activity

- the different types of training required to reach a peak conditioning cannot all be completed at the same time. The sequence of the dominant training types will therefore have to be meticulously thought up to build up the competitive performance systematically and methodically in order to bring the swimmer in maximal peak form at the right moment
- the training load the athlete can handle has limits which can only be increased by an optimal relationship between intensity and rest

The sequence of the 3 training periods may repeat itself 2 or 3 times a year depending on the number of top competitions.

By accurately scheduling the timing and duration of these 3 inseparable training periods the coach designs the optimal preparation for the entire season.

Chapter 6 BASICS OF TRAINING PLANNING

TRAINING CONCEPT AND TRAINING OUTLINE

In many countries, the sports federation fixes the *training concept* which contains general indications and instructions on training objectives, performance evolution, qualification times, compulsory training camps, high priority meetings, trials, etc.

The *training outline* is based on the training concept and provides the general guidelines for structuring the training process for a well-defined group (possibly one single swimmer) during 1 or more years. It is a rather theoretical training model and depends on:

- the sport (e.g. swimming, running, etc.)
- the sports discipline (e.g. sprint, middle or long distance, etc.)

Once the general directives are set forth, the coach can start framing and detailing the training plan. Here we will make a distinction between 2 different types of training plans (fig. 64):

1. *the individual or group training plan* which is focused on respectively one swimmer or a specific group of swimmers
2. *the training plan framed in a time perspective* . We then speak of a multi-year plan, year plan, macrocycle, mesocycle, microcycle and a single training unit plan according to the period it covers

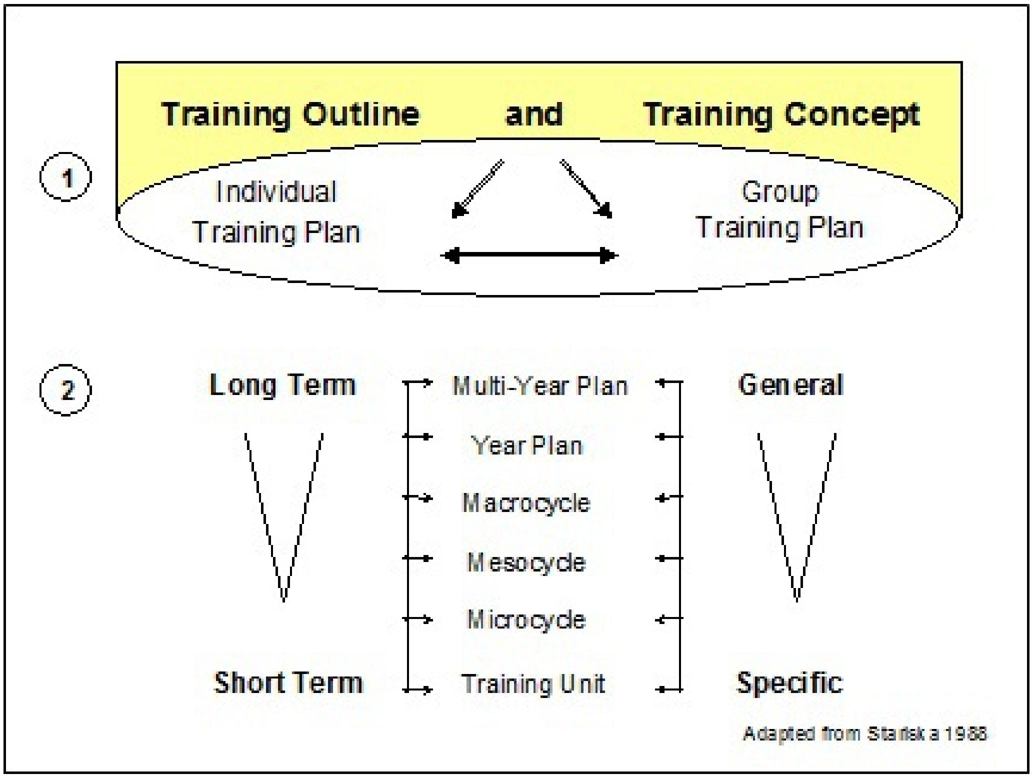


Fig. 64A training plan can be framed considering ☐ a specific target group or ☐ a time perspective.

INDIVIDUAL OR GROUP TRAINING PLAN

When the training plan is designed for one single swimmer, we logically speak of an **individual training plan** . In the case the training plan is drawn up for a group of swimmers of a more or less homogeneous performance level, we use the term **group training plan** .

In reality both training plans will be used interchangeably. Indeed, some training goals such as regeneration can certainly be completed in group. Other objectives on the contrary, such as frequency (stroke rate) training, direct themselves to the specific needs of each particular swimmer and require therefore a specific individual training plan.

The swimmer's performance level also determines the choice between an individual or group training plan. For young, even promising swimmers, preference is given to a group plan as it is still easy to put swimmers with similar performance levels together. This however does not mean that for this group the task of the coach stops with posting the workouts on the wall or writing them on the blackboard. In this case too the training plan will have to be readjusted regularly according to the physiological development of the athletes within the group. For a top level swimmer, on the other hand, a personalized training approach is absolutely necessary making the individual training plan indispensable.

TRAINING PLAN WITH A TIME PERSPECTIVE

The Multi-Year Plan

The first requirement for a continuous performance improvement over several years is to establish at an early stage a multi-year plan delineating the building-up of the conditioning over several years. This can be done for individual swimmers as well as for a specific group and starts when basic technique in the 4 strokes is mastered.

The multi-year plan consists of 3 phases, each covering several years and characterized by specific objectives (see Age Group Training) guaranteeing a progressive and durable long-term development:

* Basic training
* Build-up training
* Top level training

Basic Training *(start between 10 and 12 years of age, duration about 4 years)* In the beginning of this training phase the emphasis lies on:

* the general physical development, as broad as possible, of the beginner and age group swimmer
* stroke technique and coordination (starts/turns) which are best learned and practised at young age
* the development of the swimmer's capacity of assimilating the training load
* non swim specific activities
* learning the competition rules and how to use the training material, etc.
* the swimmer's health status. A medical screening should be conducted to detect and possibly remedy weaknesses that may negatively affect further training or are liable to affect the swimmer's health when increasing the training load

In a later stage of the basic training phase:

* more attention will be paid to the sport specific conditioning development
* the amount of training will be substantially increased, mainly by raising the number of training sessions per week while confining the duration of each training unit to 60 minutes
* the right competitive attitude will progressively be "cultivated"

Build-up Training *(start between 14 and 16 years, duration at least 3 years)* This phase of the multi-year plan:

* runs to about 3 years before peak form age, which in disciplines, where endurance and strength determine the competition performance, presumably lies between 23 and 26 years. In swimming disciplines with a high technical factor, however, peak form age is reached faster which obviously shortens the build-up phase
* emphasizes besides endurance and strength, also the mental development such as the willingness to go all-out, the dealing with setbacks or failures, the competitiveness, the concentration power, the desire for training, etc.
* can easily take up 3 to 7 years

In this phase:

* the training load is further increased by lengthening the duration of the training units. Only in a later stage and depending on the conditioning level of the swimmer the number of training units per week may again be raised (possibly even to several training units a day)
* more swim specific training and conditioning will be included. Participation in other sport activities is still allowed but only if it supports the swim specific physical development
* attention should be paid to the structure of the training process, i.e. the training periodization
* attention should also be paid to perfecting stroke techniques which the swimmer must be able to execute correctly in varying situations

Top Level Training *(start between 17 and 19 years of age, duration at least 4 years)* As opposed to the build-up phase, where the training goals were still general, broad and rough, the objectives of the top level training are:

* almost entirely determined by the specific needs of each swimmer individually
* the maximal development of all swim specific and performance determining conditioning components (developing the individual peak form). Nevertheless, the conditioning qualities that only indirectly determine the competition performance need also to reach an optimum level and be stabilized
* the perfection of the technique, now focused on the quality of execution in competition (frequency, amplitude, etc.) rather than on the correctness of the moving pattern
* the involvement of the swimmer in the planning and training processes

It is thus in this phase essential to assess:

* a profile of the swimmer's weak and strong characteristics
* the swimmer's ability to react and adapt to the various training exercises

This requires an ongoing observation of the evolution of the swimmer's conditioning level (based on training, competition and test results) and an evaluation of the training actually completed. This should indicate whether or not the swimmer has optimally adapted to the prescribed training program.

Which training information should be stated in a multi-year plan?

* long-term competition (e.g. in 2 years' time top 3 over 200 m butterfly at the National Championships) and "motivational" goals (e.g. national selections or swim camps abroad)
* long-term technique (e.g. to be able to swim in competition 100 m in all 4 strokes after one year of basic training) and conditioning objectives (e.g. a 4 mmol/l lactate swim time faster than 5:00 over 400 m freestyle at 15 years of age). The competition goals are thus

complemented with training goals (types of training exercises, intensity, volume) which become gradually more strenuous

* dry-land training. Starting strength training without proper progressive preparation greatly increases the risk for injuries; here also, it is important to consider the swimmer’s biological adaptation capacities as well as his individual trainability and regeneration capacity
* the number of training units per year which is in part determined by school or professional activities
* medical check-ups and other tests to evaluate the evolution of the conditioning, technique, etc. These tests are meant to continually adjust the training plan to the swimmer’s specific needs

Table 18 presents an example of a multi-year plan in swimming that only provides information about the training amount. An exhaustive multi-year plan should furthermore supply data related to training intensity, competitions, technique and conditioning goals.

	Basic Training				Build-up Training		Top Level Training	
Training Year	1	2	3	4	5	6	7	8
Number of training weeks per year	40	40	40	40	40	45	45	45
			(first year plan)					
			2 macrocycles of 15 to 20 weeks		2 to 3 macrocycles of 11 to 17 weeks 6 mesocycles of 5 to 8 weeks		3 to 4 macrocycles of 8 to 12 weeks 8 to 9 mesocycles of 5 to 8 weeks 45 microcycles of about 1 week	
Days of training per year	110	115	190 - 230	190 - 230	190 - 230	230	240	248
Training units/year	110	115	190 - 230	190 - 230	190 - 230	230 - 305	240 - 360	248 - 460

Tab. 18 Example of a multi-year plan in swimming (not exhaustive) adapted after Wilke and Madsen (1983).

The Year Plan

In the year plan (tab. 19) the coach will detail the training process for just one year. This plan:

- is designed in accordance with the multi-year plan
- can be framed for one single swimmer as well as for a group of swimmers
- considers the evolution of the swimmer's conditioning level and takes into account the experience/information collected from previous training periods

What information has to be stated in the year plan?

- the selection of competitions. The coach has to distinguish between top meets (Junior Olympics, nationals, Olympic trials, etc.) and test competitions (to evaluate the completed training period) that do not require specific preparation such as shaving, tapering, fine-tuning of training, etc.
- the training and competition objectives
- the training load, which is the most important element of the year plan. The coach will indicate:

    ✽ the periods of intensive or voluminous training (e.g. training camps during school breaks)
    ✽ the periods of relative rest. At this stage the coach has to build in a wave pattern in the year plan (= alternating process of training load and regeneration)
    ✽ the dominant types of training exercises for a certain period of time (mostly per mesocycle)
    ✽ the periods of technique practice
    ✽ the number of weekly training units

- the medical check-ups and other evaluation tests of the conditioning, technique, etc. (lactate tests, strength tests, etc.)
- the training evaluations

By involving the swimmer in designing the year plan, his commitment to the training work will increase.

Winter	September	October	November	December	January	February
Competitions	none	free choice (max 3)	long distance races (regional championships)	test race short distances	regional championships	nationals *[= top meet]*
General Objectives	relaxation as much as possible out of the water	priority to strength and technique training (video and other appliances). Swim training is of minor importance but increases progressively	priority to swim training, a lot of P1 and regularly some intensive training. Technique training remains important; less time spent on strength training	more race specific water and dry-land training, less but more intensive P1 training. Start with frequency (rate) training	intensive race specific water and dry-land training, P1: very intensive. Stroke frequency (rate) training	maintenance training
Tests	flexibility strength	start: technique end: lactate test	30min-test aerobic power	flexibility and strength end: lactate test	competitions	lactate in competition
Training units /week	free choice	5	6	11	9	5
Duration: AM Volume PM	free	60min: about 3000m	mid-term break 90min: 4500 - 5000m	90min: 2700 - 4500m till 120min: 6000 - 8500m	75min: 2700 - 4000m till 120min: 4500 - 7000m	90min: 3000 - 4000m
Training type	no water training except water polo	long intervals and short rest principally in freestyle	freestyle: long intervals P1: volume extensive	freestyle: extensive P1: short, much rest, moderate intensive	much contrast in intensity resistance training through races	regeneration training
Dry-land training	circuit training, Endurance strength training (30min: 2 to 3x/week)	maximal force and isokinetic (60min: 4x/week)	maximal force and isokinetic (45min: 5x/week)	endurance or explosive strength training (30min - 4x/week)	endurance or explosive strength training (30min - 3 to 4x/week)	endurance or explosive strength training (à la carte - 2x/week)

Summer	March	April	May	June	July	August
Competitions	no or only few competitions	free (ca 2 competitions)	2-3 international long course competitions	none	none	*Nationals or EC/YEC/WC/OG [= top meet]*
General objectives	first relaxation, then a lot of technique and strength training Swim training: volume and extensive	progressively more swim training, a lot of P1 and regularly some intensive training. Strength training goes on. Stroke rate training	more race specific water and dry-land training, less but more intensive P1 training. Technique training remains important; less time spent on strength training	until the exams relatively intensive work, during the exams extensive work	intensive race specific water and dry-land training, P1: very intensive. Stroke rate training	regeneration training in view of important competitions
Tests	lactate test, flexibility, strength	technique	flexibility and strength	end: lactate test		lactate in competition
Training units /week	6	12	8 - 6	8 - 6	12	8-6
Duration: AM Volume PM	90min: about 5300m	Easter holidays 90min: 4500 - 5800m	90min :2700 - 4000m 120min: 5000 - 8000m	90min: 2700 - 4000m 120min :5000 - 8000m	90min: 2800 - 4500m till 150min: 6000 - 10000m	90min: 3000 - 4000m
Training type	freestyle: long interval distances	freestyle: extensive P1: volume extensive	a lot of contrast in intensity P1: intensive through competitions	end: regeneration training	freestyle: extensive P1: first volume then intensive	regeneration training followed by transition phase
Dry-land training	maximal force and isokinetic (60min: 4x/week)	maximal force and isokinetic (45min: 5x/week)	Endurance or explosive strength training (30min: 3 to 1x/week)	maximal force and isokinetic (30min - 4x/week)	maximal force and isokinetic (45min: 4x/week)	Endurance or explosive strength training (à la carte - 2x/week)

Tab. 19 Example of a year plan for a swimmer of national level; P1= 1st stroke. In November and April: morning training only during respectively the mid-term break and the Easter Holidays.

The Macrocycle

Each peak form (i.e. temporary and stable condition allowing top performances in one or more competition(s)) is prepared in 2 distinctive periods:

* the *base training period*
* *competition training period (pre- and in-between competition training period)*

and is followed by:

* the *transition period* , which is a period of relaxation

These 3 training periods constitute what is called the macrocycle. Depending on the swimmer's conditioning profile and the competition calendar the coach will structure the macrocycle differently by varying:

* the number and duration of the training units per week
* the total training volume per week
* the volume and frequency per week of the different types of training exercises (water training as well as dry-land training)

During the whole macrocycle the evolution of the conditioning will be regularly controlled by different tests which will dictate adjustments to the training plan. It is therefore very important to choose carefully the appropriate timing for these tests when drawing up the macrocycle.

So, when periodizing, the coach splits the year plan into as many macrocycles as there are periods of top competitions. In the case where 2 or 3 top meetings are spread out over a year, the year plan will be divided into respectively 2 and 3 macrocycles. It may, however, happen that there are only 2 or 3 weeks between 2 of these top meets. In this case the coach will plan only one macrocycle to prepare for both of these 2 important top competitions.

In practice, as the objectives and the content of each macrocycle strongly depend on the swimmer's (or group's) current conditioning level and needs, the coach will only develop one macrocycle at a time. Until the coach has evaluated the training and the competition results of the previous macrocycle, he has no idea of how to improve or adjust the structure of the next macro. So, the specifics of the macrocycles to come cannot be finalized and remain only roughly described in the year plan.

The Mesocycle

Since it is not possible to train all the conditioning features necessary to deliver a top performance at the same time, the macrocycle has to be split into various smaller training periods. Each of these periods, called mesocycle, normally lasts 3 to 6 weeks and is designed to improve specific aspects of the athlete's conditioning.

Each mesocycle consists of:

* a working or high load phase; a period of voluminous and/or intensive work, whereby training volume and/or training intensity are gradually increased. This does not mean that the coach will progressively increase the intensity and/or volume of every consecutive training unit. He will instead increase the intensity and/or volume of the main training workouts or add an extra quality set (workout at an intensity higher than regeneration). Thus, the swimmer still goes through a lot of regeneration training but these low intensity exercises will either compensate for more intense and voluminous main workouts or be more often alternated with intensive work
* a regeneration phase or period of relative rest before starting the next mesocycle. The last regeneration phase of the mesocycle prior to the top competition is also called the taper phase and is meant to make the swimmer "100% competition ready" physiologically, technically as well as mentally, i.e. fully recovered (especially with respect to the neuromuscular processes) and eager to compete

The content as well as the "load structure" of each mesocycle depends on its training objectives and on the results of the conditioning tests.

Take for example a 5-week mesocycle at the beginning of a macrocycle. As the objective is the building-up of various conditioning components, the working phase will consist predominantly of aerobic capacity training occasionally spiced with anaerobic capacity sets. Training volume and intensity will be progressively increased during the first 3 weeks of the mesocycle (fig. 65). During the second part of the mesocycle (week 4 and 5, the recovery or regeneration phase) the training load will be reduced by decreasing volume as well as intensity, such as only 1 or 2 intensive workouts which will be shorter than during the working phase.

A 5-week mesocycle in the ultimate stage of the competition training period, just prior to the top meeting, will have training objectives that are quite different. Indeed, instead of building up the various conditioning components the coach will now focus on the fine-tuning. The sprinter will need maximal improvement of the anaerobic power, while the distance swimmer will accent the maximal development of aerobic power. As for both swimmers the training intensity is already high, it is no longer possible to increase simultaneously volume and intensity without any risk. So, during the working (high load) phase aiming at the specific improvement of respectively the anaerobic and aerobic power, the swimmers, and the sprinters in particular, will have to compensate for the high intensity by decreasing the training volume (fig. 66). Certain distance swimmers, with a very low anaerobic capacity, however, may be able to increase their aerobic power while increasing volume in the last part of the working phase. In this case, the coach should see to give these swimmers enough room for regeneration before starting this last mesocycle.

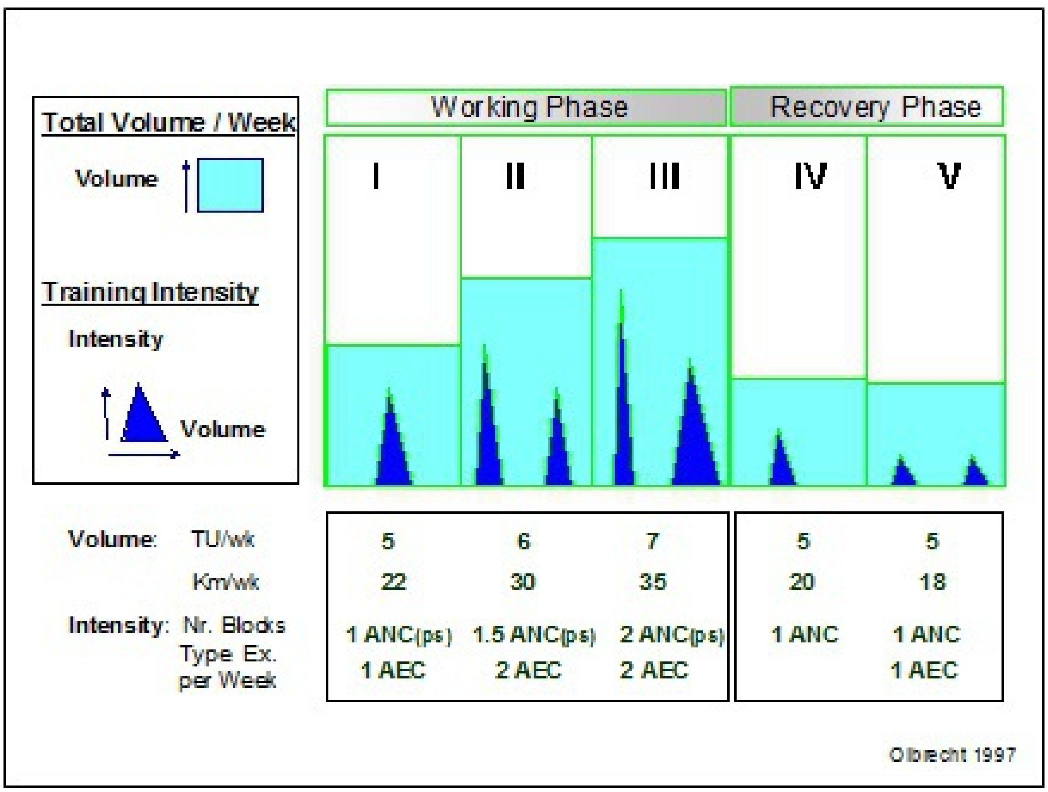


Fig. 65 Content and “load structure” of a 5-week mesocycle during the base training period (at the beginning of the macrocycle) aiming at the improvement of various conditioning components. AEC and ANC = aerobic and anaerobic capacity, ps = personal best stroke, TU/week = training units per week.

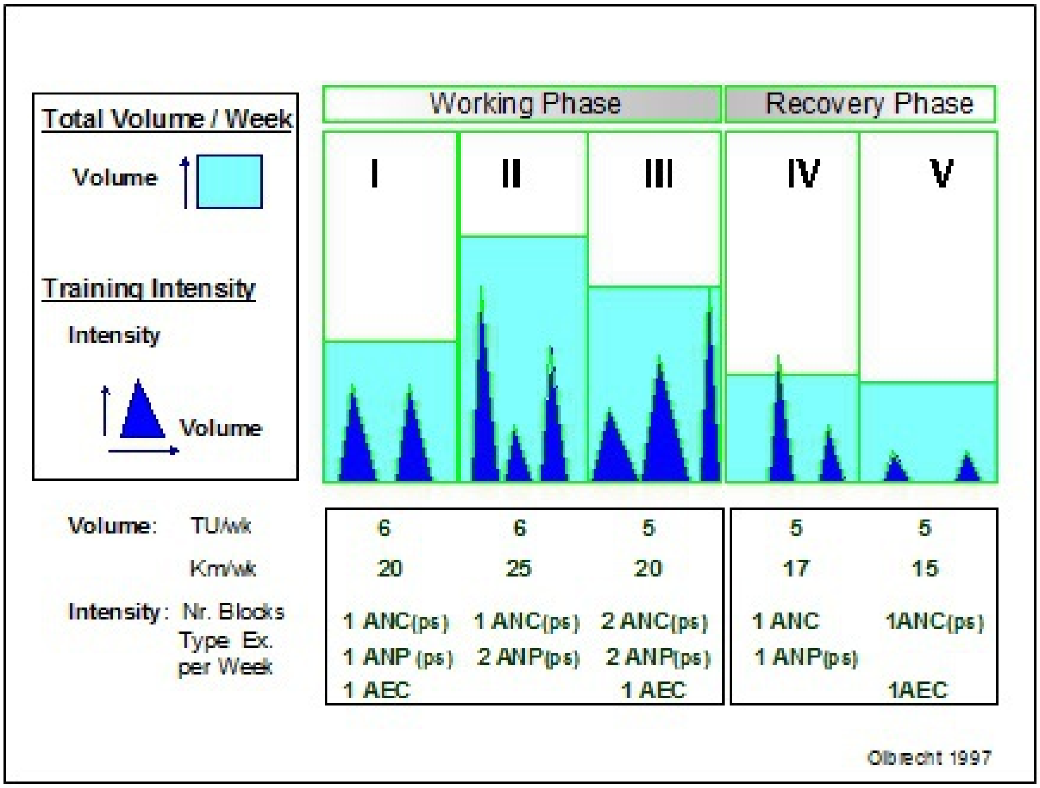


Fig. 66 Content and "load structure" of a 5-week mesocycle at the ultimate stage of the competition training period aiming at the maximal improvement of the sprinter's anaerobic power. AEC and ANC = aerobic and anaerobic capacity, ANP = anaerobic power, ps = personal best stroke, TU/week = training units per week.

Beside the 5-week mesocycle, there are, of course, various other mesocycle structures, such as the 2, 3, 4, 6, or even 7-week mesocycles. All of them, however, contain a working (high load) phase followed by a period of relative rest.

Figure 67 shows a special type of mesocycle, a 6-week mesocycle (fig. 67). According to the type of swimmer, his specialty and his needs, the first 2 weeks are spent on the improvement of the aerobic and/or anaerobic capacity and week 4 and 5 on training the aerobic or anaerobic power. Both high load phases are followed by a regeneration week (week 3 and 6). In the last week of the mesocycle (week of relative rest) some extra speed training is inserted.

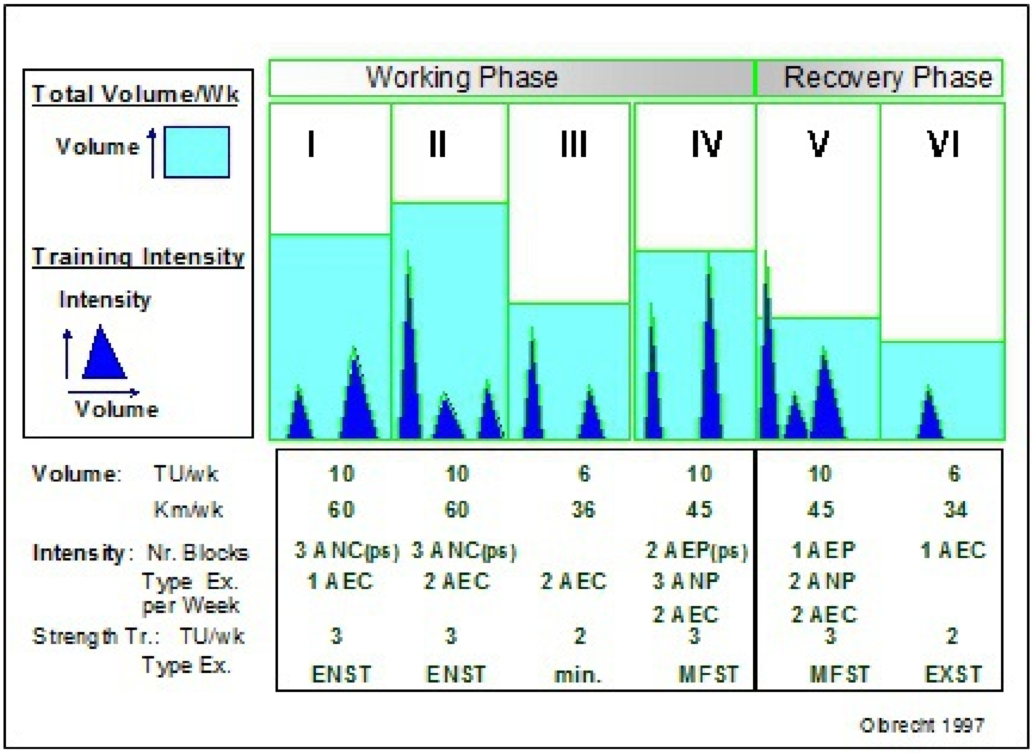


Fig. 67 Content and training load of a 6-week mesocycle for a top level swimmer including all types of training exercises (water as well as dry-land training). AEC and ANC = aerobic and anaerobic capacity, ANP = anaerobic power, ps = personal best stroke, TU/week = training units per week, MFST = maximal force, ENST = endurance force strength training, EXST = explosive force strength training, min = only a short and light strength training unit.

What is remarkable for this mesocycle is not so much the fact that it lasts 6 weeks, but rather that all types of water (aerobic and anaerobic capacities and power) and dry-land training (maximal force (MFST), endurance force strength training (ENST), explosive force strength training (EXST)) are programmed.

The wide variety of training stimuli to which the organism is exposed in a short time span makes this "load structure" quite aggressive and only suitable for top level swimmers. Indeed, the swimmer needs to be strong enough (to have enough force) and to have a sufficiently high competitive performance level to be able to "digest" this type of mesocycle. Moreover, not every top swimmer has a capacity for super-compensation that comes about "fast" enough to handle this training rhythm. So, it is clear that the

"happy few" who can support and assimilate this type of training structure, head for a promising swimming career.
This 6-week mesocycle which is strongly directed to power training, is not suitable for young swimmers who primarily require capacity training to gradually build up their conditioning so they can make continuous progress in competition.

We recommend not using this "aggressive" mesocycle structure until the last stage of the multi-year plan. In anticipation of this forceful training method, we advise coaches to apply the smoother classic 3 to 5-week mesocycle, which carries less risks of overtraining and enables the swimmer to build up his career step by step with more room for a positive training response (see Paradoxical Training Effects).

Contrary to the rather provisionally determined training objectives in the macrocycle, the goals of the mesocycle are tuned to the swimmer's current conditioning level, "trainability" and specific needs as revealed by the regularly performed conditioning tests. This individual approach is essential as not every swimmer reacts or responds in the same way to the same training stimulus. Experience has taught us, for example, that some swimmers only have to "look at" anaerobic capacity training to induce anaerobic capacity adaptations; others on the other hand, have to train a lot more only to reach the same anaerobic capacity level. Moreover, the purpose of the planned training exercises has to be considered, thought over and adjusted if necessary according to the results of each evaluation test. The final and optimal structure of the mesocycle is thus determined by:

* the evolution of the swimmer's weak and strong characteristics
* the swimmer's current "trainability" and "adaptability"

To summarize:

- The mesocycle is the key structure for the periodization of the macrocycle
- It always consists of a working phase followed by a regeneration phase
- It normally lasts 3 to 6 weeks. A mesocycle of more than 7 weeks is not very useful and can better be divided into 2 mesocycles each with a specific training goal
- The length of each mesocycle depends on the swimmer's conditioning profile, the competition calendar, the planned training camps, school commitments, etc.
- The content of the mesocycle is assessed by the coach's planning and the swimmer's conditioning level and "trainability"
- The training effect of each type of mesocycle is determined by the "dosage" (the balance) of training volume, intensity and rest

The Microcycle (Week Plan)

The microcycle provides a detailed training program for a period lasting from a few days to one week, in which case it is also called week plan (fig. 68).

The content as well as the "load structure" of the microcycle depends on the training objectives of the mesocycle in which it takes place. The training intensity and volume need to be scheduled and balanced appropriately for the super-compensation to take place even in periods of 2 or more daily training units.

The super-compensation remains the key process of the training and gets enhanced by regeneration training. If the swimmer is given too little regeneration training or relative rest, his organism will never get the chance to reach super-compensation which will in time inevitably lead to overtraining.

Moreover, the dominant training workouts too, need to be carefully and appropriately scheduled in the microcycle. Here are some tips that may help to guarantee the right timing of the important workouts in the week plan:

> * Sets or drills requiring a fine and precise locomotion, such as sprints and learning or improving technique, must be performed in a relative fresh and fit condition. Therefore, make sure not to plan this

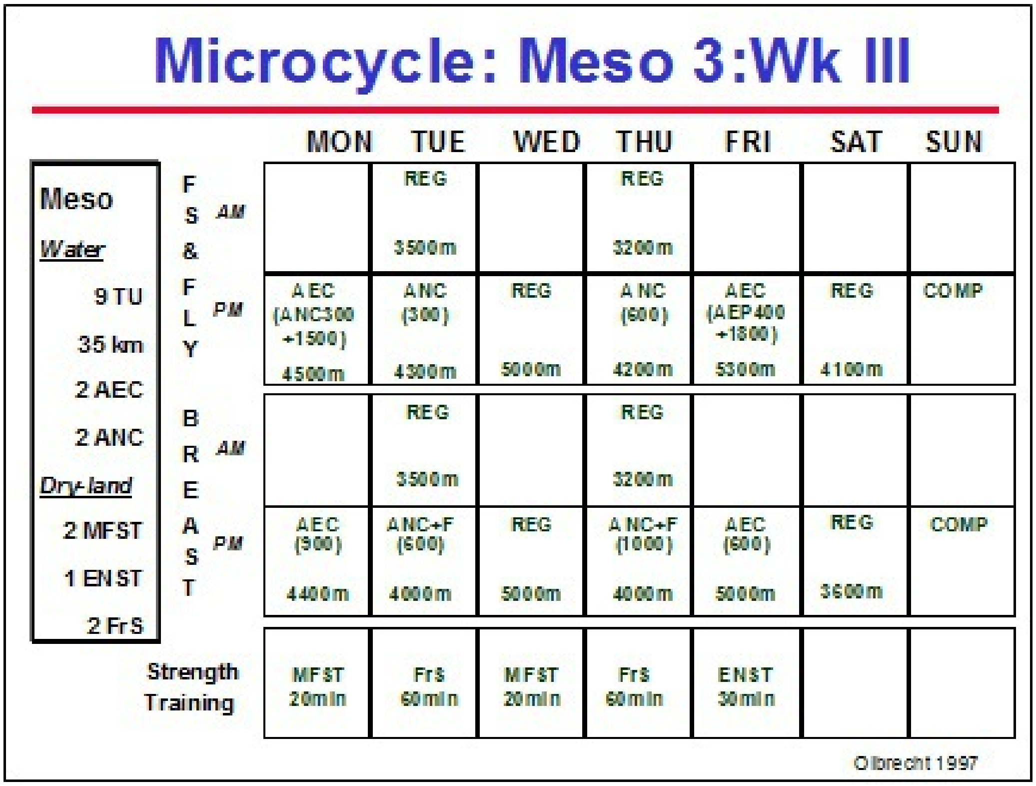
type of exercise at the end of a training unit, after a hard or demanding set or training unit

* It is advisable to avoid explosive strength training (EXST) and other “explosive” training exercises in a tired or fatigued condition, although opinions differ greatly about this

* Substantial anaerobic lactic workouts (capacity or power) should be restricted to 1, 2 or maximally 3 training sessions a week depending on the swimmer’s anaerobic capacity. They should always be alternated with low intensity training units during the base training period. In the competition training period, however, it may be useful to plan anaerobic lactic sets on 2 or 3 consecutive days for the swimmer to get used to the succession of races in a meeting over several days such as Trials, Junior/Senior Nationals, European or World championships, Olympic Games, ...

* Regeneration training is the most important training in the competition training period. In periods of more than one day of competition a week, the microcycle (week plan) will almost exclusively consist of regeneration training

In the transition period, following the competition training period, it is not necessary, even useless to design a strict microcycle. In this period, the swimmer should be allowed to physically and psychologically unwind and to exercise “à la carte”



		MON	TUE	WED	THU	FRI	SAT	SUN
FS & FLY	AM		REG 3500m		REG 3200m			
FS & FLY	PM	AEC (ANC300 +1500) 4500m	ANC (300) 4300m	REG 5000m	ANC (600) 4200m	AEC (AEP400 +1800) 5300m	REG 4100m	COMP
BREAST	AM		REG 3500m		REG 3200m			
BREAST	PM	AEC (900) 4400m	ANC+F (600) 4000m	REG 5000m	ANC+F (1000) 4000m	AEC (600) 5000m	REG 3600m	COMP
Strength Training		MFST 20min	FrS 60min	MFST 20min	FrS 60min	ENST 30min		

Fig. 68 Example of a microcycle, also called week plan, for fly and freestyle swimmers (top) and for breaststroke swimmers (below). Backstroke swimmers will join either group depending on their need of frequency training. TU = training units, REG = regeneration training, AEC and ANC = respectively aerobic and anaerobic capacity, AEP = aerobic power, F = frequency training, MFST and ENST = respectively maximal force and endurance force strength training, FrS = free sport.

The Training Unit

The different types of training plans described up to this point only provide coach and swimmer with broad guidelines on the training objectives for achieving the peak form in a gradual and structured way.

A training unit plan, on the contrary, contains all the specific details for executing each training session. As each training unit represents a link in the chain leading to a top performance, the content of this training plan can never be framed apart from the more general planning described above.

An effective training unit (session) generally consists of the following parts completed in the suggested order:

1. *The warm-up*

The goals are:

- relaxation and relief from the daily stress (e.g. school, work, etc.)
- physiological preparation (e.g. cardiovascular)
- preparation of the muscles (e.g. to induce muscle tension for the sprint work to come)

Following the warm-up there are 1 or 2 main parts made up to meet the objectives of the mesocycle:

2. *The first main part*

This part includes exercises of a high technical and coordination level requiring a well rested athlete and meant to induce conditioning adaptations and/or to improve the technical abilities (e.g. frequency (stroke rate) training, anaerobic capacity training, sprint training or technique training, ...)

3. *A short regeneration exercise*

The higher the intensity of the first main part, the more indispensable the regeneration exercise becomes

4. *The second main part*

This part includes swimming that induces conditioning adaptations but does not require high technical or coordination skills, such as long endurance repeats

5. *The cool-down*

This part of the training is very important since it starts the recovery and regeneration phase that will lead to the super-compensation. An *active* start of the regeneration not only speeds up the recovery process but also the time for super-compensation

If one training unit contains 2 main parts, the sequence of the exercises is determined by the following guidelines:

- Sets or drills that are principally technical and/or coordination directed (e.g. sprint, frequency (stroke rate) training, etc.) precede workouts intended to improve the conditioning

- Speed exercises are completed before endurance exercises or strength training

- The working muscle groups should be alternated (especially during strength training)

The usefulness of alternating the working muscle groups depends on the training goal. After an intensive training set, for example, alternating muscle groups is opportune because it relaxes the swimmer but it does not speed up the local regeneration of the muscles (physiological recovery). Indeed, experimental research has shown that the local regeneration of the stressed muscle is fastest if the muscle experiences the same type of movement as during the hard exertion but with a low intensity. Therefore, after an intensive workout, the coach will have to compromise between the swimmer's physiological regeneration (i.e. continuing the same type of exercise but at a low intensity) and his mental/psychological relaxation (i.e. alternating the working muscle groups by planning another type of workout).

Alternating the working muscle groups is also beneficial prior to a period of voluminous training in the main stroke. It is therefore advisable to have the breast- or backstroke swimmers carry out a few intervals of an endurance set in their main stroke in the weeks preceding the long and tiring sets in breast- or backstroke.

Example**

12 x 100 m	rest = 10 sec
	1, 4, 7, 10 in breaststroke, the other repeats in freestyle

During aerobic endurance training, on the other hand, it is best to reduce the alternating of the working muscle groups to a minimum as the specific nature of an aerobic endurance exercise is only guaranteed if the same muscle groups are stressed for a long period. This principle is especially important when scheduling endurance training for individual medley swimmers. Indeed, the coach should preferably have these swimmers complete large blocks in one of the 4 strokes in the base training period. For example, scheduling a 12 x 200 m set with the first 6 repeats in backstroke and the remaining 6 in freestyle is in this period better than giving a 12 x 200 m medley set. During the competition training period, however, each interval of the 12 x 200 m medley set should actually be performed in medley.

At the end of the training session it may sometimes be useful to have an evaluation together with the swimmer and to check:

* how he felt during the training unit
* what went good, not so good or just plain bad
* if the technical work was OK, etc.

This feedback is very important for possible and timely modifications of the week plan.

Example of a training unit*** Monday 22/04/96 PM
(for a breaststroke swimmer)

Warm-up: 3 x 400 m variable rest = 20 sec
(200 m freestyle/100 m backstroke/100 m breaststroke
2^{nd} 50 m = drill = 1 x arms + 3 x legs/3 x arms + 1 x legs/1 combination)
8 x 200 m Pull rest = 15 sec
(100 m freestyle/100 m medley)

Main part I:
sprint
4 x 10 m legs free choice of stroke + easy return
4 x 15 m free choice of stroke + easy return
3 x 25 m free choice of stroke with start
1 x 100 m easy

anaerobic capacity
3 x (4 x 50 m) rest = 30 sec in set/60 sec between sets
1st set: breaststroke arms + legs butterfly
2nd set: breaststroke head above water/stroke frequency (rate) 55
3rd set: breaststroke race pace 100

Regeneration:
1 x 100 m easy
12 x 50 m kick
(6 x breaststroke rest = 20 sec/6 x backstroke rest = 30 sec)

Main part II:
aerobic capacity
4 x 150 m breaststroke (1st 50 m fast) rest = 1 min
1 x 200 m easy
1 x 400 m breaststroke LA1 (very extensive)
1 x 200 m easy
3 x 400 m freestyle LA1 (very extensive)

Cool-down: 1 x 200 m easy

Summary

The following figure and table (fig. 69 and tab. 20) give an overview of the different types of training plans based on the length of time each covers. According to its time span, the content of each training plan will go from important strategies, general objectives and broad guidelines about the training process (long-term plan) to detailed schemes and directives adjusted to the swimmer’s individual and current needs (short-term plan).

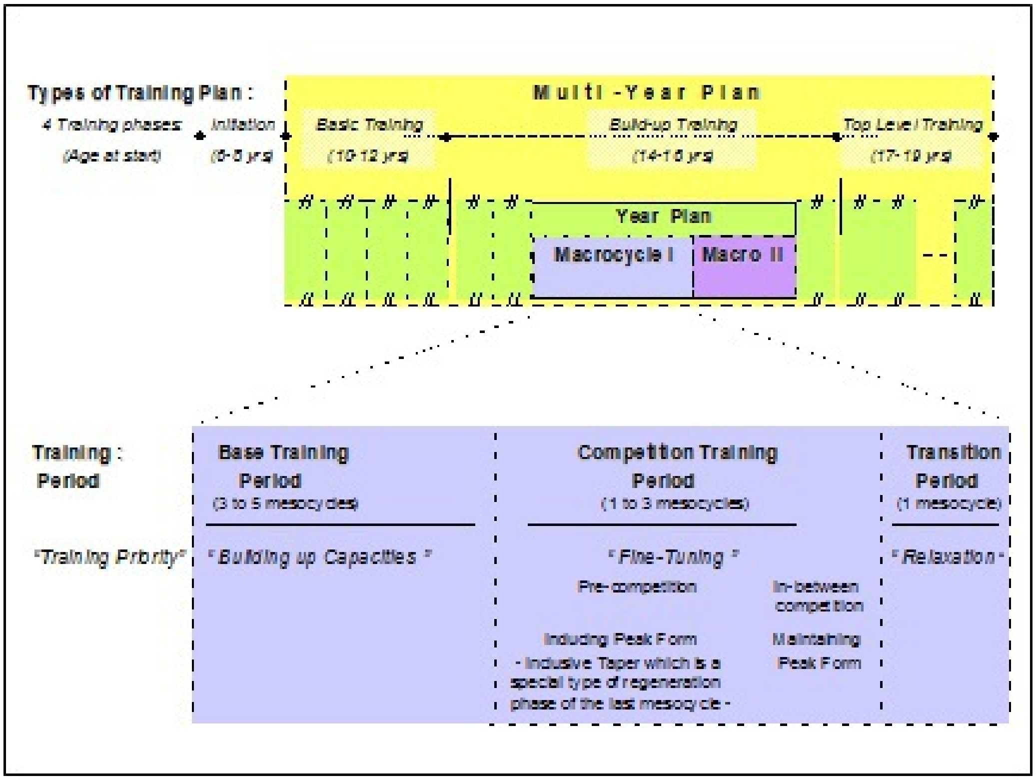


Fig. 69 Survey of the different types of training plans.

Type	Multi-Year Plan (5-8 years)	Year Plan (1 year)	Macrocycle (1/2, 1/3, 1/4 year)	Mesocycle (3-7 weeks)	Microcycle (1-7 days)	Training Unit (1 training)
Phases or Periods	- Basic training - Build-up training - Top level training	- (see periodization) ▪ base training period ▪ competition training period - form inducing - form maintaining ▪ transition period	- (see periodization) ▪ base training period ▪ competition training period - form inducing - form maintaining ▪ transition period	- Working or high load phase - Regeneration phase (=Period of relative rest)	- Alternation of intensive and regenerative Workouts	- Warm up ▪ relaxation ▪ cardiovascular ▪ preparation of the muscles - Main part(s) - Cooling down
Content	- General objectives with respect to: ▪ Competitions ▪ Technique and locomotion ▪ Training components - volume - intensity - types of swim and dry-land training ▪ Number of training units per month ▪ Motivation ▪ Tests and medical check-ups	- Detailed content of multi-year plan with respect to:: ▪ Competitions ▪ Expected race times ▪ Expected training performances ▪ Rest and training load by means of: - dominant training workouts per month - volume per month - wave pattern	- Detailed content of year plan with respect to: ▪ Number and duration of the training units per week ▪ Volume per week ▪ Frequency and volume per week of the different types of training ▪ Check-ups	- Detailed content of macrocycle with respect to: ▪ Individual adjustment of: - volume - intensity - frequency, sequence - type of training exercise based on the analysis of the performance capacity	- Detailed content of mesocycle with respect to: ▪ Individual adjustment of: - volume - intensity - frequency, sequence - type of training exercise based on the daily shape	- Detailed content of microcycle with respect to: ▪ Individual adjustment of: - volume - intensity - frequency, sequence - type of training exercise based on the daily shape

Tab. 20 Survey of the content of the different types of training plans based on the length of time each covers.

Chapter 7 BASICS OF TRAINING PERIODIZATION

Contrary to the training planning, which consists of setting general training guidelines for one or more years, the training periodization refers to the break down of the training year into different training periods with the main objective of getting the swimmer to peak at the right time.

Until recently, the periodization of the swim season was *quite* easy as the coach had only to consider 1 peak per year. The broader range of competitions available now and the evolution of swimming towards a more professional sport have resulted in swimmers having to peak more than once a year. Depending on whether the swimmer wants to peak once, twice, or even more times per year, the coach will have to consider a 1-, 2- or multi-peak periodization.

Age group swimmers generally work towards a maximum of 2 peaks per year. More yearly peaks will take away from the time that should be spent on the build-up of conditioning, which, in young swimmers, remains the most important objective to improve competitive performance at a later age.

Elite athletes can risk 4 different peaks a year, but this clearly will be at the expense of the stability of their conditioning. Plans for a 4-peak periodization should therefore be considered carefully and selectively.

The periodization of the training year starts by fixing the top competitions. The weeks preceding these major events are subsequently organized in 2 periods:

1. the *base training period* which lays the foundation for a top performance
2. the *competition training period* starting with the specific race preparation and ending with the actual event in which the swimmer wants a top performance

After the last competition, there is a wind-down period, which is called the *transition period* . During this time the athlete can relax physically as well as mentally (and so regenerate).

Each preparation for a top performance during the year must contain these 3 periods (see chapter 5, Definitions and Descriptions) with a specific load and training structure.

THE BASE TRAINING PERIOD

This period consists of 1 or more mesocycles, during which the athlete improves his conditioning, technique, etc. It has a two-fold objective:

1. to lay the foundation for reaching a top performance. From a conditioning point of view this is capacity training (aerobic and anaerobic capacity and maximal force)
2. to increase the athlete's capacity for assimilating the training load

The second goal is very important for a long-term (over several years) continuous improvement of the competitive performance. Indeed, the better the athlete's competitive performance, the more (volume as well as intensity) he will have to train to further improve it (see Declining Return on Training Investment). It is thus necessary to prepare the swimmer progressively and on time for future increases in training work. Some time must also be spent on training exercises that may not bring any short-term competition performance improvement but which will pay off in improvements later in the swimmer's career. Power training sets in water as well as on land, for example, lead quickly to top performances by young swimmers, but very often hinder improvement in competitive performance at a later age. Capacity training, however, will lead to slower improvements in competitive performance for young swimmers, but guarantees a better assimilation of any increase in training load at a later age, with the improvement in competitive performances at that stage.

The ideal preparation during the base training period covers 3 mesocycles:

Mesocycle I: *Development and completion of the general conditioning*

- The objective of this first mesocycle is to approach as close as possible the previous year's (at the same period) performance level

- The training sessions are non-specific and usually of low to moderate intensity. This period is, thus, ideal for practising other sports to build up the best possible general conditioning
- Use this mesocycle for improving technique (acquiring, correcting and automating)
- Since at this stage it is not yet necessary for training work in the water to be very specific and/or intensive, this is a good time to start (intensive) strength training. The strength training exercises do not have to be swim (movement) specific but should, for the most part address the muscle groups that are important to swimming such as triceps, latissimus dorsi, quadriceps for breaststroke swimmers, lower arm flexors, lower arm extensors and arm rotators. If the swimmer has already done strength training during the previous year, he/she proceeds to training for maximum force after a short acclimatization period. Depending on the individual need for maximum force this can be alternated with endurance strength in the form of circuit training

Mesocycle II: *Development of the specific basic abilities*

- In this mesocycle the swimmer works on improving the specific basic abilities required for his best discipline (stroke and distance)
- Training remains mainly directed towards building the aerobic capacity but, as opposed to the first mesocycle, the training program will now become much more detailed and specific. Backstroke, good butterfly and some breaststroke swimmers will start to do a lot of work in their main stroke. The intensity remains relatively low (endurance training; moderate to low intensity) but the training load becomes gradually more difficult by increasing volume
- The large volume completed during this mesocycle may somewhat repress the swimmers' anaerobic as well as sprint capacity. It is therefore advisable to insert a short anaerobic capacity training once or twice a week and to complete some sprints every 2 or 3 training units
- The level of complexity of the technique drills will be raised

- It is essential to emphasize technique training to minimize the loss in technique often caused by high volume training in the swimmer's best stroke
- Out of the water the swimmer will continue with intensive training, i.e. maximal force or a combination of maximal force and endurance strength

Mesocycle III: *Development of competition specific abilities*

- The third mesocycle is meant to maintain and stabilize the acquired basic conditioning components and to gradually prepare the swimmer for the specific competition training period that coincides with upcoming meets. At this point the swimmer starts with frequency (stroke rate) training and drills to automate technique at competition speed
- Endurance training decreases slightly and the freed up time is used for more intensive work. For 50 - 400 m swimmers the emphasis lays on anaerobic capacity training (2 to 4 times per week). The distance swimmers, on the other hand, will now complete less anaerobic capacity work but more aerobic power workouts. During this mesocycle the intensity is thus significantly higher for all the swimmers
- There will be 1 to 2 weeks of relative rest from dry-land training in order to compensate for the intensive work in the water. After this short but important period, sprinters increase explosive strength training and long distance swimmers switch to endurance strength training

The ideal base training period covers about 15 weeks, which in practice is not always feasible. Therefore, the coach will often have to reach a compromise, usually by adapting the number of cycles or their duration or by merging mesocycle I and II into 1 cycle.

THE COMPETITION TRAINING PERIOD

During this period 1 or 2 cycles are planned in order to fine tune and balance the improved conditioning components.

Experience with elite athletes has taught us that building up the conditioning usually presents no problems and can even be done very quickly. However, how long this improvement will last and how stable it will be, is proportional to the duration of the base training period. It is indeed possible to make spectacular progress in conditioning in 3 to 4 weeks but such an improvement will deteriorate very quickly. This short build-up will thus be sufficient to register a top performance during one competition, but not to repeat several top performances, even within one weekend.

Once a stable conditioning base has been established, it is essential to fine tune and balance the improved conditioning components. A “super basic endurance” (aerobic capacity), a “tremendous basic speed” and a “fabulous anaerobic capacity” are of no avail if the 2 basic capacities are not in optimal proportion. Moreover, to guarantee a top competition performance, the “super aerobic capacity” of the distance swimmer and the “super anaerobic capacity” of the sprinter must be accompanied by respectively a strong aerobic and anaerobic power.

The complexity of the competition training period is largely dependent on both the number of competitions and their distribution over this period. If the swimmer is preparing for only one race (or one meet), we speak about a *“simple competition training period”* . The entire competition training period will then be used to fine tune and balance the conditioning components specifically needed for the race the swimmer is preparing for. In this case the competition training period will only consist of the pre-competition or peak form inducing training period. The taper phase is included in this period.

The term *“complex competition training period”* is used when 2 or more consecutive events or meets (e.g. consecutive World Cups or Conference championships followed by NCAA championships) are scheduled. The competition training period will then not only consist of a peak form inducing but also of a peak form maintaining period (= in-between competition training period) as the swimmer’s optimal conditioning profile needs to be maintained to have him perform at top level in several consecutive events.

The number of mesocycles during the competition training period as well as their duration is determined by the complexity of the competition training period (simple or complex competition training period) and by the swimmer's specific needs. For example, swimmers with both a weak aerobic and anaerobic capacity, having completed a short base training period (4 to 8 weeks) and preparing for a single top competition may need only 1 mesocycle consisting of a 10 days intensive working phase and a 6 days taper phase (= regeneration phase of the last mesocycle prior to the top competition). On the other hand, swimmers with strong capacities and more consecutive events to prepare for (complex competition training period), will have to go through a much longer base training period and will need more time to induce peak form.

During the pre-competition training period, long distance swimmers will benefit from a long working (3 to 6 weeks) and a short taper phase (4 to 14 days) while sprinters will be better off with a short working (1 to 5 weeks) and a long taper phase (2 to 4 weeks).

Top level long distance swimmers can only achieve optimal aerobic power after 4 to 6 weeks of specific training. However, a mesocycle of 4 to 6 consecutive weeks that is focused on improving aerobic power presents a great risk for overtraining. It is, therefore, advisable for these swimmers to plan 2 mesocycles during the peak inducing period (pre-competition training period). The use of 2 mesocycles will also enhance the success of a short taper phase. Indeed, if the swimmer is too tired after a long working phase of aerobic power training, he will never get "100% competition ready" in only one week of taper. A required prolongation of the taper phase would on the other hand improve his anaerobic capacity and consequently deteriorate his aerobic power, which would considerably impede his competition performance on long distance events.

Likewise, top level sprinters, with both a high aerobic and anaerobic capacity, can only build up an optimal anaerobic power through 3 to 5 weeks of specific intensive anaerobic power workouts. Experience, however, has taught us that if the swimmer maintains this anaerobic power training for more than 3 weeks, basic endurance will inevitably deteriorate. It is thus advisable for these swimmers to either:

- plan a pre-competition training period of only one but long mesocycle (7 to 8 weeks). This mesocycle should consist of a working phase focused on anaerobic power training that is split into 2 periods of about 2 weeks each and separated by 3 to 5 days of extensive (low intensity) aerobic capacity training. This short period of aerobic capacity training will be preceded and followed by 1 to 2 days of regeneration. This scheme mostly allows a taper phase of only 2 weeks, or to
- plan 3 short mesocycles in the pre-competition training period; mesocycles 1 and 3 focusing on anaerobic power training and mesocycle 2, of maximally 10 days, meant to “refresh” the aerobic capacity. This scheme will permit swimmers to participate in a test competition or even in trials after the first mesocycle

During a complex competition training period, the time span between the competitions determines the choice of the different types of training. If the period between consecutive meets is more than 9 days, a small microcycle of at most 3 training days can be used to refresh the aerobic and anaerobic capacity (not power!!!). Before starting this microcycle it is also useful to plan 1 or 2 days of regeneration training. If 9 days or less are available between the competitions it is advisable to free up as much time as possible for regeneration training complemented with possibly 1 aerobic capacity training.

Research has shown that the duration of peak conditioning depends on the total duration of the peak form inducing period. Our experience, however, has taught us that 3 weeks is about the maximum period during which an athlete can be kept in peak form. As it is thus quite exceptional to be able to maintain a peak conditioning during 4 consecutive weekends, we, in this case, advise the swimmer only to compete in 3 of the 4 competitions.

The training content completed during both the peak form inducing and maintaining periods is very specific and needs to be designed individually. Only the short in-between cycle may be non-specific.

The training intensity during the peak form inducing period is very high and the rest in the main sets short. Competitions are an excellent means to build up the anaerobic power as the motivation to excel at these competitions is often greater than in training sessions. Moreover, this is also a good opportunity to get back into the competition mood and ambiance after a long preparation period. To compensate for the high intensity, the training volume must be kept low (fig. 66) and the easy work should be clearly extensive (low intensity). There will, thus, be a large contrast in intensities during this training period.

During the peak form maintaining period the coach's objective is to get and keep the swimmer fit, fresh, vigorous and full of power at the start of each competition. Training may still include a short intensive or a single high volume workout, but the time necessary for the super-compensation to take place should be unconditionally guaranteed and respected. During this period the intensity may thus still be high for a brief time, but the volume of these intensive workouts should be definitely low. The volume can also be increased but only for one day and it will have to be extensive (low intensity) and as quickly as possible compensated by sufficient regeneration work. Each intensity or volume training session must show its effects within a short-term (2 to 3 days).

Strength training continues up to about 2 weeks before the important competition in the form of endurance strength training for 200 to 1500 m swimmers and explosive strength training for 50 to 200 m swimmers. Thereafter strength training can be stopped, although some swimmers will still like to "feel or lift weights". The coach can then easily plan a "mini strength training session", but here again: it is essential for the swimmer to be able to regenerate and have 100% of his capability available at the time of the competition.

THE TRANSITION PERIOD

Not only building up to a peak conditioning (peak form) but also performing in top competitions requires a large expenditure of energy from the organism. Too often both the physical and psychological fatigue accumulate during training and competition. It is, therefore, absolutely necessary, especially for the senior swimmers, to recharge their "batteries" before starting a new season, a new preparation process. This mesocycle, also called the "active recovery mesocycle", usually lasts 2 to 4 weeks and sharply decreases the risk of overtraining in a systematic way. It is characterized by:

- the absence of competitions and intensive training
- very little stroke specific work
- all kinds of other non swim specific sport activities
- other recovery promoting activities such as sauna, massage, hydrotherapy etc.

SUMMARY

During the entire training periodization process the coach should always consider:

- the swimmer's performance level
- the swimmer's constantly evolving and changing level of conditioning, technique and mental qualities

The training plan will thus be continuously liable to modifications, adaptations and adjustments according to changes in the swimmer's conditioning levels which will be evaluated at set times and from different sides.

Moreover, the training process is multi-disciplinary. Swimming well in competition is more than the result of a good conditioning. Indeed, swimmers must also be at "top level" mentally and technically, otherwise they will perform below par despite a super level of conditioning. Very often coaches and advisers need to display great creativity and "dexterity" to reach an optimal consensus between the swimmer's physical, technical and mental preparation. The choice of a particular or specific type of workout does not always have to be scientifically motivated. Indeed the coach's personal practice, his individualized approach, his experience with his swimmer(s) may sometimes be more decisive for the composition of the training and the selection of the training exercises.

EXAMPLES

The following examples illustrate 2 different periodization plans, one with a simple competition training period for a middle distance swimmer (fig. 70) and another with a complex competition training period for a sprinter (fig. 71).

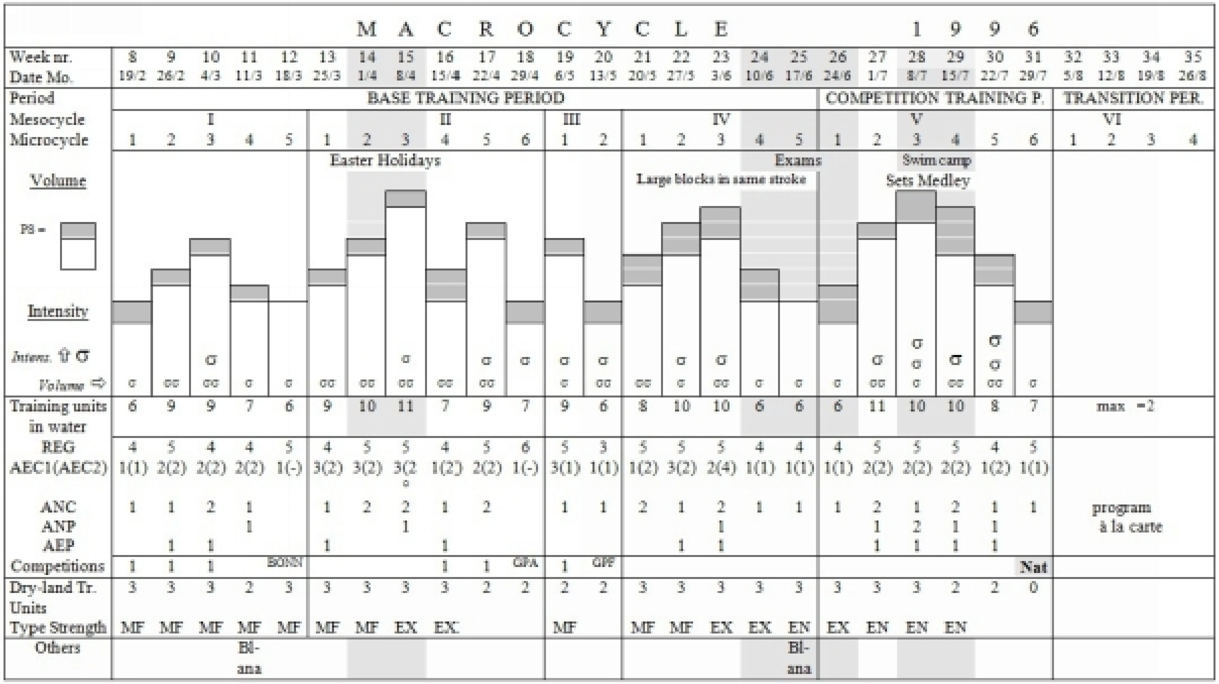


Fig. 70 Example of a periodization with a simple competition period for a middle distance swimmer (400 m freestyle and 400 m medley; PS = personal best stroke, AEC and ANC = aerobic and anaerobic capacity, AEP and ANP = aerobic and anaerobic power, EN and EX = endurance and explosive strength training, MF = maximal force). Top competition = National Championships (Nat), Bl-ana = Blood Analysis.

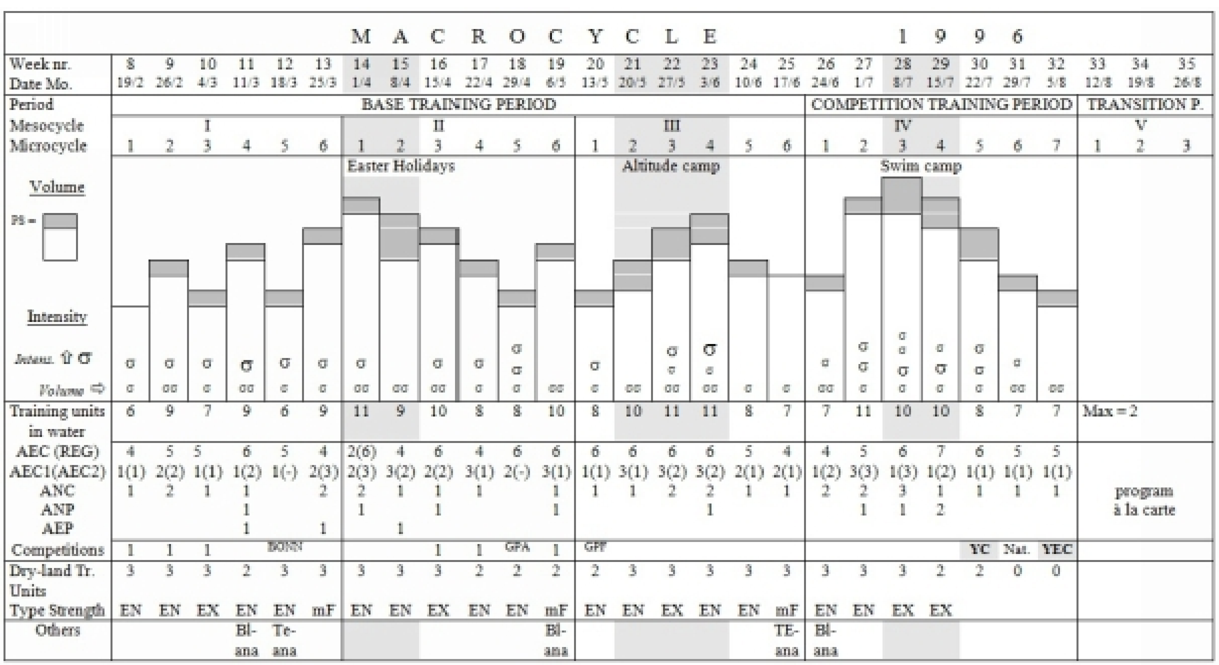


Fig. 71 Example of a periodization with a complex competition period for a sprinter (PS = personal best stroke, AEC and ANC = aerobic and anaerobic capacity, AEP and ANP = aerobic and anaerobic power, EN and EX = endurance and explosive strength training, mF = pseudo maximal force). Top competitions = Age Group National Championships (YC) and Youth European Championships (YEC). Bl-ana = Blood Analysis, TE-ana = Technique Analysis.

Chapter 8 METHODOLOGY FOR TRAINING PLANNING

Several elements, such as types of workouts, their frequency and volume, etc. are constantly recurring in the different types of training plans. These recurring elements ensure that the different types of plans are geared to and blended into one another to form a complex and rigorous unit of structured and detailed proceedings designed to improve the performance level.

In fact, a training plan is never finished. As the training process constantly needs to be adapted to the changes it induces, it is necessary to first objectively record and measure the adaptations due to the training work completed. This will enable the coach to make appropriate and evidence based changes in the training process. Although the training adaptations can somewhat be predicted from the composition of the training program, it is best to have objectively measured training results as the “human reaction” to a training process may sometimes be unforeseen. Only when the coach knows the real effect of a training program, the effectiveness of the training process can be guaranteed.

There is thus a constant and continuous interaction between the planning process and its execution. This interaction results in an uninterrupted loop of “planning”, “executing”, “measuring the training effects”, going back to the training plan and verifying whether it has to be “adjusted” for the next mesocycles. Every loop provides feedback information, which is used to plan the next cycle in order to fit in the training process with the individual trainability and so improve the training efficiency and “return”. This process is called “training steering” and will be dealt with in details later on in this part (see The Steering Principle).

In this chapter, the focus will be on the methodology for setting up training plans (fig. 72). The whole process of making training plans goes through various steps that can mainly be bundled in 2 distinctive phases:

- the design phase
- the realization phase (execution and steering)

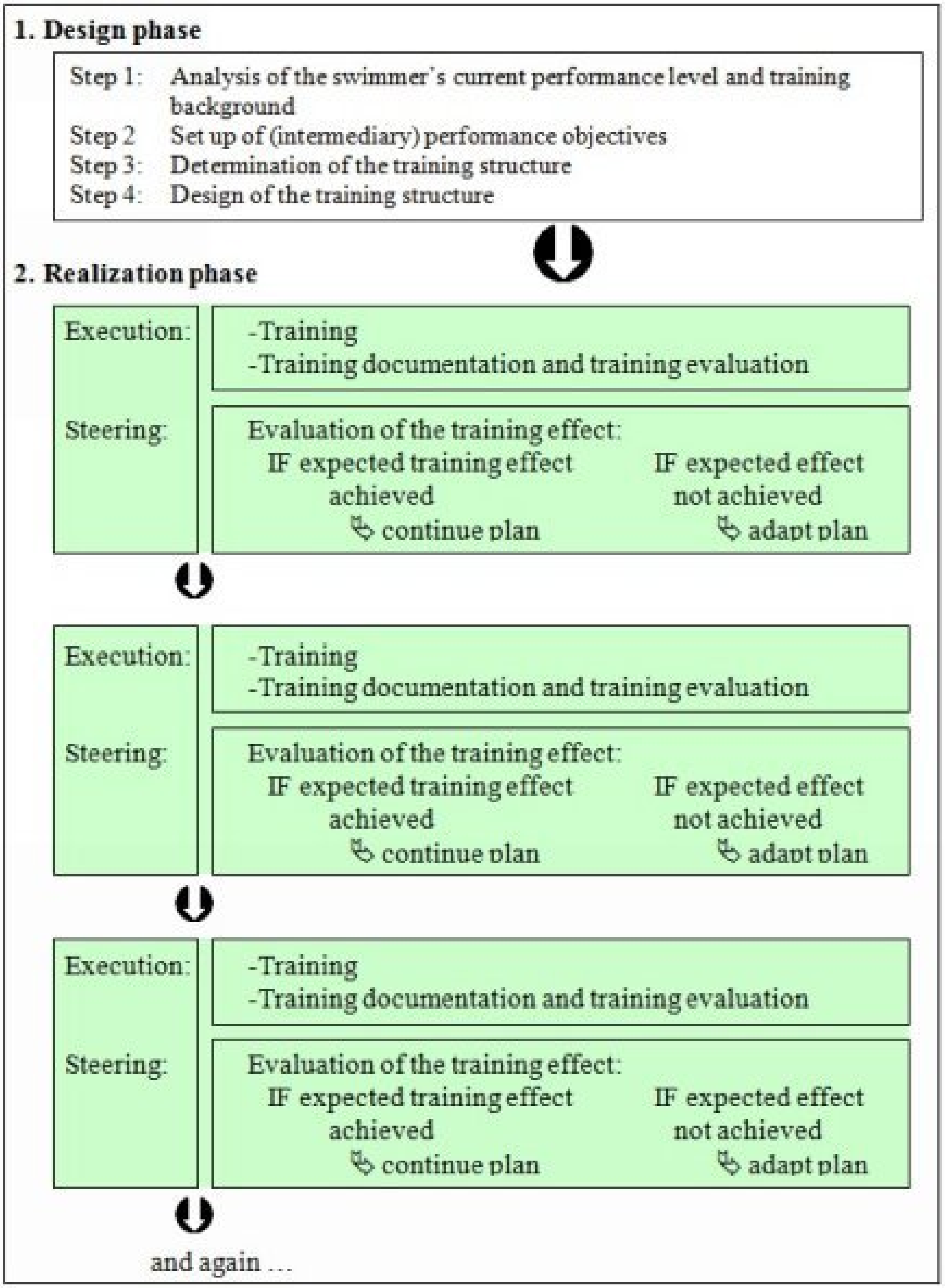


Fig. 72 The process of making training plans goes through various steps that can mainly be bundled in 2 distinctive phases: the design and realization phases.

THE DESIGN PHASE

Before training starts in a new swim season, the coach makes an overall schedule that will serve as a framework for the training process. This framework is thus a broad draft of the real training program and will need to be detailed and adapted as time goes by. The following 4 steps will guide the coach when designing his training framework:

Step 1: Analysis of the Swimmer’s Current Performance Level and Training Background

Before periodizing the first macrocycle (the time between the top competitions), i.e. fixing the base training, the competition training and the transition periods, the coach must evaluate:

- which qualities must be trained most urgently to improve the swimmer’s competitive performance
- how these qualities can best be improved during the different training periods, considering the swimmer’s individual trainability as determined by the evaluation of the training effects from previous training periods

This evaluation process ensures that no time is lost to superfluous or unnecessary types of training and that each training exercise contributes purposefully and as efficiently as possible to the improvement of the swimmer’s performance capacity. This evaluation process is based on:

1. The analysis of the training background (see chapter 8, The Realization Phase) which should provide answers to the following questions:

    - which competition and training goals were reached and which were not?
    - what went wrong during the past training year (injuries, illness, personal and/or academic problems etc.) and why?
    - did external factors disturb the training process (closing of swimming pool, exam period, other personal commitments)?
    - did the previous training work yield enough performance improvement or should another approach be tried?

2. The analysis of the current performance level which will provide answers to questions such as how strong is the swimmer's current conditioning level and how strong are his present technical, mental and tactical qualities? It will thus enable the coach to:

- assess the swimmer's strengths and weaknesses
- set priorities to different types of training

Step 2: Set up of (Intermediary) Performance Objectives

Once the coach has fixed the priorities and the types of training to get the desired improvements in conditioning, technique and/or mental qualities, he will set up specific performance objectives for the upcoming season. These objectives are distributed over the training year and must be realistic and objectively verifiable by both the coach and the swimmer. Such performance objective may be, for example, 50 m freestyle time under 26 seconds at the winter championships, or sub-minute on 100 m butterfly while correctly performing side breathing.

Steps 3 and 4: Determination and Design of the Training Structure

Once the strengths and weaknesses of the swimmer are determined and the priorities of the various types of training set, the coach will have to design the appropriate training structure. This process is liable to some rules and principles. Before describing the practical proceedings of planning (see Procedure of Periodization), we will discuss the main principles of structuring the training process.

Main Principles for Structuring the Training Process These rules and principles not only complement but also influence each other. They should thus never be considered nor incorporated into the training plan independently.

The 5 most important principles for structuring the training process are:

1. the optimal relation between work and regeneration (super-compensation)
2. the progressive build-up of the training load

3. the variation in training loads
4. the optimal timing and dosing of the swim specific and non swim specific types of training and the development of the different performance determining components
5. the durability of the development (= long-term build-up)

Each of these principles should be considered to some extent when designing the different types of training plans. The number of arrows in figure 73 represents the impact of these 5 principles on the year plan, the microcycle and the unit plan.

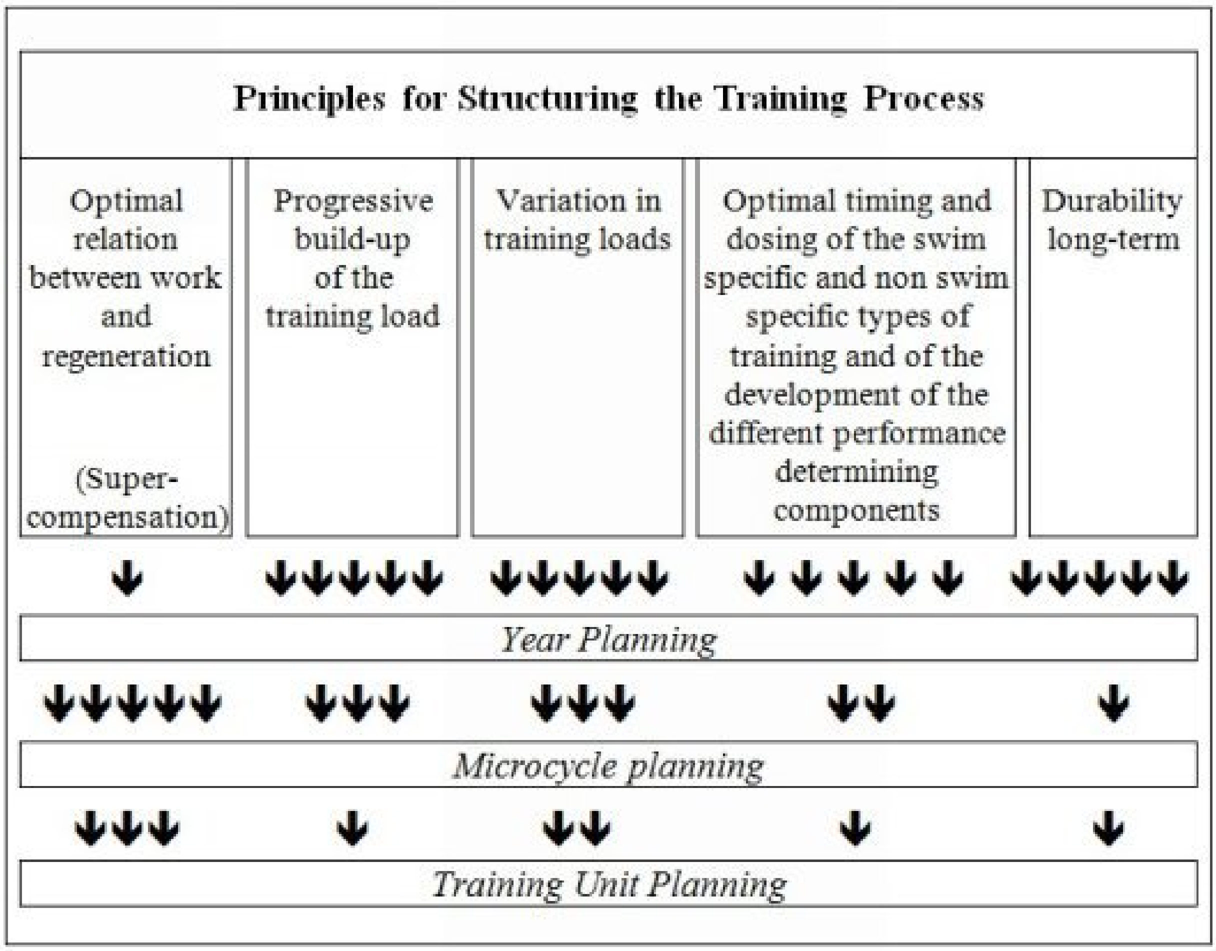


Fig. 73 The 5 main principles for structuring the training process and their influence on the different types of training plans. The number of arrows indicates their degree of importance.

Another training principle, the principle of maturity (i.e. influence of age, gender and biological maturity on trainability; see Age Group Training) has not been considered here as it does not relate to the structuring of the training plan but rather to the choice of the different types of training exercises in a later stage of the periodization.

Principle of Optimal Relation between Work and Regeneration (Super-Compensation)

The optimal relation between work and regeneration is the basic requirement for achieving a maximal super-compensation (see Super-Compensation). Depending on the type of training exercise, the timing to reach the super-compensation will vary from a few hours to several days. It is therefore very important for the coach, when designing a plan, not to consider every workout independently but to take account of:

- the training exercises recently completed
- the training workouts planned for the next days

The swimmer's conditioning should continuously improve during the several months devoted to the training process. It is therefore important to alternate high intensity and/or high volume training with extensive training so that consecutive cycles of super-compensation can take place during the entire swimming season. This will ensure a progressive build-up of the performance capacity (fig. 74: Model A).

If intensive or high volume training sessions follow each other too quickly or if the swimmer carries on with hard training in a state of overfatigue and/or when struggling with health problems, super-compensation will not get a chance to take place. His conditioning will progressively deteriorate instead of growing (fig. 74: Model B).

To estimate the duration of the relative rest period needed for super-compensation, use the data from figure 2 whereby you can either:

1. wait after each intensive or high volume training session until maximal super-compensation has been reached before starting the next long or intensive unit (fig. 74: Model A). In-between the swimmer can do easy training, or
2. neglect the timing for super-compensation and carry on with the heavy training for a short period (maximally 1.5 week for elite swimmers). After this period, during which fatigue has accumulated, it is important to insert a recovery phase (active rest)

to obtain a super-compensation effect (fig. 74: Model C) and to prevent conditioning from deteriorating (fig. 74: Model B)

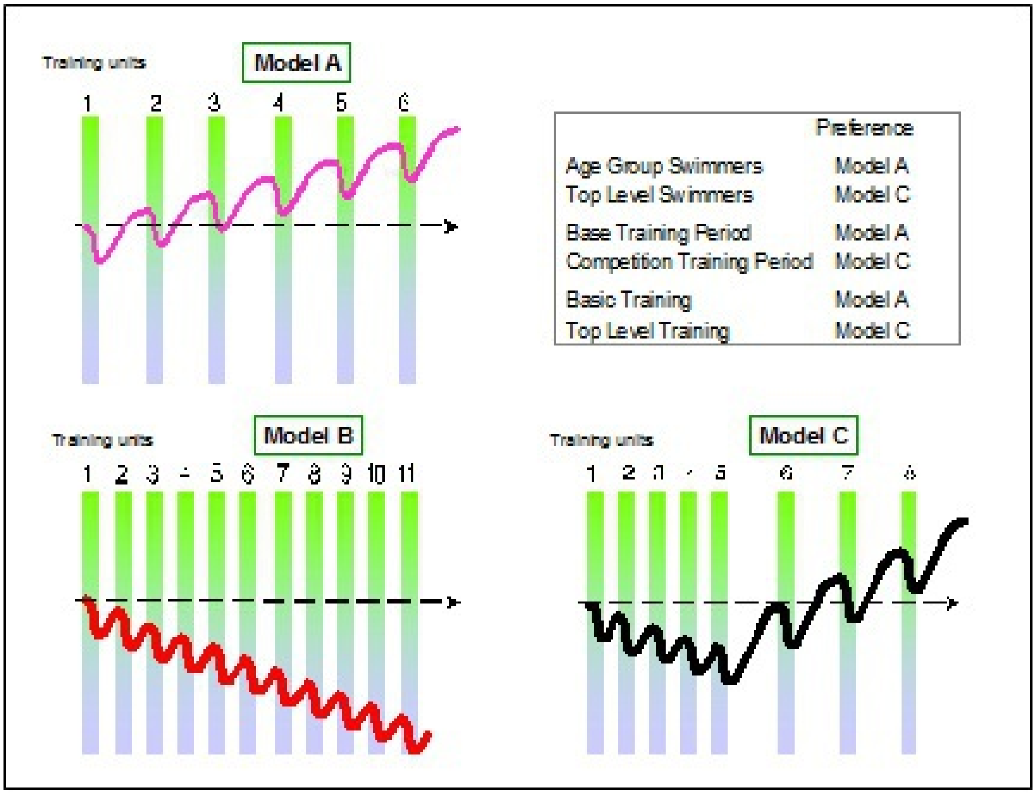


Fig. 74 The build-up of conditioning can either be achieved by A) waiting until maximal super-compensation has taken place before starting the next training session or by C) inducing a super-compensation after accumulated fatigue brought about by short consecutive hard and/or high volume training units. B) uninterrupted intense or voluminous training, without giving the organism a chance to reach super-compensation, should definitely be avoided.

We recommend that beginning competitive and age group swimmers follow the first option (Model A). These swimmers usually do not need much training (3 to 4 training units a week) to make progress which leaves them with enough room to wait for super-compensation after each training unit.

Elite swimmers, on the other hand, need increasingly more training to make further progress (principle of declining return on training investment). As a result of the amount of training units they have to go through, there will certainly not always be enough time for a complete super-compensation after every training. With these swimmers the coach will, therefore, not always respect the timing for super-compensation but will insert instead a

recovery phase only after a series of hard or voluminous training sessions (see fig. 74: Model C).
To sustain Model C without too much risk for overtraining, the swimmer needs thus quite a high level of conditioning. This training model should therefore definitely be avoided with beginning competition swimmers.

On the other hand, Model C provides the advantage to condition the body to absorb a series of short but consecutive exertions, which is very useful in preparation for a competition over several days.

> Whatever training model the coach gives his preference to, the principle of super-compensation strongly determines the number and volume of each type of training exercise used per week.
>
> However, how often a type of exercise will be planned per week will actually also depend on:
>
> - the necessity for a swimmer to improve a certain conditioning component
> - which part of the training cycle the swimmer is in

Additionally the coach can also enhance the regeneration process after certain types of training simply by including some regeneration promoting activities:

After *strength training:*
It is profitable to stretch the heavily loaded muscles - but always remain below the pain threshold -. The muscle groups that were only slightly activated, on the other hand, may be put through extensive and dynamic work.

After *anaerobic power training:*
Enough cool-down swimming stimulates the clearance of lactate and accelerates the elimination of waste products from the fatigued muscle groups.

After *all types of training:*
The type and sequence of food, which an athlete takes in after a training session, is important. He should first drink and consequently replenish his energy reserves by taking easy-digestible carbohydrates (within 20 to 30 min after the training). Then he should ingest proteins. It is also better to eat

several small meals a day than 2 or 3 large or substantial meals. Moreover, dinner should be light and low-fat so the athlete can sleep well at night.

Physiotherapeutic treatment stimulates blood circulation in the organism and/or may enhance relaxation. It has, therefore, a positive influence on regeneration. This treatment, consisting a.o. in taking a relaxing bath, a sauna (but not until at least 30 min after finishing the training unit), hot-cold showers, light massages, etc. should, however, first be discussed with a sports physician or physiotherapist.
Autogenic training and gradual muscle relaxing exercises can also further accelerate the regenerative process.

Principle of Progressive Build-up of the Training Load
This principle is very important for structuring both the macrocycle and the long-term plan (one year to multi-year plan). It is based on following assessments:

1. A training stimulus normally leads to an increase of the swimmer's competitive performance. However, after some time, that same stimulus will not sufficiently challenge the athlete's system anymore and no further adaptation will take place (see Biological Adaptations)

2. An increase in training load will then be necessary to induce further adaptation. This increase in training load can be achieved in several ways:

    - by increasing either the training volume or intensity
    - by making the intervals longer
    - by making the implementation more difficult (e.g. first small, then large hand paddles)
    - by increasing the load density (= close sequence of high volume or intensive training units)
    - by changing the training environment (e.g. altitude training)

3. The increase in training volume and intensity influences the improvement of the competitive performance most. The increase

in training volume generally stimulates the swimmer's ability to handle or assimilate the training load, which in turns is a prerequisite for a more effective increase in training intensity

4. The changes in training load must of course be in accordance with:

    - the goals set for the respective training periods in the macrocycle
    - the swimmer's current conditioning profile

5. Sudden changes in training load work more efficiently for improving competitive performance than smooth progressive changes but need nevertheless to be implemented with care as any excess may lead to injuries

6. The adaptation processes resulting from these sudden changes in training load may differ in time and amplitude. It is thus important to document each completed training session thoroughly in a training log and to evaluate carefully and as precisely as possible the induced conditioning adaptations

The training effects or adaptations to a sudden increase in training load are always delayed and are not linear proportional to the load increase (fig. 75).

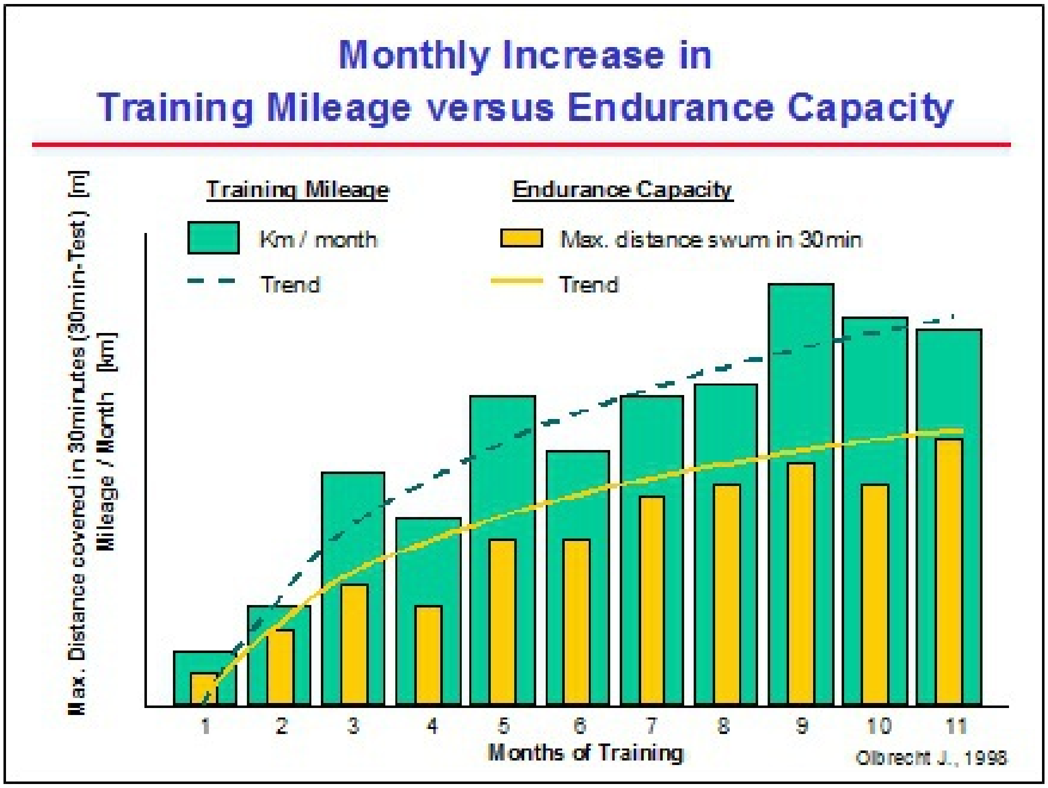


Fig. 75 Example of a sudden increase in training volume over a swimming season of 11 months and the improvement in aerobic endurance it induced, measured by means of a 30-minute test.

As described earlier, regeneration is the most important training as it induces the super-compensation after a heavy training load. The improvement in aerobic endurance, shown in figure 75, and recorded by the maximal distance swum in a 30-minute test, does not immediately follow the increase in training load but comes about a month later, after a period of decreased training volume and intensity.

Moreover, this improvement in aerobic endurance is not linear proportional to the increase in training load. Indeed, the better the aerobic endurance becomes, the less it will be improved by an increase in training load.

Principle of Variation in Training Loads

Originally this principle comes from the sports didactics but is also applicable to physiology and is based on following assessments:

1. A progressive improvement in competitive performance over a longer period of time requires, especially in the top level training,

a systematic and purposive variation in training load, training exercises and training methods. This variation ensures a homogeneous development of all the basic components determining the competition performance and will, therefore, lower the risk of performance stagnation (= no further improvement of the competition performance) considerably

2. A continued and unchanging load pattern may sometimes be used over one extra mesocycle but only to stabilize an acquired training effect. This is, for example, applicable for technique training when a certain movement needs to be automated to such an extent that it can be correctly executed even under extreme competition pressure. However, during the build-up of the conditioning, it is rarely the purpose of the training to concentrate on maintaining an acquired training adaptation. Only in the peak form maintaining period between important competitions, maintaining the acquired training effects is an important objective
3. For elite swimmers room for variation, however, is very limited as, in certain periods, they must train very specifically

Principle of Optimal Timing and Dosing of the Swim Specific and non Swim Specific Types of Training and the Development of the Different Performance Determining Components

This principle consists of a set of guidelines to assist coaches in choosing for their swimmers the right type of training at the right moment in the right proportion. It is based on both experimental evidence from different sport science disciplines and on personal experience.

The development of sport specific and performance determining conditioning components requires the right dosing and sequencing of the *different* types of training. The dosing will be determined according to the results of the conditioning tests. The sequence, on the other hand, is independent of the swimmer’s performance level but must be adhered to in the periodization process. Since the training objectives change during the macrocycle, it is clear that the sequence and dosage of the different types of training will vary accordingly.

The sequence of the different types of training has also to be build up progressively, which means from simple to complex and from non swim specific to swim specific. The time actually spent on non swim specific types of training will considerably decrease as the swimmer's competition performance improves.

Moreover, with a few exceptions, the training work will be principally voluminous and extensive (low intensity) in the beginning of the macrocycle but will progressively become more intensive and at the end of the macrocycle less voluminous.

Principle of Durability of the Development and Long-Term Build-up
This principle refers to following observations:

1. Only a continuous and consistent training program over many years and without needless interruptions will lead to the optimal development and stabilization of the competitive performance (= continuity of the training process)

2. A fast development of a particular performance determining component leads to a temporary and unstable conditioning improvement which will be quickly lost again
3. Long interruptions of the training process cause a deterioration of the acquired training adaptations
4. The rate at which conditioning components deteriorate, varies in extent and time. Indeed, the swimmer first looses aerobic power then aerobic capacity, then strength, speed, etc.
5. The success of the retraining (restoration of the lost conditioning) depends on the swimmer's age, his performance level and his training background. Elite athletes generally have the best conditioning stability and will recover quicker. Such is also the case for young swimmers and for those athletes who went through a continuous and consistent training program before the interruption
6. When emphasizing a certain type of training (= the dominant training objective) during a mesocycle the other conditioning components need to go through maintenance training

Implementation of these Principles: Procedure of Periodization Once the coach has collected the data from step 1 (Analysis of the Swimmer's Current Performance Level and Training Background) and step 2 (Set up of Intermediary Performance Objectives), he can start the actual training structuring, i.e. order the important training components in view of the systematic building of the competitive performance. During this structuring, called periodization, he should methodically and chronologically go through the following steps while constantly keeping the above mentioned important training structuring principles in mind:

1. *Selection of the Competitions where Top Performances are Expected*

The first thing to do is to select and fix the important competitions where top performances must be delivered. According to the rules of the periodization (see Basics of Training Periodization) the coach may select maximally 4 periods with top competitions

Age group swimmers, generally, work towards a maximum of 2 peaks per year. More yearly peaks will take away from the time that should be spent on the build-up of conditioning, which, in young swimmers, remains the most important and overall objective to improve competitive performance at a later age. Only elite athletes, who already have a high conditioning level can consider 4 different peaks a year. Not without due caution, however, as this clearly will be at the expense of the stability of their "form"

2. *Determination of the Number of Macrocycles per Year*

The number of peak periods will determine the number of macrocycles in the training year

3. *Fixing Training Camps (Holidays), Exams and Test Competitions*

This step influences largely both the duration and content of the different mesocycles

School vacations, for example, are very useful for boosting the volume and/or the number of training units per week, while the periods of exams are, for most swimmers, associated with a sometimes drastic reduction of the training units. For this reason the coach will, as much as possible, let voluminous training periods coincide with vacations. Intensive and/or voluminous training can also be planned until shortly before the exams with the purpose of utilizing the exam period as physical recovery phase

Swim camps also can in principle concur with the periods of voluminous training. The training camps during Christmas and Easter vacations, for example, are very important for building up the basic capacities for respectively the winter (e.g. national and other championships in the months of February or March) and summer peak (e.g. national and other championships in the months of June, July or August). During the competition period it is also useful to plan a training camp not so much to improve the basic capacities but instead to take advantage of a favorable training environment

It sometimes happens that a theoretically ideal training periodization is not always feasible in reality. Often the coach will have to find a compromise between what should ideally happen according to the above mentioned rules, and what reality permits. A classic example of this occurs in Belgium when the national championships follow the exams within a short period (4 to 5 weeks). An ideal periodization is not possible in this case, so the coach has to build up the basic capacities before the exams and settle for a short refreshment of the aerobic and anaerobic capacities when resuming training after the exam period

Eventually, it is also useful to incorporate the test competitions in the planning as soon as possible. These meets can be used as a type of training (aerobic or anaerobic power) but at the same time they offer the possibility to evaluate the training evolution

4. Subdivision of First Macrocycle in Mesocycles and Determination of the Training Content

The coach subdivides the macrocycle in different mesocycles and then determines their main training objective (= the dominant training adaptation aimed for). The more time needed to reap the benefit of a dominant training exercise, the more time will have to be allotted to it. Therefore, the largest part of the macrocycle will be spent on building up the aerobic capacity as it is the slowest to develop. Next come anaerobic capacity training and workouts to improve aerobic power. The least time is devoted to training anaerobic power as it is the fastest to adapt. Moreover, the length of each mesocycle mainly depends on the swimmer’s aerobic capacity: the better the aerobic capacity, the longer a mesocycle may last

Besides, it is also very important to first train the aerobic and anaerobic capacities and only thereafter the respective powers

In practice the coach will thus:

4.1 starting from the top competition, first fix the time needed for the power training (competition training period), then set the base training period, i.e. the time necessary to train both the aerobic and anaerobic capacities and last but not least determine the length of each mesocycle. Here the coach clearly takes into account the swimmer’s competition discipline and other individual abilities to assimilate the training

Generally, for long distance swimmers, a competition training period of 7 weeks including a taper phase of maximum 5 to 10 days will do. Over the rest of the period the coach will then distribute about 5 aerobic power training sessions, which he will make progressively harder every week

For sprinters and middle distance swimmers the competition training period varies from 5 to 10 weeks, whereby the stronger their anaerobic capacity, the longer the competition period. The taper phase (2 to 3 weeks) as well as the peak form inducing period (2 x 3 weeks with a short intermezzo of aerobic capacity training) will last longer for sprinters and middle distance swimmers with a strong anaerobic capacity than for those having a moderate anaerobic capacity. For the latter a competition training period consisting of a peak form inducing period and a taper phase of respectively 3 and 2 weeks will usually be sufficient

In addition to these general rules the coach's experience also plays a very important part in fixing the taper phase, which to lead to peak form should respond to the individual and specific needs of every swimmer

As there is no strong system to anticipate the individual "taper needs" of the swimmers, the coach will have to rely on empirical observations. He can evaluate whether the duration of the taper phase is appropriate by extending it for one more week after the top competition and concluding it with an additional competition. For swimmers who perform better on this event he should consider not only to adapt the training content of the next macrocycle but also to lengthen the taper phase. For those who do not perform so well during the second meeting, the taper phase as completed was just right or possibly too long. The coach will only be able to make a final evaluation of this last possibility by trying out a shorter taper phase in the next macrocycle

The rest of the macrocycle consists of the base training period which should last fairly long since building up both the aerobic and anaerobic capacity takes up quite some time (at least 9 and 5 weeks respectively)
Because the aerobic capacity training requires a high volume, it is obvious that the coach will select the periods with a lot of training possibilities (e.g. vacations) in order to plan and accentuate this type of training. These periods of high load, however, need to be followed by a regeneration phase for the super-compensation to take place

4.2 broadly determine the dominant training adaptations aimed for in the different mesocycles. These are not fixed individually yet, but are rather targeted on the different groups of swimmers

4.3 outline the training content. The most easiest way to do so, is to order the training workouts in 3 distinctive categories:

- exercises that are meant for the general conditioning development

- exercises aiming at a specific conditioning development and taking the type of swimmer and his competition discipline into account
- competition specific exercises

The macrocycle will start with exercises of the first category but, as it progresses, we will gradually pass to exercises of the second and finally of the third category. This sequence of training exercises in the periodization is determined by one of the 5 important principles for structuring the training process, according to which the exercises of a general nature should always precede the more specific ones

The classification of the training exercises in these 3 categories may do for the periodization of the basic training in the multi-year plan. However, it is far too general and broad for the build-up and top level training. For these training phases a more differentiated and accurate categorization is required such as classifying the exercises according to their training objective:

- development of the general/basic or more complex swim specific motor skills (kinesiological properties)
- teaching or perfecting technique or tactics
- testing to evaluate the conditioning
- preparation of competition specific efforts (loads)

This subdivision can be further differentiated by assigning to every exercise a degree of similarity to the competition effort and competition stroke (biomechanics). This leads to the following categories of training exercises (fig. 76):

GPE - General Preparatory Exercises:

- these exercises show no similarities to competition either in stroke or in the type of workload (they are called non-specific exercises)

these exercises do not directly contribute to improving competition performance. They do, however, contribute to developing the basic conditioning and coordination necessary to acquire complex adaptations from specific types of training

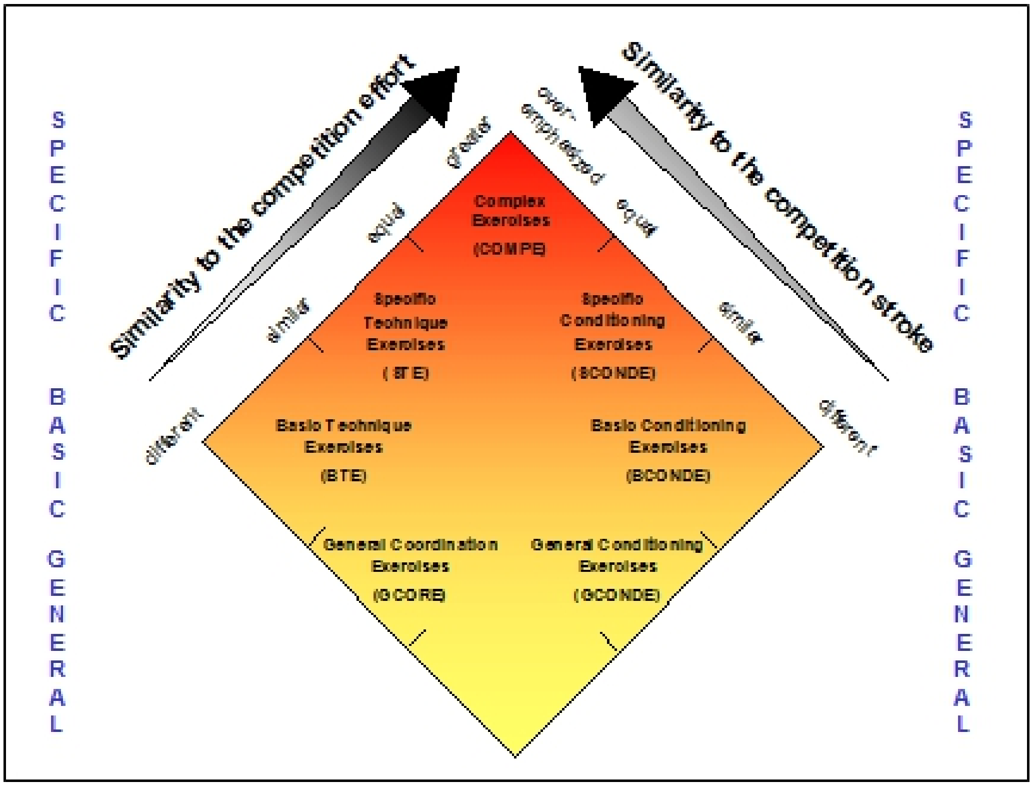


Fig. 76 Subdivision of the training exercises according to their degree of similarity to both the competition effort and competition biomechanics.

GCORE - General Coordination Exercises:

- these exercises apparently do not resemble the competition movement at all; yet they lay the foundation for a good senso-motor coordination which is required for swimming (locomotor intelligence)
- the load structure is not important as long as it does not disturb the acquisition of the locomotor intelligence

GCONDE - General Conditioning Exercises:

- these exercises differ from the competition stroke both kinetically (with respect to time and space) and dynamically (with respect to time and forces)
- these exercises, alike the GPE, do not directly contribute to improving competition performance. Instead they rather

stimulate the basic development of conditioning and coordination abilities that are necessary to acquire complex adaptations from specific types of training
- the intensity of these exercises may vary from moderate to high

BTE - Basic Technique Exercises:

- these training exercises partially show a strong similarity to the competition stroke and, as far as the type of load is concerned, address the basic conditioning properties required for swimming

BCONDE, Basic Conditioning Exercises:

- the specific competition stroke is ever more integrated in these training exercises
- these exercises are often performed progressively, with the most intensive repetition closely approaching the competition load

STE - Specific Technique Exercises:

- these training exercise or parts of it imitate the specific competition movement and automate it even under difficult coordination and/or technique conditions. They, thus, aim at a perfect technique and a stable application of this technique even in difficult situations

SCONDE - Specific Conditioning Exercises:

- these training exercise or parts of it very closely resemble the specific competition movement kinetically as well as dynamically
- these exercises are also both psychologically and physically very demanding

COMPE - Complex Exercises:

- these types of exercises have the same kinetic and dynamic character as the race but are performed under extremely difficult conditions (even more difficult than the race itself)

All these 3 different types of classifications characterize the exercises according to their external features (dynamically and kinetically). The internal processes, such as the biochemical, biological and physiological reactions, induced by these exercises are not taken into consideration

Further development in training physiology has lately added a physiological dimension to the different types of training and, what's more, now allows the coach to reckon with the specific physiological impact of the exercises on the swimmer's organism and with his ability to assimilate and adapt to the diversity of training forms. It is in the light of these most recent developments that we worked out a new physiologically based classification of the training exercises wherein we distinguish 4 categories: the aerobic and anaerobic capacity and the aerobic and anaerobic power (see Classification of Training Exercises)

In the modern approach of the training periodization we will, thus, not only consider the external features of the exercises, such as the load structure and movement patterns, but also take their physiological aspects into account. Therefore, we will, in establishing the sequence of the training workouts:

1. from a techno-motor point of view, start with the simple training exercises and progressively proceed to more complex types
2. from a physiological point of view, always have capacity training preceding power training (i.e. maximal force and aerobic and anaerobic capacity training before respectively explosive or endurance strength training and aerobic and anaerobic power training)

4.4 Once we have subdivided the macrocycle in different mesocycles, fixed, for each of these mesocycles, the training objectives and, in broad outlines, also progressively scheduled the training content, we will proportion the different types of exercises. In each mesocycle the same types of exercises may recur but it is the proportioning of these types of training workouts over a longer period (usually the mesocycle) that will determine the dominant training adaptation (= the training objective of the mesocycle)

Therefore, beside exercise characterizing factors such as volume, intensity, fraction and rest, the manipulation of which will induce particular biological adaptations and determine the type of exercise, we will now also consider the proportioning (the dosing) of the different types of exercises over a period of time, usually a mesocycle. The dosing is mostly expressed per week and refers to:

- the training frequency: how many training sessions per week
- the training volume: how many km per week
- the load frequency: how often per week do we plan one particular type of training
- the load density: how quickly do the different training exercises follow each other

Once the dosing of the different types of exercises is fixed, the coach will have to select the appropriate training methods. For workouts with a conditioning nature, such as the aerobic and anaerobic endurance and strength, he may choose between continuous swims, interval sets, all-out sets, tests, competitions, control sets in training, etc. For the more technical and tactical exercises he has global and analytical training methods at his disposal

Summary In conclusion we present a synopsis of the chronological sequence of the different steps that have to be taken in the periodization process. Beside each step we also list one of the 5 important training structuring principles that should especially be taken into consideration (tab. 21).

Chronological sequence of the different steps that have to be taken in the periodizing process	*Principle for structuring training having the greatest influence on each step*
1. Selection of top competitions	*"principle of durability of the development and long-term build-up"*
2. Determination of the number of macrocycles	*"principle of variation in training loads"*
3. Planning of exams, vacations, swim camps and test competitions	*"principle of the optimal timing and dosing of the swim specific and non swim specific types of training and of the development of the different performance determining components"*
4. Subdivision of the macrocycle into different mesocycles and determination of their training content	
4.1 fixation of the duration of the competition and base training periods and subdivision into mesocycles of: 4.1a the competition training period 4.1b the base training period	*"principle of progressive build-up of the training load"*
4.2 determination of the mesocycle objective	
4.3 selection and dosing of the different types of training exercises	*"principle of optimal relation between work and rest"*
4.4 fixation of dates for tests and evaluations (training documentation, training analysis, training adjustment and analysis of the current competitive performance)	

Tab. 21 Synopsis of the chronological sequence of the different steps that have to be taken in the periodizing process. The most important training structuring principle corresponding to each step is recorded in the right column.

THE REALIZATION PHASE

Introduction: The Individual Optimization of the Training Process

It is, however, not enough to have good intentions and write beautiful training plans on paper; these plans need also to be carried out. The final result of the completed training, in other words, the competition performance, will mainly depend on the precision and effectiveness of concretizing the plans.

It is fairly clear that a swimmer who is just transferring to the competition group cannot complete the same training as a competition swimmer who has already several years of training behind him. Also swimmers from one and the same group, which at first glance looks quite homogeneous as far as the competition performance and the main discipline are concerned, may each react very differently to identical training exercises. An individualized adaptation of the training is thus an absolute necessity if we want to reap maximum benefit from it.

For the individualization of the training 2 important aspects come forward:

1. the swimmer's training habits
2. the swimmer's individual ability to adapt (morphologically and conditionally) to the different training types

The Swimmer's Training Habits

Training adaptation, of whatever nature, can only take place if the organism is sufficiently stimulated. When creating an adequate training stimulus, we must thus take the swimmer's training habits into account.

This also implies that we must be careful with creating habits. Young swimmers whom we take up to high training volume too soon, will get used to this training rhythm but will not improve their conditioning proportionally to the work completed. If we then wish to improve these swimmers' competition performance still more, we will have to boost the volume even more. But at a certain moment boosting the number of swimming hours will start conflicting with other activities such as school commitments, etc. and, which is even worse, will lead to overtraining without ever seeing the result of the work completed.

The Swimmer's Individual Ability to Adapt (Morphologically and Conditionally) to the Different Training Types

During the macrocycle we will repeatedly evaluate the evolution of the swimmer’s competitive performance (the conditional, biomechanical and mental profile) according to the completed training. This process is very important for the continuous individualization and optimization of the training program, since it gives guidelines with respect to:

- the swimmer’s general ability to handle and assimilate training
- the appropriate intensity and volume for each type of exercise
- the necessity and priority of the different types of exercises

Indeed, based on these findings the training program for the next period is adapted to the swimmer’s current conditioning profile, his adaptation ability and individual needs. This individual optimization of the training process is necessary to get the best possible training effect. No energy is wasted on types of training or efforts that do not contribute in the broadest sense to improving the competition performance.

Yet the ideal/optimal analysis of the swimmer’s actual competitive performance needs to meet some essential conditions:

- The test must not only be swim but also stroke and discipline specific. Testing swimmers on a bicycle or treadmill in order to obtain the right information for water training is like taking the temperature with a barometer; both have to do with weather but measure something quite different

The importance of a sports specific conditioning test was already demonstrated by Holmer in 1974 through a longitudinal study of the evolution of the aerobic capacity of elite swimmers on the treadmill and in the swimming pool (fig. 77). The swimmers’ aerobic capacity measured on the treadmill remained unchanged while the one measured in the water clearly changed

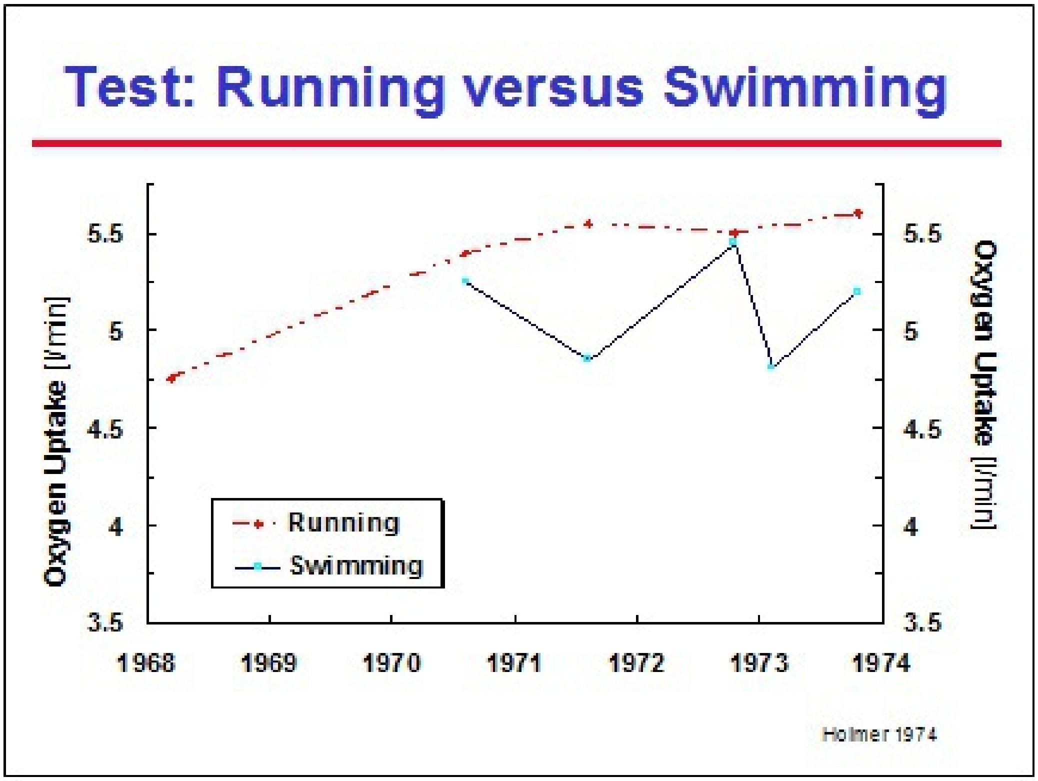


Fig. 77 Only a sport specific test procedure gives a reliable view on the sport specific condition and reliable information for the training.

- This analysis must also be carried out as precisely and completely as possible. The complexity as well as the multi-disciplinary nature of the test will increase along with the swimmer's performance level. This is necessary because, as the athlete's performance level improves, the detection of his conditioning strengths and weaknesses becomes increasingly more difficult and the guidelines for the training framing, planning and periodization need to be more precise in order to still further improve his competition performance

An overview of the different test possibilities for the determination of the athlete's current competitive performance is shown in table 22.

Analysis of the Competitive Performance					
Competition and training observations	Sport specific locomotor tests	Biomechanical tests	Medical tests	Biochemical tests	Psychological tests
	- technique - tactics	- strength - flexibility - kinetics - dynamics - neuromuscular function - biometrics - motion efficiency	- health - orthostatics - cardiovascular function - training adaptations	- energy metabolism - motion efficiency	- senso-motor system - ability to handle stress

Tab. 22 Overview of the different test possibilities for determining the athlete's current competitive performance.

Once we have diagnosed the competitive performance, we will evaluate its evolution according to the completed training. It is thus essential for the swimmer to keep a training logbook and to accurately record and document the training he has actually completed, in order to provide an overview of volume, intensity, load frequency as well as load density of the different exercise categories. The analysis of the training diary is then meant to:

* verify if the execution of the training tallies with the planning
* evaluate both the completed training and the effects it induced

Example of Training Analysis In a group of young promising swimmers, preparing for the national winter championships in the beginning of February, the evolution of the conditioning (between October and March) was ascertained by means of lactate tests, consisting of a submaximal 400 m freestyle followed, after complete rest, by a 100 m or 200 m all-out in the personal stroke. The test procedure remained the same for each swimmer during this observation period. The periodization with the distribution of volume and intensity as completed within this macrocycle is shown in figure 78.

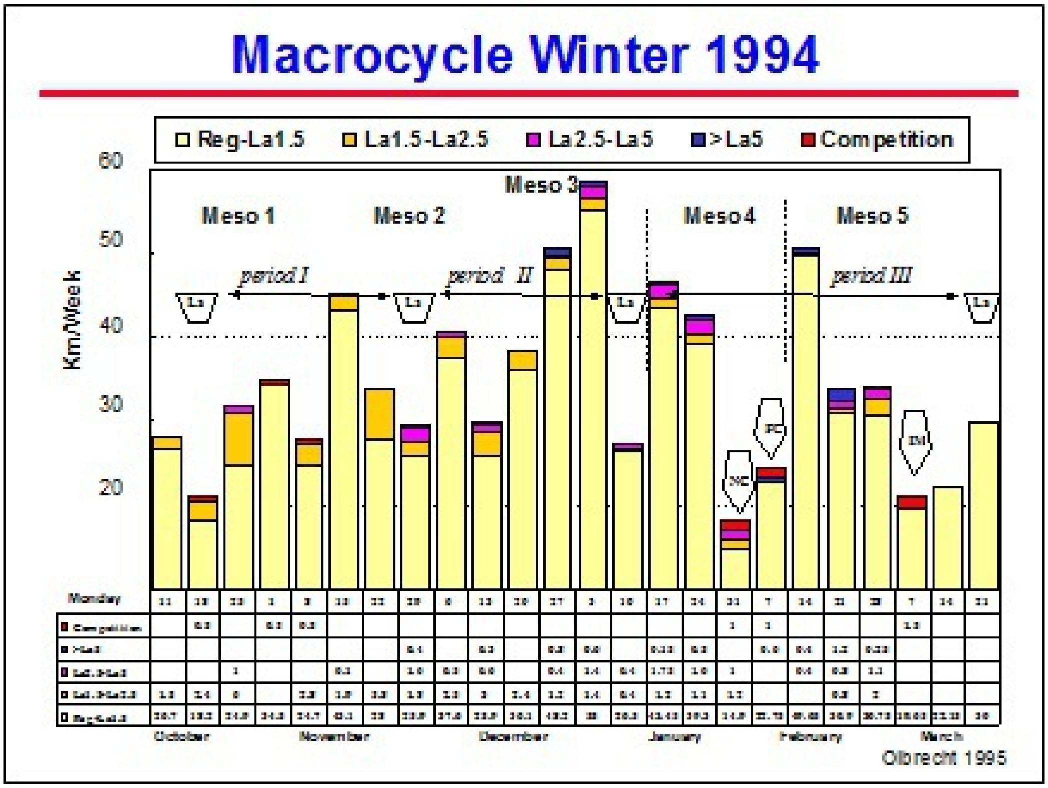


Fig. 78 Periodization with the distribution of volume and intensity as completed during the winter macrocycle.

The top part of figure 79 shows the evolution of the aerobic endurance, expressed by the 4 mmol/l lactate speed on 400 m freestyle ($V4_{400}$) and of the anaerobic performance, reflected by the highest lactate concentration after the 100 m or 200 m all-out in the personal best stroke. In the middle and at the bottom of the figure you will find the distribution of the intensive and extensive training exercises performed in-between the lactate tests.

Period I:

Training documentation: This period is characterized by a wave-shape increase in the number of kilometers/week (fig. 78). Only 10.8% of the weekly volume (= 3.5 km/week) is completed with an intensity of more than 1.5 mmol/l lactate. Exercises within the intensity range of 1.5 to 2.5 mmol/l lactate and above 2.5 mmol/l lactate were respectively proportioned to a load density of 1.4 and 0.9 training blocks/week (fig. 79). These exercises will, along with three 25 m sprints 2 to 3 times per week, make sure the anaerobic capacity is kept to standard and does not get broken down by the increase in training volume.

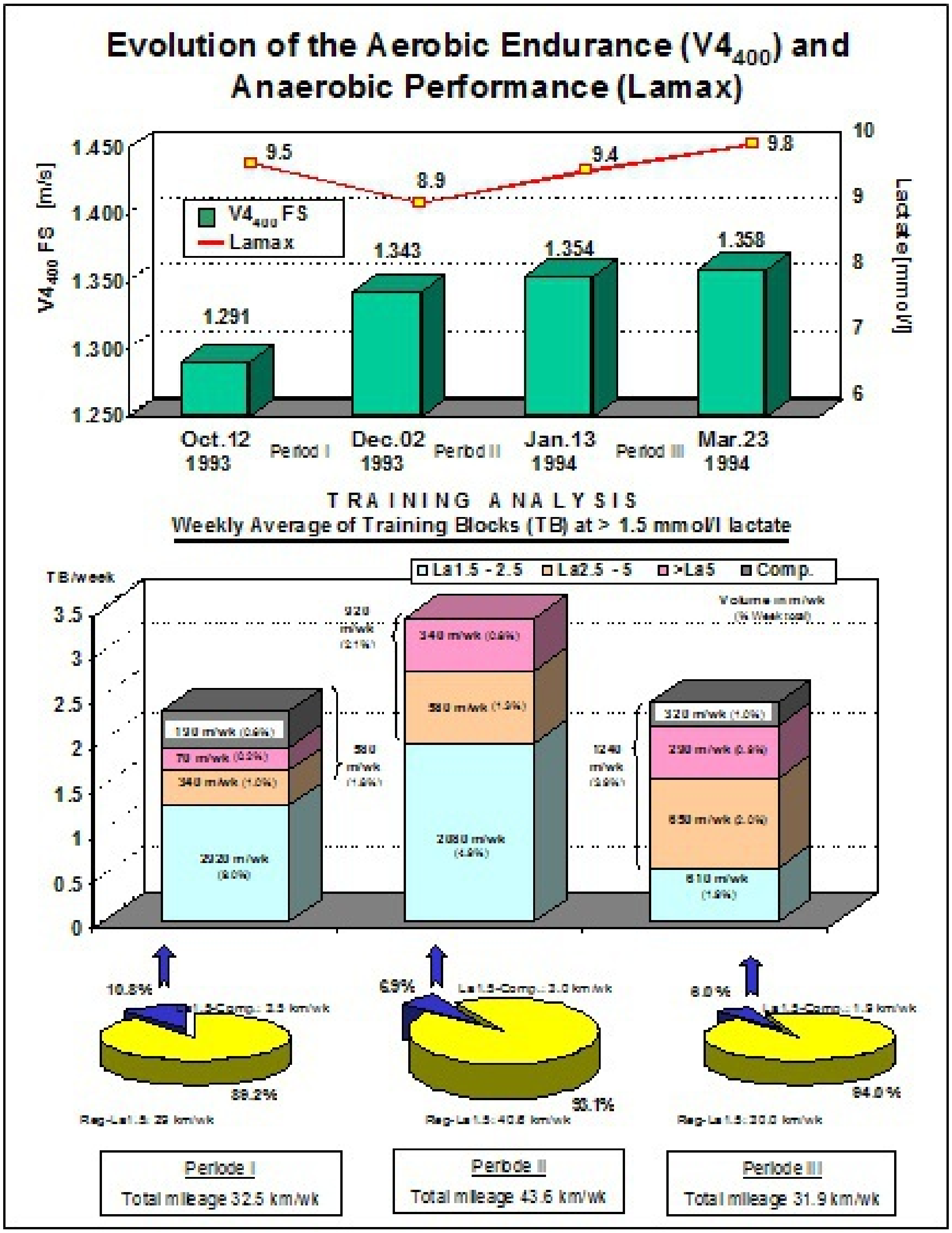


Fig. 79 Top: Evolution of the aerobic endurance, expressed by the 4 mmol/l lactate speed on 400 m freestyle ($V4_{400}$) and of the anaerobic performance, reflected by the highest lactate concentration after the 100 m or 200 m all-out in the swimmer's best stroke (Lamax). Middle and bottom: Overview of the intensive and extensive training exercises performed in-between the lactate tests.

Training effect: This training period of 2 months led to the improvement in aerobic endurance (4 mmol/l lactate speed at 400 m freestyle) averaging 12 sec. The highest lactate concentration (Lamax) after the all-out swims decreased, however, with an average of 1.2 mmol/l (fig. 79 top). This decrease could certainly not have been predicted based on the average training volume (32.5 km/week).

Training adjustment: The lower Lamax needs to be compensated for by inserting intensive training blocks (>1.5 mmol/l) more often per week. Beyond that, the fall break will be used to significantly increase the training volume.

Period II:
Training documentation: The swimmer completed, per week, about 1.7 training unit and 11.1 km more than in the previous training period. The workouts under 1.5 mmol/l lactate rose from 89.2% to 93.2% of the total weekly volume while the percentage of exercises performed at more than 1.5 mmol/l was reduced from 10.8% (period I) to 6.8% (3.0 km/week). The weekly number of training blocks within the intensity range of 1.5 to 2.5 mmol/l lactate, between 2.5 and 5 mmol/l lactate and above 5.0 mmol/l lactate was clearly higher than in the first period and rose to respectively 1.8, 0.9 and 1.4 training blocks per week. The total volume of these exercises was 920 m/week as compared to 580 m/week in period I.

Training effect: The level of the aerobic endurance was clearly preserved (in fact, the time span between the period of high volume and the test date was too short to be able to measure any improvement in aerobic endurance) and Lamax again rose to the reference value of October.

Training adjustment: From now on we will pay more attention to building up the specific competition conditioning and concentrate on power and frequency sets.

Period III:
Training documentation: Up until the winter championships training was globally less voluminous but much more intensive. The peak form period was extended by inserting a small mesocycle of high volume and almost exclusively regeneration training immediately after the Flemish championships.

Training effect: We managed to keep up the aerobic endurance quite well despite the lower training volume and more of competitions. Lamax rose slightly.

This example presents some interesting findings:

1. Increasing the training volume improves the aerobic endurance but may, if not optimally compensated by anaerobic capacity exercises, depress the maximum attainable lactate concentration

2. On average, 2 intensive blocks per week (in training or competition) will do to develop the anaerobic conditioning; this dosing, however, is very individual and requires a accurate evaluation
3. The improvement of the anaerobic conditioning is much more related to the weekly number of intensive training exercise blocks than to the volume of these exercises
4. The level of aerobic endurance can easily be maintained despite intensive training or competitions if these heavy loads are compensated for by a lot of regeneration training
5. The improvement of the aerobic capacity is stimulated by a few short intensive parts in the aerobic training

The following figure shows the training analysis of a World class female breaststroke swimmer in combination with the evolution of her conditioning and her biomechanical/motor development (fig. 80). During the winter period we already planned 2 competition peaks (Palma and World Cup), which is not particularly easy going, but here too we can observe that the aerobic endurance can be maintained despite an intensive competition period (March).

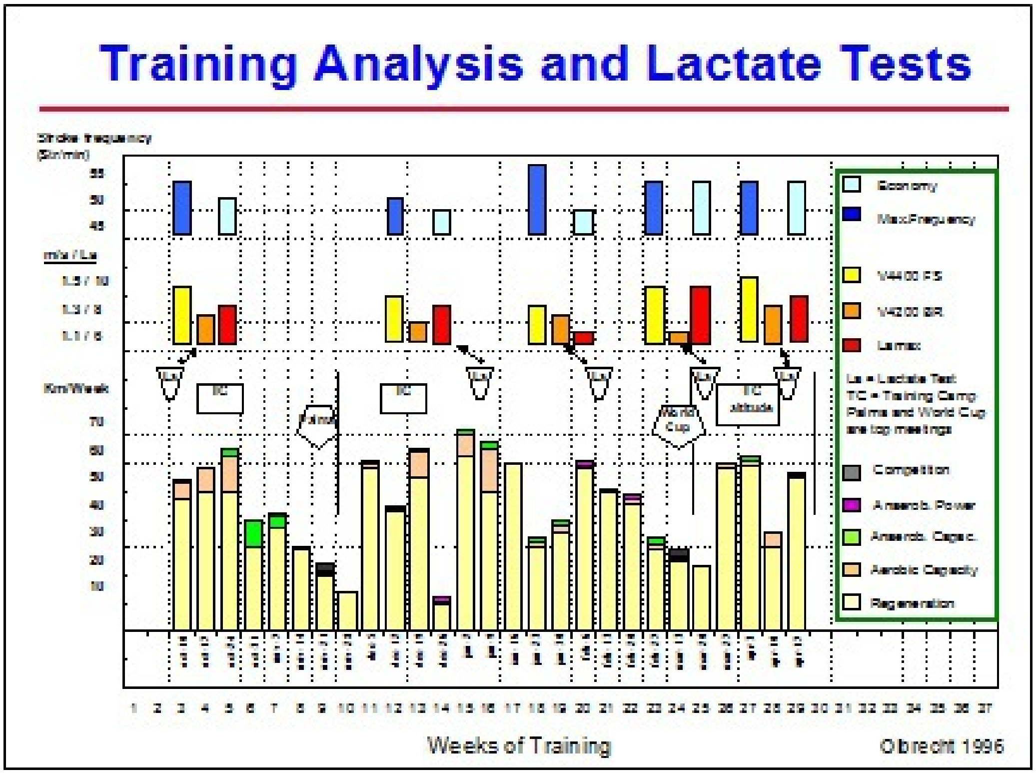


Fig. 80 Training analysis of an elite female breaststroke swimmer. Training volume and intensity are shown graphically along with the conditioning and biomechanical evolution.

Concerning the motor development we see that the efficiency of the breaststroke motion increases after voluminous training work in the personal stroke. The maximal stroke frequency (stroke rate), however, drops sharply, and, since it clearly lies below the ideal competition stroke frequency, this gain in motion efficiency does not translate into an improvement in competition performance. We will only reach an optimal relation between stroke frequency and motion efficiency after specific training exercises such as frequency training.

The Steering Principle

The training steering or monitoring starts with the analysis of the swimmer's current competitive performance. According to the appointed conditioning properties we will then:

* determine the training objectives (= what needs to be trained during the next period)

* select the most appropriate the types of training (= the types of exercises that fit best the swimmer’s present needs and trainability)
* design a training structure for a period of 5 to 15 weeks

Once this training period perfected, the swimmer’s competitive performance is evaluated again in order to get a good picture of the benefit of the completed training. Experimental research has namely clearly shown that identical training plans can lead to different training effects even for swimmers with identical conditioning and morphological properties. It is thus essential for the coach to exactly know and assess the swimmer’s individual adaptations to the different training programs.

The perfected training period is thus evaluated by repeating the test used for designing the first training structure, whereupon the coach tries to link up the measured changes in conditioning with the actual completed training program without however omitting to consider additional factors such as, for example, the swimmer’s competition results, his attendance at the training sessions, his state of health, environmental factors (exams, school commitments, etc.). The purpose of the training evaluation is by no means to reprove the coach for what did not succeed but rather to check the appropriateness of his training planning and periodization and if necessary to look for a higher yield in the next training period.

According to the results of the evaluation of the first training period the training plans for the next period will be adapted or otherwise. We call this “training steering (monitoring)”. The changes or adaptations (adjustments) to be made can vary in nature and refer to:

* the training intensity
* the training volume
* the distribution of the 4 main training exercise classes
* the composition of the exercises

The training objectives will evolve and change, but the cyclic steering principle (fig. 81) will keep recurring during the entire course of the season.

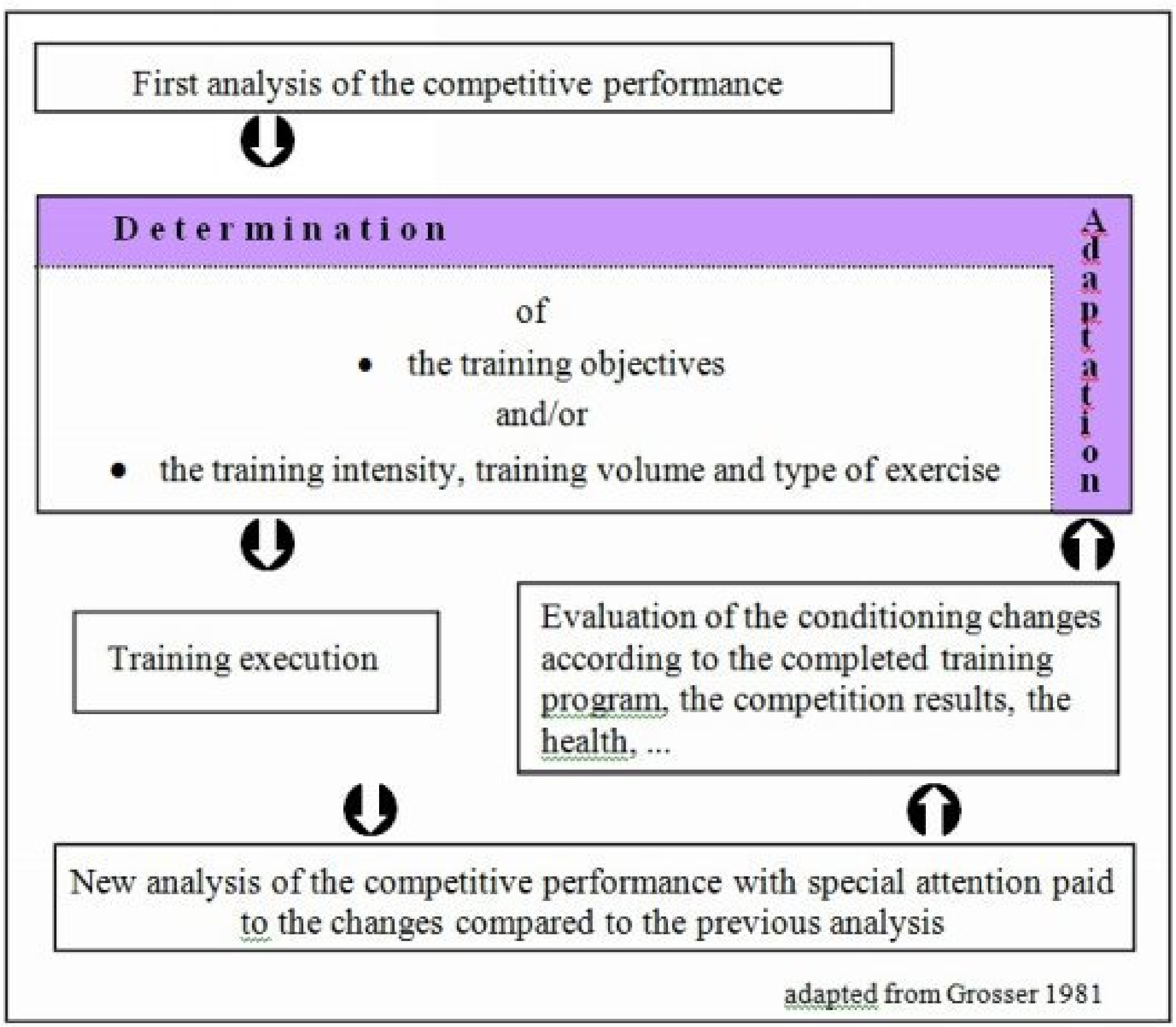


Fig. 81 Cycle of training steering (monitoring) after Grosser 1981.

Chapter 9 TEMPLATES FOR PLANNING AND PERIODIZATION

To put the coach on the track to plan and periodize his training in a systematic and well-considered way we have drawn up frameworks for scheduling:

* a multi-year plan
* a year plan
* a macrocycle with different periodizations
* a meso- and microcycle

The plans of work for the macro-, meso- and microcycle are supplemented with practical and useful guidelines and directives ensuring a gradual proceeding.

All these frames are not only helpful for the training planning and periodization but can also be used to document the most important observations and data of the effectuated training. This training documentation will thereby help to keep an overview of the completed training and provide the basis for the evaluation of the training effects.

Enjoy yourself .... and good luck!!!!

MULTI-YEAR PLAN

	Basic training					Build-up training		Top level training		
Training year:	1	2	3	4	5	6	7	8	9	10
Performance objectives										
Training weeks Number/year:										
Training days Number/year:										
Training units Number/week: Duration:										
Training twice a day Number of weeks:										
Volume km/year:										
Strength training Type: Units/week: Duration:										
Complementary sports Time spent/month: Type:										
Evaluation tests Type:										

YEAR PLAN: Year:

Winter	September	October	November	December	January	February
Competitions						
General Objectives						
Tests						
Training units/ week						
Duration AM Volume PM						
Type training						
Dry-land training						

Summer	March	April	May	June	July	August
Competitions						
General Objectives						
Tests						
Training units/ week						
Duration AM Volume PM						
Type training						
Dry-land training						

Guidelines for the periodization of a macrocycle (examples fig. 71 and 72)

Step	Reference	Assignment description
①	Nr. training week Date Monday	• Enter the date of the Monday (Monday/month; example 05/05 = Monday May 5th) of each training week
②	Competitions	• Fix then first the top competitions and then the test meets (it is always possible to add other competitions later on in case they fit the periodization)
③	Period Mesocycle Microcycle	• Indicate the periods that may affect the (course of the) training (example: *exams* = unsteady attendance; *vacations* = possibility of voluminous training program; *(compulsory) training camps* = these need to be prepared and compensated • Considering all these elements, subdivide the macrocycle. Fix first the *competition training period*, then the *transition period* and finally the *base training period*. Split then the macrocycle into different meso- and microcycles. This subdivision schedules the training objectives of the year plan in a time structure (example: in which mesocycle and for how long will we work on the aerobic capacity?) • Number subsequently the meso- and microcycles as a reference for the training log
④	Swim Training Units	• Now fill in the number of training units in the water and on dry-land
	Dry-land Training Units	Each half day of competition is considered as a training unit
⑤	REG AEC1 (AEC2) ANC AEP ANP	• Indicate per week the desired number of different types of workouts in the water and on dry-land (abbreviations see below). Since, with the exception of REG, the numbers refer to type of exercises and not to the training units containing the specific exercises, the sum of all types of exercises may differ from the number of training units(e.g. one unit may contain an ANC and an ANP set). Some Remarks: * REG = a whole training is spent on regeneration (intensity of LA1 or even slower), relaxation. It may contain <u>some</u> sprints, drills, technique
	MF EX EN	* AEC1 = both the intensive and extensive work take place during the same training unit (volume set at least 600-1200m, no drills, technique, ... in the main set) * AEC2 = the extensive work happens in the training unit following the intensive one (a competition or a unit containing a long ANC, an ANP or an AEP). Record the number of AEC2 between brackets beside the number of AEC1
⑥	Volume PS = Intensity *Intens..* ⇧ σ *Volume* ⇨	• Eventually we broadly and schematically indicate the weekly training volume and intensity. The absolute figures will be recorded later This schematic concept will help us to design a training plan which is getting progressively more difficult and to avoid "training too much too soon" causing the swimmer to be exhausted when starting the hard competition training period For each "quality training" (harder than extensive or regeneration training) we will specify the degree of intensity by means of small triangles, the height and base width of which respectively represent its intensity and volume
⑦	Others	• Use this space to record other types of training such as running, etc ... indicate visit to physician, tests or ....
	Abbreviations:	• REG = Regeneration, AEC = aerobic capacity, ANC = anaerobic capacity, AEP = aerobic power; ANP = anaerobic power training, PS = Personal best stroke, Dry-land training; MF = maximal force, EX = explosive strength training and EN = endurance strength training

MACROCYCLE .../....

Nr. Tr. week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Date Monday	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..	../..
Period Mesocycle Microcycle																												
Volume PS = Intensity *Intens.* ⇧ σ *Volume* ⇨																												
Swim Training Units REG AEC1(AEC2) ANC ANP AEP Competition																												
Dry-land Tr. Units MF EX EN																												
Others																												

Guidelines for the elaboration of a meso– and microcycle

(examples see fig. 65-68)

Step	Reference	Description of the assignment
①	**Mesocycle: . . .**	• Identify the mesocycle as it is designated in the macrocycle
②	Sw Tr.: AM Typ Volume TU PM Typ Volume TU	• The main operation is *to schedule the different types of training exercises and their respective volume* to ensure and reach the maximal benefit from the training stimulus. You will often have to find a compromise between the ideal but theoretical situation and what is practically feasible (e.g.: the availability of the swimming pool may, some days, be restrictive and limit the training possibilities). Each of the following steps has to be implemented for both the swim and dry-land training simultaneously to
	Dry Tr.: Typ Duration TU	guarantee the synergism of both types of training. Design and frame all the microcycles of the mesocycle simultaneously to keep the overview and ensure the continuity of the training content
	Others	

Procedure:

a-• Start scheduling the most important training exercises according to the objective of the macrocycle

b-• Thereupon position the less important training assignments avoiding any negative interaction between the different types of training exercises. Sprints, frequency and technique training are not explicitly mentioned in the macrocycle but need now to be specified. Sprints need to be planned in 3 of 5 training units (not too many repetitions but make sure the swimmer performs them well)

c-• Finally fill in the gaps with regeneration training

d-• Record per day the total volume of the training unit as well as of the different types of important training exercises

e-• Record any other action, remark,.... that might be important for your planning

Practical hint

It becomes now more and more important to specify the different types of exercises, especially the aerobic capacity sets

a-• At this stage you can only fix AEC1 exercises and therefore use the abbreviation AEC for this class of exercise. The AEC2 type of aerobic capacity training will come about when an intensive training unit (may be a competition) is followed by a very extensive unit. The frequency of AEC2 per week is only recorded in the weekly overview

b-• Mention between brackets the volume of the type of exercise. In case of AEC specify also the intensive bout between brackets, i.e. the type of exercise of the intensive part and its volume, complemented with the volume of the "extensive tail". If ANC is combined with frequency training, ANC(m) will be complemented with Fn, n indicating here the number of progressive swims (e.g. 4x 3x50 m progressively faster 1 to 3; n=3) and m the total volume of the set

c-• Other specifications: SPR(n) = n sprints; TE (x) = x minutes technique (e.g. starts, turns,..); AR(m) and AR(m)+PB = set of arm swimming of indicated volume m respectively without and with **P**addles and/or pull-**B**uoys; KK(m) = set of kicking

Notification examples

AEC(ANC400+1200) = an AEC containing a 400m ANC set followed by 1200m extensive work

AEC(AEP500+1500) = an AEC containing a 500m AEP set and 1500m extensive work

ANC(300) = an ANC totaling 300m

ANC(750)F3 = an ANC totaling 750m buildup as a frequency training with 3 progressive swims

TE(15) = 15 min tech.

SPR(5) = 5 sprints

Step	Reference	Description of the assignment
③	Total/Split TU Sw Tr. Km Dry Tr. TU DU	• In this *weekly overview* we indicate per week the total for: - the swim training: the number of training units (TU), AEC2, ANC and ANP blocks and competition events the training volume (Km), REG, AEC1 and AEP blocks, - the dry-land training: the number of training units (TU), their duration in min (DU), the time spent on MF, EX and EN

Abbreviations: Sw Tr. = Swim training; AM = morning training; PM = evening training; Typ = type of training exercise (REG = Regeneration (≦La1), AEC1 and 2 = aerobic capacity, ANC = anaerobic capacity, AEP = aerobic power, ANP = anaerobic power training and CMP = competition events); Dry Tr. = dry-land training; TU = training units; Km = Volume in km; DU = Duration in minutes; MF, EX and EN = respectively maximal force, explosive and endurance strength training; TE and F = Technique and Frequency training; AR and KK = Arm swimming and Kicking; SPR = Sprints; P= paddles and B = pull-buoys

Mesocycles:	Microcycle I							Microcycle II							Microcycle III						
	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Sw. Tr: AM Typ Volume																					
PM Typ Volume																					
Dry Tr: Typ Duration																					
Others																					
Total / *Split* TU		*REG*	*AEC1*	*AEC2*	*ANC*	*AEP*	*ANP*		*REG*	*AEC1*	*AEC2*	*ANC*	*AEP*	*ANP*		*REG*	*AEC1*	*AEC2*	*ANC*	*AEP*	*ANP*
Sw. Tr. Km																					
Dry Tr. TU		*MF*	*EX*	*EN*	*AR*	*KK*	*CMP*		*MF*	*EX*	*EN*	*AR*	*KK*	*CMP*		*MF*	*EX*	*EN*	*AR*	*KK*	*CMP*
DU																					

Mesocycles:	Microcycle IV							Microcycle V							Microcycle VI						
	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Sw. Tr: AM Typ Volume																					
PM Typ Volume																					
Dry Tr: Typ Duration																					
Others																					
Total / *Split* TU		*REG*	*AEC1*	*AEC2*	*ANC*	*AEP*	*ANP*		*REG*	*AEC1*	*AEC2*	*ANC*	*AEP*	*ANP*		*REG*	*AEC1*	*AEC2*	*ANC*	*AEP*	*ANP*
Sw. Tr. Km																					
Dry Tr. TU		*MF*	*EX*	*EN*	*AR*	*KK*	*CMP*		*MF*	*EX*	*EN*	*AR*	*KK*	*CMP*		*MF*	*EX*	*EN*	*AR*	*KK*	*CMP*
DU																					

GLOSSARY OF THE BASIC TERMINOLOGY AND CONCEPTS

-A-

Acidosis

The production of lactate leads to acidification, which, if it becomes too strong, will significantly hinder further energy production. The acidosis will block the activity of important enzymes that enable the glycolysis to take place. As a result, this will reduce the energy provided by the glycolysis. The athlete will then no longer have enough energy at his disposal to continue his exertion with the same intensity and he will have to slow down.

The acidosis depends strongly on the rate (= quantity per second) at which pyruvate is converted into lactate. At a low rate of conversion, the acidosis can easily be neutralized in the muscle cell. At a high rate of conversion, the buffer capacity in the muscle cell will no longer be able to counter that acidosis and the intra-muscular pH will decrease due to the accumulating H^+ ions. At this stage, the acidosis becomes so pronounced that it will reduce further glycolysis activity and, consequently, the energy production by this system.

Acidosis tolerance

This term, often used by coaches, refers to the extent a swimmer can tolerate the acidosis. The exact physiological processes behind this concept are unclear.

Adenosinemonophosphate (AMP), -diphosphate (ADP) and -triphosphate (ATP)

The muscular system's performance capacity is determined principally by the amount of energy made available for muscular activity. The more energy that is made available per second, the harder (more intensely) the muscle can operate. If, for any reason less energy is made available, the intensity of muscular activity will decrease, forcing the athlete to slow down.

The production of energy occurs in the muscle cell and is transmitted to the working muscle elements via ATP (adenosine <u>tri</u> phosphate). The working muscle captures the energy carried by ATP, converting ATP at that point into energy-poor ADP (adenosine <u>di</u> phosphate) or AMP (adenosine <u>mono</u> phosphate). ATP is present in the muscle in only very small quantities and must, therefore, be supplemented during exercise. If this does not happen, the muscle will very quickly experience an energy deficiency, forcing the athlete to interrupt or slow down the exercise until sufficient energy is again available.

Aerobic

Aerobic comes from a Greek word meaning air. It is used here to refer to the metabolic activity that requires oxygen to produce energy. We have one aerobic system as opposed to two anaerobic systems.

Aerobic - Anaerobic metabolism

Three different biochemical processes can recharge energy-poor ADP to energy-rich ATP (rephosphorylation of ADP). Two of these processes can occur without requiring oxygen and are therefore called *anaerobic metabolic systems;* one is the anaerobic alactic energy production system and the other is called the anaerobic lactic energy production system. The third biochemical process requires oxygen to generate energy-rich ATP; we call it therefore the *aerobic metabolic system* .

Aerobic capacity

Aerobic capacity is the maximum amount of oxygen one can consume per minute. It is also called VO_2 max. Some sources in literature consider VO_2 max as a parameter of power and not of capacity. With respect to its unit of measure (VO_2 max is expressed per unit of time) it can indeed be seen as a parameter of power. According to its physiological meaning however, it refers to a capacity. We join in the last argumentation and define, in line with Hollmann and Hettinger (1990), VO_2 max as a parameter of capacity.

The higher the aerobic capacity, the better the performance in competition "can" be and the faster the regeneration process after training and competition will start. The aerobic capacity is constantly changing as an athlete goes through the training cycle. It is however thought that each individual has an innate maximum aerobic capacity that is genetically determined.

Aerobic capacity training

This training is set up mainly to increase the maximum uptake of oxygen per minute (physiological adaptation induced: rise of VO_2 max). This type of exercise lays the basis for the swimmer's conditioning and is therefore not only essential for the long distance swimmer but also for the sprinter. It will thus, take up the largest part of the base training period in the periodization. During the competition training period the aerobic capacity workouts will be performed less frequently and at a lower intensity as, during this period, their only purpose is to maintain the acquired aerobic capacity and to counterbalance the intensive training work (anaerobic capacity, anaerobic power or aerobic power workouts).

Checklist for the aerobic capacity exercises:

1. high volume (for sprinters, the intervals and the total distance of the workouts are generally shorter than for long distance swimmers)

2. low intensity during the major part of the exercise but "spiced" with some intensive bouts at the beginning of the workout
3. little rest between the intervals

Aerobic conditioning

Refers to the training process that improves the aerobic capacity or power.

Aerobic endurance

Refers to the 4 mmol/l lactate speed. The contribution of aerobic capacity or aerobic power to the 4 mmol/l lactate speed, which quantifies the aerobic endurance, is not clearly defined.

Aerobic energy supply/delivery

The contribution of energy that metabolically is provided by the aerobic system is defined as aerobic energy supply or aerobic energy delivery. The aerobic capacity (VO_2 max) determines the maximal amount of energy that can be produced by the aerobic energy system during a physical activity.

Aerobic power

Aerobic power is the percentage of aerobic capacity that can be used during an athletic event. The higher this percentage, the better the aerobic power. Various metabolic factors prevent an athlete from using 100% of his aerobic capacity during the event. The lactate threshold or maximum lactate steady state is a good measure for the aerobic power. Aerobic power is directly affected by the anaerobic capacity. The higher the anaerobic capacity is, the more it becomes difficult to achieve a high aerobic power.

Aerobic power training

This training is set up mainly to maximize the use of one's aerobic capacity (physiological adaptation induced: increasing the % of VO_2 max that can be maintained during long distance exercises). This type of exercise is necessary to complete the training work. It will, therefore, be principally planned during the pre-competition training period and it should be swum in the main stroke (medley swimmers will now swim medley and freestyle sets). From a physiological point of view, the importance of this type of workout increases in accordance with the length of the competition event for which the swimmer is preparing. For 100 and 200 m races, this type of training is not a must, but for races longer than 200 m it is an absolutely essential part of the pre-competition training period.

As results from this type of training should not be expected within 3 weeks, the coach is compelled to start this type of training at least 6 weeks before the important competition event.

Checklist for the aerobic power exercises:

1. stay around the competition distance (for a 400 m swimmer the workout may be 500 m long and be repeated 2 or 3 times)
2. the rest is always short (5 to 15 sec) and the repeat distance is adjusted to enable the swimmer to maintain competition speed or just a little bit faster
3. as the swimmer gets closer to the competition, the coach will make these swim sets progressively harder by shortening the rest in proportion to the repeat distance. Example: in the beginning an aerobic power series consists mainly of 50 and 100 m intervals with 15 sec rest. In the following weeks the athlete swims longer intervals (100 and 200 m) but with the same or less rest

Aerobic threshold

Aerobic threshold is not frequently used in the popular literature but is often seen in the academic journals. Although different definitions are available, it means mostly the point where lactate starts to rise from a baseline, also called the OBLA (onset of blood lactate accumulation). It corresponds to a lower speed than the lactate threshold ($MaxLa_{ss}$ or anaerobic threshold). If an athlete exercises at an effort level above the aerobic threshold he may still achieve a lactate steady state.

Age group

Refers to young swimmers in general, without a specific range of ages.

Alactic

Refers to energy supply or an effort without production of lactate.

All-out swim

All-out swim means that the athlete provides a maximum effort. The athlete should be completely exhausted after this exercise. If an all-out swim is performed over a short period (40 - 90 seconds), lactate after this effort can be used to assess the anaerobic capacity.

Anabolic

Refers to the biological condition promoting the building-up processes.

Anaerobic

Anaerobic means without air. It is used here to refer to the metabolic activity that can produce energy without oxygen. There are two anaerobic systems as opposed to one aerobic system. The two anaerobic systems are the creatine phosphate system (also called the anaerobic alactic energy system) and the glycolytic system (also called the anaerobic lactic energy system).

Anaerobic capacity

The anaerobic capacity is the maximum amount of pyruvate that can be produced per second by the glycolysis. It is frequently called VLamax (in some literature also PLamax). Since the conversion from pyruvate into lactate and vice versa occurs very quickly, the ratio between the muscle concentrations of lactate and pyruvate is described as being 1/1. You may therefore find in literature on the subject the label “production of lactate” for what we refer to as “the production of pyruvate”. You may further find that the units of the anaerobic capacity are expressed in “mmol of lactate per liter and per second” instead of “mmol of pyruvate per liter and per second”. This may sound confusing but is in fact quite simple since they are counterparts.

The anaerobic capacity can change as the athlete goes through the training cycle. It is thought that each individual has an innate maximum anaerobic capacity that is genetically determined. However, some coaches and sports scientists have observed growth in this capacity after years of training.

Note: This term has many other meanings. The definition we provide is not common by many in the academic literature. The anaerobic capacity is also used to refer to the amount of anaerobic energy that is released during maximum activity as opposed to the highest possible rate of glycolytic energy release.

Anaerobic capacity training

This training is set up mainly to increase the capacity to break down carbohydrates anaerobically (physiological adaptation induced: rise of glycolytic rate = VLamax, PLamax or maximal lactate production rate). This type of training also plays an important part in the base training period and, to a smaller extent, in the competition training period where it is mainly used for maintenance purposes. Preferably, the anaerobic capacity exercises are inserted into the first half of the training session. This type of workout also lays the basis for the tough anaerobic power work that will come up later and must therefore certainly precede the anaerobic power sets.

Checklist for the anaerobic capacity exercises:

1. short intervals (25-50 m)
2. each repetition must be swum just a little slower than maximum pace

3. the rest must be at least as long as the exertion time span, preferably, even 2 times the exertion time
4. contrary to all other types of training, passive rest is better than active rest

A trick to control the intensity of the anaerobic capacity sets is to insert an all-out sprint early in the set:

- if the time of this sprint is more than 1.5 sec/50 m faster than the previous repetitions then the previous repeats were too slow and the remaining reps have to be swum faster
- if the time of this sprint is less than 1 sec/50 m faster than the previous repetitions then the previous swims were too fast and the remaining reps have to be swum slightly slower

Anaerobic conditioning

Refers to the training process that improves the anaerobic capacity or power.

Anaerobic energy supply/delivery

The contribution of energy that metabolically is provided by the anaerobic lactic system (glycolysis) is defined as the anaerobic energy supply or anaerobic energy delivery. The anaerobic capacity (VLamax) determines the maximal amount of energy that can be produced by the anaerobic lactic energy system during a physical activity.

Anaerobic power

The anaerobic power is the percentage of anaerobic capacity that can be used during an athletic event. Various metabolic factors prevent an athlete from using 100% of his anaerobic capacity during the entire event. For very short events the anaerobic power will near 100% of the anaerobic capacity. Anaerobic power is directly affected by the aerobic capacity, anaerobic capacity, buffering and lactate tolerance. The higher the aerobic capacity, the higher the anaerobic power will be, as the aerobic system eliminates the lactate produced.

Anaerobic power training

This training is set up mainly to maximize the use of one's anaerobic capacity (physiological adaptation induced: increasing the involved percentage of VLamax that can be maintained during a high level effort in training or in competition). This type of exercise is also used in the competition training period. The anaerobic training at the end of the base training period is meant to increase the ability to produce more lactate (anaerobic capacity training). The anaerobic power training in the competition training period is a.o. used to toughen the swimmer against acidosis. This adaptation can be completed quite quickly (2 weeks).

Checklist for the anaerobic power exercises:

1. maximum swim speed (starting with the first repetition)
2. short repeat distances; these must allow a maximum swim speed
3. very little rest (5 to 10 sec depending on the interval being used - 25 or 50 m)

The total distance of an anaerobic power set is very low (125 to 250 m). For top level swimmers, the volume can be increased to 600 m but should then be split in sets of 100 to 300 m with 10 to 20 min rest between each set.

Anaerobic threshold

The anaerobic threshold has many meanings. We use it to mean the maximum effort that can be sustained for an extended period of time without lactate steadily accumulating. We prefer the term lactate threshold or maximum lactate steady state ($MaxLa_{ss}$) instead.

-B-

Base training period

This training period is the first part of a macrocycle and is focused on building up the basic conditioning components.

Basic training phase

This phase of a swimmer's career starts between 10 and 12 years of age and lasts about 4 years. In this training phase the emphasis lays on:

- the general physical development, as broad as possible, of the beginner and age group swimmer
- stroke technique and coordination (starts/turns)
- the development of the swimmer's capacity of assimilating the training load
- non swim specific activities
- the swimmer's health status

Biological adaptation

The adaptations of the human organism follow a regular pattern. During the first 1 to 2 weeks of a new training cycle the body adapts quickly to the new stimulus. In the next few weeks the power of this same stimulus to provoke an adaptation will progressively fade to end up completely after 6 weeks. We therefore call week 1 and 2 of the adaptation process the *"fast adaptation phase"* and weeks 3 to 6 the *"stabilization phase"* .

After the body stops responding to the current training load (intensity and/or volume), it is necessary to increase it to induce further adaptation. This means that, after about 6 weeks, training should be modified by either:

1. increasing the training volume
2. increasing the training intensity
3. increasing the frequency per week of training units or intensive workouts
4. changing the proportion of workouts in each stroke (e.g. changing from freestyle to breaststroke while keeping volume and intensity the same)
5. making the training program more difficult by changing the training environment such as training at altitude or by reducing the regenerative training between intensive workouts (e.g. maintaining the same number of intensive workouts per week but with less extensive training units between them)
6. a combination of the changes in points 1-5

Why wait 6 weeks before introducing a new stimulus or a new training load? Why not change the training plan after 2 weeks when the stimulus has induced its fast adaptation? The reason is that weeks 3 to 6 are necessary to stabilize the adaptations brought about in weeks 1 and 2.

Biological maturity/age

Young swimmers of the same age may differ from a biological perspective. Some 10 year old boys may have the biological status of 8 year old boys while other male swimmers aged 10 may reach the biological status of swimmers aged 12. This example shows a different biological maturity as opposed to an identical chronological age. The biological maturity is also expressed by the biological age.

Breath control training

In this type of exercise we ask the swimmer to breathe once every x-number of strokes. Originally it was thought that this type of training, by analogy with altitude training, would restrict the availability of oxygen to be taken up by the red blood cells and in doing so, it would provide an extra stimulus for the improvement of the endurance capacity. This, however, has never been proven experimentally. Breathing a lot or a little has no effect on the swimmer's aerobic conditioning. It is therefore not appropriate to label this type of training as "hypoxia" training or to consider it as a surrogate to training at altitude. The term hypoxia training has been used erroneously for many years. It is now proper to refer to it as breath control training.

Buffering

Buffering is the ability of the body to neutralize part of the acid that is built up in the body when lactate starts to accumulate rapidly (metabolic acidosis). This can improve by training, but how much is still not well understood.

The most important system to buffer the acidosis in blood uses the plasma HCO_3^- ions (bicarbonates). HCO_3^- picks up the H^+ ions that are responsible for the acidosis, and forms H_2CO_3 that will dissociate in H_2O and CO_2. Hb and other plasma proteins contribute to a minor extent to neutralize the acidosis.

Build-up training phase

This phase of a swimmer's career starts between 14 and 16 years of age and lasts at least 3 years. This phase of the multi-year plan:

- runs to about 3 years before peak form age, which in disciplines, where endurance and strength determine the competition performance, presumably lies between 23 and 26 years. In swimming disciplines with a high technical factor, however, peak form age is reached faster which obviously shortens the build-up phase
- emphasizes besides endurance and strength, also the mental development such as the willingness to go all-out, the dealing with setbacks or failures, the competitiveness, the concentration power, the desire for training, etc.
- can easily take up 3 to 7 years

In this phase:

- the training load is further increased by lengthening the duration of the training units. Only in a later stage and depending on the conditioning level of the swimmer the number of training units per week may again be raised (possibly even to several training units a day)
- more swim specific training and conditioning will be included. Participation in other sport activities is still allowed but only if it supports the swim specific physical development
- attention should be paid to the structure of the training process, i.e. the training periodization
- attention should also be paid to perfecting stroke techniques which the swimmer must be able to execute correctly in varying situations

-C-

Capacity training exercises

The workouts that are principally meant to build up the swimmer's conditioning are labeled as capacity training exercises. The increase of the aerobic capacity (= maximal oxygen uptake, VO_2 max), the anaerobic capacity (= increase the breakdown of carbohydrates, i.e. glycolysis) as well as the increase of maximal force fall under this category.

Carbohydrates

Are organic compounds including sugars, starches, glucose and glycogen.

Catabolic

Refers to the biological condition promoting the breakdown processes.

Chronological age

See biological maturity.

Class effect

Refers to the major biological and functional adaptation(s) induced by a training exercise.

Clearance

The term "clearance" is used to describe the elimination of a substrate in the blood over time. Two separate but related processes are involved in the clearance of blood lactate:

- the process by which lactate moves from the muscles to the bloodstream. Evidence of this is seen by the rising levels of lactate in the blood as lactate leaves the muscles. This is what you would expect as lactate moves from an area of higher concentration to one of lower concentration. This is sometimes referred to as the appearance of lactate
- the process of removing lactate from the bloodstream (see elimination and lactate shuttle)

When lactate is measured in the blood, the reading is the net effect of both the appearance and elimination processes. During a steady state workout, these processes offset each other.

Competition training period

This competition training period is the second part of a macrocycle and is split in:

- a pre-competition training period or peak form inducing training period. During this period all efforts are focused on the fine-tuning and optimization of the acquired basic conditioning components (e.g. aerobic and anaerobic capacities, mental preparation, stress tolerance) in order to realize a top performance
- an in-between competition training period or peak form maintaining training period. The main objective here is the stabilization of the peak form

Conditioning profile

This mapping describes in detail the physical conditioning of the athlete. It contains at least an estimation of his aerobic and anaerobic capacity.

Continuous swimming

This type of exercise consists in swimming continuously (during at least 15 min) and is especially useful in combination with short intensive workouts to improve the aerobic capacity.

Control test

A control test, sometimes also called spot test, is a test that is used to assess some current aspects of the athlete's fitness. It is frequently used to confirm the findings of a standard lactate test, to see if an adaptation has occurred or if an athlete is exercising at the proper intensity.

Creatine phosphate system

This is one of the 2 anaerobic energy systems. It produces energy very quickly, without any lactate release but for only a very short time. The energy results from the breakdown of creatine phosphate.

-D-

Declining return on training investment

Studies have shown that the rate of improvement in conditioning is not linear to the increase of training load. As a matter of fact, the effect of an increase in training load (increase of intensity and/or volume) will be different depending upon the conditioning level of the swimmer. Indeed, the better the conditioning level, the greater the training load that will be needed to make even slight improvements. Consequently, the risks of injury and overtraining will be greater. Or, to put it differently, the better the swimmer, the less improvement a same increase in training load will produce. This phenomenon is called the principle of declining return on training investment and is applicable to all kinds of training adaptations.

Design phase

The whole process of making training plans goes through various steps that can mainly be bundled in 2 distinctive phases:

- the design phase
- the realization phase (execution and steering)

The design phase consists of 4 steps:

Step 1: Analysis of the swimmer's current performance level and training background
Step 2 Set up of (intermediary) performance objectives
Step 3: Determination of the training structure
Step 4: Design of the training structure

Detraining process

Whenever an athlete stops training, a "detraining" process sets in, inducing a relatively rapid loss of the acquired adaptation. So the detraining process means the deterioration of fitness gains towards untrained levels when training is stopped.

Drill

Drills are training exercises that aim at the automation or improvement of technical skills once the learning process is completed.

-E-

Energy supply

A working muscle needs energy to perform any activity. This energy is supplied/delivered by metabolic processes in the muscle. The higher this energy supply, the more intensely (faster) one can exercise.

Endurance capacity

See aerobic capacity.

Endurance force

The endurance force describes the ability of a muscle group to sustain a workload for a longer period of time. The higher the endurance force, the higher the workload that can be maintained during a longer period of time.

Enzymes

Enzymes are proteins that have a multitude of purposes. One of the outcomes of training is the addition or reduction of various enzymes that facilitate or block the energy production.

Extensive

The terms “extensive” and “intensive” refer to different characteristics of the training efforts depending upon whether they are used in Europe or in the US and Canada. European coaches use both terms to refer to the intensity of the training sets (resp. low intensity and high intensity) while the Americans and Canadians rather assign these terms to the volume of the exercise. In this book “extensive” and “intensive” refer to respectively “low intensity (low swimming speed)” and “high intensity (fast swimming speed)”.

-F-

Fat combustion

Fat can be used as fuel for the metabolic processes to produce energy for the muscular activity. In first instance fat is broken down to fragments that than can be combusted by means of oxygen. This process is called “fat combustion”.

Fine-tuning

Training meant to maximize the contribution of the various conditioning components and to provide an optimal synergism between them with respect to the competition the athlete is preparing for.

Fraction

See repeat distance or interval.

Frequency training

Training exercises meant to either increase the arm stroke frequency (rate) but without slipping through the water or to reduce slipping through the water at given stroke frequency. The purpose of these exercises is to improve swimming economy.

Functional adaptation

Training improves the physical performance capacity by inducing structural and functional adaptations. Structural adaptations are characterized by the creation of new or the reconstruction of existing biological components (proteins, etc.) while functional adaptations are based on a better or more efficient activity of existing biological components.

-G-

Glycogen

Glycogen is a chain of glucose molecules. Glucose is the carbohydrate used by the body for energy production. Glycogen breaks down into a substance called pyruvate and produces energy during this process; energy often referred to as anaerobic energy because it is produced without oxygen.

Glycolysis

Glycolysis comes from the Greek word “glyko” which means sugar and “lysis” meaning to break down; so the glycolysis is an anaerobic process that breaks down glycogen or glucose molecules (sugar). This process produces energy (glycolytic energy) quickly but not as fast as the creatine phosphate system.

Glycolytic capacity

See glycolytic rate.

Glycolytic rate

Refers to the amount of glycogen/glucose molecules that can be broken down per second by the glycolysis.

Graded exercise test

See progressive exercise test.

-H-

Heart rate deflection point (Conconi)

The heart rate deflection point is the point where increases in heart rate slow down compared to increases in effort level. Up to a certain point heart rates rise linearly as the effort or speed increase. Some coaches and researchers have correlated the heart rate deflection point with the lactate threshold. Its main proponent is an Italian running and cycling coach named Conconi. The heart rate deflection point is therefore sometimes also called the Conconi point.

-I-

In-between training period

See competition training period.

Individual anaerobic threshold

The individual anaerobic threshold (IAT) refers to the lactate value at which the athlete can maintain an effort for a longer period of time (at least 20 min). A lot of specific protocols have been developed in order to determine the IAT based on the lactate curve of a step test. However, the validity of these procedures is subject to a lot of controversies. It is thought that the IAT coincides with the maximum lactate steady state or lactate threshold. The individual anaerobic threshold is based on the premise that the anaerobic threshold is different for each individual as opposed to one fixed lactate value. Hence the emphasis on “individual” in the term.

Intensity

The intensity of an exercise is a measure of how fast one is swimming, cycling or running during an exercise. In strength training it is defined as the resistance the athlete has to overcome in order to complete the workout.

Intensive (intense)

The terms “extensive” and “intensive” refer to different characteristics of the training efforts depending upon whether they are used in Europe or in the US and Canada. European coaches use both terms to refer to the intensity of the training sets (resp. low intensity and high intensity) while the Americans and Canadians rather assign these terms to the volume of the exercise. In this book “extensive” and “intensive” refer to respectively “low intensity (low swimming speed)” and “high intensity (fast swimming speed)”.

Interval distance

See repeat distance.

Interval swimming

In this type of exercise the total distance is interrupted by periods of rest. Interval sets can be used:

- to improve the aerobic and anaerobic capacity and power
- as sprint training
- as frequency training (stroke rate training)

Interval sets will be composed differently depending upon whether their objective is to affect the aerobic or anaerobic processes. The balance between rest, volume and intensity of the workout will determine to what extent both metabolic systems will be activated. By changing the balance between those different training elements (rest, volume and intensity) you can gradually move through the 4 types of exercises.

-L-

Lactate

Lactate is a naturally occurring organic compound produced in everyone's body and is both a by-product of and a fuel for exercise. It is found in the muscles, the blood and various organs. The body needs it to function properly. A term that is often associated with lactate is lactic acid. They are very close chemically. We use the term lactate even though in many places lactic acid might be technically right. The use of lactate instead of lactic acid should not interfere with any interpretation. The primary source of lactate is the breakdown of a carbohydrate called glycogen.

Lactate curve

For as long as lactate has been used in sports to optimize training efficiency, the lactate curve has been considered as the starting point for describing and defining the athlete's conditioning profile. When, for example, a swimmer is asked to perform a set of progressively faster swims, lactate will be low after the first and very extensive repeat, but will rise clearly from a certain speed on (= lactate onset). The faster the subsequent swims, the higher the lactate reading will be. Plotting the swimming speed against the respective lactate values in a lactate-speed diagram and connecting all these points with a line (curve) shows the lactate curve.

Lactate elimination

The elimination is the process of removing lactate from the muscle and the bloodstream. Lactate is a very dynamic substance. When lactate is produced, it will probably leave the muscle and enter the space between muscle cells where there is a lower concentration of lactate. From there it either enters neighboring muscles or the bloodstream. It could end up in another muscle nearby or someplace else in the body.
If lactate is taken up by another muscle it will probably be turned back into pyruvate and be used for aerobic energy. Endurance training increases 1) the enzymes that readily convert lactate to pyruvate and 2) transporters that facilitate the movement of lactate back and forth across the cell membrane. Lactate can also be used by the heart as a fuel or go to the liver and be turned back into glucose and glycogen. It can move from one part of the body to another quickly. There is even evidence that some lactate turns back into glycogen in the muscles.

The lactate elimination process is also called clearance. The elimination of all lactate above resting levels is often called recovery, though this term has many other meanings in the training. Recovery of lactate to resting levels does not mean that the athlete is 100% fit. Despite the recovery of lactate he can still be fatigued and not be able to perform at his 100%.

Lactate production

Lactate production is the result of the glycolytic process (glycolysis). Therefore the glycolytic capacity (also the anaerobic capacity) is also called the lactate production capacity in older literature.

Lactate steady state

If an athlete swims, runs, bikes, rows, etc. at a constant and submaximal speed or effort for a long period (more than 20 minutes), he or she is performing a steady state workout. Lactate levels during such a workout will fluctuate at first but eventually the lactate level will settle in at a constant level. Some coaches have defined steady state workouts as those where the heart rate is constant during the workout. However, this can be very misleading as these two types of workouts do not produce the same training effect. A workout that keeps lactate levels steady will not maintain a constant heart rate.

Lactate shuttle

Lactate shuttle is the process that moves or shuttles lactate around the body. Ordinarily, a muscle which can use pyruvate for energy will get it from glycogen stored in the muscle. However, if excess lactate is available in the bloodstream or adjoining muscles, much of this lactate will be transported to the muscle and will be converted to pyruvate that can be used to produce aerobic energy.

A common misconception is that lactate is a waste product. Lactate is an important source of fuel for aerobic energy and much of it gets converted back to glucose and glycogen for future energy use. So it is definitely not a waste product.

This shuttling of the lactate to other areas of the body lowers the blood lactate levels in the producing muscles and reduces the troublesome hydrogen ions. The better the body can be trained to clear lactate from the producing muscles the longer the athlete will be able to sustain a high effort at the end of a race or during practice.

The German researcher W. Hollmann first noticed this process in the 1950’s. The term lactate shuttle was coined by George Brooks who along with several of his colleagues have researched this process thoroughly.

Lactate threshold

Lactate threshold is the maximum effort that can be sustained for an extended period without lactate steadily accumulating. It is also called the maximum lactate steady state ($MaxLa_{ss}$). Other terms that are sometimes used for this concept are OBLA (onset of blood lactate) and lactate turnpoint. These terms refer however to other characteristics of the lactate behavior than the lactate threshold.

Lactate tolerance

Lactate tolerance is the ability of the person to withstand the intense pain that is produced by high acid levels in the muscles. It is primarily a mental training as opposed to buffering which is a physical training.

While lactate tolerance is primarily associated with the pain caused by acidosis in the muscles during intense activity, this pain will ease up quickly as the intensity eases or the activity stops. The soreness in muscles the day after exercise is often blamed on lactate but has in fact nothing to do neither with lactate nor with hydrogen ions.

Lactic

Lactic refers to the presence of lactate production; e.g. lactic energy supply tells us that lactate is produced during the energy supply.

Lamax

Is the abbreviation for the highest post exercise lactate concentration.

Load density

When several high intense and/or high voluminous sets are given in a short period of time, the load density is high. The load density refers thus to the rate of occurrence of intense or voluminous workouts over a certain period of time, such as a week.

Locomotor system

The locomotor system refers to the whole complex of body components that are involved in human motion (muscles, joints, tendons, etc.).

-M-

Macrocycle

A macrocycle stretches over several months and covers a specific season such as the short course season. There are usually 2 macrocycles per year, but the competition calendar sometimes forces the coach to plan for 3 to 4 macrocycles. This is only possible for the very top level swimmers, however. For age group swimmers, it is advisable to stick to 2 macrocycles per year since more macrocycles would hurt the long-term build-up of their conditioning. Since it is not possible to train all the conditioning features necessary to deliver a top performance at the same time, the macrocycle has to be split into various smaller training periods. Each of these periods, called mesocycle, normally lasts 3 to 6 weeks and is designed to improve specific aspects of the athlete's conditioning.

Each macrocycle covers : * the *base training period*

* the *competition training period (pre- and in-between competition training period)*

and is followed by: * the *transition period* , which is a period of relaxation

Depending on the swimmer's conditioning profile and the competition calendar the coach will structure the macrocycle differently by varying:

- the number and duration of the training units per week
- the total training volume per week
- the volume and frequency per week of the different types of training exercises (water training as well as dry-land training)

Maximum force

The maximum force is the highest force that can be generated for the execution of a single movement. Depending on the movement, the muscle groups involved, etc. the maximal force measurements will quantitatively differ.

MaxLa $_{ss}$

See lactate threshold and anaerobic threshold.

Mesocycle

A mesocycle is a succession of a few microcycles; depending on the training objectives and on the conditioning level of the swimmer the mesocycle will last about 2 to 7 weeks. It always contains a phase of hard work (working or high load phase) followed by a phase of relatively easy training (regeneration phase).

- A *working or high load phase* ; a period of voluminous and/or intensive work, whereby training volume and/or training intensity are gradually increased. This does not mean that the coach will progressively increase the intensity and/or volume of every consecutive training unit. He will instead increase the intensity and/or volume of the main training workouts or add an extra quality set (workout at an intensity higher than regeneration). Thus, the swimmer still goes through a lot of regeneration training but these low intensity exercises will either compensate for more intense and voluminous main workouts or be more often alternated with intensive work
- A *regeneration phase* or period of relative rest before starting the next mesocycle. The last regeneration phase of the mesocycle prior to the top competition is also called the taper phase and is meant to make the swimmer "100% competition ready" physiologically, technically as well as mentally, i.e. fully recovered (especially with respect to the neuromuscular processes) and eager to compete

Metabolic activity - Metabolism

Refers to the biochemical activity in the cells that produces energy to live or to sustain a physical exercise.

Microcycle

The smallest training cycle is called a microcycle. It provides a detailed training program for a period lasting from a few days to one week, in which case it is also called week plan.

Mileage

Refers to the volume covered over a period of time.

Mitochondria

These are protein structures in the muscle cell, which produce aerobic energy. Endurance training increases the density of mitochondria in certain cells. Mitochondria are the part of cells that convert pyruvate into energy. The denser they are the higher the capacity of the cell to use pyruvate as fuel and produce more energy aerobically. Many sports physiologists have used the metaphor of a factory to describe mitochondria. When the mitochondria finish processing pyruvate there is a lot of energy available as well as common waste products such as water, carbon dioxide and heat.

Mitochondria are very dense in slow twitch muscle fibers (Type I) and in some fast twitch fibers (type IIa). There are far fewer mitochondria in type IIb fast twitch fibers. Type IIa fast twitch fibers have the ability to produce aerobic energy while type IIb fast twitch fibers, with very few mitochondria, produce hardly any aerobic energy. Over time, with sustained endurance training, many of the type IIb fibers will convert to type IIa fibers. Since Type IIb fast twitch fibers are usually the main source of lactate during exercise, this conversion will cause less lactate to be produced.

mmol/l

This is the basic measure of lactate. Most measures of lactate use a blood sample, though many researchers have taken muscle samples and measured the lactate in the actual muscle. There is a high correlation between blood lactate and muscle lactate. It is relatively easy to measure the lactate in blood with a portable lactate analyzer.

When a blood sample is taken, the amount of lactate is expressed as a concentration of mmol (millimoles) per liter. For example, resting blood lactate levels in humans are usually between 1.0

mmol/l and 2.0 mmol/l. Lactate levels in some athletes after major competitions have been as high as 25-30 mmol/l though levels this high are rare.

Muscle fibers

There are several different muscle fiber types but most use the term to refer to three types of muscle fibers, which make up nearly all the muscle fibers in the body. These are Slow Twitch Fibers and two types of Fast Twitch Fibers. Slow Twitch Fibers contract at a slower speed than Fast Twitch Fibers. Slow Twitch Fibers (ST) are also called Type I or Red fibers (since they are red colored.) They produce large amounts of aerobic energy. There are several fast twitch fibers but the two most important ones are Fast Twitch Oxidative (FOG) or Type IIa (red) and Fast Twitch Glycolytic (FG) or Type IIb (white). The fast twitch fibers can produce more anaerobic energy than slow twitch fibers. Type IIa can also produce high levels of aerobic energy while Type IIb produces little aerobic energy.

During everyday activity our muscles mainly use the slow twitch fibers. As exercise intensity increases additional muscle fibers are recruited which are not used often in everyday activity. Many of these are “fast twitch” fibers. Fast twitch fibers are not very good at turning pyruvate into aerobic energy. Hence a lot of this pyruvate turns into lactate. The proportion of muscle fiber types varies between people. Thus, some people produce lots of lactate and others produce very little.

-O-

Organic compound

An organic compound is composed of carbon, hydrogen and oxygen.

Oxidation capacity

Refers to the capacity to break down organic compounds by using oxygen.

Oxygen uptake

The oxygen uptake is the process that occurs in the lung when oxygen is transferred from the inspired air to the bloodstream. The more oxygen is used by the working muscles, and eventually other organs, the higher the oxygen uptake that will “refill” the bloodstream with oxygen.

-P-

Peak form

The peak form is a temporary but stable condition, which enables the swimmer to realize a top performance in his specific discipline.

Periodization

The training periodization, arranges, organizes, schedules and fixes the timing as well as the duration of each type of training, testing, etc. for one year according to the fixed objectives. The training periodization is the break down of a training year into different sequential and mutually dependent training periods (cycles) to bring the swimmer into peak form at the right moment.

Planning

The planning of training denotes the preparatory work of the coach and consists of structuring the training process in a systematic way according to the training objectives and the swimmer’s conditioning level. It is based on the coach’s experience and on his knowledge of sports sciences.

Power training exercises

The exercises used for the fine-tuning are referred to as power training workouts. They are focused on maximizing the contribution of the acquired capacities of the athlete.

Pre-competition training period

See competition training period.

Progressive exercise test

The most common test protocol to measure lactate is the progressive exercise test, also called graded exercise test or step test. For example, an athlete runs, swims, cycles or rows at progressively faster speeds, while a coach, sports physiologist or technician measures the lactate level at each speed. The athlete can also be on a stationary cycle ergometer and use power ratings instead of velocity. A rower could be rowing at a certain stroke rate or at a power rating on a rowing ergometer. When lactate is plotted against power or speed, the obtained lactate curve is respectively a lactate performance curve (LPC) or a lactate velocity curve (LVC).

Progressive tests can proceed till the athlete is completely exhausted or stop at some point where the necessary information is obtained. For example, it is common to stop a progressive test after the athlete has reached the lactate threshold or some pre-determined lactate concentration such as 4.0 mmol/l lactate.

Proteins

Are composed by many amino acids and mostly have important functions in our organism.

Pyruvate

Pyruvate is an organic compound. It is the end product of the glycolysis. When pyruvate further breaks down, it produces much more energy. This energy is called aerobic because this process uses oxygen. If pyruvate does not break down most of it turns into lactate, though small amounts may change into other substances. The chemical formula is very similar to lactate.

-R-

Realization phase

The whole process of making training plans goes through various steps that can mainly be bundled in 2 distinctive phases:

- the design phase
- the realization phase (execution and steering)

The realization phase consists of 2 parts:

Execution: - Training
- Training documentation and training evaluation

Steering: Evaluation of the training effect:

- If observed training effect = expected effect ☐ continue plan
- If observed training effect = not expected effect ☐ adapt plan

Regeneration

Regeneration is the process that starts after an exercise, that eliminates fatigue and restores or exceeds the previous performance level.

Repeat distance

An exercise is often divided into partial distances, called repeat distances or intervals, which refer to the length of each swim.

Rest

The rest refers to the amount of time allowed between intervals, sets or exercises in a training session. The rest can be active (swimming at very low intensity) or passive (no swimming at all).

Retraining
Training starts again after a period of no training.

-S-

Set
Workout consisting of a number of repeat distances or intervals.

Set distance
The set distance (volume) of an exercise is the total number of meters of the whole set.

Slope of the lactate curve
Refers to the mathematical slope of the lactate curve. Some researchers believe that the slope of the curve reflects the capacity of the athlete to sustain power for a longer period. Little research exists on this subject, and moreover it is controversial.

Explosive force
Refers to the highest force that can be generated explosively.

Sport specific
Refers to the specific characteristics of the sport. A test on for example a cycling ergometer is not a sport specific test for a swimmer.

Spot test
See control test.

Sprint interval
Is an interval workout that meets the criteria appropriate to improve sprint speed.

Standard lactate test
A standard lactate test is a fix protocol with respect to the athlete's specialty and is carried out periodically. This test is intended to provide an overall assessment of an athlete's conditioning. It is usually a kind of progressive exercise test but can also include other elements such as an all-out test or a clearance test.

Steering principle
Steering refers to the methodology a coach uses to improve training efficiency. This method consists in comparing the expected training adaptations (= objectives = what he wants to achieve) with those observed in competition, tests and training. Based on this evaluation that takes place every 4-6 weeks, the coach may decide to adjust training or to continue his original training program.

Step test
See progressive exercise test.

Strength training
Strength training is working with weights on dry-land. Strength training to increase maximum force (MFST) is considered to be capacity training while strength training to increase explosiveness (EXST) and strength training to maintain a high force involvement for a long period (ENST) are derived from maximum force training and are therefore labeled as power training. It is important to prepare the swimmers for strength training well in advance. We therefore advise at least 1 year of conditional gymnastics and 1 year of EXST with light weights before starting a real strength training program as described below. The first 2 years should, thus,

be considered as an introduction to strength training, provide an adequate basis of general strength and teach swimmers how to work correctly with weights to avoid injuries once the “serious” work begins.

Stroke rate training

See frequency training.

Structural adaptation

See functional adaptations.

Submaximal

Submaximal refers to an intensity (e.g. speed) “lower than maximal” with respect to the distance tat has to be covered. This means that a submaximal speed on a 100 m distance may be higher than the maximal speed on a 400 m race.

Super-compensation

When training the swimmer becomes tired and his physical performance drops. If he is then granted some regeneration training (training at very low intensity or active rest) and more rest between the training sessions, his physical performance will return to the starting level and several biological adaptations (normalization of the cell environment, recovery of neuromuscular stimulation processes, restoration of enzyme activity, replenishment of energy sources, etc.) will take place. After this recovery period the physical performance will further increase above the initial level. This process is called the super-compensation. The swimmer will now be able to handle the same load as before but with less strain or a more intense load with the same ease.

-T-

Technique training

For technique training there are 2 types of exercises:

- Type I: technique training swims to *teach* and *correct* the stroke pattern

For this type of training the rules are fairly strict and should be followed exactly. To be able to learn the stroke patterns correctly the swimmer has to be fresh and fit; it is therefore important:

✼ to schedule these exercises at the beginning of the training session, just after the warm-up

✼ to ensure that the rest between the intervals is long (to allow full recovery) and will be used by the coach to make corrections or give feed-back to his swimmer

✼ to keep the volume and the repeat distance short and the intensity low

- Type II: technique training swims (drills) to help *automate* the proper stroke pattern. In other words, the swimmer will acquire the stroke skills without thinking (= automation)

For the drills intended to automate the proper stroke pattern, volume and repeat distance should be made progressively longer, the rest shorter and the intensity higher. Their objective is to gradually achieve the correct form in the normal training rhythm

Threshold training

All workouts performed at the intensity corresponding to the anaerobic threshold can be used for threshold training. In these workouts, rest is mostly short and volume high.

Top level training phase

This phase of a swimmer's career starts between 17 and 19 years of age and lasts at least 4 years. As opposed to the build-up phase, where the training goals were still general, broad and rough, the objectives of the top level training are:

- almost entirely determined by the specific needs of each swimmer individually
- the maximal development of all swim specific and performance determining conditioning components (developing the individual peak form). Nevertheless, the conditioning qualities that only indirectly determine the competition performance need also to reach an optimum level and be stabilized
- the perfection of the technique, now focused on the quality of execution in competition (frequency, amplitude, etc.) rather than on the correctness of the moving pattern
- the involvement of the swimmer in the planning and training processes

It is thus in this phase essential to assess:

- a profile of the swimmer's weak and strong characteristics
- the swimmer's ability to react and adapt to the various training exercises

This requires an ongoing observation of the evolution of the swimmer's conditioning level (based on training, competition and test results) and an evaluation of the training actually completed. This should indicate whether or not the swimmer has optimally adapted to the prescribed training program.

Trainability

Refers to the ability of a swimmer to respond positively to different types of training.

Training concept

In many countries, the sports federation fixes the training concept which contains general indications and instructions on training objectives, performance evolution, qualification times, compulsory training camps, high priority meetings, trials, etc.

Training cycle

A group of training units forms a specific training period or training cycle. These cycles are characterized by the time they last and will be repeated frequently during the swimming season.

The smallest cycle is called a microcycle. In swimming a microcycle generally covers 1 training week. A mesocycle is a succession of a few microcycles, lasts about 2 to 7 weeks and contains a phase of hard work followed by a phase of relatively easy training. The duration of this period will depend upon the training objectives during that training period and on the conditioning level of the swimmer. A macrocycle consists of several mesocycles. It stretches over several months and covers a specific season such as the short course season. It covers the base, competition and transition training periods. There are usually 2 macrocycles per year, but the competition calendar sometimes forces the coach to plan for 3 to 4 macrocycles. This is only possible for the very top level swimmers, however. For age group swimmers, it is advisable to stick to 2 macrocycles per year since more macrocycles would hurt the long-term build-up of their conditioning.

Training effect

Is the response to several training units. A positive response will lead to an improvement.

Training load

Is the combination of intensity and volume the swimmer undergoes during training.

Training log

Documentation of the completed training.

Training objective

A training objective describes the training adaptations expected from training.

Training plan

A training plan is a table of contents which lists, for swimmers of every performance level, all the elements needed to help them improve their competitive performance. It usually covers more than one year but the time allocated to the different elements is still quite rough.

The key elements recorded in a training plan are, in order of importance:

- the main competitions (e.g. participation in school competitions, in international events,...)
- the main conditioning / mental / technical improvements to be achieved
- the expected evolution in performance (the key objectives for each year, e.g. make top 3 at NJO's in 2 years, break 1 minute in 100 m backstroke,...)
- a general description of the swim and dry-land training (e.g. other sports activities to complete the swim training, possibly altitude training,...)
- the number of training units (sessions) and the total mileage per week (e.g. when to start with training twice a day,...)
- the tests to support and control the effectiveness of the training (e.g. video analysis, lactate tests,...)
- non sport related commitments (sponsor or school commitments,...)

Training outline

The training outline is based on the training concept and provides the general guidelines for structuring the training process for a well-defined group (possibly one single swimmer) during 1 or more years. It is a rather theoretical training model and depends on:

- the sport (e.g. swimming, running, etc.)
- the sports discipline (e.g. sprint, middle or long distance, etc.)

Training session

An effective training session (unit) generally consists of the following parts completed in the suggested order:

1. The *warm-up*

The goals are:

- relaxation and relief from the daily stress (e.g. school, work, etc.)
- physiological preparation (e.g. cardiovascular)
- preparation of the muscles (e.g. to induce muscle tension for the sprint work to come)

Following the warm-up there are 1 or 2 main parts made up to meet the objectives of the mesocycle:

2. The *first main part* : this part includes exercises of a high technical and coordination level requiring a well rested athlete and meant to induce conditioning adaptations and/or to improve the technical abilities (e.g. frequency (stroke rate) training, anaerobic capacity training, sprint training or technique training, ...)

3. *short regeneration exercise* : the higher the intensity of the first main part, the more indispensable the regeneration exercise becomes
4. The *second main part* : this part includes swimming that induces conditioning adaptations but does not require high technical or coordination skills, such as long endurance repeats
5. The *cool-down* : this part of the training is very important since it starts the recovery and regeneration phase that will lead to the super-compensation. An *active* start of the regeneration not only speeds up the recovery process but also the time for super-compensation

Training stimulus

Is the trigger sent to the organism to induce a training effect. This trigger contains a message that is defined by the combination of intensity, volume, rest and interval of the workout. Depending on the characteristics of the message, different training effects will be induced.

Training unit

See training session.

Training volume

Is the amount of meters covered during training.

Transition period

The transition period is the third and last part of a macrocycle and consists in a period of relative rest with a temporary regression of conditioning after the peak form was reached.

Transporters

Transporters are proteins that facilitate the movement of other compounds from one part of the body to another. Lactate transporters help carry lactate over the cell walls and other membranes in the cell. They have recently become a hot topic of research since some scientists think that these transporters may affect the athletic performance.

-V-

Vegetative system

Is the complex of all organs that are under control of the autonomic nervous system, i.e. without any interference of one's own will.

Ventilatory threshold

The term ventilatory threshold (VT) has been given to the place where oxygen and carbon dioxide patterns change suddenly. It happens very close to the lactate threshold (LT) and many people assume the same conditions cause both. However, it has been shown that the two are unrelated. For healthy athletes they are close enough together to use the VT as an estimate of the LT. While oxygen consumption and carbon dioxide release data are very valuable, finding the VT has been a problem for sports scientists as the data are often not clear.

VLamax

Is the maximal lactate/pyruvate production capacity and quantifies the anaerobic capacity.

VO_2 max

Is the maximal oxygen uptake and quantifies the aerobic capacity.

Volume

A measure of how much training is performed. In strength training it is defined as the total number of repetitions, or the total amount of weight lifted in a specific period of time.

V4

The speed corresponding to a blood lactate concentration of 4 mmol/l.

-W-

Wave principle

To ensure a continuous improvement of conditioning, it is not only necessary to change the training load after approximately 6 weeks but also to provide regular periods of regeneration training for the super-compensation to take place. The structure of an effective and successful training process will therefore be characterized by a continuous alternation of periods of increased training load (training units at a higher intensity and/or volume) and periods of reduced training (regeneration training).

This alternation of intensive or high volume training and regeneration sessions is often referred to as the wave principle of training.

Workout

The exercise a coach asks to perform.

30-minute test

The 30-minute test (T30) provides a fairly accurate estimation of the swimmer's aerobic endurance and offers an acceptable alternative for the 400 m lactate test. The evaluation of the aerobic capacity by the 30-minute test is of course not as accurate as using lactate tests, but nevertheless it enables the coach to:

- measure the swimmer's aerobic endurance

- purposefully build up the training
- calculate the training intensity for the endurance exercises

This test requires the swimmer to swim as far as possible in 30 min at a steady speed. The first time the swimmer uses this type of test he can aim for times that are 1 to 2 sec/100 m slower than his 1500 m or 800 m split times.
Research has shown that the average swimming speed recorded during the 30-minute test (total distance/time in seconds) corresponds closely to the aerobic capacity determined by lactate tests. It was established that generally the longer the distance swum in 30 min, the better the aerobic endurance. The average swimming speed of the swimmers with a very strong aerobic endurance is lower than the 4 mmol/l lactate speed of the 400 m submaximal lactate test. Swimmers with a weaker aerobic endurance, on the other hand, are capable of completing the 30-minute test at a higher velocity than the 4 mmol/l lactate speed of 400 m submaximal lactate test.

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